

Forecasting Inflation with Microdata: An Adaptive Machine Learning Approach*

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July 13, 2026

Preliminary.

Abstract

Does microeconomic heterogeneity help to forecast aggregate inflation in a non-stationary environment? We develop a scan test for whether one forecast outperforms another, over an interval with unknown starting point and duration. To exploit any occasional forecasting power that the scan test detects, we design an adaptive machine learning pipeline. We encode the distribution of price changes into a high-dimensional vector, which we combine with a gradient boosted trees algorithm. We then combine this micro forecast with other benchmark forecasts, using an adaptive algorithm that makes use of the micro forecast only when it performs well. We apply the pipeline to UK microdata, with four main results. First, the micro forecast outperforms a univariate benchmark, but only in the volatile period after 2020. Second, the scan test detects periods of micro outperformance, so the micro forecast enters the combined forecast. Third, the combined forecast performs comparably to the univariate benchmark before 2020 and better at every horizon after 2020. Fourth, the value of microdata for the combined forecast materializes after 2020. We conclude that microdata are valuable for forecasting aggregate inflation, but only after large shocks.

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1 Introduction

How can we forecast the economy? Traditional forecasting approaches rely on macroeconomic data. Yet aggregate outcomes are generated by millions of microeconomic decisions about price setting, employment, consumption, and production. In recent decades, high-dimensional micro datasets tracking this behavior have become available. These data create an opportunity to use micro-level information to improve forecasts of aggregate variables. In taking advantage of this opportunity, we can answer a fundamental question in macroeconomics: does microeconomic heterogeneity matter for aggregate dynamics—and if so, when?

This paper focuses on inflation, one of the most difficult macroeconomic variables to forecast. Even sophisticated models often struggle to outperform simple univariate benchmarks (Atkeson and Ohanian, 2001). A central challenge is non-stationarity: the process governing inflation dynamics changes over time. Figure 7 illustrates this clearly for the United Kingdom and the United States over 2015–2024, with markedly different inflation behavior before and after 2020.

There is good reason to think that microdata—in our case, the millions of price changes observed each month—may improve inflation forecasts. Each price change reflects a forward-looking decision by an individual firm, and the cross section of such decisions may contain information about future aggregate inflation. In simple macroeconomic models, such as the linearized Phillips Curve with Calvo price setting, microdata does not matter for aggregate dynamics (e.g. Galí, 2015). Richer macroeconomic models contain more nuance: microdata should help forecast inflation in some environments but not others. When aggregate shocks are small, microdata do not add forecasting power conditional on macroeconomic aggregates in state-dependent models of price setting (Auclert, Rigato, Rognlie and Straub, 2024). When aggregate shocks are large, however, micro information can help predict aggregate inflation (Blanco, Boar, Jones and Midrigan, 2024). That is, microdata matters *only occasionally* for aggregate dynamics.

In this work, we introduce a method to test for and exploit the power of high dimensional microdata for forecasting inflation, in a setting in which microdata is only occasionally powerful. We develop a scan test designed to test whether one forecast occasionally outperforms a benchmark, which we apply to the microdata-based forecast. We then create an adaptive machine learning pipeline in order to exploit the occasional forecasting power of microdata, motivated by existing tools from the adaptive machine learning literature (Herbster and Warmuth, 1998; Korotin, V'yugin and Burnaev, 2020). These algorithms are well suited for high dimensional prediction in a non-stationary environment, and produce easy-to-interpret statistics that connect to macroeconomic models.

Our primary dataset consists of official UK CPI micro-price quotes. The data contain tens of thousands of monthly observations on individual prices across the economy and underlie the construction

of aggregate inflation. The sample runs from 1996 to 2024, covering both the low and stable inflation period before 2020 and the more volatile period thereafter (see Figure 7). Each observation is a price quote identified by an item code, shop, region, and shop type. We supplement these microdata with a large panel of macroeconomic predictors (Goulet Coulombe, Marcellino and Stevanović, 2021).

First, we develop a procedure for assessing whether microdata improve aggregate inflation forecasts, at least occasionally, allowing for non-stationary inflation dynamics. We develop a scan test with a composite one-sided null hypothesis that one forecast is at least as accurate as another at every point in time and at every forecasting horizon. The alternative hypothesis is that the second forecast outperforms over some contiguous interval. For each candidate interval and horizon, we compute a partially-studentized statistic based on cumulative loss differentials, and define the scan statistic as the maximum of this object over all admissible intervals and horizons. Because this maximum has a non-standard distribution, we obtain critical values using a computationally efficient Gaussian multiplier bootstrap, after first testing the loss differentials for serial dependence to determine the appropriate bootstrap regime. The interval and horizon that maximize the scan statistic point to the period and forecasting distance at which the micro forecast outperforms by the most. The scan statistic is a maximum over a large collection of candidate intervals and horizons, meaning we require the recent high-dimensional Gaussian approximation results (Chernozhukov, Chetverikov and Kato, 2013, 2017; Chang, Chen and Wu, 2024) to derive our test.

In this paper, the primary use of the scan test is forecast expert selection. The aim is to pick candidate forecasts that should be combined in order to forecast inflation, with a microdata forecast being one such candidate. However the scan test is of independent interest: There are many circumstances when one wishes to know whether one forecast outperforms another during some *ex ante* unknown length of time that is shorter than the full sample. By contrast, standard forecast comparison tests focus on average performance over the full sample (e.g. Diebold and Mariano, 1995).¹

We then develop an adaptive machine learning pipeline to exploit the occasional forecasting power of microdata. Our pipeline involves three steps. The first step encodes the raw micro-price data into a stable, high-dimensional vector of distributional statistics. This step is necessary because individual products regularly enter and exit the sample, so the raw data do not admit a fixed predictor vector. We therefore summarize, for each month and each of the 11 retained broad COICOP1 consumption categories, the cross-sectional distribution of price changes using statistics that are available in every period. These include the fraction of prices that do not change, the deciles of the distribution of non-zero price changes, and the mean non-zero price change. Stacking these statistics across categories, and collecting their lags, yields an interpretable but high-dimensional representation of the

¹The test of Giacomini and Rossi (2010) is an influential exception that we will discuss shortly.

micro-price environment, which can then be used for forecasting.

The second step acknowledges that because vector of predictors is high dimensional, conventional forecasting methods are prone to overfitting. We therefore use a machine learning method designed for high-dimensional prediction: gradient-boosted trees, implemented with XGBoost, which performs strongly in tabular forecasting settings (Chen and Guestrin, 2016; Grinsztajn, Oyallon and Varoquaux, 2022). We estimate the model on rolling windows, and develop a real time procedure for tuning the hyperparameters. We generate direct forecasts for horizons from 1 to 24 months ahead.

We contrast the microdata forecast with two benchmark forecasts from the literature. The first is a rolling-window seasonal ARIMA (SARIMA) model whose order is selected in real time using the same tuning procedure as our other forecasts, and which nests several leading univariate inflation forecasts used in the literature, including those of Atkeson and Ohanian (2001) and Stock and Watson (2007). The second benchmark applies the same rolling-window XGBoost procedure used for microdata, but with a large set of macroeconomic predictors instead, allowing us to isolate the marginal contribution of micro-price information. We are careful to train and tune both benchmarks in the same way as our microdata forecast, to avoid advantaging the latter.

The third step combines the micro, macro, and univariate forecasts, while acknowledging that each modality may have only occasional forecasting power. We use a variant of the Fixed Share algorithm of Herbster and Warmuth (1998) by Korotin, V'yugin and Burnaev (2020) which can work with multi-step ahead forecasting. The algorithm forms a weighted average of the three forecasts and updates those weights as multi-step forecast errors are realized, placing more weight on experts with better recent performance and using only information available in real time. There is a regret guarantee, valid in non-stationary environments, which bounds the algorithm's loss relative to the best sequence of experts chosen in hindsight. That is, unlike standard forecast combination methods that effectively target a single best model, Fixed Share is designed to track the best sequence of forecasts when the identity of the best forecast can change. The regret bound grows if forecasts are added that do not provide power. As such, we use the scan test to assess whether all three forecasts, micro, macro and univariate, should be included.

The Fixed Share algorithm also yields an interpretable time series of expert weights. In particular, the weight on the micro forecast measures how useful micro-price information is for forecasting inflation at a given date and horizon. This is valuable because macroeconomic models make predictions about when microdata should matter for aggregate inflation. We can therefore use the evolution of the Fixed Share weights to assess whether those theoretical predictions line up with the data. This interpretability is an important advantage of the approach, especially relative to machine learning methods that are often criticized as opaque “black boxes” (Mullainathan and Ludwig, 2024).

Overall, our empirical results show that microdata are useful for aggregate forecasting, but only

during periods of large shocks. We report four main empirical results. First, the microdata forecast outperforms the univariate benchmark only after 2020: before 2020 the univariate benchmark is better at every horizon, while after 2020 the micro forecast wins at nearly every horizon, and by the most at long horizons.

Second, the scan tests imply that the Fixed Share combined forecast should include all three modalities: micro, macro and univariate forecasts. We first test whether the macro forecast adds forecasting power relative to the univariate benchmark. We then test the micro forecast against the univariate and macro forecasts. The scan tests support adding both micro and macro forecasts.

Third, the combined forecast, which uses micro, macro and univariate information, performs comparably to the univariate benchmark before 2020 and improves on it at all 24 horizons in mean absolute error in the post-2020 window.

Fourth, the value of microdata for the combined forecast materializes after 2020. The combined forecast leans on the univariate forecast before 2020 and shifts weight toward the micro forecast afterwards, with the weight on the micro forecast peaking in 2023. A more demanding test asks whether the three-expert combined forecast outperforms, on average, a two-expert benchmark that uses only macro and univariate information. It finds no outperformance before 2020, but finds outperformance after 2020. Microdata therefore add significantly to the two-expert benchmark but only in the post-2020 high-inflation window.

Finally, we “open the black box” by investigating which features of the microdata matter for aggregate dynamics. We compute grouped Shapley values for different sets of features that map onto objects in price-setting theory.² Grouped Shapley values assess which groups contributed most to the forecast gains of the microdata. After 2020, the positive contributions come from the center and tails of the price-change distribution together with monthly dummies, while the extensive-margin block contributes essentially nothing.

Our results therefore have implications for models of inflation. If macroeconomic variables recover the history of aggregate shocks and the distributional state evolves according to a stable, estimable transition law, macro information alone should be sufficient for forecasting inflation. Microdata can improve feasible forecasts when this benchmark fails in either of two ways. First, standard macroeconomic variables may not recover the shocks that move the distributional state. Granular disturbances originating in particular firms, products, or sectors are a leading example: they can affect aggregate inflation without being fully reflected in observed aggregates (Alvarez-Blaser et al., 2025).

²For example, these sets include the extensive-margin price-adjustment block, the center of the price-change distribution, its upper and lower tails, and monthly dummies. Various models of price setting emphasize one of these margins, such as the extensive margin of price adjustment (Klenow and Kryvtsov, 2008; Nakamura and Steinsson, 2008; Blanco et al., 2024) or asymmetries and tail movements in the distribution of price changes (Ball and Mankiw, 1995; Midrigan, 2011; Vavra, 2014; Alvarez, Le Bihan and Lippi, 2016; Alvarez-Blaser, Auer, Lein and Levchenko, 2025).

Second, the mapping from the aggregate shock history to the distributional state may be unstable or difficult to estimate, particularly in periods of structural change. These failures are likely to become more consequential during periods of high and volatile inflation, when large shocks are more prevalent and price-setting nonlinearities make inflation more sensitive to the distributional state (Blanco et al., 2024). The post-2020 forecast gains are consistent with microdata revealing information that macroeconomic history alone does not reliably recover during this period.

Related literature. Our paper contributes to the literature on forecasting inflation. A classic result is that inflation is hard to forecast, meaning simple univariate models such as random walks are difficult to beat (Atkeson and Ohanian, 2001; Stock and Watson, 2007). Notwithstanding this result, models with various degrees of sophistication, using macroeconomic information, judgemental forecasts, and inflation expectations by households, forecasters and financial markets, have also been used to forecast inflation (see Faust and Wright (2013) for a review). There is a newer literature that seeks to forecast inflation using increasingly rich macroeconomic datasets combined with machine learning methods. For instance Medeiros, Vasconcelos, Veiga and Zilberman (2021) find that rich macroeconomic data and random forest algorithms lead to forecasting gains for US inflation.³ Likewise, recent work attempts to forecast inflation using aggregated statistics drawn from rich sources of microdata (e.g., Beck, Carstensen, Menz, Schnorrenberger and Wieland, 2024; Brandão-Marques, Gelos, Hoffman, Otten, Pasricha and Strauss, 2024). We develop a method that encodes the rich distributional information in microdata, uses machine learning to construct a forecast, and adaptively combines it with existing benchmark forecasts. The resulting weights are interpretable and reveal when microdata add forecasting power. In doing so, we contribute more broadly to the literature on machine learning in time series forecasting (e.g., Giannone, Lenza and Primiceri, 2021; Chi, Fan, Ghigliazza, Giannone and Wang, 2025).

There is previous work that forecasts macroeconomic variables using the distribution of microdata. A seminal, recent paper is Chang, Chen and Schorfheide (2024), which develops a “functional” vector autoregression (VAR) approach. The method involves summarizing the log density of a distribution of microdata by a low dimensional collection of objects: finite dimensional sieves with fixed basis functions and time varying coefficients. One can then estimate a vector autoregression using aggregate variables and the time varying sieve coefficients, allowing a forecast of macro variables with microdata. The main focus is on a stationary environment.⁴ Meeks and Monti (2023) develop a related approach that summarizes cross-sectional distributions with functional principal components, and links them to inflation in a linear, Phillips Curve style model. We offer a different approach: sum-

³Alternative ways to forecast with a large number of predictors, and prevent overfitting, include Bayesian shrinkage or factor analysis (e.g., Stock and Watson, 2002a; Bańbura, Giannone and Reichlin, 2010).

⁴See Lenza and Savoia (2024) for an application of this approach to firm heterogeneity in the Euro Area.

marizing microdata with a high dimensional series of statistics, which we combine with black-box forecasting algorithms and expert aggregation via the Fixed Share algorithm. Our approach is designed for a non-stationary environment.

This non-stationarity motivates us to develop a scan test for localized outperformance of the microdata forecast relative to a benchmark. Classic time series tests of forecasting performance test whether one forecast outperforms another on average, throughout the full time series (e.g. Diebold and Mariano, 1995). Instead, we test for whether there are occasional periods of time in which a chosen forecast out-performs a benchmark, without specifying *ex ante* when and for how long the forecast outperforms.

As such, our approach builds on the influential test for forecast comparisons of Giacomini and Rossi (2010). We take inspiration from their paper—which points out that comparisons across the full time series can miss localized episodes in which one forecast outperforms another—but study a different inferential problem. A first key difference in our framework is the null hypothesis. Giacomini and Rossi (2010) test whether two forecasts have equal predictive ability. We instead study a one-sided null, asserting that one forecast does not outperform the other on any relevant interval. Rejection of equal predictive ability is not sufficient for our purposes, because rejecting the null does not identify which forecast performs better.⁵ A second difference is the nature of the forecast out-performance. The tests of Giacomini and Rossi (2010) either assess equal predictive ability over intervals of unknown location but fixed length, or search for a single date after which one forecast outperforms a second; in both cases at a fixed forecast horizon. Our scan test searches for the possibility that one forecast outperforms another at an unknown start date and end date, for an unknown duration of time and at an unknown horizon. Lastly, since a forecast can outperform for an unknown interval, the scan test considers a large collection of candidate intervals. We develop an asymptotics that suits this setting. We study a high-dimensional asymptotic regime in which the minimal length of time for which the chosen forecast outperforms a benchmark grows logarithmically in the sample size; while the number of intervals grows exponentially in the sample size. For inference, we develop a high dimensional Gaussian multiplier bootstrap, which accounts for the large number of intervals and horizons that the test studies. Another novelty is partial studentization: instead of fully self-normalizing each short interval, we introduce an exponent that interpolates between unstudentized cumulative loss differentials and full local studentization. This stabilizes the scan when many short windows are searched, and we choose the exponent with simulations calibrated to the empirical loss-differential panel to avoid small-sample over-rejection.

⁵While Giacomini and Rossi (2010) entertain a one-sided alternative hypothesis, their critical values are calibrated around the case of equal predictive ability. It is not clear whether this case is the least favorable for Type-I error in a one-sided null parameter space. The main obstacle is that the relevant t-statistics are partially studentized and hence need not be monotone in the data.

Finally, we use an adaptive machine learning approach, the Fixed Share algorithm, to exploit that microdata helps to forecast inflation, but only occasionally (i.e. there is non-stationarity). A large literature studies how to combine forecasts (Bates and Granger, 1969; Timmermann, 2006), including with Bayesian model averaging approaches (e.g., Koop and Korobilis, 2012). Many methods require a full specification of the likelihood function, making it difficult to combine them with high dimensional covariates and black-box machine learning methods. The Fixed Share algorithm provides worst-case regret guarantees without distributional assumptions, in the presence of non-stationarity, and when some experts use high dimensional data and machine learning. An important previous paper also uses adaptive machine learning, namely Fouliard, Howell, Rey and Stavrakeva (2020). In addition to applying online learning in a different setting and using microdata, we combine the algorithm with a scan test and include only forecasts that provably improve the regret bound and are likely to improve the final forecast’s overall performance.

2 Data

This section describes the construction of the datasets used in the empirical analysis. We combine two main sources of information. First, we use official UK CPI micro-price quotes collected by the Office for National Statistics (ONS), which provide the micro-price panel underlying the official inflation statistics. Second, we assemble a large set of macroeconomic and financial time series which we use to forecast inflation with macro data.

2.1 Microdata on Prices

The forecasting target is monthly, headline, non-seasonally adjusted consumer price index (CPI) inflation for the United Kingdom.⁶ Our primary dataset consists of official micro-level price quotes underlying the CPI, collected by the Office for National Statistics (ONS) and spanning January 1996 to December 2024. For 1996–2016, we use the replication package of Adam and Weber (2023); for 2017 onward, we extend the dataset using price quotes and item index files from the ONS website.

Each observation is a price quote identified by four fields: a six-digit item code, an anonymised shop code, a region code for the ONS collection regions, and a shop type distinguishing chain retailers (“multiples”) from independents.⁷ For instance, an item might be a large loaf of white sliced bread, collected from a Sainsbury’s outlet in a region such as London. Each item is associated with an expenditure weight; consumer price index (CPI) inflation is calculated by aggregating individual price

⁶We measure inflation $\pi_t = 100(\text{CPI}_t/\text{CPI}_{t-1} - 1)$, using the ONS D7BT headline CPI index.

⁷Price-quote files from 2018 onward add two further shop-type codes, covering roughly 1.7% of observations.

changes using expenditure weights. There is also a flag for whether the price has a sale, as measured by the ONS price quote collector. For each observation, we calculate the monthly unit price change as

$$\Delta p_{jt} = 100 \times (\log p_{jt} - \log p_{j,t-1}),$$

the month-on-month log price change in percentage points. There are around 90,000 price quotes in the typical month.

The raw ONS panel is first cleaned using standard validity, comparability, and price filters, which we list in Appendix A.1.⁸ We organise the data at the cell level, where a cell is defined by a monthly date and a product category. The product categories are broad groupings of items at the COICOP1 level (there are 12 COICOP1 categories, such as food and non-alcoholic beverages). We balance the data by retaining only COICOP1 codes present in more than 95% of months, which leaves 11 of the 12 COICOP1 categories.⁹

Table 2 reports the percentile distribution of observations per cell, separately for all price changes and for non-zero price changes only. The difference between the two panels reflects the well-documented fact that roughly 90% of prices are unchanged at monthly frequency. We include cell counts with and without items on sale.

We validate the data in two ways. First, we reconstruct the aggregate CPI from the underlying micro quotes and verify that the implied inflation rate closely tracks the official ONS series (see Appendix A.2). Second, we benchmark the moments of the price-change distribution—mean, standard deviation, and excess kurtosis of $|\Delta p|$ —against the existing literature and find that our estimates replicate the well-known stylised facts closely (see Appendix A.3).

Our main dataset for the forecasting exercise uses all observations in the cleaned posted-price panel. For robustness, we study panels in which we treat sales prices identified by the Office of National Statistics as missing, or adjust them using a filter developed by Nakamura and Steinsson (2008) with two parameterisations. The latter filters identify likely sale episodes algorithmically and replace sale prices with the inferred regular price, rather than dropping them; Appendix A.4 describes the algorithm and its parameters in detail. The forecasting robustness exercises use the ONS-flag panel and the filter’s baseline parameterisation (Filter A); Appendices E.3 and E.4 report the results.

⁸If the same product identifier (item, shop, region, and shop type) appears more than once in a month, the public data do not say which repeated quote should be linked to observations in adjacent months. Rather than choose one quote arbitrarily, we drop the whole series; this removes roughly 6.6% of observations per quarter.

⁹The Education category (COICOP1 code 10) is the one category excluded by this filter: it appears in only 120 of the 347 sample months (February 1996 to January 2006), far below the 95% retention threshold of 330 months.

2.2 Macro Data

In addition to the micro-price data, we assemble a panel of 36 monthly macroeconomic and financial predictors. The candidate list is the union of the series in the UK Monthly Database (UKMD) of Goulet Coulombe, Marcellino and Stevanović (2021), series identified as inflation-forecasting-relevant in prior work, and additional candidates judged useful for forecasting UK CPI inflation. From this list we retain series that are monthly, cover the estimation window (January 1996, or shortly after for a few market series), and are either unrevised in practice (market prices, exchange rates, policy and market interest rates, CPI indices, claimant counts) or admit a vintage-clean construction.¹⁰ The retained panel covers actual rents, wages, labour-market slack, exchange rates, energy and food commodity prices and futures, house prices, interest rates and gilt yields, equity indices and volatility, trade, car registrations, the OECD composite leading indicator, and air-freight price indices as supply-chain pressure proxies.¹¹ Among the CPI components, only actual rents are retained, because rents are not covered by the micro-price quotes. All other CPI sub-indices are excluded, as are CPIX sub-indices, RPI components, and PPI sub-categories.¹² Wherever possible we use non-seasonally adjusted series, which avoids the look-ahead bias inherent in two-sided seasonal adjustment filters; the two aggregate trade series are available only in seasonally adjusted form.¹³ The sample runs through November 2024. The resulting dataset, detailed in Appendix Table A.3, contains the 36 monthly predictor series plus the headline, non-seasonally adjusted consumer price index (CPI), and spans the period from the start of 1996 through November 2024.

3 A Scan Test for Localized Forecast Outperformance

This section introduces a scan test. The goal is to detect localized periods in which one forecast model outperforms another, in a non-stationary environment. Traditional forecast comparisons instead focus on average performance over the full sample (e.g. Diebold and Mariano, 1995). Full sample tests are less useful in our setting. In standard macroeconomic models, microdata matters for the dynamics of aggregate inflation after large shocks to inflation, but not otherwise. As such, a forecasting model

¹⁰We exclude a series when a cleaner substitute is already included or when the series terminates early, as with most Index of Production sub-categories and discontinued benchmark rates. For average weekly earnings, we join the modern series to its predecessor to extend coverage back to 1996. We do not use GDP, firms' or households' inflation expectations, or inflation swaps, which are either not monthly or only available more recently.

¹¹The retained macro series are drawn from the ONS, the Bank of England, FRED/ALFRED, the World Bank, OECD, and market data providers. The air-freight indices are subject to revision and enter as ALFRED initial-release vintages, which begin in March 2010; earlier months use the latest available vintage.

¹²Throughout, the month- t values of these series are treated as part of the origin- t information set: forecasts dated origin t are formed after the month- t releases.

¹³The price data are not seasonally adjusted by design; the aggregate consumer price index and its microdata constituents have not been revised since 1996.

that uses microdata might outperform benchmarks only at certain times. The scan test is designed to test for this possibility.

The test is also more general and applies to any comparison of two forecast models in a non-stationary environment. The econometric theory for the test accounts for the composite null, nonstationarity, temporal dependence, and high-dimensionality. In the following section we will introduce a pipeline that uses the scan-test evidence to guide an adaptive machine learning approach, which makes use of microdata only when it at least adds localized forecasting power.

3.1 Statistical Hypotheses and Methodological Details

Roadmap. We now formalize our scan test. The aim is to decide whether a candidate forecast outperforms a second, benchmark forecast, over particular sub-periods and horizons; with rigorous statistical guarantees. As a roadmap, we first state the null and alternative hypotheses formally, then introduce a test statistic that measures statistical evidence against the null, and finally describe a Gaussian bootstrap procedure that approximates the null distribution of the statistic and converts it into a p-value. We present econometric theory that justifies the validity of our scan test. To maximize readability, we only discuss technical assumptions informally and leave all details and proofs to Appendix B.

Part I: Null and Alternative Hypotheses. Let n denote the number of target months in the evaluation sample and H the maximal forecasting horizon. Consider a benchmark method B and a competing method M . In our application, B could be a benchmark inflation forecast, such as the SARIMA model introduced below, and M could be the microdata-based inflation forecast. Writing $\hat{\pi}_{t|t-h}$ for a forecast of inflation in target month t made with information through origin month $t-h$, let $e_{t|t-h}^B = \pi_t - \hat{\pi}_{t|t-h}^B$ and $e_{t|t-h}^M = \pi_t - \hat{\pi}_{t|t-h}^M$ denote the two methods' h -step-ahead forecast errors for target month t , and define the loss differential, indexed by target month,

$$D_t^{(h)} = |e_{t|t-h}^B| - |e_{t|t-h}^M|.$$

With this sign convention, positive values of $D_t^{(h)}$ indicate that method M outperforms method B in target month t at horizon h .

The null hypothesis is

$$H_0 : \mathbb{E}[D_t^{(h)}] \leq 0 \quad \text{for all } t \in [n] := \{1, \dots, n\}, h \in [H] := \{1, \dots, H\},$$

so that under H_0 , method B is at least as good as method M across all target months and horizons. Note that it is possible for the null to be rejected in both directions, that is, for both M versus B and B versus M . If so, there are some months or forecasting horizons in which method B is better than

method M , and other months or horizons where the opposite is true.

To make rejection substantively meaningful, we consider the restricted alternative

$$H_1: \sum_{t \in [s, e]} \mathbb{E}[D_t^{(h)}] > 0 \quad \text{for some interval } [s, e] \text{ with } L_{\min} \leq e - s + 1 \leq L_{\max} \text{ and some } h \in [H].$$

That is, method M outperforms method B on average over some contiguous interval of target months of length between $L_{\min}$ and $L_{\max}$, at some horizon under study. The bounds $L_{\min}$ and $L_{\max}$ are user-specified parameters. We will discuss their choices at the end of this section. In the empirical implementation below, we use monthly data and take $H = 24$.

Part II: A Partially-Studentized Scan Statistic To test the null hypothesis against the localized alternative, we construct a scan statistic. This statistic searches over all admissible target-month intervals and forecasting horizons to determine when method M has an unusually good forecast loss relative to method B .

Let $\mathcal{I}$ be the collection of all target-month intervals $I = [s, e] \subset [n]$ with $L_{\min} \leq |I| \leq L_{\max}$. For each $I = [s, e]$, $h \in [H]$, and $D_t = (D_t^{(1)}, \dots, D_t^{(H)})^\top$, define

$$S_I^{(h)} = \sum_{t=s}^e D_t^{(h)}, \quad \bar{D}_I^{(h)} = \frac{S_I^{(h)}}{|I|}, \quad Q_I^{(h)} = \sum_{t=s}^e (D_t^{(h)} - \bar{D}_I^{(h)})^2.$$

In words, $S_I^{(h)}$ is the cumulative loss differential over the target-month interval I , $\bar{D}_I^{(h)}$ is the corresponding within-interval mean, and $Q_I^{(h)}$ is the within-interval sum of squared deviations. For a tuning exponent $\gamma \in [0, 1/2]$, define the partially-studentized statistic for interval I and horizon h as¹⁴

$$T_I^{(h)} = \frac{S_I^{(h)}}{|I|^{1/2-\gamma} (Q_I^{(h)})^\gamma}.$$

The scan statistic is defined as the double maximum

$$T^{\max} = \max_{I \in \mathcal{I}} \max_{1 \leq h \leq H} T_I^{(h)}.$$

This statistic searches jointly across target-month intervals and forecasting horizons for the most pronounced localized outperformance of method M relative to method B .

Two features of $T^{\max}$ deserve emphasis. First, the joint maximization over both intervals I and horizons h searches for localized outperformance without fixing either the interval or the forecast horizon. By searching over intervals, we do not have to specify the start month and duration of any outperformance. Extending the search to H horizons allows detection of outperformance concentrated at particular forecast horizons rather than uniformly across all of them. The cost of testing across multiple horizons will be absorbed automatically by the bootstrap. Second, partial student-

¹⁴When the denominator is numerically smaller than 10^{-12} , we set the corresponding observed or bootstrap interval statistic to zero.

tization lets the statistic interpolate between no local variance studentization ($\gamma = 0$) and full self-normalization ($\gamma = 1/2$). We will show in Appendix B that the test is asymptotically valid for any $\gamma \in [0, 1/2]$. Nonetheless, because our empirical design involves small samples, namely short intervals of length no more than 12, simulations reveal substantial over-rejection for larger values of γ . We discuss a principled approach to choosing γ below. To the best of our knowledge, partial studentization is a novel technique to address the small-sample issue for Gaussian bootstrap.

Part III: Gaussian Multiplier Bootstrap Distribution. Owing to the joint maximization over many intervals and horizons, the null distribution of $T^{\max}$ is nonstandard. We approximate it using a Gaussian multiplier bootstrap, which delivers critical values and p -values for testing H_0 against the localized alternative.

Define a draw of the bootstrap multiplier

$$\xi = (\xi_1, \dots, \xi_n)^\top \sim \mathcal{N}(0, \Theta_n)$$

which is independent of the data. Θ_n is a positive semidefinite $n \times n$ covariance matrix specified below. The role of Θ_n is to encode the temporal dependence structure imposed on the bootstrap multipliers, so its choice should mirror the dependence properties of the loss differential process under the null.

For each interval $I = [s, e]$ and horizon h , define the multiplier-weighted local scores

$$e_{t,I}^{(h)} = \xi_t (D_t^{(h)} - \bar{D}_I^{(h)}), \quad S_I^{*(h)} = \sum_{t=s}^e e_{t,I}^{(h)}.$$

The bootstrap denominator is recomputed from these local scores:

$$Q_I^{*(h)} = \sum_{t=s}^e (e_{t,I}^{(h)} - \bar{e}_I^{(h)})^2 = \sum_{t=s}^e (e_{t,I}^{(h)})^2 - \frac{(S_I^{*(h)})^2}{|I|},$$

where $\bar{e}_I^{(h)} = S_I^{*(h)} / |I|$. The bootstrap interval statistic is

$$T_I^{*(h)} = \frac{S_I^{*(h)}}{|I|^{1/2-\gamma} (Q_I^{*(h)})^\gamma}.$$

The bootstrap scan statistic is then defined as

$$T^{*,\max} = \max_{I \in \mathcal{I}} \max_{1 \leq h \leq H} T_I^{*(h)}.$$

We next take B bootstrap draws $T^{*,(1),\max}, \dots, T^{*,(B),\max}$, where each bootstrap draw involves a new, independent draw of the vector ξ , followed by recalculating $T^{*,\max}$ using the new draw. The p -value is

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{T^{*,(b),\max} > T^{\max}\}}{B + 1}.$$

The p -value is the share of bootstrap draws in which the null produces a statistic at least as extreme as the one observed in the data (after adding 1 to both the denominator and numerator as a finite

sample correction).

Choice of Θ_n For the Gaussian bootstrap to give an accurate approximation to the null distribution of the scan statistic, the joint distribution of $\{S_I^{*(h)} : L_{\min} \leq |I| \leq L_{\max}, h \in [H]\}$, conditional on the observed loss differentials, should approximate the sampling joint distribution of $\{S_I^{(h)} : L_{\min} \leq |I| \leq L_{\max}, h \in [H]\}$. Focusing on a single interval I , the covariance matrix of $S_I = (S_I^{(1)}, \dots, S_I^{(H)})^\top$ is given by

$$\sum_{t,s \in I} \text{Cov}(D_t, D_s), \quad (1)$$

while the covariance matrix of $S_I^* = (S_I^{*(1)}, \dots, S_I^{*(H)})^\top$ conditional on $(D_1, \dots, D_n)$ is given by

$$\sum_{t,s \in I} (\Theta_n)_{ts} (D_t - \bar{D}_I)(D_s - \bar{D}_I)^\top, \quad (2)$$

where $\bar{D}_I = (\bar{D}_I^{(1)}, \dots, \bar{D}_I^{(H)})^\top$. To build more intuition around the choice of Θ_n , we first consider the

Regime I where D_t are independent over time. In this case, we choose

$$\Theta_n = I_n. \quad (3)$$

In fact, under independence, the true covariance matrix of S_I and the conditional covariance matrix of S_I^* reduce to

$$\sum_{t \in I} \text{Var}(D_t), \quad \sum_{t \in I} (D_t - \bar{D}_I)(D_t - \bar{D}_I)^\top.$$

When $|I|$ is sufficiently large, the standard concentration argument implies the closeness between the two matrices (e.g. Wainwright, 2019, Chapter 6).

Next, we consider the general case where D_t are weakly dependent, or, more precisely, strongly mixing (see Assumption B.7 in Appendix B.4.1 for the formal definition). We refer to this as **Regime D**. Approximating equation (1) by equation (2) is analogous to the problem of constructing heteroscedasticity and autocorrelation consistent standard errors (Newey and West, 1987; Andrews, 1991). Following this intuition, we can choose Θ_n as

$$\Theta_n(t, u) = K\left(\frac{t - u}{b_n}\right), \quad (4)$$

where K is a kernel function and b_n is a bandwidth. Our implementation uses the quadratic spectral (QS) kernel, introduced by Andrews (1991), because it satisfies our regularity condition; see Appendix B.4.7. Unlike the standard Bartlett kernel in the Newey-West estimator (Newey and West, 1987), the QS kernel does not truncate at a finite lag and instead decays smoothly, placing nonzero weight on all lags. It is known to be optimal in a mean squared error sense for long-run variance estimation under certain conditions (Andrews, 1991).

Having chosen the QS kernel function, the remaining tuning parameter in Regime D is the bandwidth b_n , which controls how quickly dependence decays in the bootstrap covariance matrix. We

apply the optimal bandwidth derived in Andrews (1991) based on a data-driven plug-in rule based on an AR(1) approximation to the loss differentials; see also Chang, Chen and Wu (2024) in the context that is closer to our setting. Specifically, for each horizon h , we fit the autoregression

$$D_t^{(h)} = c_h + \rho_h D_{t-1}^{(h)} + \varepsilon_{h,t},$$

and obtain estimates $\hat{\rho}_h$ and $\hat{\sigma}_h^2$, where $\hat{\sigma}_h^2$ is the innovation variance. We then aggregate these horizon-specific estimates into

$$\hat{a} = \frac{\sum_{h=1}^H 4\hat{\rho}_h^2 \hat{\sigma}_h^4 (1 - \hat{\rho}_h)^{-8}}{\sum_{h=1}^H \hat{\sigma}_h^4 (1 - \hat{\rho}_h)^{-4}},$$

and set the QS bandwidth to

$$b_n = 1.3221 (\hat{a}n)^{1/5}, \quad (5)$$

where n is the time-series sample size.¹⁵ The complete procedures for Regime I and Regime D are given in Algorithms 6 and 7 in Appendix E.

Choice of γ We choose the partial-studentization exponent γ by simulation under the null, calibrated to the empirical loss-differential sequence used in the forecast comparison. For each horizon, we fix the observed sequence $\{D_t^{(h)}\}$ and generate null draws

$$D_{t,k}^{(h)} = \xi_{t,k} D_t^{(h)}, \quad \xi_{t,k} = \text{sign}(Z_{t,k}),$$

where $Z_{t,k}$ is a stationary Gaussian AR(1) process with autocorrelation ρ . This construction is useful because the random signs center the simulated loss differentials at the null of no systematic outperformance, while the multiplication by the observed $D_t^{(h)}$ preserves the empirical scale, heterogeneity across dates and horizons, and the cross-horizon co-movement in the loss-differential panel. We choose the grid for the simulation autocorrelation ρ to cover the horizon-specific AR(1) estimates $\{\hat{\rho}_h\}_{h=1}^H$ defined above. For each value of γ on a grid, each ρ , and each bootstrap regime, we simulate rejection probabilities at the nominal five-percent level. We then repeat the exercise for the final procedure that selects between Regime I and Regime D using the Ljung–Box pre-test. Across these designs, $\gamma = 0.1$ avoids over-rejection most consistently, so we use $\gamma = 0.1$ in the empirical scan tests. We include all details, including the DGP, simulation results, and robustness to the choice of γ in Appendix C.

Part IV: Pre-Test for Serial Dependence and Regime Selection. Because the appropriate choice of Θ_n depends on whether the loss differentials are serially dependent, we conduct a pre-test before

¹⁵Two numerical safeguards are applied in the implementation: the estimated AR(1) coefficients $\hat{\rho}_h$ are clipped to $[-0.99, 0.99]$ before forming $\hat{a}$, and the eigenvalues of the kernel covariance matrix Θ_n are floored at a small positive value before the multipliers are drawn.

applying the regime-specific bootstrap algorithms. This step determines whether inference should proceed under the independent noise regime (Regime I) or the dependent noise regime (Regime D).

To formally test for serial correlation, we apply the Ljung–Box test to $\{D_t^{(h)}\}$ at multiple lags. The null hypothesis is that the autocorrelations up to the tested lag are zero. Rejection of the null indicates serial correlation.

For each horizon $h \in \{1, 2, \dots, H\}$, let $r_{t,h} = D_t^{(h)}$ denote the loss differential series. Fix a maximum lag m and compute the Ljung–Box statistic

$$Q_h = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_h(k)^2}{n-k},$$

where $\hat{\rho}_h(k)$ is the sample autocorrelation of $\{r_{t,h}\}_{t=1}^n$ at lag k . We aggregate over horizons via $Q = \sum_{h=1}^H Q_h$.

To approximate the null distribution, we use a Gaussian multiplier bootstrap that preserves the cross-horizon dependence at each target month t while breaking temporal dependence. Specifically, for each bootstrap draw $b = 1, \dots, B$, sample i.i.d. multipliers $\{\xi_t^{(b)}\}_{t=1}^n$ with $\xi_t^{(b)} \sim \mathcal{N}(0, 1)$, and form the bootstrapped panel $r_{t,h}^{*(b)} = \xi_t^{(b)} r_{t,h}$ for all h . Compute $Q_h^{*(b)}$ and $Q^{*(b)} = \sum_{h=1}^H Q_h^{*(b)}$ from $\{r_{t,h}^{*(b)}\}_{t=1}^n$, and estimate the (one-sided) p -value as

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{Q^{*(b)} > Q\}}{B + 1}.$$

The null hypothesis of the Ljung–Box test corresponds to the absence of serial correlation, up to lag m , in the loss differential sequences. Under this null, any observed autocorrelation arises purely from random fluctuations. A small p -value indicates that the observed level of autocorrelation is unlikely under temporal independence, leading us to adopt the dependent noise regime (Regime D). A large p -value suggests insufficient evidence of dependence, and we proceed under the independent noise regime (Regime I).

In our application, we use $m = 12$ lags and $B = 4,999$ bootstrap draws at significance level $\delta = 0.05$. The full procedure is given in Algorithm 5 in Appendix E.

3.2 Econometric Theory of Scan Tests in Regimes I and D

We face four major challenges. First, the null hypothesis is one-sided, and Type-I error must be controlled even when some or all of the means $\mathbb{E}[D_t^{(h)}]$ are strictly negative. This is difficult because, due to partial studentization, the relevant t -statistics are not monotone in $D_t^{(h)}$. As a result, the boundary of the null, namely $\mathbb{E}[D_t^{(h)}] = 0$ for all t and h , is no longer necessarily the least favorable case. Second, the time series $D_t^{(h)}$ is non-stationary, so standard functional central limit theorems do not directly apply. Third, the statistics $D_t^{(h)}$ are highly correlated across forecasting horizons h . Fourth, the num-

ber of intervals and horizons considered by the scan test can be much larger than the sample size. In our application, for example, the occasional-outperformance scan test considers 1,449 intervals and 34,776 interval-horizon pairs with only 215 observations. This disparity motivates a high-dimensional asymptotic regime in which the number of interval-horizon pairs grows much faster than the sample size.

All four challenges call for new development in our econometric theory. We address the first challenge by introducing a new condition that controls the volatility of $\mathbb{E}[D_t^{(h)}]$. Specifically, we define an interval-wise mean-flatness index:

$$r_n := \max_{L_{\min} \leq |I| \leq L_{\max}, h \in [H]} \frac{\sum_{t \in I} (\mathbb{E}[D_t^{(h)}] - \mathbb{E}[\bar{D}_I^{(h)}])^2}{\sum_{t \in I} \mathbb{E}[(D_t^{(h)} - \mathbb{E}[D_t^{(h)}])^2]} = \max_{L_{\min} \leq |I| \leq L_{\max}, h \in [H]} \frac{\frac{1}{|I|} \sum_{t \in I} (\mathbb{E}[D_t^{(h)}] - \mathbb{E}[\bar{D}_I^{(h)}])^2}{\frac{1}{|I|} \sum_{t \in I} \text{Var}(D_t^{(h)})}.$$

The numerator measures the sample variance of $\mathbb{E}[D_t^{(h)}]$ over an interval and the denominator measures the average variance of $D_t^{(h)}$. Our theory requires r_n to be small at certain slow rates; see Assumptions B.5 and B.11 for the rate requirements for Regimes I and D, respectively. In particular, $r_n = 0$ whenever the mean is constant on each scanned interval. More generally, r_n is small insofar as the movement of $\mathbb{E}[D_t^{(h)}]$ is much slower than the intrinsic variability of $D_t^{(h)}$, which is plausible for challenging forecasting problems where $\text{Var}(D_t^{(h)})$ is large.

We address the remaining three challenges by adapting recent developments in high-dimensional distributional approximations for independent and dependent data (Chernozhukov, Chetverikov and Kato, 2013, 2017; Chang, Chen and Wu, 2024). Specifically, the cited papers derive central limit theorems for the empirical average of independent but non-identically distributed observations or a weakly dependent, nonstationary multivariate time series. The theory allows the dimension to grow exponentially in a polynomial of the sample size. While these tools are helpful in understanding the distribution of $\{S_I^{(h)} : L_{\min} \leq |I| \leq L_{\max}, h \in [H]\}$, the numerators of the t-statistics, it remains challenging to handle the partially-studentized denominators $|I|^{1/2-\gamma}(Q_I^{(h)})^\gamma$, which vary with both I and h .

We summarize our results in the following theorem.

Theorem 1 (Informal; see Appendix Theorems 11 and 15 for formal statements). *Let $\mathcal{I} = \{I \subset [n] : L_{\min} \leq |I| \leq L_{\max}\}$, let $p = H|\mathcal{I}|$ denote the number of admissible (interval, horizon) pairs, and let $c_{n,1-\alpha}^*$ be the conditional $(1-\alpha)$ -quantile of $T^{*,\max}$. Assume $\gamma \in [0, 1/2]$. If $\gamma < 1/2$, assume also that $L_{\max}/L_{\min}$ is bounded; no such bounded-ratio condition is needed when $\gamma = 1/2$. Then, under H_0 , the scan test is asymptotically valid in the sense that*

$$\limsup_{n \rightarrow \infty} \mathbb{P}(T^{\max} > c_{n,1-\alpha}^*) \leq \alpha \quad \forall \alpha \in (0, 1),$$

under either of the following assumptions, together with regularity conditions stated in Appendix B.

- (Regime I) $(D_t)_{t=1}^n$ are independent over time.
- (Regime D) $(D_t)_{t=1}^n$ are α -mixing (see Assumption B.7).

3.3 Three Practical Uses of the Scan Tests

The scan-test framework has three practical uses in our empirical analysis. The first two uses are for expert selection, and the third use is for the confirmatory analysis.

The first is *occasional outperformance*. In this use, the test asks whether a candidate forecast outperforms a benchmark over some ex ante unknown interval $I \in \mathcal{I}$ and some horizon h , with $L_{\min} \leq |I| \leq L_{\max}$. In our application, we choose $L_{\min} = 6$ and $L_{\max} = 12$.

The second use is *overall outperformance*. In this use, we set $\mathcal{I} = \{[1, n]\}$, or $L_{\min} = L_{\max} = n$. The statistic tests whether the candidate forecast improves on the benchmark over the sample as a whole at some horizon:

$$H_0 : \sum_{t=1}^n \mathbb{E}[D_t^{(h)}] \leq 0 \quad \text{for all } h = 1, \dots, H.$$

This use separates sustained full-sample gains from the localized gains targeted by the occasional-outperformance scan.

We use both occasional and overall outperformance for the expert selection. This selection determines which candidate forecasts enter our online-learning combined forecasts, which updates their weights adaptively. Starting from a benchmark forecast, the researcher screens candidate experts sequentially and includes a candidate only if it outperforms every expert selected so far. By outperformance, we mean that, for each selected expert used as the comparison forecast, either the occasional-outperformance test or the overall-outperformance test rejects the corresponding no-outperformance null at the 5% nominal level. We use these two tests because they are complementary and tend to be most powerful when the sample size is small, as in many macro forecasting problems: occasional outperformance has stronger signal strength because it targets localized gains, but it also has more noise from searching over intervals; overall outperformance has weaker signal because it smooths over time, but it also has less noise. The next section develops a pipeline that combines the experts selected by this sequential screen.

The third use is a *horizon-pooled test*. Given a user-chosen window of target months $I^* = [t_1, t_2]$ of length $T_w = t_2 - t_1 + 1$, we average the loss differential across horizons,

$$\bar{D}_t = \frac{1}{H} \sum_{h=1}^H D_t^{(h)},$$

and restrict the interval collection to $\mathcal{I} = \{I^*\}$, or $L_{\min} = L_{\max} = T_w$. The resulting statistic is the

partially-studentized mean over that window,

$$T^{\max} = \frac{S_{I^*}}{T_w^{1/2-\gamma}(Q_{I^*})^\gamma}, \quad S_{I^*} = \sum_{t=t_1}^{t_2} \bar{D}_t, \quad Q_{I^*} = \sum_{t=t_1}^{t_2} (\bar{D}_t - \bar{D}_{I^*})^2,$$

where $\bar{D}_{I^*} = S_{I^*} / T_w$. The null becomes

$$H_0 : \frac{1}{T_w} \sum_{t=t_1}^{t_2} \mathbb{E}[\bar{D}_t] \leq 0.$$

This horizon-pooled test keeps the one-sided directional null from the scan test, but targets average performance across horizons in a pre-specified period. In our empirical analysis, we use this confirmatory test to assess whether the combined-expert forecast outperforms the other benchmarks. We reuse the Gaussian multiplier bootstrap of Part III and the Regime I and Regime D specifications without modification.

3.4 Discussion: Existing Forecast Comparison Tests

Our null hypothesis differs substantially from the literature on forecast comparisons. Developing a valid test for this distinct null is one of our main methodological contributions. For one thing, we consider multiple horizons $h \in [H]$ all at once, whereas most forecast comparisons study a single horizon only. Moreover, standard forecast comparison tests assume stationarity and test whether, on average across a whole sample, one method outperforms a second (e.g. Diebold and Mariano, 1995). In effect, the null hypothesis for a given h is $\mathbb{E}[D_1^{(h)}] = \dots = \mathbb{E}[D_n^{(h)}] \leq 0$, under the assumption that $D_1^{(h)}, \dots, D_n^{(h)}$ are stationary.

An influential series of tests by Giacomini and Rossi (2010) relaxes stationarity, in order to test whether one method outperforms a second only at certain times. Our scan test shares the same motivation: forecast comparisons can miss localized episodes of relative performance. However, we target a different inferential problem. Giacomini and Rossi (2010) test local equal predictive ability, $\mathbb{E}[D_t^{(h)}] = 0$ at each month, for a fixed horizon h . They study two tests: a “Fluctuation test” that fixes the length of the interval for which the candidate forecast outperforms and searches for the best interval; and a “One-Time Reversal test”, which looks for a single point in the sample after which the candidate forecast outperforms. Their statistics are standardized by a global heteroskedasticity and autocorrelation consistent long-run variance, and their critical values come from Brownian-motion/Brownian-bridge limits under a high-level functional Central Limit Theorem with a positive scalar long-run variance.

Our scan test instead considers a composite one-sided null, $\mathbb{E}[D_t^{(h)}] \leq 0$ for all months and horizons. Therefore rejection supports the directional claim that one forecast outperforms another, somewhere. Our scan test searches jointly over an unknown starting month, an unknown ending month,

an unknown duration, and an unknown horizon, without imposing any more restrictions on when outperformance can happen. We use a high-dimensional Gaussian multiplier bootstrap to account for the search across many intervals and horizons; the bootstrap also accounts for dependence across forecast horizons. Under dependence, we do not require a single asymptotic long-run variance to exist: serial correlation is handled through the bootstrap covariance matrix Θ_n , with $\Theta_n = I_n$ under independence and a kernel covariance under dependence. Finally, the critical values in the one-sided Fluctuation test of Giacomini and Rossi (2010) are calibrated around both forecasts having equal ability. Our forecast comparison is designed to control size uniformly over the full one-sided null by using a one-sided max bootstrap critical value.

Our test is calibrated to handle the very small sample sizes that arise in the occasional outperformance application, including intervals of length 6 to 12. In Appendix C.3, we consider a suite of simulations based on our $D_t^{(h)}$ and show that existing tests substantially over-reject. In contrast, our scan test is well calibrated even for such small sample sizes.

4 Methods: Macro Forecasting with Micro Data

Section 3 developed a scan test for localized forecast outperformance. We can use the test to ask whether microdata-based forecasts occasionally improve on benchmark forecasts. This section develops a forecasting pipeline that is designed to exploit the occasional forecasting power of microdata. There are three steps. First, we encode the distribution of price changes into a stable and high dimensional vector of statistics. Second, we combine the high dimensional vector with a machine learning algorithm in order to forecast inflation. We use a gradient boosted tree algorithm, which we tune in real time. Third, if the micro forecast is indeed useful according to the scan test, we combine it with the benchmark using an adaptive machine learning method. The algorithm gives more weight to the microdata forecast when it outperforms the benchmark and downweights it otherwise. Later, in Section 6, we discuss a method to “open the black box”, letting us dissect which features of the microdata lead to its forecasting performance.

4.1 Step 1: Encoding Micro Price Data

Challenges. The aim is to forecast inflation with microdata. The raw microdata from the consumer price index, however, are not in a form that can be used directly by the prediction methods below.

The first issue is mechanical. Machine learning methods such as XGBoost require a fixed-length input vector, so each training and test observation must contain the same predictors in the same order. The raw CPI micro panel does not satisfy this requirement because products, defined as item-

store-region price quotes, regularly enter and exit the data. The typical product remains in the sample for less than a year. This turnover reflects shop behavior, for example, when a 6-pack of beer in a given grocery store is replaced by a close substitute. Because the set of observed products changes over time, using product-level price quotes directly would produce a predictor vector whose coordinates appear and disappear across months.

The second issue is statistical. Even if the raw product-level panel could be forced into a common design matrix, it would contain many noisy disaggregated predictors relative to the length of the monthly time series. A machine learning model trained on those inputs, even with regularization, risks fitting idiosyncratic product noise rather than systematic information about future inflation. More formally, statistical learning theory (e.g. Wainwright, 2019) suggests that, absent nonstationarity, forecasting accuracy can be decomposed into

$$\mathbb{E}[|Y - \hat{f}(X)|] \asymp \mathbb{E}[|\mathbb{E}[Y | X] - \hat{f}(X)|] + \sqrt{\frac{\dim(X)}{n}}, \quad (6)$$

where n is the sample size and $\dim(X)$ is the number of predictors. The first term is the approximation error, which always goes down when more predictors are included. The second term is the estimation error, which grows in the number of predictors due to overfitting. With the small sample sizes typical of inflation forecasting, adding many noisy disaggregated predictors tends to increase estimation error more than it reduces approximation error. Overfitting would be even more salient under non-stationarity, as X tends to be less predictive. While these two components cannot be easily assessed empirically in a changing environment, the guiding principle is to avoid overfitting by restricting the number of predictors.

Encoding method. The first step is therefore to encode the raw microdata into a stable vector of statistics. This encoding addresses the two issues above. It replaces unstable product coordinates with fixed monthly summaries, and it reduces the dimension of the predictor set by replacing individual product-level price changes with distributional statistics computed within broad cells. Cells are defined as the set of products in a COICOP1 category (e.g. food or clothing) in a given month. The baseline analysis uses a parsimonious summary of the cross-sectional distribution of price changes. Specifically, for each cell we calculate:

1. The fraction of observations with $\Delta p_{jt} = 0$.
2. The nine deciles (at probabilities 0.1, 0.2, ..., 0.9) of the *non-zero* price-change distribution within the cell.
3. The mean of the non-zero price-change distribution.¹⁶

¹⁶Exact formulas for all statistics are given in Appendix A.6.

The fraction of zero price changes captures the extensive margin of price adjustment. The deciles and mean summarize the distribution of realized price adjustments without using each individual quote as a separate predictor. This is the dimension-reduction step motivated by the statistical issue above: adding many disaggregated predictors can increase estimation error in a short and non-stationary monthly sample. All cell-level statistics are weighted by the expenditure weights associated with the Consumer Price Index. Subsection 5.6 considers an alternative microdata encoding that constructs these statistics without CPI expenditure weights.

These statistics give 11 micro features for each retained COICOP1 category: the fraction of observations with zero price changes, nine deciles of the non-zero price-change distribution, and the mean non-zero price change. Since the balanced sample retains 11 COICOP1 categories, stacking these features across categories, and using 12 lags, yields $11 \times 11 \times 12 = 1452$ microdata statistics at each point in time. All series are constructed at the monthly frequency and are not seasonally adjusted. For consistency with the benchmark forecasts, which also allow for seasonality, we include target-month indicators as inputs for micro forecasts so that expert comparisons and scan tests place the forecasts on comparable footing.

Augmented encoding. In Appendix F.1, we also study an augmented encoding that adds the remaining statistics from a richer price-setting summary. The additional information consists of:

1. Higher moments of the non-zero price-change distribution: variance with Bessel’s correction (m_2), and the standardised third, fourth and fifth central moments (m_3 , m_4 , m_5), where m_4 is expressed as excess kurtosis.
2. The nine deciles of time since the last reset, measured among observations whose current valid price change is zero and whose time-since-reset is defined. The clock resets on each valid non-zero price change, on the first valid price observation of a product spell, and whenever the gap between two consecutive observations of the same product exceeds 12 months.
3. The mean and the same four higher moments of this restricted time-since-change distribution.

Taken together with the baseline statistics, the augmented encoding gives 29 micro features for each retained COICOP1 category and $11 \times 29 = 319$ microdata statistics in each month, the same set of micro variables as the six-group Shapley decomposition of Appendix G.4. Consistent with the overfitting concern in the second issue above, this augmented encoding gives similar but slightly weaker forecasting performance than the baseline encoding in the final three-expert online-learning combination, with a larger gap at the standalone micro stage; full per-horizon comparisons are reported in Appendix F.1.

4.2 Step 2: Training and Tuning Individual Forecasts

The Training Framework. After this encoding, the input to the forecasting model has fixed length but remains high dimensional. Let Z_t denote the vector of microdata statistics observed in month t . For horizon $h \in \{1, \dots, 24\}$ and origin month t , $\hat{\pi}_{t+h|t}$ denotes the forecast of inflation in target month $t+h$ formed with information through t . The forecasting objective is to produce this forecast using the 12 most recent monthly values of Z_t available at t . We use a (non-linear) local projection approach: for each horizon h , we train a separate model, rather than relying on a single iterative model for all horizons, as in multi-step-ahead forecasting by VAR.

Formally, define the predictor vector available at origin month t ,

$$X_t^h = (Z_{t-11}, Z_{t-10}, \dots, Z_t, m_{t+h}),$$

which stacks the 12 monthly microdata-statistics vectors observed through month t together with $m_{t+h} \in \{0, 1\}^{11}$, the vector of eleven target-month indicators (January omitted).

Training. To limit the influence of distant history, at origin t we train the forecasting model on a rolling window containing the most recent (up to) 120 realized target months, using the pairs (X_{t-j-h}^h, π_{t-j}) for $j = 0, \dots, 119$ whenever available. We then apply the fitted model to X_t^h , which yields the forecast $\hat{\pi}_{t+h|t}$. Because training uses only data through t , this is a valid real-time h -step-ahead forecast.

The Real-time Tuning Framework. Training ML algorithms involve multiple tuning hyperparameters (e.g., learning rates) that are crucial for forecasting performance. In cross-sectional prediction problems, it is routine to carve out a random validation set outside the test data to evaluate the machine learning algorithm under different hyperparameters. For time-series forecasting, it is important to respect the temporal ordering. Thus, a random validation set is not ideal. A common exercise in the literature is to (a) fix a set of hyperparameters from a grid, (b) train the model with a rolling window, and (c) evaluate the tuning parameter based on the overall forecasting accuracy, such as the mean absolute error (MAE) or root mean-squared error (RMSE), for the entire sample period. While steps (a) and (b) are standard, step (c) incurs look-ahead bias: the hyperparameter is chosen using forecasting errors from periods that would not yet have been observed when earlier forecasts were made. Since overall accuracy depends on all time steps, this procedure may lead to overfitting and unfair comparisons.

To avoid look-ahead bias, we modify step (c) by introducing a real-time tuning framework. The idea is to score each candidate hyperparameter, at every forecast origin t , on the 48 most recent target months that have already realized, rather than on the full sample. This produces a time-varying performance metric for each candidate. Formally, we start with a grid $\mathcal{G}$ of values for the hyperparameter. For each origin t and each candidate value $\Gamma \in \mathcal{G}$, we replicate the h -step-ahead forecasting exercise

for the validation targets $\pi_{t-47}, \pi_{t-46}, \dots, \pi_t$. The forecast of a validation target π_{t-j} , $j = 0, \dots, 47$, is formed at its own origin $t - j - h$: we use the same rolling-window training procedure at that origin and apply the fitted model to X_{t-j-h}^h . This restriction ensures that each validation forecast uses only information available at its own origin, so it is a genuine h -period-ahead forecast. Averaging the 48 absolute errors gives a real-time mean absolute error (MAE) for each candidate Γ ; we choose the Γ^* that minimizes this criterion over the grid $\mathcal{G}$.

Given the selected hyperparameter Γ^* , the deployed forecast at origin t is the h -step-ahead forecast $\hat{\pi}_{t+h|t}$ obtained by applying the rolling-window training procedure above with hyperparameter Γ^* . We repeat the procedure at every origin and horizon $h \in \{1, \dots, 24\}$, so that the evaluated target months span 2007/01 through 2024/11.

Our framework works with any machine learning algorithm. Since this predictor set is large relative to the length of the monthly time series, we use gradient-boosted trees, a machine learning method suited to high-dimensional covariates and tabular forecasting applications, which outperforms other methods such as random forest or LASSO in settings similar to ours (Grinsztajn, Oyallon and Varoquaux, 2022). Gradient boosted trees build a forecast as an additive ensemble of regression trees of moderate depth. Each new tree is fitted to the residual from the previous ensemble; the extent to which each new tree modifies the ensemble is the shrinkage or “learning rate”, written v . Our implementation uses XGBoost (Chen and Guestrin, 2016). The parameter grid $\mathcal{G}$ for XGBoost is a 7×7 grid of 49 candidates: shrinkage values $v \in \{0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$ combined with tree counts $n_{\text{est}} \in \{50, 100, 200, 400, 600, 800, 1000\}$, with the tree depth fixed at the XGBoost default of six. Forecast accuracy is not sensitive to this choice: coarser and shifted grids deliver mean absolute errors within about one percent of the baseline grid.

A Computationally More Efficient Tuning Procedure. The tuning procedure described above nominally requires refitting the model 48 times for each candidate hyperparameter vector at every origin. In practice no refitting is needed, because the validation forecasts at successive origins overlap: the same forecast can later serve either as a validation forecast or as the deployed forecast. We therefore precompute, for each candidate hyperparameter $\Gamma \in \mathcal{G}$ and each horizon, a single rolling sequence of real-time forecasts: at every origin t , a model trained using the pairs (X_{t-j-h}^h, π_{t-j}) for $j = 0, \dots, 119$, whenever available, produces the forecast $\hat{\pi}_{t+h|t}^\Gamma$. Once π_{t+h} realizes, the absolute forecast error is cached. At each origin t , averaging the trailing 48 cached forecast errors for a candidate hyperparameter gives that candidate’s real-time MAE. Hyperparameter selection therefore reduces to a lookup of these trailing MAEs, and the deployed forecast is the cached $\hat{\pi}_{t+h|t}^{\Gamma^*}$. Thus, for each horizon, one rolling pass per candidate replaces the 48 refits that would otherwise be required at every origin, while preserving the same tuning criterion. Algorithms 1 and 2 in Appendix E give the pseudocode; the accom-

panying notation is collected in Table E.1.

4.3 Step 3: Combining Forecasts with Adaptive Machine Learning

Steps 1 and 2 produce a microdata-based forecast for each horizon. The remaining task is to decide whether and how this forecast should be combined with existing benchmarks, for which we develop an adaptive machine learning approach.

We study two benchmark forecasts for inflation. Our first benchmark is a univariate SARIMA model, which historically tends to forecast inflation well (Atkeson and Ohanian, 2001; Stock and Watson, 2007). However both classic and more recent work also emphasize inflation forecasts based on large numbers of macroeconomic variables (e.g. Stock and Watson, 2002*b*; Medeiros et al., 2021). Therefore our second benchmark uses a large number of macroeconomic variables, combined with the XGBoost algorithm.

We combine forecasts using an adaptive machine learning algorithm: the Fixed Share algorithm of Herbster and Warmuth (1998). The algorithm forms a weighted average of forecasts, or experts, and updates the weights as forecast errors are realized. This subsection first defines the SARIMA benchmark and the macro forecast, then describes the Fixed Share algorithm and how to select experts for inclusion.

Benchmark I: SARIMA Our first benchmark is a Seasonal Autoregressive Integrated Moving Average (SARIMA) model estimated on a rolling window, with the order selected in real time. The SARIMA order is denoted $Y = (p, d, q)(P, D, Q, M)$, with fitted parameters Θ_c^Y . We search over $p, d, q, P, D, Q \in \{0, 1\}$ and fix $M = 12$ to account for monthly seasonality. We also include a candidate that forecasts h -step ahead inflation as the trailing 12-month average, $\hat{\pi}_{t+h|t} = \frac{1}{12} \sum_{j=0}^{11} \pi_{t-j}$; which we refer to as AO-12MA, following Atkeson and Ohanian (2001).

We choose this SARIMA forecast because it nests a wide range of popular, univariate specifications for inflation forecasting in the literature. Many of these specifications are known to forecast inflation as well as, or better than, richer alternatives. Not only does our specification nest the forecast of Atkeson and Ohanian (2001), but also the IMA(1,1) forecast of Stock and Watson (2007), the “airline model” of inflation forecasting used by statistics agencies (e.g. Eurostat, 2015), and the ARMA(1,1) model of Ang, Bekaert and Wei (2007). As we shall see, this benchmark often performs excellently.

Each candidate order is fitted by maximum likelihood on the most recent 120 months of inflation data at every forecast origin. The ten-year rolling window keeps the model responsive to shifts in inflation dynamics without allowing distant observations to dominate the fit. Order selection follows the cached real-time tuning of Subsection 4.2, with the candidate orders Y taking the role of the XGBoost hyperparameter vector Γ . The full procedure is given in Algorithms 3 and 4 in Appendix E.

A SARIMA model expresses inflation as a linear function of its own lags, seasonal lags, and moving-average terms, after differencing to remove trends and seasonal unit roots.¹⁷ To produce an h -step ahead forecast, we iterate the fitted model forward from the last observed value, replacing each future innovation with zero (its conditional expectation given the history). Because the model is univariate and forecasts are iterated rather than direct, a single fit at each origin yields the forecast at every horizon, in contrast to the machine-learning experts, which train a separate model for each horizon.

Benchmark II: Macro Forecast Our second benchmark applies the rolling-window XGBoost procedure of Algorithms 1 and 2 to *macro* rather than micro features. Concretely, for a forecast formed at origin t , the predictors consist of the twelve most recent monthly observations of each macro time series described in Section 2, that is, lags $h, h+1, \dots, h+11$ of the target. All other aspects of the algorithm, including the rolling window length (120 months), hyperparameter grid, cached real-time tuning, target-month indicators, and direct h -step forecasting, are held fixed. This approach follows the high-dimensional macro forecasting literature, including Medeiros et al. (2021), who use machine learning with a large macro panel to forecast US inflation.¹⁸

Adaptive Combination of Forecasts with Fixed Share. Suppose that the scan test detects localized outperformance of a candidate expert compared to the benchmark. Then in principle, the candidate can improve forecasting performance. What remains is to find an adaptive algorithm, which makes use of the candidate when it outperforms the benchmark, but not otherwise. To this end, we combine forecasts using the Delayed Fixed Share algorithm of Korotin, V'yugin and Burnaev (2020, Algorithm 4), a variant of the Fixed Share algorithm of Herbster and Warmuth (1998).¹⁹ The delayed variant is necessary because the loss from an h -step-ahead forecast is observed only when its target is realized; updating weights before then would introduce look-ahead bias. For convenience, we refer to this delayed variant simply as Fixed Share throughout. The Fixed Share algorithm works as follows. Let K denote the number of experts in the admitted pool, indexed by $k = 1, \dots, K$. For each horizon $h \in \{1, \dots, 24\}$ and forecast origin t , let $\hat{\pi}_{t+h|t}^k$ denote the h -step-ahead forecast of expert k and π_t the realized inflation in month t . The algorithm maintains a probability vector of weights $\{w_t^{h,k}\}_{k=1}^K$ over

¹⁷Formally, a SARIMA(p, d, q)(P, D, Q, s) model takes the form $\Phi(B)\Phi_s(B^s)(1-B)^d(1-B^s)^D\pi_t = \Omega(B)\Omega_s(B^s)\varepsilon_t$, where $\Phi, \Phi_s, \Omega, \Omega_s$ are lag polynomials, B is the backshift operator, and $\varepsilon_t \sim \text{i.i.d.}(0, \sigma^2)$.

¹⁸Our implementation differs from Medeiros et al. (2021) in two respects. First, we use gradient boosted trees rather than random forests. Second, we use non-seasonally adjusted data throughout to avoid look-ahead bias, whereas Medeiros et al. (2021) use seasonally adjusted series.

¹⁹Most standard online-learning algorithms, such as multiplicative weights (MW; Cesa-Bianchi and Lugosi, 2006) and the original Fixed Share algorithm of Herbster and Warmuth (1998), assume that the outcome used to update weights is observed immediately after the forecast is made. They therefore do not directly apply to h -step-ahead forecasting with $h > 1$, where the outcome is observed only after an $(h-1)$ -step delay. A smaller branch of the literature studies online learning under delayed feedback; see, for example, Joulani, Gyorgy and Szepesvári (2013) and Korotin, V'yugin and Burnaev (2020).

experts and uses these weights to form the aggregate forecast

$$\hat{\pi}_{t+h|t}^{\text{FS}} = \sum_{k=1}^K w_t^{h,k} \hat{\pi}_{t+h|t}^k.$$

The loss on an h -step forecast is not realized until h months after the forecast is formed, so the weights must be updated using only losses that have been realized by the origin. The algorithm therefore maintains, for each horizon h , an internal weight vector $\{v_t^{h,k}\}_{k=1}^K$ over experts, indexed by the *target* month, the month whose inflation is being forecast. When the inflation rate of month t becomes available, the h -step forecasts of target t , formed at origin $t-h$, are evaluated using their absolute forecast errors $\ell_{t|t-h}^k = |\pi_t - \hat{\pi}_{t|t-h}^k|$, and the internal weights are updated in two steps. First, a Fixed Share mixing step redistributes a fraction $\alpha \in [0, 1]$ of the weight uniformly across all experts, preventing any expert from being permanently abandoned. Second, a multiplicative weights (MW) step down-weights experts in proportion to their realized absolute forecast error:

$$v_t^{h,k} \propto \left[(1-\alpha) v_{t-1}^{h,k} + \frac{\alpha}{K} \right] \exp(-\eta \ell_{t|t-h}^k), \quad \sum_{k=1}^K v_t^{h,k} = 1,$$

where $\eta > 0$ is the learning rate. The forecast weights used at origin t are obtained by propagating these internal weights through the h months separating the origin from its target, during which no further loss of horizon h is realized, so only the mixing step operates:

$$w_t^{h,k} = (1-\alpha)^h v_t^{h,k} + \frac{1 - (1-\alpha)^h}{K}.$$

The two parameters control distinct aspects of the algorithm's responsiveness. A larger η makes the multiplicative step more reactive: weight moves more quickly toward experts with lower recent losses, at the cost of responding more to short-run noise. A smaller η smooths the update and retains more of the prior weight. A larger α keeps more weight spread across experts through the mixing step, which lets previously dominated experts regain influence once their relative performance improves. The internal weights are initialized uniformly, $v_{t_0}^{h,k} = 1/K$, and remain uniform until the first h -step loss is realized, so early forecasts start from equal weights. The full path of forecasts and weights for a given η is produced by Algorithm 8 in Appendix E.

A chief attraction of the Fixed Share algorithm is its interpretability. If one expert is particularly relevant for forecasting, its weight is high. Therefore one can inspect the sequence of weights to understand when certain data modalities, such as microdata, are relevant.²⁰

²⁰The weights also have a natural Bayesian interpretation. For a fixed horizon h , let the identity of the best expert be a latent state n_s at time s . The Fixed Share algorithm considers a Markov prior on the sequence of n_s with $p(n_s = n \mid n_{s-1} = m) = (1-\alpha)\mathbf{1}\{n=m\} + \alpha/K$, which says that this latent state usually persists but can switch uniformly across experts. Let $L_{t|s} = (\ell_{t|s}^1, \dots, \ell_{t|s}^K)$ denote the vector of realized expert losses for target month t from origin s , where $\ell_{t|s}^j = \ell(\pi_t, \hat{\pi}_{t|s}^j)$. When the h -step loss for target month s is realized, exponential weighting is equivalent to the likelihood $p(L_{s|s-h} = \ell \mid n_{s-h} = n) \propto \exp\{-\eta \ell\}$, where ℓ denotes the n th component of ℓ . In this notation, $v_t^{h,n} = p(n_{t-h} = n \mid L_{h+1|1}, \dots, L_{t|t-h})$,

Regret Guarantee. The Fixed Share algorithm provides a worst-case performance guarantee, clarifying precisely what the algorithm optimizes in a non-stationary environment. The relevant object is *dynamic regret*: the gap between the cumulative loss of the online learner and the cumulative loss of an ideal benchmark sequence of experts chosen in hindsight after seeing the full sample. Formally, let

$$L_n^{\text{FS},h} = \sum_{t=1}^n |\hat{\pi}_{t|t-h}^{\text{FS}} - \pi_t|$$

denote the cumulative loss of the Fixed Share forecast over n periods. Let $L_n^{(k_0, k_1, \dots, k_S),h}$ denote the cumulative loss of the best sequence of experts $k_0, k_1, \dots, k_S$ in hindsight, when we allow the identity of the expert to switch S times over the sample. The dynamic regret of the algorithm is then defined as

$$\text{Reg}_{S,n}^h = L_n^{\text{FS},h} - \min_{k_0, k_1, \dots, k_S} L_n^{(k_0, k_1, \dots, k_S),h}.$$

The standard notion of regret is a special case with $S = 0$, which measures the comparison with the best ex-post single expert forecast (Orabona, 2019).

Let $\pi_t, \hat{\pi}_{t|t-h}^{\text{FS}}$ be bounded. Tailoring Corollary 1 of Korotin, V'yugin and Burnaev (2020) to the Fixed Share update and our setting, we can easily obtain the following result:

$$\text{Reg}_{S,n}^h = O\left(\eta nh + \frac{1}{\eta} \left\{ (S+1) \log K + S \log \frac{1}{\alpha} + (n-S-1) \log \frac{1}{1-\alpha} \right\}\right). \quad (7)$$

where K is the number of experts, η is the learning rate, α is the Fixed Share mixing parameter. Crucially, the guarantee holds for *any* sequence of outcomes. It does not require stationarity, ergodicity, or any specific model of the data generating process, which is exactly why the method is attractive for inflation forecasting in a non-stationary environment, with machine learning-based experts.

Each term on the right-hand side has a simple interpretation. The first term, ηnh , is the cost of using a positive learning rate in the exponential-weighting update; it accumulates over the n target months and grows with the forecast horizon h because feedback arrives with delay. The bracketed terms constitute the complexity cost of tracking a switching comparator: $(S+1) \log K$ reflects the number of experts available on each segment, while the two α terms price the chosen pattern of switches and non-switches. Thus, adding experts raises the regret bound through the $\log K$ penalty. As a consequence, one should include only experts whose marginal contribution is non-negligible, in the sense that the decrease in the comparator loss outweighs the increase in the $\log K$ penalty term. The last two terms imply α should be small when $S \ll n$. In particular, it is straightforward to show that the optimal α is given by $S/(n-1)$. In our empirical analysis, we set $\alpha = 0.02$, which is approximately $4/n$ at our sample size.

and the forecast-origin weight is $w_t^{h,n} = p(n_t = n \mid L_{h+1|1}, \dots, L_{t|t-h})$. We refer to Korotin, V'yugin and Burnaev (2020) for details.

To gain more insight into the impact of the learning rate η , we set α to be the optimum. Then the regret can be written as

$$\text{Reg}_{S,n}^h = O\left(\eta nh + \frac{\log(Kn)}{\eta}\right). \quad (8)$$

The tension between the two pins down the role of the learning rate: a small η yields stable weights but adapts too slowly, while a large η reallocates across experts quickly but overreacts to short-run noise.

Fixed Share can be understood as a generalization of the multiplicative weights (MW) algorithm from the online learning literature (Cesa-Bianchi and Lugosi, 2006). The MW algorithm provides a similar regret guarantee, but only relative to the single best expert in hindsight. That is too restrictive for our application: we do not expect the same forecast to be best both before and after large changes in inflation dynamics. The reason MW may fail in this environment is that it updates weights multiplicatively based on losses, so an expert that performs poorly early on sees its weight decay exponentially toward zero. Even if that expert later becomes the best, it cannot recover meaningful weight. The key innovation of Fixed Share is the mixing step governed by α : after each multiplicative update, a fraction of weight is redistributed across all experts. This keeps every expert alive, so the algorithm can track switches in the identity of the best forecast source over time.

It might be tempting to consider a more straightforward forecast which applies XGBoost to data from all three experts, and lets a machine learning (ML) algorithm perform variable selection. We will perform this exercise as robustness, but it is not our baseline. While this practice is more common in the ML literature, it is prone to overfitting—the sample size for inflation forecasting is extremely low and most predictors are not highly predictive. More formally, equation (6) suggests that the overfitting cost of blindly combining modalities could easily outweigh the benefit for challenging forecasting problems. In contrast, the Fixed Share algorithm serves as a “router” to average multiple experts which rely on a subset of predictors. It is analogous to the sparse mixture-of-experts architecture that proves to be much more effective than dense feed-forward layers in training foundation AI models (Shazeer, Mirhoseini, Maziarz, Davis, Le, Hinton and Dean, 2017; Lepikhin, Lee, Xu, Chen, Firat, Huang, Krikun, Shazeer and Chen, 2021). This approach lets the forecasting system retain a simple benchmark while allowing additional experts to receive weight only when their recent performance justifies it.

Furthermore, it is unrealistic to expect practitioners to replace a long-standing forecasting model, such as SARIMA, with a new and more black-box algorithm that has not yet been evaluated in real time, solely on the basis of backtesting performance. Our framework allows the forecaster to retain the status quo model, introduce new data modalities or machine-learning models as additional experts with zero or small initial weights, and then update these weights adaptively using the Fixed Share algorithm. The resulting weight shifts also provide a succinct and interpretable time-varying

summary of the relative performance of each expert, whether the incumbent forecast or a newly introduced one, without imposing stationarity.

Fixed Learning Rate. Our baseline fixes $\eta = 0.5$ and $\alpha = 0.02$ throughout, with the internal weights initialized uniformly at $1/K$; the combined forecasts cover target months from January 2007 onward, with each forecast formed h months earlier at its forecast origin. This specification serves as a transparent benchmark: the fixed learning rate is consistent with standard practice in the online learning literature (Herbster and Warmuth, 1998), and holding both η and α fixed reduces the implementation to a single pass of Algorithm 8 (Appendix E) with $\eta = 0.5$.

Varying Learning Rate. While a fixed $\eta = 0.5$ is our baseline, the right degree of responsiveness may itself change over time: periods of rapid structural change call for a larger η so that weights shift quickly toward accurate experts, while stable periods favor a smaller η to avoid overreacting to noise. As a robustness check, we therefore also consider a specification with a varying learning rate, which re-selects η at each forecast origin based on realized forecast performance.

The procedure works as follows. Fix a grid $\mathcal{H} = \{\eta^{(1)} < \eta^{(2)} < \dots < \eta^{(G)}\}$ of G candidate learning rates. For each $\eta^{(g)} \in \mathcal{H}$, the Fixed Share algorithm is run in parallel, producing a path of combined forecasts $\hat{\pi}_{t+h|t}^{\text{FS},(\eta^{(g)})}$. At each forecast origin t , we select the learning rate η whose path has delivered the lowest average realized error over the most recent W target months. Rather than re-evaluating all G candidates at every step, we restrict attention to the previously chosen value η_{t-1}^* and its immediate grid neighbors, reflecting the assumption that the optimal learning rate evolves gradually. Formally, let g_{t-1} denote the index of η_{t-1}^* in $\mathcal{H}$. The candidate index set at time t is

$$\mathcal{I}_t = \{\max(1, g_{t-1} - 1), g_{t-1}, \min(G, g_{t-1} + 1)\},$$

where the max and min prevent the index from falling outside the grid boundaries. Among these (at most three) candidates, we choose the one with the smallest rolling-window MAE of realized h -step errors over the most recent W target months:

$$\eta_t^* = \eta^{(g_t)}, \quad g_t = \arg\min_{g \in \mathcal{I}_t} \frac{1}{W} \sum_{s=t-W+1}^t |\pi_s - \hat{\pi}_{s|s-h}^{\text{FS},(\eta^{(g)})}|,$$

with ties kept at the incumbent value. Every term in the sum is available at the origin: the forecast of target s was formed at origin $s - h$, and π_s is realized by $t \geq s$. In our implementation, $\mathcal{H}$ consists of $G = 21$ values $\{0.25, 0.30, \dots, 1.25\}$ in steps of 0.05, with rolling window $W = 60$ months and initial value $\eta_0 = 0.5$, the fixed-rate baseline. The complete procedure is described in Algorithms 8 and 9 in Appendix E.²¹

²¹We have considered other algorithms for updating η , such as AdaHedge (de Rooij, van Erven, Grünwald and Koolen, 2014); however they performed poorly in our setting.

Scan Test for Expert Selection. As discussed above, to maximize the performance gain from the Fixed Share algorithm, we need to choose experts judiciously such that the improvement in the comparator loss $L_n^{(k_0, k_1, \dots, k_S), h}$ exceeds the $\log K$ penalty increase. We therefore use a sequential expert inclusion procedure based on the scan test introduced in Section 3. The idea is to start with a benchmark and add a new expert only if it outperforms every expert selected so far, in the sense that either the occasional-outperformance test or the overall-outperformance test (Subsection 3.3) rejects the corresponding no-outperformance null at the 5% nominal level.

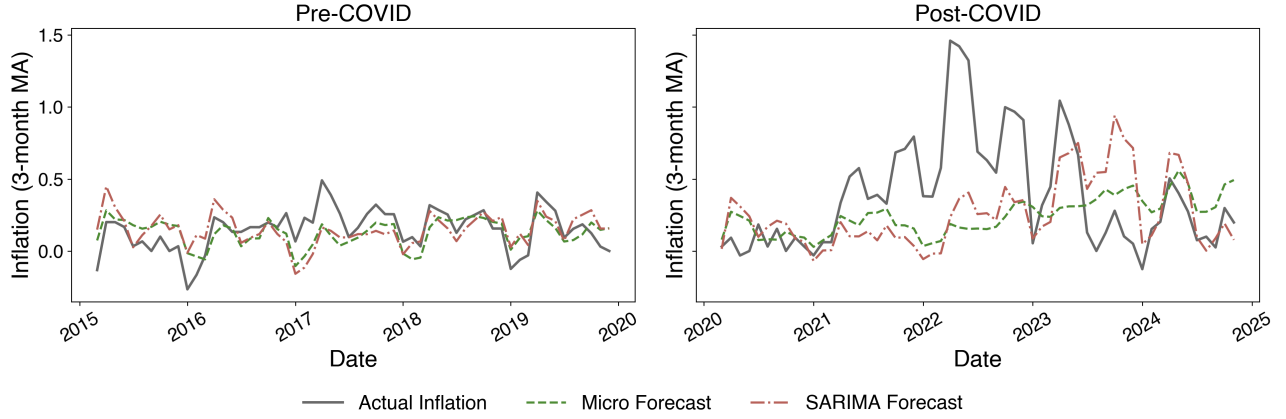
In practice, the most common forecast is given by the univariate SARIMA model, which will always be considered as the first expert. We consider the macro forecast as the second expert if it outperforms the univariate forecast. The micro model described in Sections 4.1 and 4.2 joins the pool of experts only if it outperforms both the univariate and the macro forecasts, each used as the comparison forecast. Effectively, we set a rather high bar for the usage of microdata. As a consequence, the inclusion of the microdata-based expert suggests strong statistical evidence that the microdata bring new information that is helpful for inflation forecasting.

In our expert-selection scan tests, we use $B = 4,999$ bootstrap draws applied to a panel of $n = 215$ monthly observations spanning January 2007 through November 2024, across $H = 24$ forecast horizons. The occasional-outperformance scan uses $L_{\min} = 6$ and $L_{\max} = 12$, giving $\sum_{L=6}^{12} (n - L + 1) = 1,449$ candidate intervals and $H \times 1,449 = 34,776$ interval-horizon pairs; the overall-outperformance test uses the single full-sample interval, $L_{\min} = L_{\max} = n$.

5 Results: Forecasting Inflation with Microdata

We now present the paper’s four main empirical results. First, the micro forecast outperforms the univariate benchmark only in the post-2020 window. Before then, the univariate benchmark performs better at every horizon; afterward, the micro forecast wins at nearly every horizon, with the largest gains at long horizons. Second, applying scan tests, we find that the Fixed Share combined forecast should include all three modalities: micro, macro and univariate forecasts. Third, this combined forecast performs comparably to the univariate benchmark before 2020 and better than it at every horizon after 2020. Fourth, we confirm the time-varying value of microdata. The combined forecast leans on the univariate forecast before 2020 and shifts weight toward the micro forecast afterward, especially at short and medium horizons. The full three-expert combined forecast also outperforms the two-expert benchmark that excludes microdata, on average, only in the post-2020 window.

Figure 1: Micro vs. Univariate Forecast, 12 Months Ahead



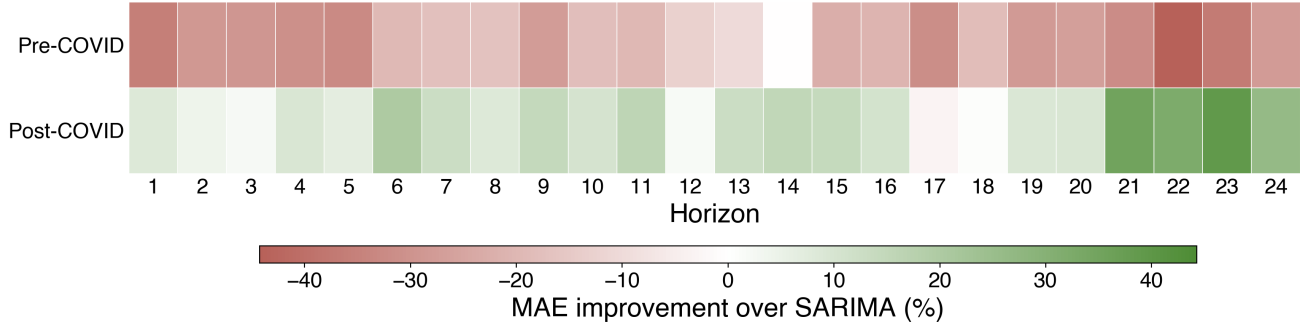
Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the micro 12-month-ahead forecast, and the univariate (SARIMA) 12-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 12$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

5.1 Micro Forecasts versus the Univariate Benchmark

We now present the first result: the micro forecast outperforms the univariate benchmark, but only after 2020. Figure 1 visualizes it. We plot the 12 month ahead forecast of monthly inflation, for both the univariate forecast and the microdata forecast, as well as actual inflation in the corresponding period. For interpretability, we take a three month moving average of all three series. Clearly, before 2020 the univariate forecast performs excellently. The microdata forecast has notably larger losses, especially in 2018-2019. However, after 2020 the micro forecast performs better. Neither forecast captures the rise in inflation in 2021-22, which according to both models was due to unforecastable shocks. However the microdata-based model does a better job of forecasting the subsequent propagation of these shocks, during the decline in inflation of 2023-24. The micro model predicts a gradual decline, whereas the univariate model predicts that inflation will be persistently high. At the end of the sample, as inflation becomes less volatile again, the univariate model again starts to perform better.

We then identify the forecast horizons that account for the micro model’s post-2020 outperformance. Figure 2 presents a heatmap. The x axis is the horizon of the forecast; the two rows are the pre-2020 and post-2020 evaluation windows, and each cell reports the percent improvement in the mean absolute error (MAE) of the micro forecast relative to the univariate forecast. Evidently, the micro forecast underperforms at all horizons before 2020; whereas after 2020 the microdata outperforms at 23 of the 24 horizons, and by the most at long horizons, where the improvement reaches about 40 percent. Tables 3 and 4 confirm these results by reporting the univariate-forecast MAE and the relative MAE improvement of the micro forecast; Appendix Figures D.1–D.3 plot the time series of forecasts at $h = 1, 6$, and 24.

Figure 2: Micro Forecast Improvement Relative to Univariate



Notes: Cells report the percentage improvement in mean absolute error (MAE) of the micro forecast relative to the univariate forecast by horizon $h = 1, \dots, 24$ and evaluation window. Positive values indicate lower MAE for the micro forecast than for the univariate forecast; negative values indicate the reverse. Green shading denotes positive improvements and red shading denotes negative improvements. Numerical values are reported in Table 4.

The occasional-outperformance scan test confirms the first result. It tests the null hypothesis that microdata do not outperform the univariate forecast at any horizon or during any time period. The Ljung–Box pre-test rejects serial independence for this comparison ($p = 0.034$), so the critical values use the Regime D bootstrap. The test rejects, with a bootstrap p -value of 0.041. The scan test therefore provides statistical evidence that microdata occasionally outperform the univariate forecast.

5.2 Selecting Experts with the Scan Test

We now present the second result: scan tests imply that the Fixed Share combined forecast should include all three experts: the micro, macro and univariate forecasts. As discussed in Section 3, we select experts sequentially. Starting from the univariate SARIMA benchmark, a candidate forecast is admitted only if it outperforms every expert selected so far, where outperformance means that either the occasional-outperformance test or the overall-outperformance test rejects the corresponding no-outperformance null at the 5% level. We first test whether the macro forecast outperforms the univariate benchmark. We then test the micro forecast separately against the univariate benchmark and the macro forecast.

Appendix G.1 contains details on the comparison between the macro forecast and the univariate benchmark. Before 2020, the macro forecast has a worse MAE than the univariate benchmark at all 24 horizons. After 2020, the macro forecast improves on the univariate forecast at 19 of the 24 horizons, by 10.5% on average, with the largest percent gain in the MAE reaching 41.9% at $h = 23$. The scan test confirms that the macro expert adds forecasting power: the bootstrap p -values are 0.037 for occasional outperformance and 0.024 for overall outperformance.²² We therefore add the macro forecast

²²For this comparison, the Ljung–Box pre-test does not reject serial independence ($p = 0.223$), so both tests use the

to the selected expert set.

We next apply the same sequential rule to the micro forecast. As shown in the preceding subsection, the occasional-outperformance test rejects the null of no occasional outperformance of the micro forecast relative to the univariate benchmark ($p = 0.041$). Against the macro forecast,²³ the overall-outperformance test rejects the null of no overall outperformance of the micro forecast ($p = 0.026$).²⁴ Thus, the micro forecast passes the sequential screen against both previously selected experts, through occasional outperformance relative to the univariate forecast and overall outperformance relative to the macro forecast. The sequential rule therefore admits the micro forecast, and all three experts enter the combined forecast.

5.3 Combining Experts with Fixed Share

We combine the three forecasts, micro, macro and univariate, using Fixed Share. This exercise leads to our third result: the combined forecast performs comparably to the univariate forecast before 2020, and better than it at every horizon after 2020.

Our baseline implementation holds the learning rate fixed at $\eta = 0.5$, with $K = 3$ experts and $\alpha = 0.02$, and combined forecasts covering target months from January 2007 onward. As a robustness check, we also run the varying-learning-rate version of Algorithm 9, described in Subsection 4.3.²⁵ Results are reported for the pre-2020 (2015–2019) and post-2020 (2020–2024) evaluation windows.

Figure 3, top panel, plots the 12-month-ahead forecast from the Fixed Share combination, together with the three expert forecasts. The adaptive combination performs well because it leans on the univariate forecast when that benchmark is useful and shifts toward the micro forecast when microdata add predictive content. Before 2020, it stays close to the univariate forecast: in 2017–18, for instance, the combined and univariate forecasts nearly coincide. During the 2023 disinflation, when the micro forecast is most useful, the combined forecast adaptively breaks away from the univariate forecast, which overpredicts inflation, and moves toward the micro forecast.

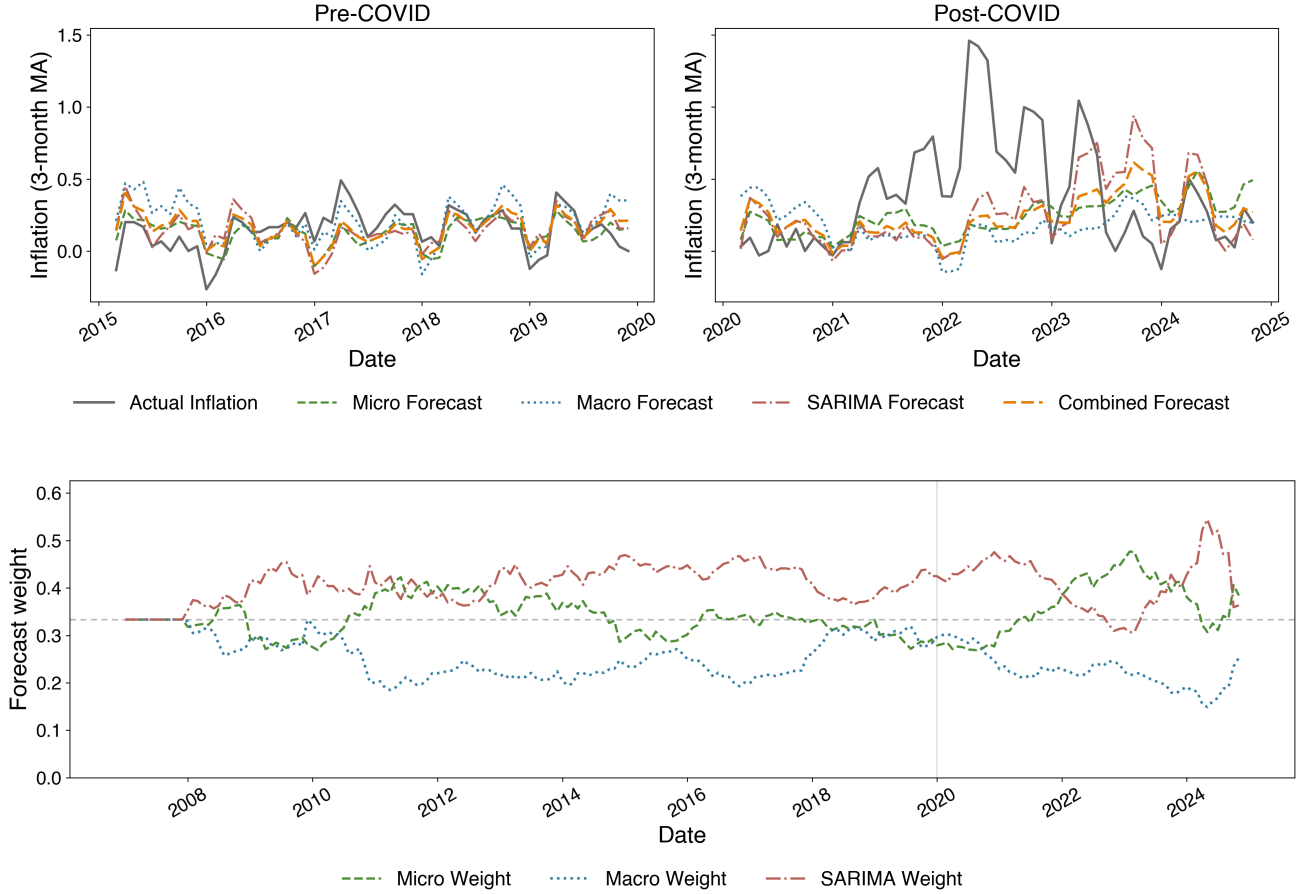
Regime I bootstrap.

²³For the comparison with the macro forecast, the Ljung–Box pre-test does not reject serial independence ($p = 0.226$), so the corresponding tests use the Regime I bootstrap.

²⁴Against the univariate forecast, the overall-outperformance test does not reject the null that the micro forecast fails to improve on the univariate forecast over the full sample at any horizon ($p = 0.152$). Against the macro forecast, the occasional-outperformance test does not reject the null that the micro forecast fails to improve on the macro forecast over any admissible interval and horizon ($p = 0.258$). These results illustrate the tradeoff discussed in Section 3.3. The occasional test targets a stronger localized signal but incurs more noise from searching over intervals. The overall test smooths over time, reducing noise but also weakening the signal.

²⁵This version re-selects the learning rate at each origin using the neighbor-only rolling search with $W = 60$ months. Its accuracy is nearly indistinguishable from the fixed-rate baseline: it has lower MAE at roughly half of the horizons and higher MAE at the rest, and the average gap is within 0.3 percent in both windows (Appendix G.3, Table G.3). We therefore report the fixed-rate baseline throughout.

Figure 3: Combined Forecast and Combination Weights, 12 Months Ahead



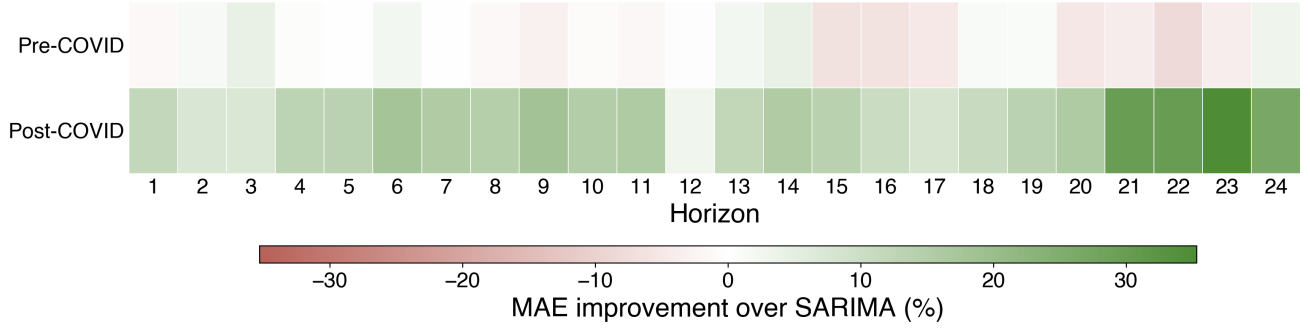
Notes: The top panel plots 3-month trailing moving averages of realized inflation, the three expert forecasts (micro, macro, and univariate), and the Fixed Share combined forecast for the pre-2020 (2015–2019) and post-2020 (2020–2024) windows; dates on the x-axis are target months, so a forecast dated t was formed at origin $t - 12$. The bottom panel plots the combination weights on the three experts over the full sample, with the dashed horizontal line at the equal weight $1/3$ and the vertical line marking 2020.

We confirm the advantages of the combined forecast in the heatmap of Figure 4. The heatmap shows that prior to 2020, the combined forecast has an MAE comparable to the univariate model: the average gap across horizons is 1.2 percent, and no shortfall exceeds 9 percent. After 2020, the combined forecast has a lower MAE at every horizon, by 16 percent on average and by around 30 percent at the longest horizons. Table 5 reports the underlying MAEs of the combination and of each expert, by horizon and window.

5.4 The Time Varying Value of Microdata

We now present our fourth result, which confirms the time varying value of microdata. One way to see the value of microdata is to inspect the weights of the Fixed Share combination, which are plotted

Figure 4: Combined Forecast Improvement Relative to Univariate



Notes: Cells report the percentage MAE improvement of the three-expert Fixed Share combination relative to the univariate forecast by forecast horizon $h = 1, \dots, 24$ and evaluation window. The three experts are the micro forecast, the macro forecast, and the univariate forecast. Positive values indicate lower MAE for the combination than for the univariate forecast; negative values indicate the reverse. Numerical values are reported in Table 5.

for the 12-month-ahead forecast in the bottom panel of Figure 3. The weight on microdata is below its equal-weight benchmark of $1/3$ in the years before 2020. After 2020 it rises, and it peaks in 2023, the period in which the micro forecast is most informative about the disinflation. The pattern is horizon specific: the average micro weight roughly doubles after 2020 at $h = 6$, from 0.21 to 0.41, rises from 0.32 to 0.36 at $h = 12$, and drifts slightly down at the longest horizons.²⁶

We next show the time varying value of microdata by comparing the MAE of the three-expert combined forecast to the two-expert benchmark. Figure 5 presents a heatmap. Before 2020, adding the micro forecast leaves the MAE essentially unchanged on average: it helps modestly at some short and medium horizons and hurts at the longest ones. After 2020, adding the micro forecast lowers MAE at 19 of the 24 horizons, by 2.9 percent on average and by up to 9.5 percent.

We use the alternative, more demanding application of the scan test described in Subsection 3.3. Specifically, we test the three-expert combined forecast against the two-expert benchmark that combines the univariate and macro forecasts, without microdata. The question is whether adding the microdata expert improves forecast accuracy *on average*, rather than only at some point in time. We apply the horizon-pooled test described in Subsection 3.3 separately before and after 2020 (for 2015–19 and 2020–24). Specifically, we study the horizon-averaged loss differential

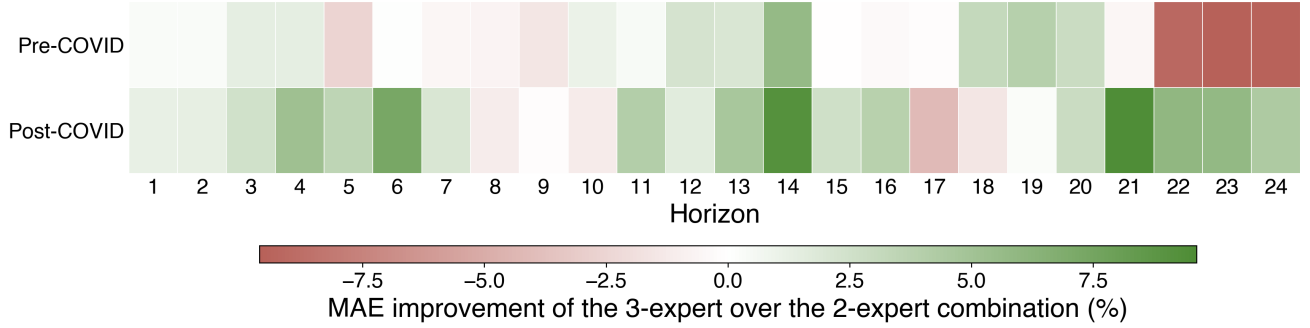
$$\bar{D}_t = \frac{1}{H} \sum_{h=1}^H D_t^{(h)} = \frac{1}{H} \sum_{h=1}^H \left(|\varepsilon_{t,\text{two-expert}}^{(h)}| - |\varepsilon_{t,\text{three-expert}}^{(h)}| \right),$$

with $H = 24$. The baseline regime for each window is chosen by the multiplier-bootstrap Ljung–Box diagnostic computed on that window.

For the pre-2020 window, the Ljung–Box test does not reject ($p = 0.301$), so the baseline uses

²⁶The corresponding weight paths at other horizons are reported in Appendix D.

Figure 5: Improvement from Adding the Micro Forecast, Relative to Two Experts



Notes: Cells report the percentage MAE improvement of the three-expert Fixed Share combination relative to the two-expert Fixed Share benchmark by forecast horizon $h = 1, \dots, 24$ and evaluation window. The three-expert combination uses the micro forecast, the macro forecast, and the univariate forecast; the two-expert benchmark uses only the macro and univariate forecasts. Positive values indicate lower MAE after adding the micro forecast; negative values indicate higher MAE. Numerical values are reported in Table 6.

Regime I. The test fails to reject the null of no average improvement over the two-expert benchmark, with a Regime I bootstrap p -value of 0.548. Adding the micro expert therefore does not improve on the two-expert forecast on average before 2020.

After 2020, microdata is valuable for forecasting. For the post-COVID window, the Ljung–Box test does not reject serial independence ($p = 0.709$), so the baseline uses Regime I. The test rejects decisively, with a bootstrap p -value of 0.001. The full three-expert combination therefore adds significantly to the two-expert benchmark on average after 2020.

This exercise moves beyond the previous uses of the scan test, for expert selection. Those tests showed that microdata improved upon the macro and univariate forecasts for at least some horizon and interval. Now, we have shown a stronger claim: the three-expert combined forecast, including microdata, improves upon the corresponding combination without microdata, on average after 2020.

5.5 Combined Three-Expert Forecast vs. a Single XGBoost Forecast

A natural alternative to the combined three-expert forecast is a single XGBoost forecast built from one pooled nonlinear regression. This pooled forecast uses the full feature set across the three experts: the 121 micro distributional statistics, the macro covariates (the rents CPI permutation), and lagged inflation, for 158 pre-lag predictors in total. The comparison isolates the value of using Fixed Share to combine experts, relative to a pooled XGBoost forecast based on the same information.

The single XGBoost forecast performs much worse before 2020, and moderately worse afterwards. Before 2020, the percent MAE gaps between the combined three-expert forecast and the single XGBoost forecast are +25.7 (on average for horizons 1-6), +8.2 (horizons 7-12), +20.1 (horizons 13-18), and +25.1 (horizons 19-24) percentage points, averaging a 19.8 percentage-point gap. In the

post-2020 window, the gaps narrow to +8.4, +3.2, +5.2, and -0.6 percentage points, averaging a 4.1 percentage-point gap. The combined three-expert forecast retains the edge through $h = 20$; the single XGBoost forecast pulls ahead only at the longest horizons ($h = 21$ to 24), by margins comparable to the three-expert forecast’s own lead at shorter horizons. The contrast highlights the advantages of the Fixed Share algorithm, which exploits the univariate forecast when it is useful, and adaptively switches to the micro forecast at the relevant time. The pre-2020 weakness of the single XGBoost forecast is consistent with the overfitting concern formalized in equation (6): when the microdata signal is limited, putting all predictors into one pooled XGBoost regression can raise estimation error rather than improve the forecast.

5.6 Robustness to Alternative Microdata Encodings

The baseline micro forecast incorporates three choices in encoding microdata: the 11-feature encoding, expenditure-weighted within-category statistics, and price quotes that include sales. We evaluate each choice against an alternative that appears to perform worse; the detailed results are reported in Appendix F.

Augmented encoding. We replace the 11-feature encoding with the full 29-feature augmented encoding, which adds the variance (with Bessel’s correction), the standardised third and fifth central moments and the excess kurtosis of the non-zero price-change distribution, the nine deciles of the time-since-last-change distribution, and five moments of the same distribution (Appendix F.1). The three-expert combined forecast is nearly unchanged: relative to the baseline, the augmented-encoding three-expert combined forecast’s MAE performance deteriorates by 2.8 percentage points after 2020 and 0.3 percentage points before 2020, averaging across forecast horizons. The standalone micro forecast’s MAE performance deteriorates more relative to the baseline: by 2.5 percentage points before 2020 and 7.3 percentage points after 2020. The deterioration in MAE performance is likely due to overfitting from the higher-dimensional encoding.

Unweighted statistics. Replacing expenditure-weighted statistics with equal-weighted statistics in the encoding (Appendix F.2) deteriorates the standalone micro forecast’s MAE performance relative to the baseline in both windows: by 1.6 percentage points in the pre-2020 window and by 3.3 percentage points in the post-2020 window, averaging across forecast horizons. The three-expert combined forecast is more stable: its MAE performance is essentially unchanged before 2020 and deteriorates by 0.9 percentage points after 2020.

Prices excluding sales. Dropping ONS-sale-flagged quotes (Appendix F.3) improves the standalone micro forecast’s MAE performance relative to the baseline by 2.3 percentage points in the pre-2020

window, but deteriorates that performance by 8.9 percentage points in the post-2020 window. The three-expert combined forecast is more stable: its MAE performance is essentially unchanged before 2020 and deteriorates by 2.6 percentage points after 2020, again leaving the baseline preferred. Sale-driven price changes therefore carry useful predictive signal during the post-COVID inflation surge, even though they add noise in stable times.

Sales-price imputation. Replacing temporary sale prices with the regular price inferred by the Nakamura and Steinsson (2008) filter, parameterisation A (Appendix F.4), deteriorates the standalone micro forecast’s MAE performance relative to the baseline by 8.9 percentage points in the pre-2020 window and 6.3 percentage points in the post-2020 window. In the pre-2020 window the imputation step is more harmful than dropping the flagged quotes outright, injecting noise of its own even when sales carry little signal; in the post-2020 window it is somewhat less harmful than dropping the quotes, though both alternatives still underperform the baseline. The three-expert combined forecast is less affected: its MAE performance is essentially unchanged before 2020 and deteriorates by 2.1 percentage points after 2020.

6 Opening the Black Box: Grouped Shapley Decompositions

Machine learning methods are sometimes criticized as a “black box” (e.g. Mullainathan and Ludwig, 2024). We propose to “open” the black box by investigating which features of the microdata matter for its forecasting power. In particular, we use a grouped Shapley decomposition to ask which micro variables account for the incremental value of the micro expert relative to the two-expert benchmark combining only the univariate and macro forecasts.

Construction. We partition the micro variables, computed within each COICOP1 category and then stacked across categories (except the monthly dummies, which are common across categories), into five groups:

1. the fraction of zero price changes;
2. the mean plus the middle three deciles of the price-change distribution;
3. the top three deciles of the price-change distribution;
4. the bottom three deciles of the price-change distribution;
5. monthly dummies.

Our choice of grouping is useful because it maps the forecasting exercise into objects that price-setting theory treats as economically meaningful. The first group captures the extensive margin of price adjustment, a central empirical moment in studies of price rigidity (Klenow and Kryvtsov, 2008; Nakamura and Steinsson, 2008; Blanco et al., 2024). The second group captures the center of the price-change distribution through the mean and middle deciles, which summarize typical repricing magnitudes. The third and fourth groups isolate the upper and lower tails of the price-change distribution, allowing the decomposition to distinguish unusually large price increases from unusually large price decreases or discounts. These tail groups are natural objects for theories that emphasize asymmetry, state dependence, and granular shocks (Ball and Mankiw, 1995; Midrigan, 2011; Vavra, 2014; Alvarez, Le Bihan and Lippi, 2016; Alvarez-Blaser et al., 2025). The fifth and last group consists of monthly dummies, which absorb the seasonal structure in inflation. The benchmarks already account for seasonality: the macro expert includes month indicators, while the univariate SARIMA benchmark explicitly allows for monthly seasonal dynamics. The monthly-dummy group therefore should not be read as adding a seasonal adjustment that is missing from the competing forecasts. The relevant comparison is whether seasonal timing improves forecasts when it is combined with micro-price information, beyond what the macro and univariate forecasts already extract from seasonality.

Grouped Shapley values reveal the marginal contribution of each group to the overall forecasting power of the microdata, in a way that accounts for the interactions and nonlinearity of the machine learning algorithm. We calculate grouped Shapley values as follows. For each horizon h and nonempty coalition S of grouped micro variables, we estimate the micro expert using only the groups in S and combine the resulting forecast with the macro and univariate experts using the same Fixed Share procedure as in Subsection 4.3. Let $\text{MAE}_{S,h}$ denote the mean absolute error of this three-expert forecast, and let $\text{MAE}_{\text{baseline},h}$ denote the mean absolute error of the two-expert alternative. We define the coalition value as the signed percentage reduction in mean absolute error,

$$v_h(S) = 100 \times \frac{\text{MAE}_{\text{baseline},h} - \text{MAE}_{S,h}}{\text{MAE}_{\text{baseline},h}}.$$

For the empty coalition, we set $v_h(\emptyset) = 0$: it corresponds to the benchmark without any micro inputs.

For each horizon, we evaluate all 2^5 coalitions of these groups, compute the corresponding values $v_h(S)$, and then assign each group its Shapley value, that is, its average marginal contribution across all possible orders in which the groups can enter the model. Formally, if $\mathcal{G}$ denotes the set of the five groups and $g \in \mathcal{G}$ is one particular group, its Shapley value at horizon h is

$$\phi_{g,h} = \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|! (|\mathcal{G}| - |S| - 1)!}{|\mathcal{G}|!} \left[v_h(S \cup \{g\}) - v_h(S) \right].$$

This expression averages the marginal contribution of group g over all possible subsets S that exclude it, with the standard combinatorial weights. Positive Shapley values indicate that a group lowers fore-

cast errors relative to the two-expert online-learning benchmark on average, while negative values indicate that the group worsens forecast performance. The unit of $\phi_{g,h}$ is percent decrease in MAE compared to the two-expert benchmark MAE. For example, $\phi_{g,h} = 1$ means that, at horizon h , group g accounts for an average MAE improvement equal to one percent of the two-expert benchmark MAE, averaged over all coalitions of the other groups. Because $v_h(\emptyset) = 0$, the Shapley values add up exactly: the group contributions sum to $v_h(\mathcal{G})$, the full three-expert combination’s percentage MAE improvement over the two-expert benchmark.

Computation. Re-tuning the micro expert for every coalition would multiply the full hyperparameter search by 2^5 . We avoid this by freezing the hyperparameters: each coalition’s micro expert is re-trained at every origin on its feature subset, but reuses the hyperparameter sequence that the full micro expert already selected in real time (Subsection 4.2). Freezing preserves the no-look-ahead structure of the pipeline, because each reused hyperparameter was selected from information available at its origin, and it reduces the cost of the full decomposition to roughly 1.3 times one training run of the micro expert. Algorithms 10 and 11 in Appendix E give the complete procedure.

Results. We apply the grouped Shapley decomposition to measure which groups of micro variables account for the MAE gain from adding the micro expert to the two-expert benchmark that combines only the univariate and macro forecasts.

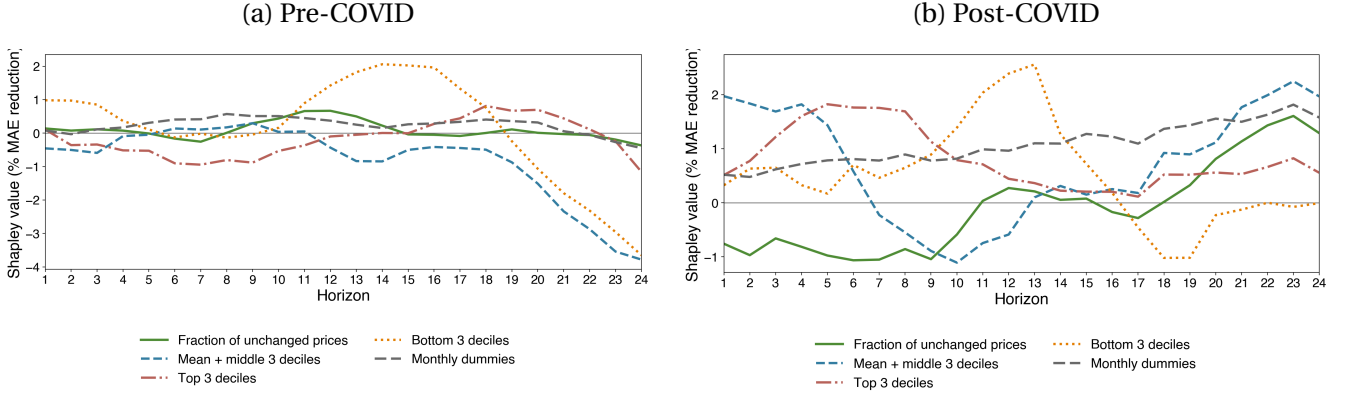
Table 1: Grouped Shapley Values for the Micro Expert, Averaged over 24 Horizons

Feature group	Pre-2020	Post-2020
Fraction of zero price changes	0.0859	-0.1215
Mean + middle 3 deciles	-0.7840	0.6796
Top 3 deciles	-0.1960	0.7720
Bottom 3 deciles	0.2602	0.5342
Monthly dummies	0.2360	1.0435

Notes: Entries are grouped Shapley contributions of each feature group to the MAE gain from adding the micro expert to the two-expert online-learning benchmark, which combines the univariate and macro forecasts. Values are in percent of benchmark MAE and are averaged over horizons $h = 1, \dots, 24$ within each evaluation window. Positive entries reduce MAE; negative entries increase it.

Table 1 reports Shapley values averaged over horizons $h = 1, \dots, 24$. In the pre-COVID window, every contribution is small: the mean plus middle-three-deciles group is the most negative at -0.78 , the top-three-deciles group is mildly negative at -0.20 , and the bottom three deciles, monthly dummies, and the fraction of zero price changes are mildly positive, at 0.26 , 0.24 , and 0.09 . Thus, before COVID, no group delivers a large positive contribution. This mirrors the broader finding that microdata add little forecasting power in a stable inflation environment.

Figure 6: Grouped Shapley Values by Horizon



Notes: The figure reports 5-group Shapley contributions for the incremental value of the micro forecast in the three-expert combined forecast relative to the two-expert benchmark that combines the univariate and macro forecasts. Values are smoothed across adjacent horizons using a 6-horizon moving average. Positive values indicate reductions in MAE from the corresponding feature group; negative values indicate increases in MAE. The horizontal line marks a zero contribution.

In the post-COVID window, the positive contributions come from the mean plus middle three deciles at 0.68, the top three deciles at 0.77, the bottom three deciles at 0.53, and monthly dummies at 1.04.²⁷ The contribution from the group for the fraction of zero price changes is essentially zero on average, at -0.12 . The five contributions sum to 2.91 percent, the three-expert combination's average post-2020 MAE gain over the two-expert benchmark (Table 6).

Figure 6 reports the same decomposition at the horizon level: at each horizon h , the plotted value is the average of the per-horizon grouped Shapley values over a six-horizon local window, using the available adjacent horizons. Pre-COVID, the contributions are small at short and medium horizons. The lower tail is positive only at short and medium horizons and turns negative later, while the losses concentrate at the longest horizons, where the mean plus middle-three-deciles group and the bottom three deciles become sharply more negative, both falling below -3 by $h = 24$. Post-COVID, the bottom three deciles peak at medium horizons, around $h = 11$ to 14 , and remain positive through roughly $h = 16$ before turning negative at longer horizons; the top three deciles contribute most at short and medium horizons before fading near the end of the horizon range; the mean plus middle-three-deciles group is strongest at the shortest horizons and again from about $h = 20$ onward; and monthly dummies are positive at every horizon and grow toward the long end. The fraction of zero price changes is negative at short horizons and positive at the longest ones, netting out to essentially zero on average.

²⁷The positive Shapley value for monthly dummies does not mean that the benchmarks lack seasonal controls. It means that, within the micro forecast, seasonal timing adds predictive content when combined with micro-price statistics.

7 Implications for Models of Inflation

Overall, our results suggest that microdata matters for aggregate inflation dynamics, but only during periods of high and volatile inflation such as after 2020. We briefly discuss implications of our findings for models of price setting.

Consider first a benchmark. Let μ_t denote the cross-sectional distribution of firms' pricing states relevant for future inflation (e.g., price gaps in standard menu-cost models), let $u_{-\infty:t}$ denote the history of aggregate shocks through date t . Under a stable transition law, this distributional state can be written as

$$\mu_t = \Gamma(u_{-\infty:t}).$$

If the history of observed macro variables recovers $u_{-\infty:t}$ and Γ is consistently estimable from aggregate data, then μ_t can be recovered from macro information alone. In this benchmark, observing microdata separately cannot improve the optimal forecast of future inflation.

The familiar first-order result is a local version of this benchmark. In standard menu-cost models, a first-order approximation is accurate when aggregate shocks are small (Auclert et al., 2024). Aggregate inflation then admits a linear vector moving-average representation,

$$\pi_t = \mu_\pi + \sum_{l=0}^{\infty} \Psi_l u_{t-l},$$

where μ_π denotes steady-state inflation, u_t is the vector of aggregate innovations, and $\{\Psi_l\}$ are impulse response coefficients. If macro variables recover current and past aggregate shocks, this representation implies that microdata do not add predictive content.

Microdata can improve feasible forecasts when this benchmark fails in one of two ways. First, the observed macro variables may not recover the shocks $u_{-\infty:t}$ that move the distributional state. A leading case is granular shocks. Disturbances to large firms, products, or sectors need not wash out in a finite economy and can therefore affect aggregate inflation. Micro-price distributions may contain information about these disturbances beyond what appears in standard aggregates (Alvarez-Blaser et al., 2025). Dispersion, skewness, or sectoral relative-price shocks can likewise change the distributional state and future inflation without being summarized by the observed macro variables (Ball and Mankiw, 1995; Vavra, 2014).

Second, the mapping Γ from the shock history to the distributional state may not be consistently estimable. Changes in policy, regimes, structural parameters, or the initial cross-sectional state can make this mapping unstable, so that it cannot be recovered from past data. Even when the mapping is stable, the combination of the inversion from macro variables to shocks, the transition law for the distributional state, and the law of motion for inflation may be difficult to estimate in a short sample, especially in a large-shock region that is rarely observed before 2020. Statistics of observed micro-

level price changes can provide additional information about the underlying distributional state and thereby improve a feasible forecast.

Periods of high and volatile inflation such as after 2020 can lead to both of these failures. The disturbances key to those episodes, for example shocks to supply chains, are often granular in origin, so the observed macro variables miss a larger share of the relevant shocks. Such episodes also coincide with changes in policy and in the structure of the economy, which can make the mapping from the shock history to the distributional state unstable, while the data needed to re-estimate it remain scarce. Price-setting nonlinearity amplifies both failures: in menu-cost models, the frequency of price adjustment and inflation pass-through respond nonlinearly to shock size, so inflation becomes more sensitive to the distributional state after large shocks (Blanco et al., 2024). The observed distribution of price changes can supply part of the information lost to unrecovered shocks or to an unstable or poorly estimated mapping. Consistent with this discussion, the incremental value of the micro expert appears after the large inflationary shocks shown in Figure 7, rather than during the stable 2015–2019 period.

8 Conclusion

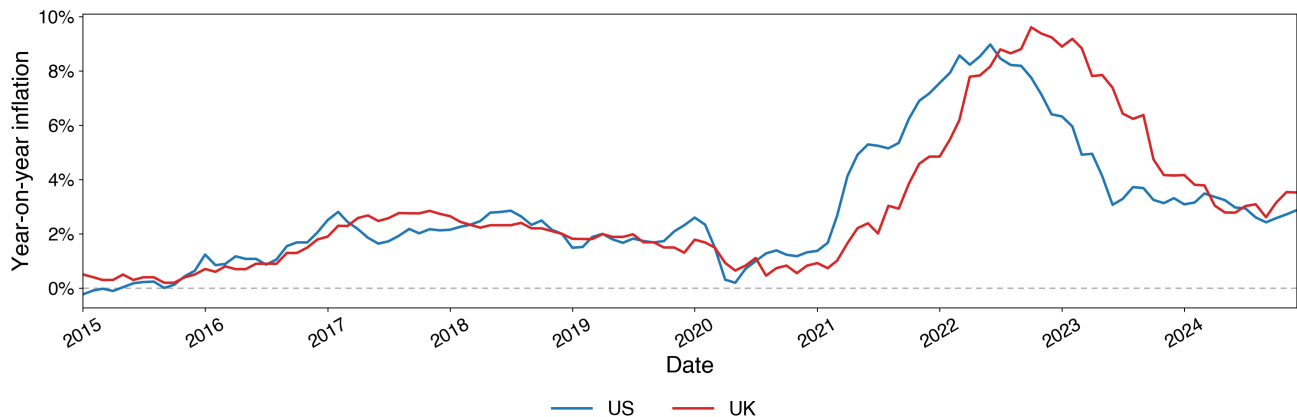
This paper asks whether microdata matters for aggregate inflation dynamics. Standard models, as well as the non-stationary dynamics of inflation, suggest that microdata might be important only occasionally. Therefore we start by developing a new scan test, which assesses whether one forecast outperforms another at an unknown horizon and time period. This scan test can be used to detect the occasional forecasting power of microdata for inflation, but also used more broadly. We then develop an adaptive machine learning based pipeline to take advantage of this occasional forecasting power. There are three steps, which we apply to microdata underlying the UK consumer price index. First, we encode the distribution of price changes into a high dimensional vector of statistics. Second, we combine this vector with a machine learning algorithm, gradient boosted trees, tuned by a cached real-time procedure, in order to forecast inflation. Third, we combine the resulting forecast with the benchmarks through the Fixed Share algorithm, which raises the weight on microdata when it is useful for forecasting and lowers it otherwise. Microdata enter the combination only after the scan test detects localized outperformance beyond the existing experts.

There are four main empirical results. First, the microdata forecast outperforms the univariate benchmark only after 2020; before 2020 the univariate benchmark performs better at every horizon. Second, the scan tests imply that the Fixed Share combined forecast should include all three modalities: micro, macro and univariate forecasts. Third, the combined forecast performs comparably to the univariate benchmark before 2020 and improves on it for all 24 horizons. Fourth, the value of micro-

data for the combined forecast materializes after 2020. We also “open the black box” using grouped Shapley values to study why microdata help forecast inflation. This decomposition suggests that the positive contributions come from the center and tails of the price-change distribution together with monthly dummies, while the extensive margin of price adjustment contributes little.

Our findings connect to models of inflation dynamics. The finding that microdata improve forecasts only during high and volatile inflation is consistent with benchmark models of price setting, in which microdata carry additional predictive content when shocks are large. Several directions for future work emerge naturally. The same pipeline of distributional encoding, scan testing, and adaptive combination could be applied to forecast other macroeconomic aggregates, such as GDP growth or employment, using firm-level or household-level microdata. Extending the analysis to other countries would help assess whether the forecasting value of microdata during high and volatile inflation is a general feature of inflation dynamics.

Figure 7: Headline CPI Inflation in the United States and United Kingdom, 2015–2024



Notes: Year-on-year headline CPI inflation for the United States (blue) and United Kingdom (red), January 2015 to December 2024. Data source: Federal Reserve Economic Data (FRED).

Tables

Table 2: COICOP1 Cell Counts (Observations per Cell)

Treatment	p5	p10	p25	p50	p75	p90	p95
<i>Panel A: All price changes</i>							
Drops ONS sales	316	1,297	2,731	6,809	9,526	15,747	20,503
Includes sales	343	1,310	2,800	7,288	11,111	17,609	21,676
<i>Panel B: Non-zero price changes only</i>							
Drops ONS sales	23	55	211	423	739	1,600	2,778
Includes sales	45	89	264	788	1,597	2,813	4,328

Notes: This table reports percentiles of the distribution of cell counts across all (date, COICOP1 code) cells, separately for all price observations (Panel A) and for non-zero price changes only (Panel B). The “Drops ONS sales” panel sets ONS-sale-flagged prices to missing, sets the following observation’s price change to missing, reruns short-spell deletion, and recomputes time-since-change. NS-filtered datasets have cell counts equal to the “includes sales” figures, since observations are adjusted rather than dropped.

Table 3: Univariate Forecast Mean Absolute Error Before and After 2020

h	Pre 2020 MAE	Post 2020 MAE
1	0.1364	0.3915
2	0.1373	0.3842
3	0.1424	0.3985
4	0.1512	0.3957
5	0.1476	0.3975
6	0.1503	0.4268
7	0.1479	0.4764
8	0.1532	0.4843
9	0.1474	0.5070
10	0.1547	0.4504
11	0.1479	0.4883
12	0.1556	0.4235
13	0.1825	0.4753
14	0.1820	0.4878
15	0.1651	0.4893
16	0.1655	0.4768
17	0.1605	0.4722
18	0.1694	0.4736
19	0.1792	0.4720
20	0.1670	0.4712
21	0.1656	0.6530
22	0.1693	0.6208
23	0.1686	0.7061
24	0.1778	0.5808

Notes: The table reports mean absolute error (MAE) for the rolling-window univariate forecast at forecast horizons $h = 1, \dots, 24$. The pre-2020 evaluation window is 2015–2019 and the post-2020 evaluation window is 2020–2024. Errors are computed against the unsmoothed monthly inflation target.

Table 4: Micro percent MAE improvement over SARIMA, pre- and post-COVID.

h	Pre-COVID % impr.	Post-COVID % impr.
1	-35.67	8.49
2	-28.92	4.46
3	-29.10	2.14
4	-30.97	9.67
5	-32.52	6.73
6	-19.51	20.31
7	-17.33	13.00
8	-17.05	8.63
9	-27.49	14.63
10	-18.19	10.43
11	-19.84	16.04
12	-12.81	1.79
13	-10.32	12.95
14	-0.13	15.28
15	-22.62	14.27
16	-21.06	11.03
17	-31.47	-3.33
18	-18.48	0.82
19	-28.04	9.07
20	-26.65	9.34
21	-32.13	35.42
22	-44.26	32.27
23	-36.87	39.33
24	-27.88	26.46

Table 5: Fixed-Share Combination Mean Absolute Error Before and After 2020

h	Pre 2020				Post 2020			
	FS	Micro	Macro	SARIMA	FS	Micro	Macro	SARIMA
1	0.1385	0.1850	0.1708	0.1364	0.3444	0.3583	0.3607	0.3915
2	0.1348	0.1770	0.1678	0.1373	0.3548	0.3670	0.3962	0.3842
3	0.1363	0.1838	0.1629	0.1424	0.3690	0.3899	0.4137	0.3985
4	0.1502	0.1980	0.1679	0.1512	0.3436	0.3575	0.4080	0.3957
5	0.1478	0.1956	0.1559	0.1476	0.3429	0.3707	0.4035	0.3975
6	0.1467	0.1796	0.1681	0.1503	0.3505	0.3401	0.4221	0.4268
7	0.1481	0.1736	0.1631	0.1479	0.4019	0.4145	0.4072	0.4764
8	0.1557	0.1793	0.1835	0.1532	0.4130	0.4425	0.4040	0.4843
9	0.1523	0.1879	0.1843	0.1474	0.4145	0.4328	0.4079	0.5070
10	0.1562	0.1829	0.1876	0.1547	0.3819	0.4035	0.4050	0.4504
11	0.1507	0.1773	0.1763	0.1479	0.4107	0.4100	0.4622	0.4883
12	0.1561	0.1756	0.1878	0.1556	0.4116	0.4160	0.4656	0.4235
13	0.1783	0.2014	0.1921	0.1825	0.4170	0.4137	0.4582	0.4753
14	0.1741	0.1822	0.2004	0.1820	0.4122	0.4132	0.4755	0.4878
15	0.1761	0.2024	0.2042	0.1651	0.4213	0.4195	0.4340	0.4893
16	0.1763	0.2003	0.2079	0.1655	0.4266	0.4242	0.4569	0.4768
17	0.1686	0.2110	0.1928	0.1605	0.4332	0.4879	0.4360	0.4722
18	0.1668	0.2007	0.1868	0.1694	0.4221	0.4697	0.4303	0.4736
19	0.1771	0.2294	0.2034	0.1792	0.4077	0.4292	0.4244	0.4720
20	0.1762	0.2115	0.2062	0.1670	0.3964	0.4272	0.4172	0.4712
21	0.1725	0.2188	0.1988	0.1656	0.4569	0.4217	0.4339	0.6530
22	0.1831	0.2442	0.1877	0.1693	0.4332	0.4205	0.4090	0.6208
23	0.1754	0.2307	0.1782	0.1686	0.4569	0.4284	0.4105	0.7061
24	0.1719	0.2273	0.1907	0.1778	0.4267	0.4271	0.4113	0.5808

Notes: Mean absolute error of the patient fixed-share combination (FS, $\alpha = 0.02$, $\eta = 0.5$, $\rho = 0$, grid=fine) against its experts (Micro, Macro, SARIMA), by forecast horizon. The pre-2020 window is 2015–2019 and the post-2020 window is 2020–2024.

Table 6: Adding the Micro Expert: Three- vs Two-Expert Fixed Share

h	Pre 2020			Post 2020		
	3-expert	2-expert	Δ (%)	3-expert	2-expert	Δ (%)
1	0.1385	0.1390	0.32	0.3444	0.3487	1.24
2	0.1348	0.1352	0.33	0.3548	0.3596	1.32
3	0.1363	0.1382	1.39	0.3690	0.3786	2.54
4	0.1502	0.1523	1.37	0.3436	0.3624	5.19
5	0.1478	0.1440	-2.63	0.3429	0.3553	3.48
6	0.1467	0.1469	0.12	0.3505	0.3782	7.34
7	0.1481	0.1472	-0.56	0.4019	0.4104	2.08
8	0.1557	0.1546	-0.70	0.4130	0.4082	-1.18
9	0.1523	0.1500	-1.51	0.4145	0.4139	-0.15
10	0.1562	0.1579	1.12	0.3819	0.3773	-1.22
11	0.1507	0.1513	0.40	0.4107	0.4283	4.11
12	0.1561	0.1598	2.30	0.4116	0.4183	1.59
13	0.1783	0.1820	2.04	0.4170	0.4374	4.67
14	0.1741	0.1848	5.78	0.4122	0.4539	9.18
15	0.1761	0.1760	-0.04	0.4213	0.4325	2.58
16	0.1763	0.1757	-0.35	0.4266	0.4439	3.89
17	0.1686	0.1683	-0.16	0.4332	0.4158	-4.18
18	0.1668	0.1723	3.20	0.4221	0.4156	-1.55
19	0.1771	0.1844	3.95	0.4077	0.4087	0.26
20	0.1762	0.1814	2.88	0.3964	0.4083	2.91
21	0.1725	0.1715	-0.59	0.4569	0.5050	9.53
22	0.1831	0.1678	-9.13	0.4332	0.4607	5.97
23	0.1754	0.1600	-9.62	0.4569	0.4850	5.79
24	0.1719	0.1570	-9.48	0.4267	0.4463	4.39

Notes: Mean absolute error of the 3-expert patient fixed-share combination (micro + macro + SARIMA) and the 2-expert macro + SARIMA combination ($\alpha = 0.02$, $\eta = 0.5$, $\rho = 0$, grid=fine), by horizon. Δ is the percent MAE improvement from adding the micro expert (positive = the 3-expert combination is better). Pre-2020 = 2015–2019, post-2020 = 2020–2024.

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Appendix for “Forecasting Inflation with Microdata: An Adaptive Machine Learning Approach”

This appendix collects supporting material for the main text. Appendix A details the construction and cleaning of the micro-price and macro datasets. Appendix B develops the econometric theory of the scan tests, and Appendix C reports the simulation studies behind their finite-sample calibration, including the choice of the partial-studentization exponent and a comparison with the Giacomini–Rossi fluctuation test. Appendix D contains additional figures, and Appendix E collects the notation and pseudocode for all algorithms. Appendix F examines robustness to alternative microdata encodings, aggregation schemes, and data filters, and Appendix G presents additional empirical results.

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A Data Appendix

This appendix gathers implementation details on data construction that are useful for replication but are not central to the main exposition. We organise the material into six subsections: cleaning of the ONS micro-price panel (Appendix A.1), aggregation and CPI replication (Appendix A.2), benchmarking of the price-change distribution (Appendix A.3), treatment of temporary sales (Appendix A.4), macro predictors (Appendix A.5), and exact formulas for distributional statistics (Appendix A.6).

A.1 Price Quotes Data Cleaning

A.1.1 Cleaning

We follow Adam and Weber (2023) and apply the steps below in order.

1. *Sample and region filters.* We drop January 1996, since it is a partial initial month and cannot be used to compute lagged price changes. Since the monthly price-change variable requires a valid consecutive-month lagged price, the encoded micro-feature panel begins in March 1996 rather than January 1996. Thus, while the raw quote panel spans January 1996 through December 2024, the encoded monthly micro-feature panel spans March 1996 through December 2024. We also remove observations with invalid region code 0 and exclude Region 1 (“Catalogue collections”), which covers items collected centrally, such as shelter, university tuition fees, and rail fares, for which only item-level indices are published; following Blanco et al. (2024), we exclude these observations. Region 13 (“NI”) is Northern Ireland; these observations (1,357,662 rows, 4.31% of the cleaned panel) are retained.
2. *Validity filter.* Retain observations with valid current prices. For observations that are not flagged as non-comparable substitutions, we also require the base price to be valid according to the Office for National Statistics codes. For non-comparable substitutions, identified by the N or Z flags, we retain the observation for price-change encoding but set its base price to missing before any base-price-dependent calculation.
3. *Duplicate removal.* If a product identifier appears more than once in any given period, the entire series is dropped. This deliberately conservative rule removes roughly 6.6% of otherwise valid observations per quarter (ranging between 5.1% and 8.3%). Most colliding cells show distinct prices, consistent with the ONS caveat that distinct outlets can share a shop code, so a per-month resolution rule would have to guess which quote to keep; we drop the series instead.
4. *Price rounding.* All prices are rounded to two decimal places.

5. *Implausible prices.* Zero prices, zero base prices, and prices below £0.001 are removed.
6. *Substitution spells.* Comparable substitutions, identified by the C or X flags, non-comparable substitutions, identified by the N or Z flags, and permanent pack-size, weight, or volume changes, identified by the W flag, start new product spells. Therefore the substitution month is treated as the first observation of a new spell: its contemporaneous price change is missing, while the following month's price change is computed relative to the new product whenever the next observation is consecutive.
7. *Short-spell removal.* Rather than splitting product series at gaps, we store prices in a time $\times$ product matrix where missing months are represented as NaN; gaps therefore propagate naturally as missing observations. Single isolated observations (with NaN on both sides) are set to missing, and any product column with fewer than two non-missing observations is dropped. This rule conditions on the adjacent month $t+1$: whether an isolated month- t quote is kept can depend on the presence of a quote in month $t+1$, so the cleaned month- t panel embeds one month of look-ahead relative to a strictly real-time vintage.²⁸

At the initial cleaning checkpoint, the validity, duplicate, price, and flag-based filters reduce the raw ONS file from 42,013,725 to 33,053,579 price-quote rows, a reduction of 21.33%. After the remaining baseline filters, including region exclusion, substitution spell splitting, short-spell deletion, COICOP1 balancing, and final price-change recomputation, the final baseline panel contains 31,001,584 non-missing price observations. Relative to the raw ONS file, this corresponds to an effective removal share of 26.21%. The final number of valid non-missing price-change observations used to construct the encoded micro features is 27,847,570, corresponding to 33.72% fewer observations than the raw price-quote count.

A.2 Aggregation and CPI Replication

Aggregation proceeds in four steps.

²⁸Adam and Weber (2023) instead split each product series into new sub-series at any gap exceeding one month. Their stricter rule removes 3.3% of price-quote observations and 23% of product series.

A.2.1 Step 1 – Stratum index

Within each item i and stratum cell k , the price quotes are aggregated into a stratum-level price index using a shop-weighted geometric mean of log price relatives to the February base:

$$\text{stratum index}_{t,k} = 100 \times \exp \left(\frac{\sum_{j \in k} w_j^{\text{shop}} \cdot \log(p_{jt}/p_{j0})}{\sum_{j \in k} w_j^{\text{shop}}} \right), \quad (\text{A.1})$$

where j runs over all products in stratum k , p_{j0} is the February base price carried in the ONS base-price field, and w_j^{shop} is the shop weight. Observations with missing base prices, including retained N/Z non-comparable substitution rows, are excluded from this Jevons calculation. They are retained only for the construction of product-level price-change features. Shop weights are set to NaN wherever the corresponding price is missing, so the denominator always sums only over observed products. If a stratum cell code is zero (no sub-item stratification), a uniform stratum weight of 100 is applied.

A.2.2 Step 2 – Item index

Stratum indices are combined into the item index by a weighted arithmetic mean using stratum weights w_k^{stratum} :

$$\text{item index}_{t,i} = \frac{\sum_k w_{k,t}^{\text{stratum}} \cdot \text{stratum index}_{t,k}}{\sum_k w_{k,t}^{\text{stratum}}}. \quad (\text{A.2})$$

Each price quote j in stratum k , item i , COICOP1 category c , and month t receives an implicit product weight

$$\omega_{j,t} = \underbrace{\frac{w_j^{\text{shop}}}{\sum_{j' \in k} w_{j'}^{\text{shop}}}}_{\text{share within stratum}} \times \underbrace{\frac{w_{k,t}^{\text{stratum}}}{\sum_{k' \in i} w_{k',t}^{\text{stratum}}}}_{\text{share within item}} \times \underbrace{\frac{w_{i,t}^{\text{item}}}{\sum_{i' \in c} w_{i',t}^{\text{item}}}}_{\text{share within COICOP1}}. \quad (\text{A.3})$$

The first two factors give the price quote's proportional contribution to the item index; multiplying by the third factor extends $\omega_{j,t}$ to the quote's proportional contribution within its COICOP1 category in month t , so that $\sum_{j \in c,t} \omega_{j,t} = 1$ for each category-month cell. This three-factor weight is the basis for all expenditure-weighted price-change features used in the forecasting exercise (Appendix A.6). Single-stratum items receive share within item = 1; items whose stratum weights are all zero in a month are likewise assigned a uniform within-item weight $1/|\{k' : k' \in i\}|$.

A.2.3 Step 3 – Quality screen

Each reconstructed item index is compared to the corresponding official ONS item series over their joint sample. An item is deemed *suitable* if the RMSE of the log-level difference between the two series is below 2% and there are no gaps in the reconstructed series. Of the 1,318 reconstructed items, 944 are suitable under this screen over the full joint sample.

A.2.4 Step 4 – CPI aggregation and chaining

The reconstructed headline index is a weighted average of suitable item indices, with item weights normalised to sum to one in each period:

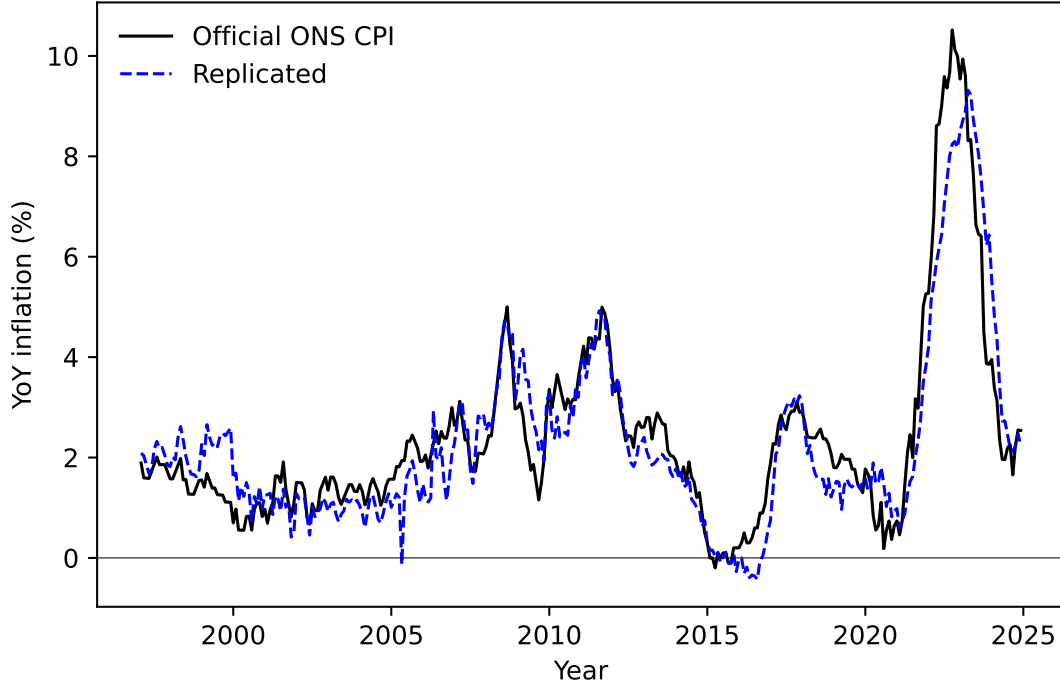
$$\text{CPI}_t^{\text{unchained}} = \sum_i \tilde{w}_{i,t}^{\text{item}} \cdot \text{item index}_{t,i}, \quad \tilde{w}_{i,t}^{\text{item}} = \frac{w_{i,t}^{\text{item}}}{\sum_{i'} w_{i',t}^{\text{item}}}. \quad (\text{A.4})$$

Because each annual index is expressed relative to the February base of that year, the unchained series restarts at 100 each February. The resulting chained index tracks the official ONS CPI closely over 1996–2016. The post-2016 extension is somewhat noisier, likely reflecting major ONS weight revisions introduced in 2017.

To assess the quality of the aggregation more transparently, we compare the inflation rate implied by our reconstructed CPI with the corresponding official ONS inflation series, for the baseline panel with all posted prices.

Figure A.1 plots the corresponding year-on-year inflation series. The replicated series closely tracks the official ONS benchmark, indicating that the sequence of cleaning, weighting, item-index construction, and chaining reproduces the aggregate inflation dynamics well. This figure therefore serves as an important validation exercise for the data construction pipeline: it shows that the resulting micro-price panel remains consistent with the official CPI at the aggregate level.

Figure A.1: Official ONS Inflation and Replicated CPI Inflation: All Posted Prices



Notes: The figure compares official UK ONS non-seasonally-adjusted year-on-year CPI inflation with the rate implied by the reconstructed chained index built from the baseline micro-price panel using all posted prices. The sample runs from February 1997 to December 2024. The solid black line denotes official CPI inflation and the dashed blue line denotes the replicated series. Over the joint sample, the root mean squared difference is 0.76 percentage points and the correlation is 0.92.

A.3 Benchmarking the Price-Change Distribution

To validate the pipeline we compute moments of $|\Delta p_t|$ following the protocol of Blanco et al. (2024). Each non-zero price change is standardised at the item level by its weighted within-item mean and standard deviation, using the CPI product weight $\omega_{j,t}$ defined in Appendix A.2. The top and bottom 2% of the standardised distribution are trimmed, zero price changes are excluded, and the reported mean and standard deviation are weighted averages of $|\Delta p|$ in original (proportion) units, while the reported kurtosis is the ω -weighted raw Pearson kurtosis of the standardised price change. These aggregate statistics are not used in estimation; they serve as a check that the cleaning, weighting, and aggregation are correctly implemented.

Table A.1 reports the resulting moments across five sales treatments and lists published estimates from the literature.

The most direct comparison is the UK row of Blanco et al. (2024), who use the same ONS micro-price data, weight observations by item-level CPI expenditure weights, standardise at the item level, and trim the top and bottom 2% of the standardised distribution. Our NS Filter A and NS Filter B rows,

Table A.1: Moments of $|\Delta p|$ across Sales Treatments and Literature (2% Trim)

	Mean $ \Delta p $	Std dev $ \Delta p $	Kurtosis
<i>This paper</i>			
All prices, no trim	0.157	0.177	7.75
All prices (2% trim)	0.141	0.150	2.90
ONS sales dropped	0.088	0.099	3.23
NS Filter A	0.115	0.138	3.24
NS Filter B	0.127	0.145	3.13
<i>Literature</i>			
Klenow and Kryvtsov (2008)	0.113	–	–
Nakamura and Steinsson (2008)	0.085	–	–
Blanco et al. (2024)	0.110	–	4.02
Karadi and Reiff (2019)	0.099	–	3.98
Blanco et al. (2024) (UK)	0.153	0.205	3.81

Notes: All “This paper” moments are computed on the baseline cleaned ONS micro-price panel using the CPI product weight $\omega_{j,t}$ defined in Appendix A.2. Mean and standard deviation of $|\Delta p|$ are reported in original (proportion) units; kurtosis is the raw (Pearson) kurtosis of price changes after item-level standardisation, matching the protocol of Blanco et al. (2024). The 2% trim drops the top and bottom 2% of the item-standardised distribution. For the trimmed rows, the baseline (with sales) uses all valid price quotes; the “ONS sales dropped” row sets ONS S-flagged observations to missing; “NS Filter A” and “NS Filter B” apply the Nakamura and Steinsson (2008) sale filter with the parameters of Table A.2. Nakamura and Steinsson (2008) report the median $|\Delta p|$ rather than the mean. The literature values use different datasets, different sale-handling, and (in some cases) different standardisation rules; see Appendix A.3 for a discussion of comparability.

which remove temporary sales by the Nakamura and Steinsson (2008) filter, are the closest like-for-like analogues: they imply a mean $|\Delta p|$ of 0.115–0.127 and kurtosis of 3.13–3.24, bracketing the Blanco et al. (2024) UK values of 0.153 and 3.81. The remaining gap reflects differences in the sale-detection rule (the V-shape filter used in Blanco et al. (2024) is less aggressive than NS Filter A), differences in the item-coverage and base-price cleaning, and a slightly different sample period.

The remaining literature rows are listed for context only. Klenow and Kryvtsov (2008), Nakamura and Steinsson (2008), the US row of Blanco et al. (2024), and Karadi and Reiff (2019) use US scanner or CPI microdata under distinct sample-construction, weighting, and standardisation rules, and are not directly comparable to our ONS replication. The kurtosis values in the literature panel are raw Pearson kurtosis as reported in those papers, so a value near 3 indicates a near-normal standardised distribution and a value near 1 a near two-point distribution in the sense of Golosov and Lucas (2007).

A.4 Treatment of Sales

We consider three sale treatments. The first is an ONS-sale-dropped panel. For this panel, observations with the ONS S flag are not used as valid prices: their price, log price, and contemporaneous price change are set to missing. Since the price change in the month after a sale depends on the sale

price as the lagged price, the following observation's price change is also set to missing within each product series. After this operation, we rerun the short-spell deletion rule, setting isolated observations to missing and dropping product series with fewer than two valid prices. Finally, we recompute the time-since-change variable from the resulting filtered price-change series.

The ONS S flag may incompletely capture temporary price reductions. We therefore also construct two sale-adjusted panels, following Nakamura and Steinsson (2008). The filter operates product by product on the price level series. It is governed by three integer parameters: a short rebound window JJ , a minimum number of distinct price points KK , and a longer look-ahead window LL . At each period t with observed price p_t , a *regular price* r_t is assigned via the following ordered rules:

1. **Continuity.** $p_t = r_{t-1}$: set $r_t = r_{t-1}$.
2. **Price increase.** $p_t > r_{t-1}$: set $r_t = p_t$.
3. **Drop with rebound.** $p_t < r_{t-1}$ and $p_{t+s} = r_{t-1}$ for some $s \leq JJ$: classify period t as a sale, set $r_t = r_{t-1}$.
4. **Drop, no rebound, sufficient variation.** If the price does not rebound within JJ periods but the following LL observations contain at least KK distinct price values, treat the drop as permanent and set $r_t = p_t$.
5. **Drop, no rebound, modal-price rule.** If the maximum price in the LL -period window reappears later in the series, adopt it as the regular price.
6. **Default.** Set $r_t = p_t$.

The output is the adjusted series $\{r_t\}$ and a binary indicator equal to one when $r_t \neq p_t$. We identify sales using the following two parameterisations:

Table A.2: Nakamura–Steinsson Filter Parameters

	Rebound window JJ	Min. distinct prices KK	Look-ahead LL
Filter A (baseline)	3	3	3
Filter B (conservative)	1	1	3

Notes: Filter A is the baseline sales-filter parameterisation used in the robustness exercises. Filter B is a more conservative parameterisation. The columns report the rebound window, minimum number of distinct prices in a spell, and look-ahead horizon used by the Nakamura–Steinsson sale-filter algorithm. For Filter B the look-ahead parameter is inert: with $KK = 1$ the modal-price branch that uses LL is never reached, so Filter B produces identical sale flags for any value of LL (the canonical parameterisation sets $L = 1$).

A.5 Macro Predictors

Table A.3 lists the baseline macroeconomic and financial predictors used in the macro forecasting exercise.

Table A.3: Macro Panel Series by Category

Variable	Description	Source	Tr.
<i>Consumer prices</i>			
CPI_ALL_D7BT*	CPI index 00 all items	ONS	$\Delta \ln$
CPI_RENTS_D7CE	CPI index 04.1 actual rents for housing	ONS	$\Delta \ln$
<i>Wages and labour-market slack</i>			
AWE_NSA_TOTAL_PAY_COMPOSITE	ONS AWE whole-economy NSA total-pay composite: MD9M historic level through 1999 and KA5Q current index from 2000	ONS	$\Delta \ln$
CLAIMS	Claimant Count, K02000001 UK, people, NSA, thousands	ONS	$\Delta \ln$
CLAIMS_RATE	Claimant Count rate, K02000001 UK, people, NSA, percentage	ONS	level
<i>Exchange rates</i>			
GBP_BROAD	Broad effective exchange rate index, sterling, Jan 2005=100	BoE	$\Delta \ln$
GBP_CAN	GBP/CAD spot rate, monthly average	BoE	$\Delta \ln$
GBP_EUR	GBP/EUR spot rate, monthly average	BoE	$\Delta \ln$
GBP_JAP	GBP/JPY spot rate, monthly average	BoE	$\Delta \ln$
GBP_US	GBP/USD spot rate, monthly average	BoE	$\Delta \ln$
<i>Energy, food and commodity prices and futures</i>			
COMP_P	World Bank primary commodity price index	World Bank	$\Delta \ln$
CRUDE_GAS_FUT	Natural gas futures front-month (FN1 or UK/European gas future depending vendor)	ICE	$\Delta \ln$
CRUDE_OIL_FUT	Brent crude oil front-month futures (CO1)	ICE	$\Delta \ln$
FOODFUT	Wheat futures index (INFAWHP)	Bloomberg	$\Delta \ln$
OIL_PRICE	Crude oil prices: Brent Europe, dollars per barrel, monthly NSA	FRED	$\Delta \ln$
<i>House prices</i>			
HALIFAX_HPI_NSA	Halifax UK house price index, non-seasonally adjusted	Halifax	$\Delta \ln$
NATIONWIDE_HPI_NSA	Nationwide UK house price index, non-seasonally adjusted	Nationwide	$\Delta \ln$

Continued on next page

(continued)

Variable	Description	Source	Tr.
<i>Interest and mortgage rates</i>			
BANK_RATE	Official Bank Rate, monthly average	BoE	level
BGS_10yrs_yld	10-year gilt nominal par yield	BoE	level
BGS_20yrs_yld	20-year gilt nominal par yield	BoE	level
BGS_5yrs_yld	5-year gilt nominal par yield	BoE	level
MORT_FRATE_2YRS	2-year fixed mortgage rate, 75% LTV	BoE	level
MORT_FRATE_5YRS	5-year fixed mortgage rate, 75% LTV	BoE	level
SONIA	Sterling overnight index average rate	FRED/BoE	level
<i>Financial markets and uncertainty</i>			
FTSE250	FTSE 250 index	FTSE	$\Delta \ln$
FTSE_ALL	FTSE All Share index	FTSE	$\Delta \ln$
SP500	S&P 500 index	S&P	$\Delta \ln$
UK_focused_equity	iShares MSCI United Kingdom ETF (EWU)	iShares	$\Delta \ln$
VIX	CBOE Volatility Index	CBOE	Δ
<i>Trade</i>			
ONS_TOTAL_TRADE_EXPORTS_REV_TRI [†]	UK trade revisions triangle: total trade exports, current prices, seasonally adjusted	ONS	$\Delta \ln$
ONS_TOTAL_TRADE_IMPORTS_REV_TRI [†]	UK trade revisions triangle: total trade imports, current prices, seasonally adjusted	ONS	$\Delta \ln$
<i>Supply-chain pressure</i>			
AIR_FREIGHT_IN_ASIA_IC1312	Inbound Price Index (International Services): Air Freight for Asia	FRED/ALFRED	$\Delta \ln$
AIR_FREIGHT_IN_EUROPE_IC1311	Inbound Price Index (International Services): Air Freight for Europe	FRED/ALFRED	$\Delta \ln$
AIR_FREIGHT_OUT_ASIA_IS2312	Outbound Price Index (International Services): Air Freight for Asia	FRED/ALFRED	$\Delta \ln$
AIR_FREIGHT_OUT_EUROPE_IS2311	Outbound Price Index (International Services): Air Freight for Europe	FRED/ALFRED	$\Delta \ln$
<i>Activity and sentiment</i>			
ACEA_UK_CAR_REG_NSA	New passenger car registrations, United Kingdom	OECD	$\Delta \ln$
CLI	OECD composite leading indicator for UK	OECD	level

Notes: This table lists the 37 series of the macro panel. “Tr.” is the stationarity transformation applied before estimation: $\Delta \ln = 100 \times \log$ difference; level = untransformed; Δ = first difference of the level (VIX). All series are non-seasonally adjusted except the two trade series ([†]), which are only available seasonally adjusted and enter as the first-estimate row of the ONS trade revisions triangles. *CPI_ALL_D7BT is the history of the target index itself and is never used as a predictor.

A.6 Exact Formulas for Distributional Statistics

This subsection gives the exact definitions of the statistics used to encode the microdata, described in Section 4. Let c index a COICOP1 category and t a calendar month. A cell (c, t) contains all retained price observations j in category c during month t ; let $\mathcal{J}_{ct}$ denote this set, with $N_{ct} = |\mathcal{J}_{ct}|$. Price changes Δp_{jt} are defined as in Section 2. Some observations in $\mathcal{J}_{ct}$ may have missing Δp_{jt} , for example when a valid consecutive-month price change cannot be computed.

Weighted price-change statistics. Each price observation carries the CPI product weight $\omega_{j,t}$ defined in Equation (A.3). For each cell, denote the total weight and the total weight on zero and non-zero price-change observations by

$$W_{ct} = \sum_{j \in \mathcal{J}_{ct}} \omega_{j,t}, \quad \mathcal{J}_{ct}^0 = \{j \in \mathcal{J}_{ct} : \Delta p_{jt} = 0\}, \quad W_{ct}^0 = \sum_{j \in \mathcal{J}_{ct}^0} \omega_{j,t},$$

$$\mathcal{J}_{ct}^+ = \{j \in \mathcal{J}_{ct} : \Delta p_{jt} \neq 0 \text{ and } \Delta p_{jt} \text{ is observed}\}, \quad W_{ct}^+ = \sum_{j \in \mathcal{J}_{ct}^+} \omega_{j,t}.$$

The weighted fraction of zero price changes is

$$f_{ct}^0 = \frac{W_{ct}^0}{W_{ct}}.$$

The nine deciles $q_{k,ct}$ for $k \in \{0.1, 0.2, \dots, 0.9\}$ are the ω -weighted empirical quantiles of $\{\Delta p_{jt}\}_{j \in \mathcal{J}_{ct}^+}$, defined as the smallest observed value q such that

$$\frac{1}{W_{ct}^+} \sum_{j \in \mathcal{J}_{ct}^+ : \Delta p_{jt} \leq q} \omega_{j,t} \geq k.$$

The five moments use the Kish effective sample size

$$n_{\text{eff},ct}^+ = \frac{(W_{ct}^+)^2}{\sum_{j \in \mathcal{J}_{ct}^+} \omega_{j,t}^2},$$

which equals n_{ct}^+ when all weights within the cell are equal and shrinks toward 1 as the weights concentrate on a few observations. Let

$$\bar{m}_{ct} = \frac{1}{W_{ct}^+} \sum_{j \in \mathcal{J}_{ct}^+} \omega_{j,t} \Delta p_{jt}, \quad s_{ct} = \sqrt{\frac{n_{\text{eff},ct}^+}{n_{\text{eff},ct}^+ - 1} \cdot \frac{1}{W_{ct}^+} \sum_{j \in \mathcal{J}_{ct}^+} \omega_{j,t} (\Delta p_{jt} - \bar{m}_{ct})^2}.$$

Then

$$\begin{aligned}
m_{1,ct} &= \bar{m}_{ct}, \\
m_{2,ct} &= s_{ct}^2, \\
m_{k,ct} &= \frac{1}{W_{ct}^+ s_{ct}^k} \sum_{j \in \mathcal{J}_{ct}^+} \omega_{j,t} (\Delta p_{jt} - \bar{m}_{ct})^k, \quad k = 3, 5, \\
m_{4,ct} &= \frac{1}{W_{ct}^+ s_{ct}^4} \sum_{j \in \mathcal{J}_{ct}^+} \omega_{j,t} (\Delta p_{jt} - \bar{m}_{ct})^4 - 3.
\end{aligned}$$

Here m_1 is the ω -weighted mean, m_2 is the Kish/Bessel-corrected weighted sample variance, m_3 is weighted skewness, m_4 is weighted excess kurtosis (so that a normal distribution has $m_4 = 0$), and m_5 is the weighted standardised fifth central moment. The higher-order moments m_3 – m_5 use the Bessel-corrected standard deviation s_{ct} in the denominator but divide the weighted raw central moments by W_{ct}^+ in the numerator. The unweighted robustness check in Appendix E2 replaces $\omega_{j,t}$ throughout with uniform weights, in which case W_{ct}^+ collapses to n_{ct}^+ and $n_{\text{eff},ct}^+$ to n_{ct}^+ , and the formulas above reduce to the standard unweighted analogues.

Comparison with Alvarez, Le Bihan and Lippi (2016) kurtosis. Alvarez, Le Bihan and Lippi (2016) report raw (non-excess) kurtosis, denoted $\text{Kur}(\Delta p)$, which equals $m_4 + 3$ in our notation. Under normality $\text{Kur} = 3$; under a Golosov and Lucas (2007) two-point distribution $\text{Kur} = 1$; under Calvo $\text{Kur} = 6$. Their sufficient-statistic result for cumulative output is $M = \frac{\delta}{6\epsilon} \text{Kur}(\Delta p) / N(\Delta p)$, where $N(\Delta p)$ is the frequency of price changes. After correcting for cross-sectional heterogeneity and measurement error, Alvarez, Le Bihan and Lippi (2016) estimate $\text{Kur}(\Delta p) \approx 4$ in U.S. data, corresponding to $m_4 \approx 1$ in our convention.

Alvarez, Le Bihan and Lippi (2016) compute kurtosis on price changes that have been standardised at the UPC-outlet (cell) level, with very small changes discarded before pooling. Our COICOP1-month forecasting features defined above use $\omega_{j,t}$ -weighted within-cell moments without prior item-level standardisation, while the validation exercise in Appendix A.3 follows the protocol of Blanco et al. (2024), which applies $\omega_{j,t}$ weighting together with item-level standardisation and a 2% trim of the standardised distribution. Both of our pipelines differ from Alvarez, Le Bihan and Lippi (2016) in either the standardisation step or the trimming rule, so the resulting kurtosis values are not directly comparable to $\text{Kur}(\Delta p) \approx 4$.

Time-since-change statistics. For each product j in month t , after comparable and non-comparable substitution events have been split into separate product spells, define the time since the last reset as

$$\delta_{jt} = t - t_{jt}^*,$$

where t_{jt}^* is the most recent reset month $t' \leq t$. Reset months are defined as follows: (i) the first valid price observation of a product spell; (ii) any valid consecutive-month price-change observation with $\Delta p_{jt'} \neq 0$; and (iii) any observation following a gap of more than 12 months within the same product spell. Thus the entry month of a product spell has $\delta_{jt} = 0$ even though its price change is missing. Because time-since-change features are computed only from observations with zero valid price changes, entry months do not enter these moments.

For the time-since-change features, we use observations with zero valid price changes and a defined time-since-reset. Let

$$\mathcal{J}_{ct}^{0,\delta} = \{j \in \mathcal{J}_{ct} : \Delta p_{jt} = 0 \text{ and } \delta_{jt} \text{ is non-missing}\}, \quad W_{ct}^{0,\delta} = \sum_{j \in \mathcal{J}_{ct}^{0,\delta}} \omega_{j,t}.$$

The nine deciles and five moments of $\{\delta_{jt}\}_{j \in \mathcal{J}_{ct}^{0,\delta}}$ are computed by the same $\omega_{j,t}$ -weighted formulas as above, substituting δ_{jt} for Δp_{jt} , $\mathcal{J}_{ct}^{0,\delta}$ for $\mathcal{J}_{ct}^+$, and $W_{ct}^{0,\delta}$ for W_{ct}^+ .

B Econometric Theory of Scan Tests

B.1 Setup

B.1.1 Review of the scan test under simplified notation

We use triangular-array notation and suppress the row index. Fix a deterministic exponent $\gamma \in [0, 1/2]$. The exponent γ is part of the test specification and is not selected from the data. We use $\alpha \in (0, 1)$ below for the nominal test level. Let $[q] := \{1, \dots, q\}$, let $\mathcal{J}$ be a collection of intervals in $[n]$, and define

$$L_{\min} := \min_{I \in \mathcal{J}} |I|, \quad L_{\max} := \max_{I \in \mathcal{J}} |I|, \quad A := \mathcal{J} \times [H], \quad p := |A| = H|\mathcal{J}|. \quad (\text{B.1})$$

Because $p \leq n^2$,

$$\log(pn) \asymp \log n. \quad (\text{B.2})$$

For $a \in A$, write $a = (I, h)$ for its associated tuple, and set

$$S_a := \sum_{t \in I} D_t^{(h)}, \quad \bar{D}_a := \frac{S_a}{|I|}, \quad Q_a := \sum_{t \in I} \left(D_t^{(h)} - \bar{D}_a \right)^2. \quad (\text{B.3})$$

$$T_a := \frac{S_a}{|I|^{1/2-\gamma} Q_a^\gamma}, \quad T_{\max} := \max_{a \in A} T_a. \quad (\text{B.4})$$

For $\gamma = 0$, we define $q^0 = 1$ for every $q \geq 0$, so $T_a = S_a$. For $\gamma > 0$, a ratio with a zero denominator is defined to be zero. The one-sided null is

$$H_0: \quad \mathbb{E}[D_t^{(h)}] \leq 0 \quad \text{for every } t \in [n], h \in [H]. \quad (\text{B.5})$$

Let $\xi = (\xi_1, \dots, \xi_n)^\top \sim N(0, \Theta_n)$ independently of the data, where Θ_n is positive semidefinite with unit diagonal. Regime I uses $\Theta_n = I_n$; Regime D uses the kernel covariance matrix defined in Assumption B.9. For $a = (I, h) \in A$, write

$$S_a^* := \sum_{t \in I} \xi_t \left(D_t^{(h)} - \bar{D}_a \right), \quad \bar{D}_a^* := \frac{S_a^*}{|I|}, \quad Q_a^* := \sum_{t \in I} \left\{ \xi_t \left(D_t^{(h)} - \bar{D}_a \right) - \bar{D}_a^* \right\}^2.$$

The bootstrap statistic is

$$T_a^* := \frac{S_a^*}{|I|^{1/2-\gamma} (Q_a^*)^\gamma}, \quad T_{\max}^* := \max_{a \in A} T_a^*. \quad (\text{B.6})$$

The conditional critical value for the bootstrap statistic above is

$$c_{n,1-\alpha}^* := \inf\{x : \mathbb{P}^*(T_{\max}^* \leq x) \geq 1 - \alpha\}. \quad (\text{B.7})$$

Here $\mathbb{P}^*$ denotes probability conditional on the data; conditional stochastic orders are understood in outer probability. The test rejects when $T_{\max} > c_{n,1-\alpha}^*$.

B.1.2 Other notation used for proofs

Write

$$\mu_t^{(h)} := \mathbb{E}[D_t^{(h)}], \quad d_t^{(h)} := D_t^{(h)} - \mu_t^{(h)}. \quad (\text{B.8})$$

For $a = (I, h) \in A$, write

$$S_a^\circ := \sum_{t \in I} d_t^{(h)}. \quad (\text{B.9})$$

Define

$$\bar{d}_a := \frac{1}{|I|} \sum_{t \in I} d_t^{(h)}, \quad d_{t,a} := d_t^{(h)} - \bar{d}_a. \quad (\text{B.10})$$

$$\bar{\mu}_a := \frac{1}{|I|} \sum_{t \in I} \mu_t^{(h)}, \quad \mu_{t,a} := \mu_t^{(h)} - \bar{\mu}_a. \quad (\text{B.11})$$

$$q_a := \sum_{t \in I} d_{t,a}^2, \quad v_a := \sum_{t \in I} \mathbb{E}[d_t^{(h)2}]. \quad (\text{B.12})$$

$$q_a^\circ := q_a - v_a = \sum_{t \in I} d_{t,a}^2 - v_a, \quad C_a := \sum_{t \in I} d_{t,a} \mu_{t,a}. \quad (\text{B.13})$$

The centered comparison statistics are

$$T_{\max}^\circ := \max_{a \in A} \frac{S_a^\circ}{|I|^{1/2-\gamma} Q_a^\gamma}. \quad (\text{B.14})$$

The interval-wise mean-flatness index is

$$v_{a,\mu} := \sum_{t \in I} \mu_{t,a}^2, \quad r_n := \max_{a=(I,h) \in A} \frac{v_{a,\mu}}{v_a}. \quad (\text{B.15})$$

It controls the shape of the mean within an interval, not its level. In particular, arbitrary constant negative shifts leave r_n equal to zero.

For positive deterministic sequences a_n and b_n , $a_n = \Omega(b_n)$ means $b_n = O(a_n)$, and $a_n = \omega(b_n)$ means $b_n = o(a_n)$.

B.1.3 Common assumptions

Assumption B.1. *The centered coordinates $d_t^{(h)} = D_t^{(h)} - \mu_t^{(h)}$ are uniformly sub-Gaussian: for a constant $B_0 < \infty$,*

$$\sup_{t \in [n], h \in [H]} \|d_t^{(h)}\|_{\psi_2} \leq B_0.$$

The Orlicz norms are defined in Section B.5.

Assumption B.2. For constants $0 < \sigma_-^2 \leq \sigma_+^2 < \infty$,

$$\sigma_-^2 \leq \mathbb{E}[d_t^{(h)2}] \leq \sigma_+^2 \quad \text{for every } t \in [n], h \in [H].$$

Assumption B.3. For a constant $C_L < \infty$,

$$\left(\frac{L_{\max}}{L_{\min}} \right)^{1-2\gamma} \leq C_L.$$

For fixed $\gamma < 1/2$, this requires $L_{\max}/L_{\min} = O(1)$. For $\gamma = 1/2$, it imposes no restriction.

B.2 Preliminary results

B.2.1 Lemmas used for both regimes

The following algebraic decomposition is used in both regimes to compare the observed denominator with its oracle variance scale.

Lemma 2. For every $a = (I, h) \in A$,

$$\frac{Q_a - v_a}{v_a} = \frac{q_a^\circ}{v_a} + 2 \frac{C_a}{v_a} + \frac{v_{a,\mu}}{v_a}.$$

Proof. Because $\sum_{t \in I} \mu_{t,a} = 0$,

$$D_t^{(h)} - \bar{D}_a = d_{t,a} + \mu_{t,a}.$$

Expanding the square and using the definition of C_a gives

$$Q_a = q_a + 2C_a + v_{a,\mu}.$$

Also,

$$q_a = \sum_{t \in I} \left(d_t^{(h)} - \bar{d}_a \right)^2 = \sum_{t \in I} d_t^{(h)2} - |I| \bar{d}_a^2,$$

and the definition (B.13) gives $q_a - v_a = q_a^\circ$. The displayed identity follows by subtracting v_a and dividing by v_a . $\square$

The next lemma gives the deterministic perturbation bound for changing the variance normalizer in a power-normalized statistic.

Lemma 3. Let $(s_a, V_a, \tilde{V}_a)_{a \in A}$ be real numbers with $V_a, \tilde{V}_a > 0$, and set

$$\Delta_V := \max_{a \in A} \left| \frac{V_a}{\tilde{V}_a} - 1 \right|.$$

On $\{\Delta_V \leq 1/2\}$,

$$\max_{a \in A} \left| \frac{s_a}{|I|^{1/2-\gamma} V_a^\gamma} - \frac{s_a}{|I|^{1/2-\gamma} \tilde{V}_a^\gamma} \right| \leq C\gamma \Delta_V \max_{a \in A} \frac{|s_a|}{|I|^{1/2-\gamma} \tilde{V}_a^\gamma},$$

where C is universal.

Proof. For $\gamma = 0$ the two sides on the left are identical. For $\gamma > 0$, write $u_a = V_a/\tilde{V}_a$. On $\{\Delta_V \leq 1/2\}$, $u_a \in [1/2, 3/2]$ for every a . The function $u \mapsto u^{-\gamma}$ has derivative bounded by $C\gamma$ on this interval, and therefore $|u_a^{-\gamma} - 1| \leq C\gamma|u_a - 1|$. Multiplying by $|s_a|/(|I|^{1/2-\gamma} \tilde{V}_a^\gamma)$ and taking maxima proves the claim. $\square$

The next proposition provides the distribution-function perturbation bound used when one statistic is uniformly close to another.

Proposition 4. Let X and Y be real-valued random variables on a probability space with probability measure P . For every $\eta > 0$,

$$\sup_{x \in \mathbb{R}} |P(X \leq x) - P(Y \leq x)| \leq P(|X - Y| > \eta) + \sup_{x \in \mathbb{R}} P(|Y - x| \leq \eta).$$

Proof. For every $x \in \mathbb{R}$,

$$\{X \leq x\} \subset \{Y \leq x + \eta\} \cup \{|X - Y| > \eta\},$$

and

$$\{Y \leq x - \eta\} \subset \{X \leq x\} \cup \{|X - Y| > \eta\}.$$

The first inclusion gives

$$P(X \leq x) - P(Y \leq x) \leq P(|X - Y| > \eta) + P(x < Y \leq x + \eta),$$

and the second inclusion gives

$$P(Y \leq x) - P(X \leq x) \leq P(|X - Y| > \eta) + P(x - \eta < Y \leq x).$$

Taking the supremum over x proves the result. $\square$

The next lemma converts convergence in probability into a deterministic error threshold.

Lemma 5. *If $Z_n \geq 0$ and $Z_n = o_p(1)$, then there exists a deterministic sequence $\rho_n \downarrow 0$ such that*

$$\mathbb{P}(Z_n > \rho_n) \rightarrow 0.$$

Proof. For each integer $j \geq 1$, choose N_j large enough that $\mathbb{P}(Z_n > 1/j) \leq 1/j$ whenever $n \geq N_j$, and take the sequence (N_j) increasing. Set $\rho_n = 1/j$ for $N_j \leq n < N_{j+1}$. Then $\rho_n \downarrow 0$ and $\mathbb{P}(Z_n > \rho_n) \leq \rho_n$ for all sufficiently large n . $\square$

B.2.2 Bounding the tail probability of $T_{\max}$ by $T_{\max}^\circ$

The following lemma gives the pathwise comparison between the feasible statistic and its centered counterpart under the null.

Lemma 6. *Under (B.5),*

$$T_{\max} \leq T_{\max}^\circ.$$

Proof. Under (B.5), for every $a = (I, h) \in A$,

$$\sum_{t \in I} \mu_t^{(h)} \leq 0.$$

Therefore

$$S_a = S_a^\circ + \sum_{t \in I} \mu_t^{(h)} \leq S_a^\circ \quad \text{for every } a \in A.$$

The two coordinates T_a and the corresponding coordinate in $T_{\max}^\circ$ have the same realized denominator $|I|^{1/2-\gamma} Q_a^\gamma$, so the coordinatewise inequality implies $T_{\max} \leq T_{\max}^\circ$. $\square$

The next lemma gives the common tail-probability comparison between $T_{\max}^\circ$ and $\mathbb{T}_{\max}^\circ$.

Lemma 7. *Set*

$$\Delta_Q := \max_{a \in A} \left| \frac{Q_a}{v_a} - 1 \right|.$$

If

$$\mathbb{P}(\Delta_Q > 1/2) + \mathbb{P}(C\gamma\Delta_Q \bar{\mathbb{T}}_{\max}^\circ > \eta_n) \rightarrow 0$$

for a universal constant $C < \infty$, and

$$\sup_{x \in \mathbb{R}} \mathbb{P}(|\mathbb{T}_{\max}^\circ - x| \leq \eta_n) \rightarrow 0,$$

then

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(T_{\max}^\circ > x) - \mathbb{P}(\mathbb{T}_{\max}^\circ > x)| = \sup_{x \in \mathbb{R}} |\mathbb{P}(T_{\max}^\circ \leq x) - \mathbb{P}(\mathbb{T}_{\max}^\circ \leq x)| = o(1).$$

Proof. On $\{\Delta_Q \leq 1/2\}$, $Q_a, v_a > 0$ for every $a \in A$. Lemma 3, applied with $s_a = S_a^\circ$, $V_a = Q_a$, and $\tilde{V}_a = v_a$, gives

$$\max_{a \in A} \left| \frac{S_a^\circ}{|I|^{1/2-\gamma} Q_a^\gamma} - \mathbb{T}_a^\circ \right| \leq C\gamma\Delta_Q \bar{\mathbb{T}}_{\max}^\circ.$$

Using $|\max_a x_a - \max_a y_a| \leq \max_a |x_a - y_a|$ gives

$$|T_{\max}^\circ - \mathbb{T}_{\max}^\circ| \leq C\gamma\Delta_Q \bar{\mathbb{T}}_{\max}^\circ.$$

Thus $\mathbb{P}(|T_{\max}^\circ - \mathbb{T}_{\max}^\circ| > \eta_n) \rightarrow 0$ under the displayed assumptions. Proposition 4, applied with $X = T_{\max}^\circ$ and $Y = \mathbb{T}_{\max}^\circ$, proves the distribution-function claim. The tail-probability claim is identical because tail probabilities are complements of distribution functions. $\square$

B.2.3 Approximating the bootstrap distribution by a fixed-denominator version

The proofs use the following auxiliary fixed-denominator bootstrap statistics

$$T_{0,a}^* := \frac{S_a^*}{|I|^{1/2-\gamma} Q_a^\gamma}, \quad T_{0,\max}^* := \max_{a \in A} T_{0,a}^*. \quad (\text{B.16})$$

Conditionally on the data, $(T_{0,a}^*)_{a \in A}$ is Gaussian. The actual bootstrap statistic implemented in (B.6) is not Gaussian, because the same multipliers enter its numerator and denominator. But we will show that the two versions have the same asymptotic bootstrap distribution.

The following lemma is used in both regimes. It permits nonconstant means through the index r_n and therefore does not impose a bound on the level of the mean. It gives a uniform concentration bound for the random bootstrap denominator and compares the random- and fixed-denominator bootstrap distribution functions.

Lemma 8. *Let $k_n = \Omega(1)$ be a deterministic sequence, and suppose Assumptions B.2 and B.3 hold. Suppose that, for some universal constants $0 < c < C < \infty$, with probability $1 - o(1)$ over data, uniformly over $a = (I, h) \in A$,*

$$Q_a \geq c|I|, \quad (\text{B.17})$$

$$\max_{t \in I} (d_t^{(h)})^2 \leq C \log n, \quad \sum_{t \in I} (d_t^{(h)})^4 \leq C|I|, \quad (\text{B.18})$$

$$c \leq \text{Var}^* \left(\frac{S_a^*}{\sqrt{Q_a}} \right) \leq C. \quad (\text{B.19})$$

Suppose also that Θ_n has unit diagonal and

$$\max_t \sum_{u=1}^n |\Theta_n(t, u)| \leq Ck_n, \quad \max_t \sum_{u=1}^n \Theta_n(t, u)^2 \leq Ck_n. \quad (\text{B.20})$$

Then, in probability,

$$\Delta_Q^* := \max_{a \in A} \left| \frac{Q_a^*}{Q_a} - 1 \right| = O_{P^*} \left(\sqrt{\frac{k_n \log n}{L_{\min}}} + \frac{k_n (\log n)^2}{L_{\min}} + k_n r_n \log n \right). \quad (\text{B.21})$$

If

$$\frac{k_n (\log n)^3}{L_{\min}} \rightarrow 0, \quad k_n r_n (\log n)^2 \rightarrow 0, \quad (\text{B.22})$$

then

$$\sup_{x \in \mathbb{R}} |\mathbb{P}^*(T_{\max}^* \leq x) - \mathbb{P}^*(T_{0, \max}^* \leq x)| = o_p(1). \quad (\text{B.23})$$

Proof. Fix $a = (I, h)$. Since the bootstrap sample mean over I is $\bar{D}_a^*$,

$$Q_a^* = \left\{ \sum_{t \in I} \left(D_t^{(h)} - \bar{D}_a \right)^2 \xi_t^2 - |I| (\bar{D}_a^*)^2 \right\}.$$

Define

$$w_{t,a} := \frac{(D_t^{(h)} - \bar{D}_a)^2}{Q_a}.$$

Because $\sum_{t \in I} w_{t,a} = 1$ and $\Theta_n(t, t) = 1$, the exact identity

$$\frac{Q_a^*}{Q_a} = 1 + \sum_{t \in I} w_{t,a} (\xi_t^2 - 1) - \frac{(S_a^*)^2}{|I| Q_a} \quad (\text{B.24})$$

holds. Consequently,

$$\Delta_Q^* \leq \max_{a=(I,h) \in A} \left| \sum_{t \in I} w_{t,a} (\xi_t^2 - 1) \right| + \frac{1}{L_{\min}} \max_a \left(\frac{S_a^*}{\sqrt{Q_a}} \right)^2. \quad (\text{B.25})$$

The bounds in (B.18) imply

$$\max_{t \in I} d_{t,a}^2 \leq C \log n, \quad \sum_{t \in I} d_{t,a}^4 \leq C |I|$$

by $|\bar{d}_a| \leq \max_{t \in I} |d_t^{(h)}|$ and Jensen's inequality. Since $D_t^{(h)} - \bar{D}_a = d_{t,a} + \mu_{t,a}$, Assumption B.2, (B.15),

and (B.17)–(B.18) imply

$$\max_{t \in I} w_{t,a} \leq C \left(\frac{\log n}{|I|} + r_n \right), \quad (\text{B.26})$$

$$\sum_{t \in I} w_{t,a}^2 \leq C \left(\frac{1}{|I|} + r_n^2 \right). \quad (\text{B.27})$$

Indeed, $\sum_{t \in I} \mu_{t,a}^2 = v_{a,\mu} \leq r_n v_a \leq C r_n |I|$ by (B.15) and Assumption B.2. Hence $\max_{t \in I} \mu_{t,a}^2 \leq C r_n |I|$, and the inequality $(u + v)^2 \leq 2(u^2 + v^2)$ gives

$$\max_{t \in I} \left(D_t^{(h)} - \bar{D}_a \right)^2 \leq C(\log n + r_n |I|).$$

Dividing by $Q_a \geq c|I|$ yields (B.26). Similarly,

$$\sum_{t \in I} \left(D_t^{(h)} - \bar{D}_a \right)^4 \leq 8 \sum_{t \in I} d_{t,a}^4 + 8 \sum_{t \in I} \mu_{t,a}^4, \quad \sum_{t \in I} \mu_{t,a}^4 \leq \left(\sum_{t \in I} \mu_{t,a}^2 \right)^2 \leq C r_n^2 |I|^2.$$

Together with $\sum_{t \in I} d_{t,a}^4 \leq C|I|$, this gives

$$\sum_{t \in I} \left(D_t^{(h)} - \bar{D}_a \right)^4 \leq C|I| + C r_n^2 |I|^2.$$

Dividing by $Q_a^2 \geq c|I|^2$ gives (B.27).

Write $\xi_I = \Theta_I^{1/2} Z_I$, where $Z_I \sim N(0, I_{|I|})$. Then

$$\sum_{t \in I} w_{t,a} (\xi_t^2 - 1) = Z_I^\top B_a Z_I - \text{tr}(B_a), \quad B_a := \Theta_I^{1/2} \text{diag}(w_{t,a} : t \in I) \Theta_I^{1/2}.$$

The trace equals one because

$$\text{tr}(B_a) = \sum_{t \in I} w_{t,a} \Theta_n(t, t) = \sum_{t \in I} w_{t,a} = 1.$$

Using $2w_{t,a}w_{u,a} \leq w_{t,a}^2 + w_{u,a}^2$, (B.20), and (B.27),

$$\begin{aligned}
\|B_a\|_F^2 &= \sum_{t,u \in I} w_{t,a}w_{u,a}\Theta_n(t,u)^2 \\
&\leq \frac{1}{2} \sum_{t,u \in I} \{w_{t,a}^2 + w_{u,a}^2\}\Theta_n(t,u)^2 \\
&= \sum_{t \in I} w_{t,a}^2 \sum_{u \in I} \Theta_n(t,u)^2 \\
&\leq Ck_n \sum_{t \in I} w_{t,a}^2 \leq Ck_n \left(\frac{1}{|I|} + r_n^2 \right).
\end{aligned}$$

Furthermore, (B.20) and (B.26) give

$$\|B_a\|_{\text{op}} \leq Ck_n \left(\frac{\log n}{|I|} + r_n \right).$$

The Gaussian quadratic-form inequality in Proposition 26 therefore yields, for $u \geq 1$,

$$\mathbb{P}^* \left(\left| \sum_{t \in I} w_{t,a}(\xi_t^2 - 1) \right| > C \left[\sqrt{k_n u \left(\frac{1}{|I|} + r_n^2 \right)} + k_n u \left(\frac{\log n}{|I|} + r_n \right) \right] \right) \leq 2e^{-u}.$$

By (B.2), taking u proportional to $\log n$ and applying a union bound gives, after absorbing $r_n \sqrt{k_n \log n}$ into $k_n r_n \log n$ because $k_n = \Omega(1)$ and $\log n \rightarrow \infty$,

$$\max_{a=(I,h) \in A} \left| \sum_{t \in I} w_{t,a}(\xi_t^2 - 1) \right| = O_{P^*} \left(\sqrt{\frac{k_n \log n}{L_{\min}}} + \frac{k_n (\log n)^2}{L_{\min}} + k_n r_n \log n \right) \quad (\text{B.28})$$

in probability over the data. By (B.2), (B.19), and Proposition 33 applied conditionally on the data,

$$\max_a \left| \frac{S_a^*}{\sqrt{Q_a}} \right| = O_{P^*}(\sqrt{\log n}) \quad (\text{B.29})$$

in probability over the data. Equations (B.25)–(B.29) and the implication

$$\frac{\log n}{L_{\min}} = o\left(\frac{k_n (\log n)^2}{L_{\min}}\right),$$

which follows from $k_n = \Omega(1)$ and $\log n \rightarrow \infty$, prove (B.21).

If $\gamma = 0$, then $T_{\max}^* = T_{0,\max}^*$ identically. Suppose $\gamma > 0$. Lemma 3, applied with $s_a = S_a^*$, $V_a = Q_a^*$, and $\tilde{V}_a = Q_a$, gives

$$|T_{\max}^* - T_{0,\max}^*| \leq C\gamma \Delta_Q^* \max_a |T_{0,a}^*|. \quad (\text{B.30})$$

Moreover,

$$T_{0,a}^* = \left(\frac{Q_a}{|I|} \right)^{1/2-\gamma} \frac{S_a^*}{\sqrt{Q_a}}.$$

Equations (B.17)–(B.18), the definition of r_n , Assumption B.2, and (B.22) imply with probability $1 - o(1)$,

$$c \leq \min_{a=(I,h) \in A} \left(\frac{Q_a}{|I|} \right)^{1/2-\gamma} \leq \max_{a=(I,h) \in A} \left(\frac{Q_a}{|I|} \right)^{1/2-\gamma} \leq C.$$

Indeed, (B.17) gives $Q_a/|I| \geq c$. For the upper bound,

$$Q_a = \sum_{t \in I} (d_{t,a} + \mu_{t,a})^2 \leq 2 \sum_{t \in I} d_{t,a}^2 + 2\nu_{a,\mu}.$$

By Cauchy–Schwarz and (B.18), $\sum_{t \in I} d_{t,a}^2 \leq |I|^{1/2} (\sum_{t \in I} d_{t,a}^4)^{1/2} \leq C|I|$. Also $\nu_{a,\mu} \leq r_n \nu_a \leq Cr_n |I|$, and (B.22) with $k_n = \Omega(1)$ and $\log n \rightarrow \infty$ gives $r_n = o(1)$. Hence $Q_a/|I| \leq C$ uniformly. Since $1/2 - \gamma \in [0, 1/2]$,

$$c \leq \left(\frac{Q_a}{|I|} \right)^{1/2-\gamma} \leq C$$

uniformly. Therefore, with probability $1 - o(1)$ over data,

$$c \leq \min_{a \in A} \text{Var}^*(T_{0,a}^*) \leq \max_{a \in A} \text{Var}^*(T_{0,a}^*) \leq C, \quad (\text{B.31})$$

because

$$\text{Var}^*(T_{0,a}^*) = \left(\frac{Q_a}{|I|} \right)^{1-2\gamma} \text{Var}^*\left(\frac{S_a^*}{\sqrt{Q_a}} \right)$$

and (B.19) holds. On this event, $(T_{0,a}^*)_{a \in A}$ is conditionally Gaussian with uniformly bounded conditional ψ_2 norms. Proposition 33, applied conditionally on the data with $m = p$, gives

$$\max_a |T_{0,a}^*| = O_{P^*}(\sqrt{\log p}) = O_{P^*}(\sqrt{\log n}) \quad (\text{B.32})$$

in probability over the data, where the last bound uses (B.2). The two conditions in (B.22) imply $\Delta_Q^* \log n = o_{P^*}(1)$ in probability over the data. Combining (B.22), (B.32), and (B.30) gives

$$|T_{\max}^* - T_{0,\max}^*| = o_{P^*}((\log n)^{-1/2}) \quad (\text{B.33})$$

in probability over the data. Choose a deterministic sequence $\eta_n = o((\log n)^{-1/2})$ such that

$$\mathbb{P}^*(|T_{\max}^* - T_{0,\max}^*| > \eta_n) = o_p(1), \quad (\text{B.34})$$

which is possible by (B.33). Proposition 24, applied conditionally on the data to $T_{0,\max}^*$ using (B.31), gives

$$\sup_{x \in \mathbb{R}} \mathbb{P}^*(|T_{0,\max}^* - x| \leq \eta_n) = o_p(1). \quad (\text{B.35})$$

Lemma 7, applied conditionally on the data with $P_n = \mathbb{P}^*$, $X_n = T_{\max}^*$, and $Y_n = T_{0,\max}^*$, together with (B.34) and (B.35), proves (B.23). $\square$

B.2.4 Gaussian approximation of the fixed-denominator bootstrap distribution

Let

$$\omega_a := \left(\frac{v_a}{|I|} \right)^{1/2-\gamma}, \quad \mathbb{T}_a^\circ := \frac{S_a^\circ}{|I|^{1/2-\gamma} v_a^\gamma} = \omega_a \frac{S_a^\circ}{\sqrt{v_a}}.$$

Define

$$\mathbb{T}^\circ := (\mathbb{T}_a^\circ)_{a \in A}, \quad \mathbb{T}_{\max}^\circ := \max_{a \in A} \mathbb{T}_a^\circ, \quad \bar{\mathbb{T}}_{\max}^\circ := \max_{a \in A} |\mathbb{T}_a^\circ|. \quad (\text{B.36})$$

Let

$$\Sigma_\circ := \text{Cov}(\mathbb{T}^\circ), \quad G_\circ \sim N(0, \Sigma_\circ).$$

This definition includes all serial cross-covariances in Regime D. Under Regime I independence it reduces to

$$(\Sigma_\circ)_{ab} = \frac{\sum_{t \in I_a \cap I_b} \mathbb{E}[d_t^{(h_a)} d_t^{(h_b)}]}{|I_a|^{1/2-\gamma} |I_b|^{1/2-\gamma} v_a^\gamma v_b^\gamma}.$$

Write

$$G_{\circ, \max} := \max_{a \in A} G_{\circ, a}, \quad q_{\circ, n, u} := \inf\{x \in \mathbb{R} : \mathbb{P}(G_{\circ, \max} \leq x) \geq u\}.$$

The following lemma shows that the fixed-denominator bootstrap approximation follows from consistency of its conditional covariance matrix. For matrices indexed by A , write

$$\|\Sigma\|_{\max} := \max_{a, b \in A} |\Sigma_{ab}|.$$

Lemma 9. *Let*

$$\hat{\Sigma}_{ab} := \text{Cov}^*(T_{0,a}^*, T_{0,b}^*), \quad a, b \in A.$$

Suppose that, for constants $0 < c < C < \infty$,

$$c \leq \min_{a \in A} (\Sigma_\circ)_{aa} \leq \max_{a \in A} (\Sigma_\circ)_{aa} \leq C,$$

and, with probability $1 - o(1)$ over data,

$$c \leq \min_{a \in A} \widehat{\Sigma}_{aa}, \quad \max_{a \in A} \widehat{\Sigma}_{aa} \leq C.$$

If

$$\|\widehat{\Sigma} - \Sigma_{\circ}\|_{\max} = o_p((\log p)^{-2}), \quad (\text{B.37})$$

then

$$\sup_{x \in \mathbb{R}} |\mathbb{P}^*(T_{0,\max}^* \leq x) - \mathbb{P}(G_{\circ,\max} \leq x)| = o_p(1).$$

Proof. Let

$$\Delta_{\Sigma} := \|\widehat{\Sigma} - \Sigma_{\circ}\|_{\max}.$$

The covariance consistency assumption gives a deterministic sequence $\varepsilon_n \downarrow 0$ such that

$$\varepsilon_n (\log p)^2 \rightarrow 0, \quad \mathbb{P}(\Delta_{\Sigma} > \varepsilon_n) \rightarrow 0.$$

On the event where $\Delta_{\Sigma} \leq \varepsilon_n$ and the displayed variance bounds for $\widehat{\Sigma}$ hold, the conditional law of $(T_{0,a}^*)_{a \in A}$ is centered Gaussian with covariance $\widehat{\Sigma}$, and $G_{\circ}$ is centered Gaussian with covariance $\Sigma_{\circ}$. Proposition 25 yields

$$\sup_{x \in \mathbb{R}} |\mathbb{P}^*(T_{0,\max}^* \leq x) - \mathbb{P}(G_{\circ,\max} \leq x)| \leq C \varepsilon_n^{1/3} \{1 \vee \log(p/\varepsilon_n)\}^{2/3}.$$

The right-hand side is $o(1)$ because $\varepsilon_n (\log p)^2 \rightarrow 0$ and $\varepsilon_n \{\log(1/\varepsilon_n)\}^2 \rightarrow 0$. Since the event has probability tending to one, the desired conclusion follows. $\square$

The next lemma transfers oracle Gaussian approximation and bootstrap approximation into asymptotic size control.

Lemma 10. Fix $\alpha \in (0, 1)$. Under (B.5), suppose that the conditions of Lemmas 8 and 9 hold. Suppose also that

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(T_{\max}^{\circ} \leq x) - \mathbb{P}(G_{\circ,\max} \leq x)| = o(1).$$

Then

$$\limsup_{n \rightarrow \infty} \mathbb{P}(T_{\max} > c_{n,1-\alpha}^*) \leq \alpha.$$

Proof. Lemmas 8 and 9 imply

$$Z_n := \sup_{x \in \mathbb{R}} |\mathbb{P}^*(T_{\max}^* \leq x) - \mathbb{P}(G_{\circ,\max} \leq x)| = o_p(1).$$

Lemma 5 gives a deterministic sequence $\rho_n \downarrow 0$ such that $\mathbb{P}(Z_n > \rho_n) \rightarrow 0$. Let

$$F_n^*(x) := \mathbb{P}^*(T_{\max}^* \leq x), \quad E_n := \left\{ \sup_{x \in \mathbb{R}} |F_n^*(x) - \mathbb{P}(G_{\circ, \max} \leq x)| \leq \rho_n \right\}.$$

Then $\mathbb{P}(E_n^c) \rightarrow 0$. On E_n , $F_n^*(c_{n, 1-\alpha}^*) \geq 1 - \alpha$, and hence

$$\mathbb{P}(G_{\circ, \max} \leq c_{n, 1-\alpha}^*) \geq 1 - \alpha - \rho_n.$$

For all sufficiently large n , $\rho_n < 1 - \alpha$, so

$$c_{n, 1-\alpha}^* \geq q_{\circ, n, 1-\alpha-\rho_n} \quad \text{on } E_n.$$

Lemma 6 gives $T_{\max} \leq T_{\max}^\circ$ under (B.5). Therefore

$$\mathbb{P}(T_{\max} > c_{n, 1-\alpha}^*) \leq \mathbb{P}(T_{\max}^\circ > q_{\circ, n, 1-\alpha-\rho_n}) + \mathbb{P}(E_n^c).$$

Let

$$\Delta_{\text{cdf}, n} := \sup_{x \in \mathbb{R}} |\mathbb{P}(T_{\max}^\circ \leq x) - \mathbb{P}(G_{\circ, \max} \leq x)|.$$

By assumption, $\Delta_{\text{cdf}, n} \rightarrow 0$, so

$$\mathbb{P}(T_{\max}^\circ > q_{\circ, n, 1-\alpha-\rho_n}) \leq \mathbb{P}(G_{\circ, \max} > q_{\circ, n, 1-\alpha-\rho_n}) + \Delta_{\text{cdf}, n} \leq \alpha + \rho_n + \Delta_{\text{cdf}, n}.$$

Taking the limit superior proves the claim. □

B.2.5 Roadmap of the proofs

The final goal in each regime is

$$\limsup_{n \rightarrow \infty} \mathbb{P}(T_{\max} > c_{n, 1-\alpha}^*) \leq \alpha \quad \forall \alpha \in (0, 1).$$

By Lemma 10, it is enough to show

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(T_{\max}^\circ \leq x) - \mathbb{P}(G_{\circ, \max} \leq x)| \rightarrow 0$$

and to verify the conditions of Lemmas 8 and 9. The display is reduced by Lemma 7 to proving

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(T_{\max}^\circ \leq x) - \mathbb{P}(G_{\circ, \max} \leq x)| \rightarrow 0. \tag{B.38}$$

The bootstrap part is reduced by Lemmas 8 and 9 to proving the covariance consistency condition

$$\|\hat{\Sigma} - \Sigma_\circ\|_{\max} = o_p((\log p)^{-2}). \quad (\text{B.39})$$

All remaining work in Regimes I and D is therefore to verify the centered Gaussian approximation (B.38), the covariance consistency condition (B.39), and the denominator and anti-concentration conditions required by Lemmas 7, 8, and 9.

B.3 Regime I Validity

B.3.1 Additional assumptions for Regime I

We consider the following additional assumptions for Regime I.

Assumption B.4. *The vectors $D_1, \dots, D_n$ are independent across t .*

Assumption B.5. *The interval-wise mean is asymptotically flat:*

$$r_n = o((\log n)^{-2}).$$

Assumption B.6. *The minimal interval length grows at least polynomially in n :*

$$L_{\min} = \omega((\log n)^7).$$

Remark 1. Assumption B.5 constrains the *shape* of the mean on each scanned interval, not its level. If $\mu_t^{(h)} \equiv -c_h$ is constant in t , then $r_n = 0$ for every n regardless of the magnitude of c_h .

B.3.2 Main theorem

Theorem 11. *Fix the exponent $\gamma \in [0, 1/2]$. Assume Assumptions B.1, B.2, B.3, and B.4–B.6. Let $c_{n,1-\alpha}^*$ be the conditional $(1 - \alpha)$ -quantile in (B.6). Then the Gaussian multiplier bootstrap with $\Theta_n = I_n$ is asymptotically conservative: under (B.5),*

$$\limsup_{n \rightarrow \infty} \mathbb{P}(T_{\max} > c_{n,1-\alpha}^*) \leq \alpha \quad \forall \alpha \in (0, 1).$$

Following the roadmap in Section B.2.5, it remains to prove (B.38) and (B.39) and to verify the conditions in Lemmas 7–9. In what follows, Section B.3.3 gives preliminary results, Sections B.3.4 and B.3.5 prove (B.38) and (B.39), respectively, and Section B.3.6 completes the proof of Theorem 11 by

verifying the conditions in Lemmas 7–9. For Sections B.3.4–B.3.6, we assume Assumptions B.1, B.2, B.3, and B.4–B.6 all hold.

B.3.3 Preliminary bounds

The three lemmas in this subsection supply the Regime I inputs needed for the roadmap. Lemma 12 controls $\bar{\mathbb{T}}_{\max}^\circ$ and the intersection sums $S_b^\circ/\sqrt{v_b}$; these bounds control sample-centering terms in Lemma 13, in the covariance expansion for (B.39), and in the verification of Lemma 7. Lemma 13 gives $\Delta_Q = o_p(1)$; this is used to compare $T_{\max}^\circ$ with $\mathbb{T}_{\max}^\circ$, to lower bound Q_a in Lemma 8, and to control the denominator perturbation in (B.39). Lemma 14 verifies the maximum and fourth-moment condition (B.18) required by Lemma 8.

Lemma 12. *Under Assumptions B.1, B.2, and B.4,*

$$\bar{\mathbb{T}}_{\max}^\circ = O_p(\sqrt{\log n}).$$

Moreover, if

$$\mathcal{K} := \{I \cap J : I, J \in \mathcal{I}, I \cap J \neq \emptyset\},$$

then

$$\max_{\substack{b=(K,h): \\ K \in \mathcal{K}, h \in [H]}} \frac{|S_b^\circ|}{\sqrt{v_b}} = O_p(\sqrt{\log n}).$$

Proof. Fix $a = (I, h)$. By Assumptions B.1, B.4, and Proposition 34,

$$\left\| S_a^\circ \right\|_{\psi_2} \leq CB_0 \sqrt{|I|}$$

for a constant $C < \infty$ depending only on universal constants. Since

$$v_a \geq \sigma_-^2 |I|,$$

we obtain the uniform bound

$$\left\| \frac{S_a^\circ}{\sqrt{v_a}} \right\|_{\psi_2} \leq \frac{CB_0}{\sigma_-}.$$

Hence, for some constants $c_1, c_2 > 0$,

$$\mathbb{P}\left(\left| \frac{S_a^\circ}{\sqrt{v_a}} \right| > u\right) \leq c_1 e^{-c_2 u^2} \quad (u \geq 0)$$

uniformly in $a \in A$. Since $\mathbb{T}_a^\circ = \omega_a S_a^\circ / \sqrt{v_a}$ and Assumption B.2 bounds ω_a uniformly above, there exist

constants $c'_1, c'_2 > 0$ such that

$$\mathbb{P}(|\mathbb{T}_a^\circ| > u) \leq c'_1 e^{-c'_2 u^2} \quad (u \geq 0)$$

uniformly in $a \in A$. A union bound over the p coordinates and the choice $u = M\sqrt{\log n}$, with M large enough, prove the first claim.

For the second claim, note that $|\mathcal{K}| \leq |\mathcal{J}|^2$, hence

$$H|\mathcal{K}| \leq H|\mathcal{J}|^2 = \frac{p^2}{H} \leq p^2.$$

For each $b = (K, h)$ with $K \in \mathcal{K}$ and $h \in [H]$, Assumptions B.1 and B.4, Proposition 34, and $v_b \geq \sigma_-^2 |K|$ give $\|S_b^\circ / \sqrt{v_b}\|_{\psi_2} \leq C$. Applying the union bound over $H|\mathcal{K}| \leq p^2$ variables gives the same $\sqrt{\log n}$ order. $\square$

The next lemma shows that the observed denominator uniformly approximates the oracle variance scale in Regime I.

Lemma 13. *Under Assumptions B.1, B.2, and B.4–B.6,*

$$\Delta_Q := \max_{a \in A} \left| \frac{Q_a}{v_a} - 1 \right| = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + r_n \right).$$

In particular, $\Delta_Q = o_p(1)$ and $\mathbb{P}(\Delta_Q > 1/2) \rightarrow 0$.

Proof. By Lemma 2, it suffices to control the three terms in the decomposition of $(Q_a - v_a)/v_a$ uniformly over $a \in A$.

Step 1: control of $\frac{q_a^\circ}{v_a}$. By (B.13),

$$q_a^\circ = \sum_{t \in I} \left(d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}] \right) - |I| \bar{d}_a^2.$$

By Assumption B.1 and Proposition 35, $d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}]$ is uniformly sub-exponential. Hence Bernstein's inequality yields, for every fixed $a = (I, h)$ and every $u \geq 1$,

$$\mathbb{P} \left(\left| \sum_{t \in I} \left(d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}] \right) \right| > C(\sqrt{|I|u} + u) \right) \leq 2e^{-u}$$

for some constant $C < \infty$. Since $p = |A| \leq n^2$, the choice $u = x + 2 \log n$ gives $pe^{-u} \leq e^{-x}$. Taking this u and applying a union bound over $a \in A$ gives, for every $x \geq 1$, with probability at least $1 - 2e^{-x}$,

$$\max_{a=(I,h) \in A} \frac{\left| \sum_{t \in I} (d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}]) \right|}{v_a} \leq C \max_{a \in A} \left\{ \sqrt{\frac{x + 2 \log n}{|I_a|}} + \frac{x + 2 \log n}{|I_a|} \right\}.$$

Since $|I_a| \geq L_{\min}$, this proves

$$\max_{a=(I,h) \in A} \frac{\left| \sum_{t \in I} (d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}]) \right|}{v_a} = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + \frac{\log n}{L_{\min}} \right).$$

Assumption B.2 gives

$$\frac{v_a}{|I|} \geq \sigma_-^2, \quad \omega_a = \left(\frac{v_a}{|I|} \right)^{1/2-\gamma} \geq (\sigma_-^2)^{1/2-\gamma}, \quad \omega_a^{-1} \leq C.$$

Since $S_a^\circ / \sqrt{v_a} = \omega_a^{-1} \mathbb{T}_a^\circ$, Lemma 12 gives

$$\max_{a \in A} \frac{(S_a^\circ)^2}{|I_a| v_a} \leq \frac{C(\bar{\mathbb{T}}_{\max}^\circ)^2}{L_{\min}} = O_p \left(\frac{\log n}{L_{\min}} \right).$$

Combining the last two displays with the decomposition of q_a° at the start of this step gives

$$\max_{a \in A} \frac{|q_a^\circ|}{v_a} = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + \frac{\log n}{L_{\min}} \right).$$

Step 2: control of $\frac{C_a}{v_a}$. Since $\sum_{t \in I} \mu_{t,a} = 0$, the random variable C_a in (B.13) satisfies $C_a = \sum_{t \in I} \mu_{t,a} d_t^{(h)}$. Conditionally on the deterministic coefficients $\mu_{t,a}$, it is a weighted sum of the independent centered sub-Gaussian variables $\{d_t^{(h)} : t \in I\}$. By Proposition 34,

$$\|C_a\|_{\psi_2} \leq C B_0 \left(\sum_{t \in I} \mu_{t,a}^2 \right)^{1/2} = C B_0 v_{a,\mu}^{1/2}.$$

Therefore, for every fixed a and every $u \geq 0$,

$$\mathbb{P}(|C_a| > C v_{a,\mu}^{1/2} \sqrt{u}) \leq 2e^{-u}.$$

A union bound over $a \in A$ yields directly

$$\max_{a \in A} \frac{|C_a|}{v_a} = O_p \left(\max_{a \in A} \frac{\sqrt{v_{a,\mu} \log n}}{v_a} \right).$$

Since $v_{a,\mu} \leq r_n v_a$ and $v_a \geq \sigma_-^2 L_{\min}$,

$$\max_{a \in A} \frac{|C_a|}{v_a} = O_p \left(\sqrt{\frac{r_n \log n}{L_{\min}}} \right).$$

Step 3: control of $\frac{v_{a,\mu}}{v_a}$. By definition (B.15) of r_n ,

$$\max_{a \in A} \frac{v_{a,\mu}}{v_a} = r_n.$$

Assumptions B.5 and B.6 imply

$$\frac{\log n}{L_{\min}} = o\left(\sqrt{\frac{\log n}{L_{\min}}}\right), \quad \sqrt{\frac{r_n \log n}{L_{\min}}} = O\left(\sqrt{\frac{\log n}{L_{\min}}}\right).$$

Collecting the three bounds proves

$$\Delta_Q = O_p\left(\sqrt{\frac{\log n}{L_{\min}}} + r_n\right),$$

as claimed. □

The next lemma gives uniform maximum and fourth-moment bounds for the centered variables $d_t^{(h)} = D_t^{(h)} - \mu_t^{(h)}$ in Regime I.

Lemma 14. *Under Assumptions B.1, B.4, and B.6,*

$$\max_{a=(I,h) \in A} \max_{t \in I} (d_t^{(h)})^2 = O_p(\log n), \quad \max_{a=(I,h) \in A} \frac{1}{|I|} \sum_{t \in I} (d_t^{(h)})^4 = O_p(1).$$

Proof. Assumption B.1, (B.2), the union bound over $nH \leq np$, and Proposition 33 give

$$\max_{t \in [n], h \in [H]} |d_t^{(h)}| = O_p(\sqrt{\log n}).$$

This proves the first conclusion.

Uniform sub-Gaussianity implies

$$\sup_{t,h} \mathbb{E}[(d_t^{(h)})^4] < \infty, \quad \sup_{t,h} \mathbb{P}((d_t^{(h)})^4 > u) \leq 2e^{-c\sqrt{u}}.$$

Let $X_t^{(h)} = (d_t^{(h)})^4 - \mathbb{E}[(d_t^{(h)})^4]$. For the variance term V in (B.77), where $\varphi_M(x) = (x \wedge M) \vee (-M)$,

$$\begin{aligned} \text{Cov}\left(\varphi_M(X_s^{(h)}), \varphi_M(X_t^{(h)})\right) &= 0 \quad (s \neq t), \\ \sup_{M,t,h} \text{Var}\left(\varphi_M(X_t^{(h)})\right) &\leq \sup_{t,h} \mathbb{E}[(X_t^{(h)})^2] \leq C \sup_{t,h} \mathbb{E}[(d_t^{(h)})^8] < \infty. \end{aligned}$$

Hence $V = O(1)$ uniformly. Proposition 31 may therefore be applied with $\gamma_1 = 1$ and $\gamma_2 = 1/2$. Taking

$y = Cm$ in (B.78) gives, for an interval of length m ,

$$\mathbb{P}\left(\left|\sum_{t \in I} X_t^{(h)}\right| > Cm\right) \leq C(m+1)e^{-cm^{1/3}} + Ce^{-cm}. \quad (\text{B.40})$$

Assumption B.6 gives $(\log n)^7 / L_{\min} \rightarrow 0$, which implies $\log n = o(L_{\min}^{1/3})$. A union bound over A in (B.40) proves the result. $\square$

B.3.4 Proof of the centered Gaussian approximation

For $t \in [n]$ and $a = (I, h) \in A$, define

$$Z_{t,a} := \frac{\sqrt{n} \mathbf{1}\{t \in I\} d_t^{(h)}}{|I|^{1/2-\gamma} \nu_a^\gamma}. \quad (\text{B.41})$$

Let $Z_t := (Z_{t,a})_{a \in A}$. By the definition of $\mathbb{T}^\circ$ in (B.36),

$$\mathbb{T}^\circ = \frac{1}{\sqrt{n}} \sum_{t=1}^n Z_t.$$

Set

$$B_n^I := 1 \vee B_0(1 \vee \sigma_-^{-1}) \sqrt{\frac{n}{L_{\min}}}.$$

Assumption B.2 gives

$$\sigma_-^2 |I| \leq \nu_a = \sum_{u \in I} \mathbb{E}[d_u^{(h)2}] \leq \sigma_+^2 |I|.$$

In particular,

$$|I|^{1/2-\gamma} \nu_a^\gamma \geq |I|^{1/2-\gamma} (\sigma_-^2 |I|)^\gamma = \sigma_-^{2\gamma} |I|^{1/2}, \quad \sigma_-^{-2\gamma} \leq 1 \vee \sigma_-^{-1},$$

because $0 \leq \gamma \leq 1/2$. Therefore

$$\|Z_{t,a}\|_{\psi_2} \leq \frac{B_0 \sqrt{n}}{|I|^{1/2-\gamma} \nu_a^\gamma} \leq B_0(1 \vee \sigma_-^{-1}) \sqrt{\frac{n}{|I|}} \leq B_n^I \quad \forall t, a.$$

The standard embedding of Orlicz norms gives $\|Z_{t,a}\|_{\psi_1} \leq C \|Z_{t,a}\|_{\psi_2} \leq CB_n^I$. Thus the condition $\max_{t,a} \|Z_{t,a}\|_{\psi_1} \leq B$ in Proposition 23 holds with $B = CB_n^I$, and

$$\frac{(CB_n^I)^2 \log^7(pn)}{n} = C^2 \frac{(B_n^I)^2 \log^7(pn)}{n}.$$

Finally,

$$\text{Var}\left(\frac{1}{\sqrt{n}} \sum_{t=1}^n Z_{t,a}\right) = \text{Var}(\mathbb{T}_a^\circ) = \left(\frac{v_a}{|I|}\right)^{1-2\gamma} \quad \forall a \in A.$$

The same bound from Assumption B.2 gives

$$(\sigma_-^2)^{1-2\gamma} \leq \left(\frac{v_a}{|I|}\right)^{1-2\gamma} \leq (\sigma_+^2)^{1-2\gamma} \quad \forall a \in A,$$

so the coordinate variances are uniformly bounded above and away from zero. By Assumption B.6,

$$\frac{(B_n^I)^2 (\log(pn))^7}{n} \lesssim \frac{(\log n)^7}{L_{\min}} \rightarrow 0,$$

where $\log(pn) \asymp \log n$ by (B.2) and the final convergence is Assumption B.6. The conditions of Proposition 23 are therefore satisfied. Thus,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}(\mathbb{T}_{\max}^\circ \leq x) - \mathbb{P}(G_{\circ, \max} \leq x) \right| \rightarrow 0. \quad (\text{B.42})$$

This is (B.38).

B.3.5 Proof of the fixed-denominator bootstrap covariance

For $a = (I, h)$ and $b = (J, g)$, set $K := I \cap J$. Recall from (B.10) and (B.11) that $D_t^{(h)} - \bar{D}_a = d_{t,a} + \mu_{t,a}$ for $t \in I$. By direct calculation,

$$\hat{\Sigma}_{ab} = \frac{\hat{\Omega}_{ab}}{|I_a|^{1/2-\gamma} |I_b|^{1/2-\gamma} Q_a^\gamma Q_b^\gamma}.$$

Here

$$\hat{\Omega}_{ab} := \sum_{t \in K} \left(D_t^{(h)} - \bar{D}_a \right) \left(D_t^{(g)} - \bar{D}_b \right).$$

The target covariance entry is

$$(\Sigma_\circ)_{ab} = \frac{\Omega_{ab}}{|I_a|^{1/2-\gamma} |I_b|^{1/2-\gamma} v_a^\gamma v_b^\gamma}, \quad \Omega_{ab} := \sum_{t \in K} \mathbb{E}[d_t^{(h)} d_t^{(g)}].$$

Hence

$$\hat{\Sigma}_{ab} - (\Sigma_\circ)_{ab} = \frac{\hat{\Omega}_{ab} - \Omega_{ab}}{|I_a|^{1/2-\gamma} |I_b|^{1/2-\gamma} Q_a^\gamma Q_b^\gamma} \quad (\text{B.43})$$

$$+ (\Sigma_\circ)_{ab} \left\{ \left(\frac{Q_a}{v_a} \right)^{-\gamma} \left(\frac{Q_b}{v_b} \right)^{-\gamma} - 1 \right\}. \quad (\text{B.44})$$

We bound the two terms uniformly.

Step 1: bounding (B.44). On $\{\Delta_Q \leq 1/2\}$, for every $a, b \in A$,

$$\frac{Q_a}{v_a}, \frac{Q_b}{v_b} \in [1/2, 3/2], \quad \left| \left(\frac{Q_a}{v_a} \right)^{-\gamma} \left(\frac{Q_b}{v_b} \right)^{-\gamma} - 1 \right| \leq C \left(\left| \frac{Q_a}{v_a} - 1 \right| + \left| \frac{Q_b}{v_b} - 1 \right| \right) \leq C \Delta_Q.$$

Cauchy–Schwarz and Assumption B.2 give

$$\begin{aligned} |\Omega_{ab}| &\leq \left(\sum_{t \in K} \mathbb{E}[d_t^{(h)2}] \right)^{1/2} \left(\sum_{t \in K} \mathbb{E}[d_t^{(g)2}] \right)^{1/2} \leq \sqrt{v_a v_b}, \\ |(\Sigma_\circ)_{ab}| &\leq \left(\frac{v_a}{|I|} \right)^{1/2-\gamma} \left(\frac{v_b}{|J|} \right)^{1/2-\gamma} \leq C. \end{aligned}$$

Therefore

$$\max_{a,b} \left| (\Sigma_\circ)_{ab} \left\{ \left(\frac{Q_a}{v_a} \right)^{-\gamma} \left(\frac{Q_b}{v_b} \right)^{-\gamma} - 1 \right\} \right| \leq C \Delta_Q = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + r_n \right), \quad (\text{B.45})$$

by Lemma 13.

Step 2: bounding (B.43). Expand

$$\hat{\Omega}_{ab} = \sum_{t \in K} d_{t,a} d_{t,b} + \sum_{t \in K} d_{t,a} \mu_{t,b} + \sum_{t \in K} d_{t,b} \mu_{t,a} + \sum_{t \in K} \mu_{t,a} \mu_{t,b}. \quad (\text{B.46})$$

We analyze the four pieces.

Bounding the first term of (B.46). A direct expansion gives

$$\sum_{t \in K} d_{t,a} d_{t,b} - \Omega_{ab} = \hat{\omega}_{ab}^\circ - \bar{d}_b S_{(K,h)}^\circ - \bar{d}_a S_{(K,g)}^\circ + |K| \bar{d}_a \bar{d}_b,$$

where

$$\hat{\omega}_{ab}^\circ := \sum_{t \in K} \left(d_t^{(h)} d_t^{(g)} - \mathbb{E}[d_t^{(h)} d_t^{(g)}] \right).$$

By Assumption B.1, $d_t^{(h)}$ and $d_t^{(g)}$ are both sub-Gaussian. By Proposition 35, $d_t^{(h)} d_t^{(g)}$ is sub-exponential; Bernstein's inequality and a union bound over the p^2 pairs (a, b) yield

$$\max_{a,b} \frac{|\hat{\omega}_{ab}^\circ|}{\sqrt{v_a v_b}} = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + \frac{\log n}{L_{\min}} \right).$$

Also, Assumption B.2 gives $\omega_b^{-1} \leq C$, and $S_b^\circ / \sqrt{v_b} = \omega_b^{-1} \mathbb{T}_b^\circ$. Hence Lemma 12 gives

$$\max_b \frac{|\bar{d}_b|}{\sqrt{v_b}} = \max_b \frac{|S_b^\circ|}{|I_b| \sqrt{v_b}} \leq \frac{C \bar{\mathbb{T}}_{\max}^\circ}{L_{\min}} = O_p \left(\frac{\sqrt{\log n}}{L_{\min}} \right),$$

and

$$\max_{a,b} \frac{|S_{(K,h)}^\circ|}{\sqrt{v_a}} = O_p(\sqrt{\log n}), \quad \max_{a,b} \frac{|S_{(K,g)}^\circ|}{\sqrt{v_b}} = O_p(\sqrt{\log n}),$$

so the two sample-mean terms are both

$$O_p\left(\frac{\log n}{L_{\min}}\right).$$

Finally,

$$\max_{a,b} \frac{|K| |\bar{d}_a \bar{d}_b|}{\sqrt{v_a v_b}} \leq \max_{a,b} \frac{|K|}{|I_a| |I_b|} \frac{|S_a^\circ|}{\sqrt{v_a}} \frac{|S_b^\circ|}{\sqrt{v_b}} \leq \frac{C(\bar{\mathbb{T}}_{\max}^\circ)^2}{L_{\min}} = O_p\left(\frac{\log n}{L_{\min}}\right),$$

where the last line follows from Lemma 12. Therefore

$$\max_{a,b} \frac{|\sum_{t \in K} d_{t,a} d_{t,b} - \Omega_{ab}|}{\sqrt{v_a v_b}} = O_p\left(\sqrt{\frac{\log n}{L_{\min}}} + \frac{\log n}{L_{\min}}\right). \quad (\text{B.47})$$

Bounding the second and third terms of (B.46). Consider $\sum_{t \in K} d_{t,a} \mu_{t,b}$; the other term is symmetric.

Then

$$\sum_{t \in K} d_{t,a} \mu_{t,b} = \sum_{t \in K} d_t^{(h)} \mu_{t,b} - \bar{d}_a \sum_{t \in K} \mu_{t,b}.$$

The first term is a weighted sum of the independent centered sub-Gaussian variables $\{d_t^{(h)} : t \in K\}$ with coefficient vector $(\mu_{t,b})_{t \in K}$, hence is sub-Gaussian with scale at most $CB_0(\sum_{t \in K} \mu_{t,b}^2)^{1/2} \leq CB_0 v_{b,\mu}^{1/2}$. For fixed (a, b) and $u \geq 1$, (B.15) and $v_a \geq \sigma_-^2 L_{\min}$ give

$$\mathbb{P}\left(\frac{|\sum_{t \in K} d_t^{(h)} \mu_{t,b}|}{\sqrt{v_a v_b}} > C \sqrt{\frac{r_n u}{L_{\min}}}\right) \leq 2e^{-u}.$$

Since there are at most $p^2 \leq n^4$ pairs (a, b) , taking $u = x + 4 \log n$ gives

$$\max_{a,b} \frac{|\sum_{t \in K} d_t^{(h)} \mu_{t,b}|}{\sqrt{v_a v_b}} = O_p\left(\sqrt{\frac{r_n \log n}{L_{\min}}}\right).$$

For the second term, Cauchy–Schwarz yields

$$\left|\sum_{t \in K} \mu_{t,b}\right| \leq \sqrt{|K| v_{b,\mu}} \leq \sqrt{|I_a| v_{b,\mu}}.$$

Assumption B.2 gives $\omega_a^{-1} \leq C$ and $S_a^\circ / \sqrt{v_a} = \omega_a^{-1} \mathbb{T}_a^\circ$. Thus Lemma 12 gives

$$\frac{|\bar{d}_a| |\sum_{t \in K} \mu_{t,b}|}{\sqrt{v_a v_b}} \leq \frac{|S_a^\circ|}{\sqrt{v_a}} \frac{1}{\sqrt{|I_a|}} \sqrt{\frac{v_{b,\mu}}{v_b}} \leq C \bar{\mathbb{T}}_{\max}^\circ \sqrt{\frac{r_n}{L_{\min}}} = O_p \left(\sqrt{\frac{r_n \log n}{L_{\min}}} \right).$$

Hence

$$\max_{a,b} \frac{|\sum_{t \in K} d_{t,a} \mu_{t,b}|}{\sqrt{v_a v_b}} = O_p \left(\sqrt{\frac{r_n \log n}{L_{\min}}} \right), \quad (\text{B.48})$$

and the same bound holds for $\sum_{t \in K} d_{t,b} \mu_{t,a}$.

Bounding the last term of (B.46). By Cauchy–Schwarz and the definition of r_n ,

$$\max_{a,b} \frac{|\sum_{t \in K} \mu_{t,a} \mu_{t,b}|}{\sqrt{v_a v_b}} \leq \max_{a,b} \sqrt{\frac{\sum_{t \in K} \mu_{t,a}^2}{v_a}} \sqrt{\frac{\sum_{t \in K} \mu_{t,b}^2}{v_b}} \leq r_n. \quad (\text{B.49})$$

Combining (B.47), (B.48), and (B.49) yields

$$\max_{a,b} \frac{|\hat{\Omega}_{ab} - \Omega_{ab}|}{\sqrt{v_a v_b}} = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + r_n \right).$$

On $\{\Delta_Q \leq 1/2\}$, Assumption B.3 gives

$$\max_{a,b} \frac{|\hat{\Omega}_{ab} - \Omega_{ab}|}{|I_a|^{1/2-\gamma} |I_b|^{1/2-\gamma} Q_a^\gamma Q_b^\gamma} \leq C \max_{a,b} \frac{|\hat{\Omega}_{ab} - \Omega_{ab}|}{\sqrt{v_a v_b}} = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + r_n \right).$$

Together with (B.45) and the decomposition (B.43)–(B.44), this proves

$$\|\hat{\Sigma} - \Sigma_\circ\|_{\max} = O_p \left(\sqrt{\frac{\log n}{L_{\min}}} + r_n \right).$$

Finally, Assumptions B.5 and B.6 imply

$$\left(\sqrt{\frac{\log n}{L_{\min}}} + r_n \right) (\log n)^2 = \frac{(\log n)^{5/2}}{\sqrt{L_{\min}}} + r_n (\log n)^2 \rightarrow 0.$$

Thus,

$$\|\hat{\Sigma} - \Sigma_\circ\|_{\max} = o_p((\log n)^{-2}).$$

B.3.6 Verifying conditions of the reduction lemmas

We verify the hypotheses of the three reduction lemmas used in the roadmap for Regime I. We first verify Lemma 7. For $\gamma = 0$, the two denominators are identical. Uniformly over $\gamma \in [0, 1/2]$, Lemmas 13 and 12 give

$$C\gamma\Delta_Q\bar{\mathbb{T}}_{\max}^\circ = O_p\left(\left(\sqrt{\frac{\log n}{L_{\min}}} + r_n\right)\sqrt{\log n}\right).$$

Assumptions B.5 and B.6 imply

$$\left(\sqrt{\frac{\log n}{L_{\min}}} + r_n\right)\log n = \frac{(\log n)^{3/2}}{\sqrt{L_{\min}}} + r_n \log n \rightarrow 0.$$

Choose a deterministic sequence $M_n \uparrow \infty$ sufficiently slowly that, with

$$\begin{aligned}\eta_n &:= M_n \left(\sqrt{\frac{\log n}{L_{\min}}} + r_n\right)\sqrt{\log n}, \\ \eta_n &\rightarrow 0, \quad \eta_n \sqrt{1 \vee \log(p/\eta_n)} \rightarrow 0.\end{aligned}$$

Then

$$\mathbb{P}(\Delta_Q > 1/2) + \mathbb{P}(C\gamma\Delta_Q\bar{\mathbb{T}}_{\max}^\circ > \eta_n) \rightarrow 0. \quad (\text{B.50})$$

Since $\Sigma_\circ = \text{Cov}(\mathbb{T}^\circ)$ and $\mathbb{T}^\circ$ is defined in (B.36), Regime I independence gives, for $a = (I, h)$,

$$(\Sigma_\circ)_{aa} = \text{Var}(\mathbb{T}_a^\circ) = \left(\frac{v_a}{|I|}\right)^{1-2\gamma}.$$

Assumption B.2 therefore yields

$$(\sigma_-^2)^{1-2\gamma} \leq (\Sigma_\circ)_{aa} \leq (\sigma_+^2)^{1-2\gamma} \quad \forall a \in A.$$

Proposition 24 therefore implies

$$\sup_{x \in \mathbb{R}} \mathbb{P}(|G_{\circ, \max} - x| \leq \eta_n) \rightarrow 0. \quad (\text{B.51})$$

For every $x \in \mathbb{R}$, (B.38) gives

$$\mathbb{P}(|\mathbb{T}_{\max}^\circ - x| \leq \eta_n) \leq \mathbb{P}(|G_{\circ, \max} - x| \leq \eta_n) + 2 \sup_{z \in \mathbb{R}} |\mathbb{P}(\mathbb{T}_{\max}^\circ \leq z) - \mathbb{P}(G_{\circ, \max} \leq z)|.$$

Thus (B.38) and (B.51) imply

$$\sup_{x \in \mathbb{R}} \mathbb{P}(|\mathbb{T}_{\max}^{\circ} - x| \leq \eta_n) \rightarrow 0.$$

Together with (B.50), this verifies the conditions of Lemma 7.

We next verify Lemma 8 with $k_n = 1$. Lemma 13 and $v_a \geq \sigma_-^2 |I|$ give $Q_a \geq c|I|$ uniformly with probability $1 - o(1)$ over data. Lemma 14 verifies (B.18). Since $\Theta_n = I_n$,

$$\text{Var}^* \left(\frac{S_a^*}{\sqrt{Q_a}} \right) = \frac{\sum_{t \in I} (D_t^{(h)} - \bar{D}_a)^2}{Q_a} = 1,$$

which verifies (B.19), and (B.20) holds with $k_n = 1$. Finally, Assumptions B.5 and B.6 give

$$\frac{(\log n)^3}{L_{\min}} \rightarrow 0, \quad r_n (\log n)^2 \rightarrow 0.$$

Thus (B.22) holds.

It remains to verify Lemma 9. Under Regime I independence, $(\Sigma_{\circ})_{aa} = (v_a/|I|)^{1-2\gamma}$, so Assumption B.2 gives diagonal entries bounded above and away from zero. Equation (B.39) gives $\|\hat{\Sigma} - \Sigma_{\circ}\|_{\max} = o_p((\log n)^{-2})$, and hence the same diagonal bounds hold for $\hat{\Sigma}$ with probability $1 - o(1)$ over data. This verifies all conditions of Lemma 9.

We now complete the proof of Theorem 11. Sections B.3.4 and B.3.5 prove (B.38) and (B.39), respectively, and the preceding paragraphs verify the conditions of Lemmas 7–9. Therefore the conditions of Lemma 10 hold. Applying Lemma 10 proves the theorem.

B.4 Regime D Validity

B.4.1 Additional assumptions for Regime D

We impose the following assumptions.

Assumption B.7. *The centered vector sequence $d_t = (d_t^{(1)}, \dots, d_t^{(H)})$, $t = 1, \dots, n$, is geometrically α -mixing: there exist constants $K_1 > 1$, $K_2 > 0$, and $\gamma_{\text{mix}} > 0$ such that*

$$\alpha(\ell) := \sup_{1 \leq k \leq n-\ell} \sup_{A \in \sigma(d_1, \dots, d_k), B \in \sigma(d_{k+\ell}, \dots, d_n)} |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)| \leq K_1 e^{-K_2 \ell^{\gamma_{\text{mix}}}} \quad \forall \ell \geq 1.$$

Assumption B.8. *The centered oracle coordinates are nondegenerate: there exist constants $0 < c_- \leq c_+ < \infty$ such that*

$$c_- \leq \text{Var} \left(\frac{S_a^{\circ}}{\sqrt{v_a}} \right) \leq c_+ \quad \forall a \in A.$$

Assumption B.9. *The kernel is symmetric, continuously differentiable, satisfies $K(0) = 1$,*

$$\sup_{x \in \mathbb{R}} |K'(x)| < \infty,$$

and obeys

$$|K(x)| \leq C_K(1 + |x|)^{-\vartheta} \quad \forall x \in \mathbb{R}$$

for some constants $C_K < \infty$ and $\vartheta > 1$. The bandwidth satisfies

$$b_n \asymp n^\rho, \quad 0 < \rho < \frac{\vartheta - 1}{3\vartheta - 2}.$$

The matrices

$$\Theta_n(t, u) := K\left(\frac{t - u}{b_n}\right), \quad 1 \leq t, u \leq n, \quad (\text{B.52})$$

are positive semidefinite.

Remark 2. We prove in Section B.4.7 that the quadratic spectral kernel introduced by Andrews (1991) satisfies Assumption B.9. Section B.4.8 further shows that the test remains valid in Regime D with the data-driven bandwidth used in the main text.

Assumption B.10. *Let*

$$B_n := 1 \vee \frac{B_0 \sqrt{n}}{\sigma - \sqrt{L_{\min}}}.$$

Assume

$$(\log n)^{3-\gamma_{\text{mix}}} = o(n^{\gamma_{\text{mix}}/3}),$$

and

$$(B_n)^{2/3} \frac{(\log n)^{(1+2\gamma_{\text{mix}})/(3\gamma_{\text{mix}})}}{n^{1/9}} + B_n \frac{(\log n)^{7/6}}{n^{1/9}} \rightarrow 0.$$

Let $c_1 = c_1(\rho, \vartheta) > 0$ and $c_2 = c_2(2, \gamma_{\text{mix}}, \vartheta) > 0$ denote the constants delivered by Proposition 28 with its kernel parameters specialized to $\gamma_1 = 2$ and $\gamma_2 = \gamma_{\text{mix}}$. Assume

$$(B_n)^2 n^{-c_1} (\log n)^{c_2+2} \rightarrow 0, \quad (B_n)^2 n^{-\rho} (\log n)^2 \rightarrow 0.$$

In addition, $b_n(\log n)^2/n \rightarrow 0$.

Assumption B.11. *The interval-wise mean is asymptotically flat:*

$$b_n r_n (\log n)^2 \rightarrow 0.$$

Assumption B.12. *The bandwidth, dimension, mixing exponent, and minimum interval length satisfy*

$$\begin{aligned} \frac{(\log n)^5}{L_{\min}} &\rightarrow 0, & \frac{b_n(\log n)^3}{L_{\min}} &\rightarrow 0, \\ \frac{(\log n)^{1+2/\gamma_{\text{mix}}}}{L_{\min}} &\rightarrow 0, & \frac{(\log n)^{2+1/\gamma_{\text{mix}}}}{L_{\min}} &\rightarrow 0. \end{aligned}$$

When $\gamma_{\text{mix}} = 1$, they are implied by the first condition.

Remark 3. When $\gamma_{\text{mix}} = 1$ and the quadratic spectral kernel is used, Section B.4.7 gives $\vartheta = 2$ and $b_n \asymp n^\rho$ with $0 < \rho < 1/4$. Take $\rho = 1/4 - \epsilon$ for a fixed $0 < \epsilon < 1/4$. Proposition 29 gives

$$c_1(1/4 - \epsilon, 2) = \frac{4\epsilon}{3}, \quad c_2(2, 1, 2) = \frac{10}{3}, \quad c_2^\sharp(2, 1, 2) = \frac{2}{3}.$$

Since $p \leq n^2$, we use the improved exponent $c_2^\sharp = 2/3$ in Assumption B.10. The definition in Assumption B.10 satisfies

$$1 \vee \frac{B_0 \sqrt{n}}{\sigma_- \sqrt{L_{\min}}} \asymp \sqrt{\frac{n}{L_{\min}}},$$

and Assumption B.10 reduces to

$$\frac{n^{7/18}(\log n)^{7/6}}{\sqrt{L_{\min}}} \rightarrow 0, \quad \frac{n^{1-4\epsilon/3}(\log n)^{8/3}}{L_{\min}} \rightarrow 0, \quad \frac{n^{3/4+\epsilon}(\log n)^2}{L_{\min}} \rightarrow 0. \quad (\text{B.53})$$

Indeed, $(\log n)^2 = o(n^{1/3})$ and $n^{-3/4-\epsilon}(\log n)^2 \rightarrow 0$ hold automatically. Also, $L_{\min} \leq n$ gives the omitted high-dimensional CLT term

$$\frac{n^{2/9}(\log n)/L_{\min}^{1/3}}{n^{7/18}(\log n)^{7/6}/L_{\min}^{1/2}} = \left(\frac{L_{\min}}{n \log n} \right)^{1/6} \rightarrow 0.$$

The first condition in (B.53) is implied by the other two. Indeed, those two conditions require $L_{\min}$ to dominate polynomial powers with exponents $1 - 4\epsilon/3$ and $3/4 + \epsilon$, respectively. Since

$$\max\{1 - 4\epsilon/3, 3/4 + \epsilon\} \geq 6/7,$$

the resulting polynomial slack is stronger than the requirement $L_{\min} \gg n^{7/9}(\log n)^{7/3}$ imposed by the first condition. Assumption B.11 becomes

$$n^{1/4-\epsilon} r_n (\log n)^2 \rightarrow 0, \quad (\text{B.54})$$

and Assumption B.12 reduces to

$$\frac{n^{1/4-\epsilon}(\log n)^3}{L_{\min}} \rightarrow 0,$$

which is implied by the first condition in (B.53) and also implies the other three length conditions in Assumption B.12. Combining (B.53) and (B.54), Assumptions B.10–B.12 hold if

$$L_{\min} = \omega(\max\{n^{1-4\epsilon/3}(\log n)^{8/3}, n^{3/4+\epsilon}(\log n)^2\}), \quad r_n = o\left(\frac{1}{n^{1/4-\epsilon}(\log n)^2}\right).$$

In Section 3.1, we choose $b_n \asymp n^{1/5}$, giving $\epsilon = 1/20$. In this case, the condition reduces to

$$L_{\min} = \omega(n^{14/15}(\log n)^{8/3}), \quad r_n = o\left(\frac{1}{n^{1/5}(\log n)^2}\right).$$

B.4.2 Main theorem

Theorem 15. *Fix the exponent $\gamma \in [0, 1/2]$. Assume Assumptions B.1, B.2, B.3, B.7–B.9, and B.10–B.12. Let $c_{n,1-\alpha}^*$ be the conditional $(1 - \alpha)$ -quantile in (B.6). Then the Gaussian multiplier bootstrap with covariance $\Theta_n(t, u) = K((t - u)/b_n)$ is asymptotically conservative: under (B.5),*

$$\limsup_{n \rightarrow \infty} \mathbb{P}(T_{\max} > c_{n,1-\alpha}^*) \leq \alpha \quad \forall \alpha \in (0, 1).$$

Following the roadmap in Section B.2.5, we are left to prove (B.38) and (B.39) and verify the conditions in Lemmas 7–9. In the following, Section B.4.3 provides preliminary results, Section B.4.4 proves (B.38), Section B.4.5 proves (B.39), and Section B.4.6 completes the proof of Theorem 15 by verifying the conditions in Lemmas 7–9. For Sections B.4.4–B.4.6, we assume Assumptions B.1, B.2, B.3, B.7–B.9, and B.10–B.12 all hold.

B.4.3 Preliminary bounds

Lemma 16. *Under Assumption B.9, there exists $C_{\Theta} < \infty$ such that*

$$\max_{1 \leq t \leq n} \sum_{u=1}^n |\Theta_n(t, u)| \leq C_{\Theta} b_n, \quad \max_{1 \leq t \leq n} \sum_{u=1}^n \Theta_n(t, u)^2 \leq C_{\Theta} b_n \quad \text{for all } n.$$

Consequently,

$$|x^{\top} \Theta_n y| \leq C_{\Theta} b_n \|x\|_2 \|y\|_2 \quad \forall x, y \in \mathbb{R}^n.$$

Proof. Fix t . Since $|K(x)| \leq C_K(1 + |x|)^{-\vartheta}$ with $\vartheta > 1$,

$$\sum_{u=1}^n |\Theta_n(t, u)| \leq \sum_{u=1}^n C_K \left(1 + \frac{|t - u|}{b_n}\right)^{-\vartheta} \leq C_K \sum_{m \in \mathbb{Z}} \left(1 + \frac{|m|}{b_n}\right)^{-\vartheta}.$$

Split the last sum into $|m| \leq b_n$ and $|m| > b_n$. The first part contributes $O(b_n)$. For the second part,

$$\sum_{|m| > b_n} \left(1 + \frac{|m|}{b_n}\right)^{-\vartheta} \leq C b_n \sum_{k \geq 1} (1+k)^{-\vartheta} = O(b_n)$$

because $\vartheta > 1$. This proves the row-sum bound. For the squared row-sum bound,

$$\sum_{u=1}^n \Theta_n(t, u)^2 \leq C \sum_{m \in \mathbb{Z}} \left(1 + \frac{|m|}{b_n}\right)^{-2\vartheta}.$$

The terms with $|m| \leq b_n$ contribute $O(b_n)$, while the terms with $|m| > b_n$ are bounded by $C b_n \sum_{k \geq 1} (1+k)^{-2\vartheta} = O(b_n)$ because $2\vartheta > 1$. The quadratic-form bound follows from

$$|x^\top \Theta_n y| \leq \|x\|_2 \|\Theta_n y\|_2 \leq \|x\|_2 \left(\max_t \sum_u |\Theta_n(t, u)| \right) \|y\|_2.$$

□

Lemma 17. *Under Assumptions B.1, B.2, B.3, B.7–B.9, and B.10,*

$$\bar{\mathbb{T}}_{\max}^\circ = O_p(\sqrt{\log n}).$$

Proof. Define

$$Z_{t,a}^\circ := \frac{\sqrt{n} \mathbf{1}\{t \in I\} d_t^{(h)}}{|I|^{1/2-\gamma} v_a^\gamma}, \quad a = (I, h) \in A.$$

Then

$$\mathbb{T}^\circ = \frac{1}{\sqrt{n}} \sum_{t=1}^n Z_t^\circ.$$

As in Section B.3.4,

$$\|Z_{t,a}^\circ\|_{\psi_2} \leq B_0(1 \vee \sigma_-^{-1}) \sqrt{\frac{n}{L_{\min}}} \lesssim B_n \quad \forall t, a.$$

The sequence inherits the same geometric mixing rate. Furthermore,

$$\text{Var}(\mathbb{T}_a^\circ) = \left(\frac{v_a}{|I|}\right)^{1-2\gamma} \text{Var}\left(\frac{S_a^\circ}{\sqrt{v_a}}\right),$$

so Assumptions B.8 and B.2 bound every coordinate variance above and away from zero. Hence Proposition 27 applies under Assumption B.10 and gives

$$\sup_{A \in \mathcal{R}^p} \left| \mathbb{P}(\mathbb{T}^\circ \in A) - \mathbb{P}(G_\circ \in A) \right| \rightarrow 0.$$

Applying this with $A = [-x, x]^p$ yields

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}(\bar{\mathbb{T}}_{\max}^{\circ} \leq x) - \mathbb{P}\left(\max_{a \in A} |G_{\circ, a}| \leq x\right) \right| \rightarrow 0.$$

Proposition 33 applied to $(G_{\circ, a})_{a \in A}$ gives

$$\max_{a \in A} |G_{\circ, a}| = O_p(\sqrt{\log n}),$$

and hence $\bar{\mathbb{T}}_{\max}^{\circ} = O_p(\sqrt{\log n})$. □

Lemma 18. *Assume Assumptions B.1, B.2, and B.7–B.12. Then there exist constants $C_A, c_A, \eta_0 \in (0, \infty)$, depending only on $(B_0, K_1, K_2, \gamma_{\text{mix}})$, such that for every interval $I \subset [n]$ and every $x \geq 1$ satisfying*

$$x \leq c_A |I|^{\gamma_{\text{mix}}/(2+\gamma_{\text{mix}})},$$

one has

$$\mathbb{P}\left(\left|\sum_{t \in I} \left\{(d_t^{(h)})^2 - \mathbb{E}[(d_t^{(h)})^2]\right\}\right| > C_A \sqrt{|I|x}\right) \leq 2e^{-x}.$$

Proof. Fix an interval I and let s be its left endpoint. Define

$$Z_r := (d_{s+r-1}^{(h)})^2 - \mathbb{E}[(d_{s+r-1}^{(h)})^2], \quad r = 1, \dots, |I|.$$

Since each Z_r is a measurable function of D_{s+r-1} , the sequence $(Z_r)_{r=1}^{|I|}$ inherits geometric strong mixing condition (Assumption B.7). After possibly decreasing the exponent constant, the prefactor K_1 can be absorbed into the exponential rate, so there exists $\tilde{K}_2 > 0$, depending only on $(K_1, K_2, \gamma_{\text{mix}})$, such that

$$\alpha_Z(\ell) \leq \exp(-\tilde{K}_2 \ell^{\gamma_{\text{mix}}}) \quad \text{for all } \ell \geq 1.$$

By Assumption B.1 and Proposition 35,

$$\|(d_t^{(h)})^2\|_{\psi_1} \leq B_0^2 \implies \|(d_t^{(h)})^2 - \mathbb{E}[(d_t^{(h)})^2]\|_{\psi_1} \leq 2B_0^2.$$

Therefore there exists $b_0 < \infty$, depending only on B_0 , such that

$$\sup_{1 \leq r \leq |I|} \mathbb{P}(|Z_r| > u) \leq \exp(1 - u/b_0) \quad \text{for all } u > 0.$$

Thus Proposition 31 applies with

$$\gamma_1 = \gamma_{\text{mix}}, \quad \gamma_2 = 1, \quad \gamma = \left(\frac{1}{\gamma_{\text{mix}}} + 1 \right)^{-1} = \frac{\gamma_{\text{mix}}}{1 + \gamma_{\text{mix}}}.$$

Consequently,

$$\frac{\gamma}{2 - \gamma} = \frac{\gamma_{\text{mix}}}{2 + \gamma_{\text{mix}}}.$$

Hence there exist constants $C_0, C_1, C_2, \eta_0 \in (0, \infty)$, depending only on $(B_0, K_1, K_2, \gamma_{\text{mix}})$, such that for every $y \geq C_0(\log |I|)^{\eta_0}$,

$$\mathbb{P} \left(\max_{1 \leq j \leq |I|} \left| \sum_{r=1}^j Z_r \right| \geq y \right) \leq (|I| + 1) \exp \left(-\frac{y^\gamma}{C_1} \right) + \exp \left(-\frac{y^2}{C_2 + C_2 |I| V} \right).$$

Let $q(u)$ denote the tail quantile in Proposition 31,

$$q(u) := \inf \{ x \geq 0 : \sup_r \mathbb{P}(|Z_r| > x) \leq u \}.$$

Since $\sup_r \mathbb{P}(|Z_r| > x) \leq \exp(1 - x/b_0)$, for $0 < u \leq 1$,

$$x \geq b_0 \log(e/u) \implies \sup_r \mathbb{P}(|Z_r| > x) \leq u.$$

Thus $q(u) \leq b_0 \log(e/u)$. Proposition 32 therefore gives

$$\int_0^{2\alpha_Z(k)} q(u)^2 du \lesssim \alpha_Z(k) \log^2(e/\alpha_Z(k)).$$

Because $\alpha_Z(k) \leq \exp(-\tilde{K}_2 k^{\gamma_{\text{mix}}})$,

$$\sum_{k=0}^{\infty} \int_0^{2\alpha_Z(k)} q(u)^2 du \lesssim \sum_{k=0}^{\infty} \alpha_Z(k) \log^2(e/\alpha_Z(k)) \lesssim \sum_{k=0}^{\infty} e^{-\tilde{K}_2 k^{\gamma_{\text{mix}}}} (1 + k^{\gamma_{\text{mix}}})^2 < \infty.$$

Hence the variance term V in Proposition 31 is bounded uniformly in $|I|$, so $V = O(1)$. Since the full interval sum is one of the partial sums, there exists $C < \infty$ such that, for every $y \geq C_0(\log |I|)^{\eta_0}$,

$$\mathbb{P} \left(\left| \sum_{t \in I} \left\{ (d_t^{(h)})^2 - \mathbb{E}[(d_t^{(h)})^2] \right\} \right| \geq y \right) \leq (|I| + 1) \exp \left(-\frac{y^\gamma}{C_1} \right) + \exp \left(-\frac{y^2}{C|I|} \right). \quad (\text{B.55})$$

Now set $y = A\sqrt{|I|x}$ with $A > 0$ to be chosen. Because $x \geq 1$, one has $y \geq A\sqrt{|I|}$, which dominates $C_0(\log |I|)^{\eta_0}$ for all sufficiently large $|I|$; the finitely many small values of $|I|$ are absorbed into the final

constant C_A . Also, from

$$x \leq c_A |I|^{\gamma/(2-\gamma)} = c_A |I|^{\gamma_{\text{mix}}/(2+\gamma_{\text{mix}})},$$

it follows that $x \lesssim |I|^{\gamma/2} x^{\gamma/2}$. Thus the first term in (B.55) satisfies, by choosing A large enough and then c_A small enough,

$$(|I| + 1) \exp\left(-\frac{A^\gamma |I|^{\gamma/2} x^{\gamma/2}}{C_1}\right) \leq e^{-x}.$$

For the second term in (B.55),

$$\exp\left(-\frac{A^2 x}{C}\right) \leq e^{-x}$$

for A chosen large enough. Combining the two bounds gives the claimed inequality after renaming A as C_A . $\square$

Lemma 19. *Under Assumptions B.1, B.2, and B.7–B.12,*

$$\Delta_Q = O_p\left(\sqrt{\frac{\log n}{L_{\min}}} + \sqrt{r_n}\right).$$

In particular, $\Delta_Q = o_p(1)$ and $\mathbb{P}(\Delta_Q > 1/2) \rightarrow 0$.

Proof. Lemma 2 expresses both $q_a - v_a$ and $Q_a - q_a$ in terms of q_a° , C_a , and $v_{a,\mu}$. We control these three terms uniformly.

Step 1: control of $\frac{q_a^\circ}{v_a}$. By (B.13),

$$q_a^\circ = \sum_{t \in I} \left(d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}] \right) - |I| \bar{d}_a^2.$$

By Lemma 18, there exist constants $C_A, c_A > 0$ such that, for every fixed $a = (I, h)$ and every $x \geq 1$ with

$$x + 2 \log n \leq c_A |I|^{\gamma_{\text{mix}}/(2+\gamma_{\text{mix}})},$$

one has

$$\mathbb{P}\left(\left|\sum_{t \in I} \left(d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}]\right)\right| > C_A \sqrt{|I|(x + 2 \log n)}\right) \leq 2e^{-(x + 2 \log n)}.$$

Assumption B.12 implies

$$\log n = o\left(L_{\min}^{\gamma_{\text{mix}}/(2+\gamma_{\text{mix}})}\right),$$

so the restriction is eventually satisfied uniformly over all intervals for any constant x . A union bound over the p coordinates, retaining the coordinate-specific normalization, gives

$$\max_{a=(I,h) \in A} \frac{\left|\sum_{t \in I} \left(d_t^{(h)2} - \mathbb{E}[d_t^{(h)2}]\right)\right|}{v_a} = O_p\left(\max_{a \in A} \sqrt{\frac{\log n}{|I_a|}}\right) = O_p\left(\sqrt{\frac{\log n}{L_{\min}}}\right).$$

Assumption B.2 gives

$$\frac{\nu_a}{|I|} \geq \sigma_-^2, \quad \omega_a = \left(\frac{\nu_a}{|I|} \right)^{1/2-\gamma} \geq (\sigma_-^2)^{1/2-\gamma}, \quad \omega_a^{-1} \leq C.$$

Since $S_a^\circ / \sqrt{\nu_a} = \omega_a^{-1} \mathbb{T}_a^\circ$, Lemma 17 gives

$$\max_{a \in A} \frac{(S_a^\circ)^2}{|I| \nu_a} \leq \frac{C(\mathbb{T}_{\max}^\circ)^2}{L_{\min}} = O_p\left(\frac{\log n}{L_{\min}}\right).$$

Combining the last two displays with the decomposition of q_a° at the start of this step gives

$$\max_{a \in A} \frac{|q_a^\circ|}{\nu_a} = O_p\left(\sqrt{\frac{\log n}{L_{\min}}} + \frac{\log n}{L_{\min}}\right). \quad (\text{B.56})$$

Step 2: control of $\frac{C_a}{\nu_a}$. The equality $\sum_{t \in I} \mu_{t,a} = 0$ gives $C_a = \sum_{t \in I} d_{t,a} \mu_{t,a}$. Hence Cauchy–Schwarz yields the pathwise bound

$$\frac{|C_a|}{\nu_a} \leq \left(\frac{q_a}{\nu_a} \right)^{1/2} \left(\frac{\nu_{a,\mu}}{\nu_a} \right)^{1/2} \leq \sqrt{r_n} \left(\frac{q_a}{\nu_a} \right)^{1/2}.$$

Equation (B.56) shows that $\max_a q_a / \nu_a = 1 + o_p(1)$. Therefore

$$\max_{a \in A} \frac{|C_a|}{\nu_a} = O_p(\sqrt{r_n}). \quad (\text{B.57})$$

Step 3: control of $\frac{\nu_{a,\mu}}{\nu_a}$. By definition (B.15) of r_n ,

$$\max_{a \in A} \frac{\nu_{a,\mu}}{\nu_a} = r_n. \quad (\text{B.58})$$

Combining (B.56), (B.57), and (B.58) gives

$$\max_{a \in A} \left| \frac{Q_a - \nu_a}{\nu_a} \right| = O_p\left(\sqrt{\frac{\log n}{L_{\min}}} + \frac{\log n}{L_{\min}} + \sqrt{r_n} + r_n\right),$$

Assumption B.12 implies $\log n / L_{\min} = o(\sqrt{\log n / L_{\min}})$, and Assumption B.11, together with $b_n \rightarrow \infty$, implies $r_n = o(1)$ and hence $r_n = O(\sqrt{r_n})$. Thus the displayed bound on $\max_{a \in A} |(Q_a - \nu_a) / \nu_a|$ gives the asserted rate for Δ_Q . Assumptions B.11 and B.12 make

$$\sqrt{\frac{\log n}{L_{\min}}} + \sqrt{r_n} \rightarrow 0,$$

proving the stated probability conclusions. $\square$

The next lemma gives uniform maximum and fourth-moment bounds for the centered variables $d_t^{(h)} = D_t^{(h)} - \mu_t^{(h)}$ in Regime D.

Lemma 20. *Under Assumptions B.1, B.7, and B.12,*

$$\max_{a=(I,h) \in A} \max_{t \in I} (d_t^{(h)})^2 = O_p(\log n), \quad \max_{a=(I,h) \in A} \frac{1}{|I|} \sum_{t \in I} (d_t^{(h)})^4 = O_p(1).$$

Proof. The first claim follows from Assumption B.1, (B.2), $nH \leq np$, and Proposition 33, exactly as in Lemma 14. It remains to prove the fourth-moment bound.

Set $X_t^{(h)} = (d_t^{(h)})^4 - \mathbb{E}[(d_t^{(h)})^4]$. Uniform sub-Gaussianity gives a semiexponential tail with exponent 1/2:

$$\sup_{t,h} \mathbb{P}(|X_t^{(h)}| > u) \leq C \exp(-c\sqrt{u}), \quad u > 0.$$

The sequence inherits the mixing coefficients in Assumption B.7. Apply Proposition 31 with

$$\gamma_1 = \gamma_{\text{mix}}, \quad \gamma_2 = \frac{1}{2}, \quad \gamma_4 := \left(\frac{1}{\gamma_{\text{mix}}} + 2 \right)^{-1}.$$

The fourth powers have uniformly bounded second moments and their quantile envelope obeys $q(u) \lesssim \log^2(e/u)$. Consequently, Proposition 32 gives

$$\int_0^{2\alpha_X(k)} q(u)^2 du \lesssim \alpha_X(k) \log^4(e/\alpha_X(k)).$$

By Assumption B.7, $\alpha_X(k) \leq \exp(-\tilde{K}_2 k^{\gamma_{\text{mix}}})$ for a constant $\tilde{K}_2 > 0$. Hence

$$\sum_{k>0} \int_0^{2\alpha_X(k)} q(u)^2 du \lesssim \sum_{k>0} e^{-\tilde{K}_2 k^{\gamma_{\text{mix}}}} (1 + k^{\gamma_{\text{mix}}})^4 < \infty.$$

Together with the uniform boundedness of $\mathbb{E}[(X_t^{(h)})^2]$, equation (B.79) gives $V = O(1)$ uniformly over I and h . Taking $y = Cm$ in (B.78) gives

$$\mathbb{P} \left(\left| \sum_{t \in I} X_t^{(h)} \right| > Cm \right) \leq C(m+1)e^{-cm^{\gamma_4}} + Ce^{-cm}. \quad (\text{B.59})$$

Because $1/\gamma_4 = 2 + 1/\gamma_{\text{mix}}$, Assumption B.12 implies $\log n = o(L_{\min}^{\gamma_4})$. A union bound over the p interval-coordinate pairs in (B.59) proves the second claim. $\square$

B.4.4 Proof of the centered Gaussian approximation

As in Lemma 17, set

$$Z_{t,a}^\circ := \frac{\sqrt{n} \mathbf{1}\{t \in I\} d_t^{(h)}}{|I|^{1/2-\gamma} v_a^\gamma}. \quad (\text{B.60})$$

Then

$$\mathbb{T}^\circ = \frac{1}{\sqrt{n}} \sum_{t=1}^n Z_t^\circ.$$

The sequence inherits the same geometric mixing rate, and Assumptions B.8 and B.3 yield

$$c \leq \text{Var}\left(\frac{1}{\sqrt{n}} \sum_{t=1}^n Z_{t,a}^\circ\right) \leq C \quad \forall a \in A.$$

By Assumptions B.1 and B.2,

$$\|Z_{t,a}^\circ\|_{\psi_2} \leq B_0(1 \vee \sigma^{-1}) \sqrt{\frac{n}{L_{\min}}} \leq B_n, \quad \forall t, a. \quad (\text{B.60a})$$

First, we assume $p = \Omega(n^\kappa)$ for some $\kappa > 0$. Equation (B.60a) verifies the sub-Gaussian bound in Proposition 27; the variance bounds above verify its nondegeneracy condition, and Assumption B.7 gives the required geometric mixing. The first two rate conditions in Assumption B.10 are exactly the CLT rate conditions in Proposition 27. Hence, with $S_n = \mathbb{T}^\circ$ and $G = G_\circ$, the one-sided rectangle $(-\infty, x]^p$ gives the event $\{\mathbb{T}_{\max}^\circ \leq x\}$ for each x . Then Proposition 27 implies

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}(\mathbb{T}_{\max}^\circ \leq x) - \mathbb{P}(G_{\circ, \max} \leq x) \right| \rightarrow 0. \quad (\text{B.61})$$

This is (B.38).

It remains to cover the case in which $p = o(n^\kappa)$ for every $\kappa > 0$. Proposition 27 requires a polynomial dimension. Set

$$\tilde{A} := A \times [m_n], m_n := \left\lceil \frac{n^{1/2}}{p} \right\rceil, \quad \tilde{p} := |\tilde{A}| = m_n p,$$

and let $R_n : \mathbb{R}^p \rightarrow \mathbb{R}^{\tilde{p}}$ be the coordinate-replication map

$$(R_n z)_{(a,j)} := z_a, \quad (a, j) \in A \times [m_n].$$

The duplicated sequence is

$$\tilde{Z}_t^\circ := R_n Z_t^\circ, \quad t = 1, \dots, n,$$

so only the coordinate index is enlarged; the time index is not replicated or reordered. Equivalently,

for $\tilde{a} = (a, j) \in \tilde{A}$,

$$\tilde{Z}_{t,\tilde{a}}^\circ := Z_{t,a}^\circ, \quad \tilde{\mathbb{T}}_{\tilde{a}}^\circ := \mathbb{T}_a^\circ.$$

Thus

$$\tilde{\mathbb{T}}_{\max}^\circ := \max_{\tilde{a} \in \tilde{A}} \tilde{\mathbb{T}}_{\tilde{a}}^\circ = \max_{a \in A} \mathbb{T}_a^\circ = \mathbb{T}_{\max}^\circ.$$

Since $p = o(n^{1/2})$, we have $n^{1/2} \leq \tilde{p} < n^{1/2} + p \asymp n^{1/2}$, so the duplicated array satisfies $\tilde{p} = \Omega(n^{1/3})$, the dimension requirement in Proposition 27. For $\tilde{a} = (a, j)$,

$$\max_{t,\tilde{a}} \|\tilde{Z}_{t,\tilde{a}}^\circ\|_{\psi_2} = \max_{t,a} \|Z_{t,a}^\circ\|_{\psi_2}, \quad \text{Var}\left(\frac{1}{\sqrt{n}} \sum_{t=1}^n \tilde{Z}_{t,\tilde{a}}^\circ\right) = \text{Var}\left(\frac{1}{\sqrt{n}} \sum_{t=1}^n Z_{t,a}^\circ\right).$$

Moreover, for any $B \subset [n]$,

$$\sigma(\tilde{Z}_t^\circ : t \in B) = \sigma(R_n Z_t^\circ : t \in B) = \sigma(Z_t^\circ : t \in B),$$

because $Z_{t,a}^\circ = \tilde{Z}_{t,(a,1)}^\circ$. Hence, with the usual definition

$$\alpha_Z(\ell) := \sup_t \alpha(\sigma(Z_s^\circ : s \leq t), \sigma(Z_s^\circ : s \geq t + \ell)),$$

we have $\alpha_{\tilde{Z}}(\ell) = \alpha_Z(\ell)$ for every ℓ . Moreover $\log \tilde{p} \asymp \log n$. Hence the first two displays in Assumption B.10 imply the two rate conditions in Proposition 27 with dimension $\tilde{p}$:

$$(\log \tilde{p})^{3-\gamma_{\text{mix}}} = o(n^{\gamma_{\text{mix}}/3}), \quad B_n^{2/3} \frac{(\log \tilde{p})^{(1+2\gamma_{\text{mix}})/(3\gamma_{\text{mix}})}}{n^{1/9}} + B_n \frac{(\log \tilde{p})^{7/6}}{n^{1/9}} \rightarrow 0.$$

Let $\tilde{G}_\circ$ be Gaussian with the covariance matrix of $\tilde{\mathbb{T}}^\circ$. Proposition 27 applied to the duplicated array gives

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}(\tilde{\mathbb{T}}_{\max}^\circ \leq x) - \mathbb{P}(\tilde{G}_{\circ, \max} \leq x) \right| \rightarrow 0.$$

Construct $\tilde{G}_\circ$ by duplicating the coordinates of $G_\circ$; then $\tilde{G}_{\circ, \max} = G_{\circ, \max}$ pathwise. Therefore (B.61), and hence (B.38), also holds in the sub-polynomial case.

B.4.5 Proof of the fixed-denominator bootstrap covariance

For $a = (I, h)$ and $b = (J, g)$, the conditional covariance matrix of the fixed-denominator bootstrap satisfies

$$\hat{\Sigma}_{ab} = \frac{\sum_{t \in I} \sum_{u \in J} \Theta_n(t, u) \left(D_t^{(h)} - \bar{D}_a \right) \left(D_u^{(g)} - \bar{D}_b \right)}{|I|^{1/2-\gamma} |J|^{1/2-\gamma} Q_a^\gamma Q_b^\gamma}. \quad (\text{B.62})$$

Use Z_t° from (B.60) and set $\bar{Z}^\circ := n^{-1} \sum_{t=1}^n Z_t^\circ$. Let

$$\hat{\Sigma}^\circ := \frac{1}{n} \sum_{t=1}^n \sum_{u=1}^n \Theta_n(t, u) (Z_t^\circ - \bar{Z}^\circ) (Z_u^\circ - \bar{Z}^\circ)^\top.$$

The array $(Z_t^\circ)_{t \leq n}$ is centered, inherits the geometric mixing rate from Assumption B.7, and satisfies $\max_{t,a} \|Z_{t,a}^\circ\|_{\psi_2} \leq B_n$ by (B.60a). Recall that

$$\Sigma_\circ = \text{Cov}(\mathbb{T}^\circ) = \text{Cov}\left(\frac{1}{\sqrt{n}} \sum_{t=1}^n Z_t^\circ\right).$$

Therefore Proposition 28, applied to $(Z_t^\circ)_{t \leq n}$, yields

$$\|\hat{\Sigma}^\circ - \Sigma_\circ\|_{\max} = O_p(B_n^2 n^{-c_1} (\log p)^{c_2}) + O(B_n^2 n^{-\rho}).$$

Because $\log p \leq \log n$, the condition $B_n^2 n^{-c_1} (\log n)^{c_2+2} \rightarrow 0$ makes the first term $o_p((\log n)^{-2})$, while $B_n^2 n^{-\rho} (\log n)^2 \rightarrow 0$ makes the second term $o((\log n)^{-2})$. Hence

$$\|\hat{\Sigma}^\circ - \Sigma_\circ\|_{\max} = o_p((\log n)^{-2}). \quad (\text{B.63})$$

It remains to compare the feasible covariance in (B.62) with the globally centered oracle covariance $\hat{\Sigma}^\circ$. For $a = (I, h)$, equations (B.10)–(B.11) give

$$D_t^{(h)} - \bar{D}_a = d_{t,a} + \mu_{t,a}, \quad t \in I, \quad (\text{B.64})$$

Define

$$w_{t,I} := \frac{\mathbf{1}\{t \in I\}}{|I|} - \frac{1}{n}, \quad t \in [n].$$

Then (B.60) gives

$$Z_{t,a}^\circ - \bar{Z}_a^\circ = \frac{\sqrt{n}}{|I|^{1/2-\gamma} \nu_a^\gamma} \{\mathbf{1}\{t \in I\} d_{t,a} + S_a^\circ w_{t,I}\}. \quad (\text{B.65})$$

Also,

$$\sum_{t \in I} (d_{t,a} + \mu_{t,a})^2 = Q_a, \quad \sum_{t \in I} \mu_{t,a}^2 = \nu_{a,\mu}, \quad \sum_{t=1}^n (S_a^\circ)^2 w_{t,I}^2 \leq \frac{(S_a^\circ)^2}{|I|}. \quad (\text{B.66})$$

To see the denominator bound, write $x_c := Q_c / \nu_c$. On $\{\Delta_Q \leq 1/2\}$, the definition of Δ_Q gives $x_c \in [1/2, 3/2]$ for every $c \in A$. Since $f(x) := x^{-\gamma}$ has

$$\sup_{1/2 \leq x \leq 3/2} |f'(x)| = \sup_{1/2 \leq x \leq 3/2} \gamma x^{-\gamma-1} \leq C,$$

the mean value theorem yields, uniformly in c ,

$$\max_{c \in A} \left| \left(\frac{\nu_c}{Q_c} \right)^\gamma - 1 \right| = \max_{c \in A} |x_c^{-\gamma} - 1| \leq C \max_{c \in A} |x_c - 1| \leq C\Delta_Q. \quad (\text{B.67})$$

For any $x, y \in \mathbb{R}^n$, Lemma 16 gives

$$\left| \sum_{t=1}^n \sum_{u=1}^n \Theta_n(t, u) x_t y_u \right| \leq C b_n \|x\|_2 \|y\|_2. \quad (\text{B.68})$$

To obtain an entrywise bound for $\widehat{\Sigma}_{ab} - \widehat{\Sigma}_{ab}^\circ$, (B.62) and (B.65) give

$$\begin{aligned} \widehat{\Sigma}_{ab} &= \frac{1}{|I|^{1/2-\gamma} |J|^{1/2-\gamma} Q_a^\gamma Q_b^\gamma} \sum_{t=1}^n \sum_{s=1}^n \Theta_n(t, s) \mathbf{1}\{t \in I\} (d_{t,a} + \mu_{t,a}) \\ &\quad \times \mathbf{1}\{s \in J\} (d_{s,b} + \mu_{s,b}), \\ \widehat{\Sigma}_{ab}^\circ &= \frac{1}{|I|^{1/2-\gamma} |J|^{1/2-\gamma} \nu_a^\gamma \nu_b^\gamma} \sum_{t=1}^n \sum_{s=1}^n \Theta_n(t, s) \{ \mathbf{1}\{t \in I\} d_{t,a} + S_a^\circ w_{t,I} \} \\ &\quad \times \{ \mathbf{1}\{s \in J\} d_{s,b} + S_b^\circ w_{s,J} \}. \end{aligned}$$

Define the same oracle-centered covariance with the feasible denominator:

$$\begin{aligned} \widetilde{\Sigma}_{ab}^\circ &:= \frac{1}{|I|^{1/2-\gamma} |J|^{1/2-\gamma} Q_a^\gamma Q_b^\gamma} \sum_{t=1}^n \sum_{s=1}^n \Theta_n(t, s) \{ \mathbf{1}\{t \in I\} d_{t,a} + S_a^\circ w_{t,I} \} \\ &\quad \times \{ \mathbf{1}\{s \in J\} d_{s,b} + S_b^\circ w_{s,J} \}. \end{aligned}$$

Then

$$\widetilde{\Sigma}_{ab}^\circ = (Q_a/\nu_a)^{-\gamma} (Q_b/\nu_b)^{-\gamma} \widehat{\Sigma}_{ab}^\circ.$$

On $\{\Delta_Q \leq 1/2\}$, (B.67) gives

$$\left| (Q_a/\nu_a)^{-\gamma} (Q_b/\nu_b)^{-\gamma} - 1 \right| \leq C\Delta_Q.$$

Also, (B.63) and $\|\Sigma_\circ\|_{\max} \leq C$ give $\|\widehat{\Sigma}^\circ\|_{\max} = O_p(1)$. Hence, with probability $1 - o(1)$,

$$\max_{a,b \in A} \left| \widetilde{\Sigma}_{ab}^\circ - \widehat{\Sigma}_{ab}^\circ \right| \leq C\Delta_Q,$$

which controls the denominator replacement.

For the numerator difference $\widehat{\Sigma}_{ab} - \widetilde{\Sigma}_{ab}^\circ$, the $\mathbf{1}\{t \in I\}d_{t,a}\mathbf{1}\{s \in J\}d_{s,b}$ term cancels:

$$\begin{aligned} & \mathbf{1}\{t \in I\}(d_{t,a} + \mu_{t,a})\mathbf{1}\{s \in J\}(d_{s,b} + \mu_{s,b}) \\ & - \{\mathbf{1}\{t \in I\}d_{t,a} + S_a^\circ w_{t,I}\}\{\mathbf{1}\{s \in J\}d_{s,b} + S_b^\circ w_{s,J}\} \\ & = \mathbf{1}\{t \in I\}\mathbf{1}\{s \in J\}(\mu_{t,a}d_{s,b} + d_{t,a}\mu_{s,b} + \mu_{t,a}\mu_{s,b}) \\ & - \mathbf{1}\{t \in I\}d_{t,a}S_b^\circ w_{s,J} - S_a^\circ w_{t,I}\mathbf{1}\{s \in J\}d_{s,b} \\ & - S_a^\circ S_b^\circ w_{t,I}w_{s,J}. \end{aligned}$$

Equations (B.66) and (B.67), and the definition (B.15), give, on $\{\Delta_Q \leq 1/2\}$,

$$\begin{aligned} \frac{\|(\mathbf{1}\{t \in I\}d_{t,a})_{t=1}^n\|_2}{\sqrt{v_a}} & \leq \left\{2\frac{Q_a}{v_a} + 2\frac{v_{a,\mu}}{v_a}\right\}^{1/2} \leq C, \quad \frac{\|(\mathbf{1}\{t \in I\}\mu_{t,a})_{t=1}^n\|_2}{\sqrt{v_a}} = \sqrt{\frac{v_{a,\mu}}{v_a}}, \\ \frac{\|(S_a^\circ w_{t,I})_{t=1}^n\|_2}{\sqrt{v_a}} & \leq \frac{|S_a^\circ|}{\sqrt{|I|v_a}} \leq \frac{|S_a^\circ|}{\sqrt{v_a}}. \end{aligned}$$

By (B.68), uniformly in $a = (I, h)$ and $b = (J, g)$,

$$\begin{aligned} \left|\widehat{\Sigma}_{ab} - \widetilde{\Sigma}_{ab}^\circ\right| & \leq C \frac{b_n}{\sqrt{|I||J|}} \left[\frac{|S_a^\circ|}{\sqrt{v_a}} + \frac{|S_b^\circ|}{\sqrt{v_b}} + \sqrt{\frac{v_{a,\mu}}{v_a}} + \sqrt{\frac{v_{b,\mu}}{v_b}} \right. \\ & \quad \left. + \left(\frac{|S_a^\circ|}{\sqrt{v_a}} + \sqrt{\frac{v_{a,\mu}}{v_a}} \right) \left(\frac{|S_b^\circ|}{\sqrt{v_b}} + \sqrt{\frac{v_{b,\mu}}{v_b}} \right) \right] \end{aligned} \quad (\text{B.69})$$

By Assumption B.2, (B.36), and (B.15),

$$\max_{a \in A} \frac{|S_a^\circ|}{\sqrt{v_a}} = \max_{a=(I,h) \in A} \left(\frac{|I|}{v_a} \right)^{1/2-\gamma} |\mathbb{T}_a^\circ| \leq C \bar{\mathbb{T}}_{\max}^\circ, \quad \max_{a \in A} \sqrt{\frac{v_{a,\mu}}{v_a}} \leq \sqrt{r_n}.$$

Since $|I|, |J| \geq L_{\min}$, (B.69) yields

$$\max_{a,b \in A} \left| \widehat{\Sigma}_{ab} - \widetilde{\Sigma}_{ab}^\circ \right| \leq C \frac{b_n}{L_{\min}} \left\{ \bar{\mathbb{T}}_{\max}^\circ + \sqrt{r_n} + (\bar{\mathbb{T}}_{\max}^\circ + \sqrt{r_n})^2 \right\}.$$

The triangle inequality and the denominator-replacement bound give

$$\begin{aligned} \|\widehat{\Sigma} - \widehat{\Sigma}^\circ\|_{\max} & \leq \max_{a,b \in A} \left| \widehat{\Sigma}_{ab} - \widetilde{\Sigma}_{ab}^\circ \right| + \max_{a,b \in A} \left| \widetilde{\Sigma}_{ab}^\circ - \widehat{\Sigma}_{ab}^\circ \right| \\ & \leq C \Delta_Q + C \frac{b_n}{L_{\min}} \left\{ \bar{\mathbb{T}}_{\max}^\circ + \sqrt{r_n} + (\bar{\mathbb{T}}_{\max}^\circ + \sqrt{r_n})^2 \right\}. \end{aligned}$$

Lemma 19, Lemma 17, and Assumptions B.11 and B.12 imply

$$\Delta_Q = o_p((\log n)^{-2}), \quad \frac{b_n}{L_{\min}} \{ \bar{\mathbb{T}}_{\max}^\circ + \sqrt{r_n} + (\bar{\mathbb{T}}_{\max}^\circ + \sqrt{r_n})^2 \} = o_p((\log n)^{-2}).$$

Therefore

$$\|\hat{\Sigma} - \hat{\Sigma}^\circ\|_{\max} = o_p((\log n)^{-2}). \quad (\text{B.70})$$

Combining (B.63) and (B.70) proves (B.39).

B.4.6 Verifying conditions of the reduction lemmas

We first verify Lemma 7. For $\gamma = 0$, $T_{\max}^\circ = \mathbb{T}_{\max}^\circ$. For $\gamma > 0$, Lemmas 19 and 17 imply

$$C\gamma\Delta_Q\bar{\mathbb{T}}_{\max}^\circ = O_p\left(\left\{\sqrt{\frac{\log n}{L_{\min}}} + \sqrt{r_n}\right\}\sqrt{\log n}\right).$$

Moreover,

$$\left\{\sqrt{\frac{\log n}{L_{\min}}} + \sqrt{r_n}\right\}\log n \lesssim \frac{(\log n)^{3/2}}{\sqrt{L_{\min}}} + \sqrt{r_n}\log n \rightarrow 0,$$

where Assumption B.12 controls the first term, while $b_n r_n (\log n)^2 \rightarrow 0$ and $b_n \rightarrow \infty$ control the second term. Choose $M_n \uparrow \infty$ sufficiently slowly that

$$\eta_n := M_n \left\{\sqrt{\frac{\log n}{L_{\min}}} + \sqrt{r_n}\right\}\sqrt{\log n} \rightarrow 0, \quad \eta_n \sqrt{1 \vee \log(p/\eta_n)} \rightarrow 0.$$

Then

$$\mathbb{P}(\Delta_Q > 1/2) + \mathbb{P}(C\gamma\Delta_Q\bar{\mathbb{T}}_{\max}^\circ > \eta_n) \rightarrow 0. \quad (\text{B.71})$$

Since $\Sigma_\circ = \text{Cov}(\mathbb{T}^\circ)$ and $\mathbb{T}^\circ$ is defined in (B.36), for $a = (I, h)$,

$$(\Sigma_\circ)_{aa} = \text{Var}(\mathbb{T}_a^\circ) = \left(\frac{v_a}{|I|}\right)^{1-2\gamma} \text{Var}\left(\frac{S_a^\circ}{\sqrt{v_a}}\right).$$

Assumptions B.2 and B.8 therefore give constants $0 < c < C < \infty$ such that

$$c \leq (\Sigma_\circ)_{aa} \leq C \quad \forall a \in A.$$

Proposition 24 gives

$$\sup_{x \in \mathbb{R}} \mathbb{P}(|G_{\circ, \max} - x| \leq \eta_n) \rightarrow 0. \quad (\text{B.72})$$

For every $x \in \mathbb{R}$, (B.61) gives

$$\mathbb{P}(|\mathbb{T}_{\max}^\circ - x| \leq \eta_n) \leq \mathbb{P}(|G_{\circ, \max} - x| \leq \eta_n) + 2 \sup_{z \in \mathbb{R}} |\mathbb{P}(\mathbb{T}_{\max}^\circ \leq z) - \mathbb{P}(G_{\circ, \max} \leq z)|.$$

Equations (B.61) and (B.72) therefore imply

$$\sup_{x \in \mathbb{R}} \mathbb{P}(|\mathbb{T}_{\max}^\circ - x| \leq \eta_n) \rightarrow 0.$$

Together with (B.71), this verifies the conditions of Lemma 7.

We next verify Lemma 8 with $k_n = b_n$. Lemma 19 and $v_a \geq \sigma_-^2 |I|$ verify (B.17), and Lemma 20 verifies (B.18). The row conditions (B.20) follow from Lemma 16. The conditional variance condition (B.19) follows from (B.39). The diagonal bound established above gives constants $0 < c_0 < C_0 < \infty$ such that

$$c_0 \leq \min_{a \in A} (\Sigma_\circ)_{aa} \leq \max_{a \in A} (\Sigma_\circ)_{aa} \leq C_0.$$

Also, by definition of $\widehat{\Sigma}$ in Lemma 9,

$$\widehat{\Sigma}_{aa} = \text{Cov}^*(T_{0,a}^*, T_{0,a}^*) = \text{Var}^*(T_{0,a}^*).$$

Equation (B.39) implies

$$\mathbb{P}(\|\widehat{\Sigma} - \Sigma_\circ\|_{\max} \leq c_0/2) \rightarrow 1.$$

On this event,

$$\min_{a \in A} \text{Var}^*(T_{0,a}^*) = \min_{a \in A} \widehat{\Sigma}_{aa} \geq \min_{a \in A} (\Sigma_\circ)_{aa} - \|\widehat{\Sigma} - \Sigma_\circ\|_{\max} \geq c_0/2,$$

and

$$\max_{a \in A} \text{Var}^*(T_{0,a}^*) = \max_{a \in A} \widehat{\Sigma}_{aa} \leq \max_{a \in A} (\Sigma_\circ)_{aa} + \|\widehat{\Sigma} - \Sigma_\circ\|_{\max} \leq C_0 + c_0/2.$$

Renaming constants gives, with probability $1 - o(1)$ over data,

$$c \leq \min_{a \in A} \text{Var}^*(T_{0,a}^*) \leq \max_{a \in A} \text{Var}^*(T_{0,a}^*) \leq C.$$

Since Lemma 19 and Assumption B.2 make $Q_a/|I|$ uniformly bounded above and away from zero with probability $1 - o(1)$ over data,

$$\text{Var}^*\left(\frac{S_a^*}{\sqrt{Q_a}}\right) = \left(\frac{|I|}{Q_a}\right)^{1-2\gamma} \text{Var}^*(T_{0,a}^*)$$

is also uniformly bounded above and away from zero. Thus (B.19) holds. Assumptions B.11 and B.12

imply

$$\frac{b_n(\log n)^3}{L_{\min}} \rightarrow 0, \quad b_n r_n (\log n)^2 \rightarrow 0.$$

Thus (B.22) holds.

It remains to verify Lemma 9. By the diagonal-bound argument used to verify (B.19), the diagonal entries of $\Sigma_\circ$ and $\widehat{\Sigma}$ satisfy the two nondegeneracy conditions in Lemma 9. Together with (B.39), this verifies all conditions of Lemma 9.

We now complete the proof of Theorem 15. Section B.4.4 proves (B.38), Section B.4.5 proves (B.39), and the preceding paragraphs verify the conditions of Lemmas 7–9. Therefore the conditions of Lemma 10 hold. Applying Lemma 10 proves the theorem.

B.4.7 Quadratic spectral kernel satisfies Assumption B.9

Lemma 21. *Let*

$$K_{\text{QS}}(x) := \begin{cases} \frac{25}{12\pi^2 x^2} \left\{ \frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5) \right\}, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Then K_{QS} is symmetric and continuously differentiable on $\mathbb{R}$, satisfies $K_{\text{QS}}(0) = 1$, and obeys

$$\sup_{x \in \mathbb{R}} |K'_{\text{QS}}(x)| < \infty, \quad |K_{\text{QS}}(x)| \leq C_{\text{QS}}(1 + |x|)^{-2} \quad \forall x \in \mathbb{R}$$

for some constant $C_{\text{QS}} < \infty$. Consequently, Assumption B.9 holds for the quadratic spectral kernel with $\vartheta = 2$ whenever

$$b_n \asymp n^\rho, \quad 0 < \rho < \frac{1}{4}.$$

Proof. Symmetry is immediate because K_{QS} depends on x only through x^2 and the even functions $\sin(z)/z$ and $\cos(z)$. For $x \neq 0$, the formula is smooth. Set $c := 6\pi/5$. Using the Taylor expansions

$$\frac{\sin(cx)}{cx} = 1 - \frac{c^2 x^2}{6} + \frac{c^4 x^4}{120} + O(x^6), \quad \cos(cx) = 1 - \frac{c^2 x^2}{2} + \frac{c^4 x^4}{24} + O(x^6),$$

we obtain

$$\frac{\sin(cx)}{cx} - \cos(cx) = \frac{c^2 x^2}{3} - \frac{c^4 x^4}{30} + O(x^6).$$

Therefore,

$$K_{\text{QS}}(x) = 1 - \frac{18\pi^2}{125} x^2 + O(x^4) \quad (x \rightarrow 0).$$

Hence K_{QS} extends continuously at 0 with value 1, and the expansion also gives $K'_{\text{QS}}(0) = 0$; thus $K_{\text{QS}} \in C^1(\mathbb{R})$.

The positive-semidefiniteness condition follows from an explicit Fourier representation. With $c = 6\pi/5$, define

$$f_c(u) := \frac{3}{4c} \left(1 - \frac{u^2}{c^2}\right) \mathbf{1}_{\{|u| \leq c\}}.$$

This is a nonnegative density, and direct integration gives

$$K_{\text{QS}}(x) = \int_{-c}^c e^{iux} f_c(u) du = \frac{3}{c^2 x^2} \left\{ \frac{\sin(cx)}{cx} - \cos(cx) \right\}. \quad (\text{B.73})$$

Because f_c has compact support, differentiation under the integral is valid and gives

$$K'_{\text{QS}}(x) = \int_{-c}^c iue^{iux} f_c(u) du, \quad \sup_{x \in \mathbb{R}} |K'_{\text{QS}}(x)| \leq \int_{-c}^c |u| f_c(u) du < \infty.$$

Therefore, for arbitrary real numbers $v_1, \dots, v_n$,

$$\sum_{t,u=1}^n v_t v_u K_{\text{QS}}\left(\frac{t-u}{b_n}\right) = \int_{-c}^c \left| \sum_{t=1}^n v_t e^{iut/b_n} \right|^2 f_c(u) du \geq 0.$$

Thus every matrix in (B.52) is positive semidefinite.

For the tail bound, again with $c = 6\pi/5$, for $x \neq 0$ we have

$$|K_{\text{QS}}(x)| \leq \frac{25}{12\pi^2 x^2} \left(\left| \frac{\sin(cx)}{cx} \right| + |\cos(cx)| \right) \leq \frac{25}{12\pi^2 x^2} \left(1 + \frac{1}{c|x|} \right).$$

In particular, for $|x| \geq 1$,

$$|K_{\text{QS}}(x)| \leq \frac{25}{12\pi^2} \left(1 + \frac{1}{c} \right) |x|^{-2} \leq 4 \frac{25}{12\pi^2} \left(1 + \frac{1}{c} \right) (1 + |x|)^{-2}.$$

On the compact set $[-1, 1]$, continuity implies $\sup_{|x| \leq 1} |K_{\text{QS}}(x)| < \infty$. Enlarging the constant if necessary yields

$$|K_{\text{QS}}(x)| \leq C_{\text{QS}} (1 + |x|)^{-2} \quad \forall x \in \mathbb{R}.$$

Thus Assumption B.9 is satisfied with $\vartheta = 2$. The bandwidth restriction in that assumption then becomes

$$0 < \rho < \frac{\vartheta - 1}{3\vartheta - 2} = \frac{1}{4}.$$

□

B.4.8 Extension of Theorem 15 to data-dependent bandwidths

This subsection shows that the test remains valid in Regime D with the data-driven bandwidth used in the main text. The implementation uses the quadratic spectral kernel of Andrews (1991). For each horizon h , it fits

$$D_t^{(h)} = c_h + \rho_h D_{t-1}^{(h)} + \epsilon_{h,t},$$

forms

$$\hat{a} := \frac{\sum_{h=1}^H 4\hat{\rho}_h^2 \hat{\sigma}_h^4 (1 - \hat{\rho}_h)^{-8}}{\sum_{h=1}^H \hat{\sigma}_h^4 (1 - \hat{\rho}_h)^{-4}}, \quad \hat{b}_n := 1.3221(\hat{a}n)^{1/5}. \quad (\text{B.74})$$

The resulting multiplier covariance is

$$\hat{\Theta}_n(t, u) := K_{\text{QS}}\left(\frac{t - u}{\hat{b}_n}\right).$$

The next corollary shows that Theorem 15 continues to work for the data-driven stepsize in (B.74) under mild convergence of $\hat{a}$.

Corollary 22. *Suppose that Assumptions B.1, B.2, B.3, B.7–B.8, and B.10–B.12 hold with b_n replaced by $b_n^\circ = 1.3221(a_0 n)^{1/5}$ for some constant $a_0 > 0$. If, for some constant $\epsilon > 0$,*

$$\hat{a} - a_0 = o_p(n^{-1/5-\epsilon}),$$

then the conclusion of Theorem 15 remains valid when the bootstrap statistic is computed using the Andrews bandwidth $\hat{b}_n = 1.3221(\hat{a}n)^{1/5}$.

Proof. Let

$$\Theta_n^\circ(t, u) := K_{\text{QS}}\left(\frac{t - u}{b_n^\circ}\right).$$

The condition on $\hat{a}$ gives

$$\frac{\hat{b}_n}{b_n^\circ} = \left(\frac{\hat{a}}{a_0}\right)^{1/5} = 1 + o_p(n^{-1/5-\epsilon}), \quad |\hat{b}_n - b_n^\circ| = o_p(n^{-\epsilon}),$$

and therefore, with probability $1 - o(1)$ over data,

$$cb_n^\circ \leq \hat{b}_n \leq Cb_n^\circ.$$

The explicit quadratic spectral formula and the expansion at zero in the proof of Lemma 21 give

$$|K'_{\text{QS}}(x)| \leq C(1 + |x|)^{-2} \quad \forall x \in \mathbb{R}.$$

On the event $cb_n^\circ \leq \hat{b}_n \leq Cb_n^\circ$, the mean value theorem gives

$$\begin{aligned} & \max_{1 \leq t \leq n} \sum_{u=1}^n \left| K_{\text{QS}}\left(\frac{t-u}{\hat{b}_n}\right) - K_{\text{QS}}\left(\frac{t-u}{b_n^\circ}\right) \right| \\ & \leq C \left| \frac{1}{\hat{b}_n} - \frac{1}{b_n^\circ} \right| \sum_{m=1}^{n-1} m \left(1 + \frac{m}{b_n^\circ}\right)^{-2} \\ & \leq C |\hat{b}_n - b_n^\circ| \log n = o_p((\log n)^{-2}), \end{aligned}$$

because $|\hat{b}_n - b_n^\circ| \log n = o_p((\log n)^{-2})$.

For any positive bandwidth b , the Fourier representation (B.73) gives positive semidefiniteness of the matrix $K_{\text{QS}}((t-u)/b)$, and $K_{\text{QS}}(0) = 1$ gives unit diagonal. Thus $\hat{\Theta}_n$ is a valid covariance matrix for the multipliers.

The row-sum argument in Lemma 16 gives, with probability $1 - o(1)$ over data,

$$\max_t \sum_u |\hat{\Theta}_n(t, u)| \leq C\hat{b}_n \leq Cb_n^\circ, \quad \max_t \sum_u \hat{\Theta}_n(t, u)^2 \leq C\hat{b}_n \leq Cb_n^\circ. \quad (\text{B.75})$$

It remains to verify the fixed-denominator covariance condition (B.37) in Lemma 9; by (B.2), in Regime D it is enough to show $\|\hat{\Sigma}(\hat{\Theta}_n) - \Sigma_\circ\|_{\max} = o_p((\log n)^{-2})$. For any covariance matrix Θ , define

$$\hat{\Sigma}_{ab}(\Theta) := \frac{\sum_{t \in I_a} \sum_{u \in I_b} \Theta(t, u) \left(D_t^{(h_a)} - \bar{D}_a\right) \left(D_u^{(h_b)} - \bar{D}_b\right)}{|I_a|^{1/2-\gamma} |I_b|^{1/2-\gamma} Q_a^\gamma Q_b^\gamma}.$$

Let

$$v_{a,t} := \frac{\mathbf{1}\{t \in I_a\} \left(D_t^{(h_a)} - \bar{D}_a\right)}{|I_a|^{1/2-\gamma} Q_a^\gamma}.$$

Then

$$\hat{\Sigma}_{ab}(\hat{\Theta}_n) - \hat{\Sigma}_{ab}(\Theta_n^\circ) = v_a^\top (\hat{\Theta}_n - \Theta_n^\circ) v_b.$$

Moreover,

$$\|v_a\|_2 = \left(\frac{Q_a}{|I_a|}\right)^{1/2-\gamma}.$$

Lemma 19 and Assumption B.2 imply $\max_a \|v_a\|_2 = O_p(1)$. Since $\hat{\Theta}_n - \Theta_n^\circ$ is symmetric, Schur's bound gives

$$\|\hat{\Theta}_n - \Theta_n^\circ\|_{\text{op}} \leq \max_{1 \leq t \leq n} \sum_{u=1}^n |\hat{\Theta}_n(t, u) - \Theta_n^\circ(t, u)| = o_p((\log n)^{-2}).$$

Hence

$$\begin{aligned}
\max_{a,b \in A} |\hat{\Sigma}_{ab}(\hat{\Theta}_n) - \hat{\Sigma}_{ab}(\Theta_n^\circ)| &= \max_{a,b \in A} |\nu_a^\top (\hat{\Theta}_n - \Theta_n^\circ) \nu_b| \\
&\leq \left(\max_{a \in A} \|\nu_a\|_2 \right)^2 \|\hat{\Theta}_n - \Theta_n^\circ\|_{\text{op}} \\
&= O_p(1) o_p((\log n)^{-2}) = o_p((\log n)^{-2}).
\end{aligned}$$

The proof in Section B.4.5 applied with the deterministic benchmark Θ_n° gives

$$\|\hat{\Sigma}(\Theta_n^\circ) - \Sigma_\circ\|_{\max} = o_p((\log n)^{-2}).$$

Combining the bounds for $\|\hat{\Sigma}(\hat{\Theta}_n) - \hat{\Sigma}(\Theta_n^\circ)\|_{\max}$ and $\|\hat{\Sigma}(\Theta_n^\circ) - \Sigma_\circ\|_{\max}$ yields

$$\|\hat{\Sigma}(\hat{\Theta}_n) - \Sigma_\circ\|_{\max} = o_p((\log n)^{-2}). \quad (\text{B.76})$$

The centered Gaussian approximation requirement (B.38) is unchanged because the statistic defined in (B.36) has coordinates

$$\mathbb{T}_a^\circ = \frac{S_a^\circ}{|I|^{1/2-\gamma} \nu_a^\gamma}, \quad \Sigma_\circ = \text{Cov}(\mathbb{T}^\circ),$$

contains no Θ_n . Hence the proof of (B.38) in Section B.4.4 applies without change.

For Lemma 8 with $k_n = b_n^\circ$, (B.17), (B.18), and (B.22) are the Section B.4.6 bounds with $b_n = b_n^\circ$, and (B.20) follows from (B.75). For (B.19), the diagonal entries of $\Sigma_\circ$ are bounded above and away from zero, and (B.76) gives, with probability $1 - o(1)$ over data,

$$c \leq \min_{a \in A} \hat{\Sigma}_{aa}(\hat{\Theta}_n) \leq \max_{a \in A} \hat{\Sigma}_{aa}(\hat{\Theta}_n) \leq C.$$

Since $\hat{\Sigma}_{aa}(\hat{\Theta}_n) = \text{Var}^*(T_{0,a}^*)$, $Q_a/|I| \asymp 1$ with probability $1 - o(1)$ over data, and

$$\text{Var}^*\left(\frac{S_a^*}{\sqrt{Q_a}}\right) = \left(\frac{|I|}{Q_a}\right)^{1-2\gamma} \text{Var}^*(T_{0,a}^*),$$

(B.19) follows. Finally, (B.76) verifies (B.37) with $\hat{\Sigma} = \hat{\Sigma}(\hat{\Theta}_n)$. Therefore the verification in Section B.4.6 and Lemma 10 apply with $\hat{\Theta}_n$ in place of Θ_n° , proving the same size conclusion as Theorem 15. $\square$

B.5 Miscellany

This section states the key technical results our proof relies on for completeness.

The next proposition is the independent high-dimensional Gaussian approximation of Chernozhukov, Chetverikov and Kato (2017, Theorem 2.1 and Proposition 2.1).

Proposition 23. Let $X_1, \dots, X_n$ be independent centered random vectors in $\mathbb{R}^p$, and let

$$S_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i, \quad \Sigma := \text{Cov}(S_n), \quad G \sim N(0, \Sigma).$$

Assume that, for some constants $b > 0$ and $B_n \geq 1$,

$$\min_{1 \leq j \leq p} \Sigma_{jj} \geq b, \quad \max_{1 \leq i \leq n, 1 \leq j \leq p} \|X_{ij}\|_{\psi_1} \leq B_n.$$

Then there exists a constant $C = C(b) < \infty$ such that

$$\sup_{A \in \mathcal{R}^p} |\mathbb{P}(S_n \in A) - \mathbb{P}(G \in A)| \leq C \left(\frac{B_n^2 \log^7(pn)}{n} \right)^{1/6},$$

where $\mathcal{R}^p$ denotes the class of coordinatewise hyperrectangles in $\mathbb{R}^p$. In particular, if $B_n^2 \log^7(pn) = o(n)$, then the left-hand side converges to 0.

The next proposition is the Gaussian anti-concentration bound of Chernozhukov, Chetverikov and Kato (2015, Theorem 3 and Corollary 1).

Proposition 24. Let $G = (G_1, \dots, G_p)^\top$ be a centered Gaussian vector such that

$$0 < \underline{\sigma} \leq \sqrt{\text{Var}(G_j)} \leq \bar{\sigma} < \infty \quad \forall j \in [p].$$

Then there exists a constant $C < \infty$, depending only on $\underline{\sigma}$ and $\bar{\sigma}$, such that for every $\varepsilon > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{P} \left(\left| \max_{1 \leq j \leq p} G_j - x \right| \leq \varepsilon \right) \leq C \varepsilon \left\{ 1 \vee \sqrt{\log(p/\varepsilon)} \right\}.$$

In particular, if $\varepsilon_n \downarrow 0$ and $\varepsilon_n \sqrt{1 \vee \log(p/\varepsilon_n)} \rightarrow 0$, then

$$\sup_{x \in \mathbb{R}} \mathbb{P} \left(\left| \max_{1 \leq j \leq p} G_j - x \right| \leq \varepsilon_n \right) \rightarrow 0.$$

The same conclusion holds with $\max_{1 \leq j \leq p} |G_j|$ in place of $\max_{1 \leq j \leq p} G_j$ after replacing p by $2p$.

The next proposition is the Gaussian comparison inequality from Chernozhukov, Chetverikov and Kato (2015, Theorem 2).

Proposition 25. Let $X = (X_1, \dots, X_p)^\top$ and $Y = (Y_1, \dots, Y_p)^\top$ be centered Gaussian vectors with covariance matrices Σ^X and Σ^Y . Assume that

$$0 < \underline{\sigma} \leq \sqrt{\text{Var}(X_j)}, \sqrt{\text{Var}(Y_j)} \leq \bar{\sigma} < \infty \quad \forall j \in [p],$$

and define

$$\Delta := \|\Sigma^X - \Sigma^Y\|_{\max}.$$

Then there exists a constant $C < \infty$, depending only on $\underline{\sigma}$ and $\bar{\sigma}$, such that

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\max_{1 \leq j \leq p} X_j \leq x \right) - \mathbb{P} \left(\max_{1 \leq j \leq p} Y_j \leq x \right) \right| \leq C \Delta^{1/3} \{1 \vee \log(p/\Delta)\}^{2/3},$$

with the convention that the right-hand side equals 0 when $\Delta = 0$. In particular, if $\Delta_n = o((\log p)^{-2})$, then the Kolmogorov distance between the two Gaussian maxima converges to 0.

The next proposition gives the Gaussian quadratic-form concentration bound from Vershynin (2018, Theorem 6.2.1).

Proposition 26. *Let $Z \sim N(0, I_m)$ and let B be a deterministic symmetric $m \times m$ matrix. There is a universal constant $C < \infty$ such that, for every $u \geq 1$,*

$$\mathbb{P}(|Z^\top B Z - \text{tr}(B)| > C\{\|B\|_F \sqrt{u} + \|B\|_{\text{op}} u\}) \leq 2e^{-u}.$$

The next proposition is the dependent high-dimensional CLT from Chang, Chen and Wu (2024, Theorem 1).

Proposition 27. *Let $X_1, \dots, X_n$ be centered random vectors in $\mathbb{R}^p$, and let*

$$S_n := \frac{1}{\sqrt{n}} \sum_{t=1}^n X_t, \quad \Xi := \text{Cov}(S_n), \quad G \sim N(0, \Xi).$$

Assume that $\{X_t\}$ is geometrically α -mixing with

$$\alpha(\ell) \leq K_1 e^{-K_2 \ell^{\gamma_{\text{mix}}}} \quad (\ell \geq 1)$$

for some constants $K_1 > 1$, $K_2 > 0$, and $\gamma_{\text{mix}} > 0$, that $p = \Omega(n^\kappa)$ for some $\kappa > 0$, and that

$$\max_{t \in [n], j \in [p]} \|X_{t,j}\|_{\psi_2} \leq B_n, \quad \min_{1 \leq j \leq p} \Xi_{jj} \geq b > 0.$$

If, in addition,

$$(\log p)^{3-\gamma_{\text{mix}}} = o(n^{\gamma_{\text{mix}}/3})$$

and

$$B_n^{2/3} \frac{(\log p)^{(1+2\gamma_{\text{mix}})/(3\gamma_{\text{mix}})}}{n^{1/9}} + B_n \frac{(\log p)^{7/6}}{n^{1/9}} \rightarrow 0,$$

then

$$\sup_{B \in \mathcal{R}^p} |\mathbb{P}(S_n \in B) - \mathbb{P}(G \in B)| \rightarrow 0.$$

In particular,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\max_{1 \leq j \leq p} S_{n,j} \leq x \right) - \mathbb{P} \left(\max_{1 \leq j \leq p} G_j \leq x \right) \right| \rightarrow 0.$$

The next proposition gives the kernel covariance consistency bound from Chang, Chen and Wu (2024, Theorem 11(i)), specialized to $\gamma_1 = 2$.

Proposition 28. *Let $X_1, \dots, X_n$ be centered random vectors in $\mathbb{R}^p$. Suppose*

$$\max_{t,j} \|X_{t,j}\|_{\psi_{\gamma_1}} \leq B_n$$

for some $\gamma_1 \geq 1$, and suppose their strong-mixing coefficients obey

$$\alpha_X(\ell) \leq K_1 e^{-K_2 \ell^{\gamma_2}}, \quad \ell \geq 1,$$

for constants $K_1, K_2, \gamma_2 > 0$. Let K satisfy the smoothness and decay conditions in Assumption B.9, and define

$$\bar{X} := \frac{1}{n} \sum_{t=1}^n X_t, \quad \hat{\Xi}_n := \sum_{j=-n+1}^{n-1} K\left(\frac{j}{b_n}\right) \hat{H}_j,$$

where

$$\hat{H}_j := \frac{1}{n} \sum_{t=j+1}^n (X_t - \bar{X})(X_{t-j} - \bar{X})^\top \quad \text{if } j \geq 0,$$

$$\hat{H}_j := \frac{1}{n} \sum_{t=-j+1}^n (X_{t+j} - \bar{X})(X_t - \bar{X})^\top \quad \text{if } j < 0,$$

and $\Xi := \text{Cov}(n^{-1/2} \sum_{t=1}^n X_t)$. Assume $p > n^\kappa$ for some constant $\kappa > 0$ and $b_n \asymp n^\rho$. Then there exist constants $c_1 > 0$ depending only on (ρ, ϑ) and $c_2 > 0$ depending only on $(\gamma_1, \gamma_2, \vartheta)$ such that

$$\|\hat{\Xi}_n - \Xi\|_{\max} = O_p\{B_n^2 n^{-c_1} (\log p)^{c_2}\} + O(B_n^2 n^{-\rho}).$$

The next proposition makes admissible exponents in the covariance consistency bound of Chang, Chen and Wu (2024) explicit for Regime D.

Proposition 29. *Under the assumptions of Proposition 28, specialize $\gamma_1 = 2$ and $\gamma_2 = \gamma_{\text{mix}}$. Then an admissible choice of the exponents in that proposition is*

$$c_1(\rho, \vartheta) = \frac{\vartheta - 1 - (3\vartheta - 2)\rho}{2\vartheta - 1}, \quad c_2(2, \gamma_{\text{mix}}, \vartheta) = 2 + \frac{1}{\gamma_{\text{mix}}} + \frac{1}{2\vartheta - 1}.$$

If, in addition, $p \leq n^2$, c_2 can be improved into

$$c_2^\sharp(2, \gamma_{\text{mix}}, \vartheta) = \frac{\vartheta}{2\vartheta - 1},$$

in the sense that

$$\|\hat{\Xi}_n - \Xi\|_{\max} = O_p\left\{B_n^2 n^{-c_1(\rho, \vartheta)} (\log p)^{\vartheta/(2\vartheta-1)}\right\} + O(B_n^2 n^{-\rho}).$$

For the quadratic spectral kernel, $\vartheta = 2$, this becomes

$$c_1(\rho, 2) = \frac{1-4\rho}{3}, \quad c_2^\sharp(2, \gamma_{\text{mix}}, 2) = \frac{2}{3},$$

whereas the generic single-envelope choice is $c_2(2, \gamma_{\text{mix}}, 2) = 7/3 + 1/\gamma_{\text{mix}}$.

Proof. Chang, Chen and Wu (2024, Theorem 11(i)) state only the existence of c_1 and c_2 , so these exponents are not uniquely defined. We derive the admissible choices above from its proof. Write $\Delta_{n,r} := \|\hat{\Xi}_n - \Xi\|_{\max}$, $\ell_p := \log p$, $x_+ := \max\{x, 0\}$, and

$$\gamma_*^{-1} := 2 + \left(\frac{2}{\gamma_1} - 1\right)_+ + \frac{1}{\gamma_2}.$$

To make Chang, Chen and Wu (2024, equation (S.52)) explicit, define the population lag matrices

$$H_j := \begin{cases} \frac{1}{n} \sum_{t=j+1}^n \mathbb{E}(X_t X_{t-j}^\top), & j \geq 0, \\ \frac{1}{n} \sum_{t=-j+1}^n \mathbb{E}(X_{t+j} X_t^\top), & j < 0, \end{cases} \quad \Xi^\star := \sum_{j=-n+1}^{n-1} K(j/b_n) H_j.$$

Since the X_t are centered, grouping the double covariance sum by its lag gives

$$\Xi = \frac{1}{n} \sum_{s=1}^n \sum_{t=1}^n \mathbb{E}(X_s X_t^\top) = \sum_{j=-n+1}^{n-1} H_j.$$

Consequently, adding and subtracting the deterministic oracle target $\Xi^\star$ yields the exact decomposition

$$\begin{aligned} \hat{\Xi}_n - \Xi &= (\hat{\Xi}_n - \Xi^\star) + (\Xi^\star - \Xi) \\ &= \sum_{j=-n+1}^{n-1} K(j/b_n) (\hat{H}_j - H_j) + \sum_{j=-n+1}^{n-1} \{K(j/b_n) - 1\} H_j. \end{aligned}$$

Thus Chang, Chen and Wu (2024, equation (S.52)) is the max-norm triangle inequality

$$\Delta_{n,r} \leq \underbrace{\left\| \sum_{j=-n+1}^{n-1} K(j/b_n)(\hat{H}_j - H_j) \right\|_{\max}}_{\mathcal{S}_n \text{ (stochastic estimation error)}} + \underbrace{\left\| \sum_{j=-n+1}^{n-1} \{K(j/b_n) - 1\} H_j \right\|_{\max}}_{\mathcal{B}_{n,K} \text{ (kernel bias)}}.$$

The bias term is deterministic. Writing $H_j = (H_{j,a,b})_{a,b \in [p]}$, symmetry of K and $K(0) = 1$ give

$$\mathcal{B}_{n,K} \leq \max_{a,b \in [p]} \sum_{j=1}^{n-1} |K(j/b_n) - 1| (|H_{j,a,b}| + |H_{-j,a,b}|).$$

In the unified notation used in Chang, Chen and Wu (2024, proof of Theorem 11(i)–(ii)), Davydov's inequality and the mixing condition imply

$$|H_{\pm j,a,b}| \lesssim B_n^2 \exp\{-CL_n^{-r_2} j^{r_2}\}, \quad j \geq 1,$$

while the bounded derivative of K implies, by the mean-value theorem, $|K(j/b_n) - 1| \lesssim j/b_n$. Therefore

$$\begin{aligned} \mathcal{B}_{n,K} &\lesssim \frac{B_n^2}{b_n} \sum_{j=1}^{n-1} j \exp\{-C(j/L_n)^{r_2}\} \\ &\lesssim B_n^2 b_n^{-1} L_n^2. \end{aligned}$$

This is Chang, Chen and Wu (2024, equation (S.53)). For the α -mixing case, their specialization is $L_n = 1$ and $r_2 = \gamma_2$; hence $\mathcal{B}_{n,K} = O(B_n^2 b_n^{-1}) = O(B_n^2 n^{-\rho})$.

The stochastic part is also explicit. For $j \geq 0$,

$$\begin{aligned} \hat{H}_j - H_j &= \frac{1}{n} \sum_{t=j+1}^n \{X_t X_{t-j}^\top - \mathbb{E}(X_t X_{t-j}^\top)\} \\ &\quad - \left(\frac{1}{n} \sum_{t=j+1}^n X_t \right) \bar{X}^\top - \bar{X} \left(\frac{1}{n} \sum_{t=j+1}^n X_{t-j} \right)^\top + \frac{n-j}{n} \bar{X} \bar{X}^\top. \end{aligned}$$

After multiplication by $K(j/b_n)$ and summation over j , Chang, Chen and Wu (2024, equation (S.54)) labels the max norms of these four contributions as I_1, I_2, I_3, I_4 . Chang, Chen and Wu (2024, equation (S.55)) then splits the leading centered cross-product term I_1 at a lag M , with lags $j \leq M$ controlled by a mixing concentration inequality and lags $j > M$ controlled by the polynomial decay $|K(j/b_n)| \lesssim (j/b_n)^{-\vartheta}$. These steps give the aggregate bound

$$\begin{aligned} \Delta_{n,r} &= O(B_n^2 n^{-\rho}) + O_p\left(B_n^2 n^{\vartheta\rho} M^{1-\vartheta} \ell_p^{2/\gamma_1}\right) \\ &\quad + O_p\left(B_n^2 n^{(2\rho-1)/2} M^{1/2} \ell_p^{1/2}\right) + O_p\left(B_n^2 n^{\rho-1} M \ell_p^{1/\gamma_*}\right), \end{aligned}$$

where M is the lag-truncation point. In the proof of Chang, Chen and Wu (2024, Theorem 11(i), immediately after equation (S.55)), they choose

$$M \asymp n^{\{1-2\rho+2\vartheta\rho\}/(2\vartheta-1)} \ell_p^{(4-\gamma_1)/\{\gamma_1(2\vartheta-1)\}}.$$

Substitution shows that the first two stochastic terms in the preceding display have common rate

$$B_n^2 n^{-c_A} \ell_p^{d_A}, \quad c_A := \frac{\vartheta-1-(3\vartheta-2)\rho}{2\vartheta-1}, \quad d_A := \frac{\gamma_1(\vartheta-1)+2}{\gamma_1(2\vartheta-1)},$$

whereas the last stochastic term has rate

$$B_n^2 n^{-c_B} \ell_p^{d_B}, \quad c_B := \frac{2(\vartheta-1)-(4\vartheta-3)\rho}{2\vartheta-1}, \quad d_B := \gamma_*^{-1} + \frac{4-\gamma_1}{\gamma_1(2\vartheta-1)}.$$

Since

$$c_B - c_A = \frac{(\vartheta-1)(1-\rho)}{2\vartheta-1} > 0,$$

one admissible single-envelope choice in Proposition 28 is

$$c_1 = c_A, \quad c_2 = \max\{d_A, d_B\}.$$

The bandwidth restriction $\rho < (\vartheta-1)/(3\vartheta-2)$ is exactly the condition $c_1 > 0$.

For the Regime D specialization $\gamma_1 = 2$ and $\gamma_2 = \gamma_{\text{mix}}$, one has $d_B > d_A$ and

$$d_A = \frac{\vartheta}{2\vartheta-1}, \quad d_B = 2 + \frac{1}{\gamma_{\text{mix}}} + \frac{1}{2\vartheta-1}.$$

It follows that the single-envelope choice is

$$c_1 = c_A = \frac{\vartheta-1-(3\vartheta-2)\rho}{2\vartheta-1}, \quad c_2 = d_B = 2 + \frac{1}{\gamma_{\text{mix}}} + \frac{1}{2\vartheta-1},$$

which proves the first assertion.

For the sharper assertion, retain the two stochastic terms separately:

$$\Delta_{n,r} = O_p(B_n^2 n^{-c_A} \ell_p^{d_A}) + O_p(B_n^2 n^{-c_B} \ell_p^{d_B}) + O(B_n^2 n^{-\rho}).$$

Under the assumptions of Proposition 28, $p = \Omega(n^\kappa)$; if also $p \leq n^2$, then $\ell_p \asymp \log n$. Since $c_B - c_A > 0$,

$$\frac{n^{-c_B} \ell_p^{d_B}}{n^{-c_A} \ell_p^{d_A}} = n^{-(c_B-c_A)} \ell_p^{d_B-d_A} \longrightarrow 0.$$

The second stochastic term can therefore be absorbed into the first, so the sharper leading exponent is $c_2^\sharp = d_A = \vartheta/(2\vartheta - 1)$. Finally, setting $\vartheta = 2$ gives

$$c_1 = \frac{1-4\rho}{3}, \quad c_2^\sharp = \frac{2}{3}, \quad c_2 = \frac{7}{3} + \frac{1}{\gamma_{\text{mix}}},$$

as claimed. $\square$

The next proposition gives the bootstrap high-dimensional CLT comparison from Chang, Chen and Wu (2024, Theorem 10(i)).

Proposition 30. *Let $X_1, \dots, X_n$ be centered random vectors in $\mathbb{R}^p$, and define*

$$S_{n,x} := \frac{1}{\sqrt{n}} \sum_{t=1}^n X_t, \quad \Xi := \text{Cov}(S_{n,x}).$$

Let $G \sim N(0, \Xi)$. Given a random covariance estimator $\hat{\Xi}_n$, let $\hat{G}$ be a conditionally Gaussian random vector satisfying

$$\hat{G} \mid X_1, \dots, X_n \sim N(0, \hat{\Xi}_n).$$

For a class $\mathcal{A}$ of Borel subsets of $\mathbb{R}^p$, define

$$\rho_n(\mathcal{A}) := \sup_{A \in \mathcal{A}} |\mathbb{P}(S_{n,x} \in A) - \mathbb{P}(G \in A)|, \quad \hat{\rho}_n(\mathcal{A}) := \sup_{A \in \mathcal{A}} |\mathbb{P}(S_{n,x} \in A) - \mathbb{P}(\hat{G} \in A \mid X_1, \dots, X_n)|.$$

Let $\mathcal{A}^{\text{re}}$ denote the class of all hyper-rectangles in $\mathbb{R}^p$, namely all sets of the form

$$A = \prod_{j=1}^p [a_j, b_j], \quad -\infty \leq a_j \leq b_j \leq \infty.$$

These sets include one-sided rectangles by allowing infinite endpoints. For each $j \in [p]$, let

$$V_{n,j} := \text{Var}\left(\frac{1}{\sqrt{n}} \sum_{t=1}^n X_{t,j}\right).$$

Assume that there exists a constant $K_3 > 0$ such that $\min_{j \in [p]} V_{n,j} \geq K_3$. Further, assume that $p \geq n^\kappa$ for some constant $\kappa > 0$. Then

$$\hat{\rho}_n(\mathcal{A}^{\text{re}}) \lesssim \rho_n(\mathcal{A}^{\text{re}}) + \|\hat{\Xi}_n - \Xi\|_{\max}^{1/3} (\log p)^{2/3},$$

where $\lesssim$ hides a universal positive constant.

The next proposition is the strong-mixing Bernstein inequality from Merlevède, Peligrad and Rio

(2011, Theorem 1 and Remark 1).

Proposition 31. *Let $(X_j)_{j \geq 1}$ be centered real-valued random variables, and define their strong-mixing coefficients by*

$$\alpha_X(\ell) := \sup_{k \geq 1} \sup_{\substack{A \in \sigma(X_1, \dots, X_k), \\ B \in \sigma(X_{k+\ell}, X_{k+\ell+1}, \dots)}} |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)|, \quad \ell \geq 1.$$

Assume that there exist constants $c > 0$, $b > 0$, $\gamma_1 > 0$, and $\gamma_2 \in (0, \infty]$ such that

$$\alpha_X(n) \leq \exp(-cn^{\gamma_1}) \quad \text{for all } n \geq 1,$$

and

$$\sup_{j \geq 1} \mathbb{P}(|X_j| > t) \leq \exp(1 - (t/b)^{\gamma_2}) \quad \text{for all } t > 0.$$

Let

$$\gamma := \left(\frac{1}{\gamma_1} + \frac{1}{\gamma_2} \right)^{-1}, \quad \varphi_M(x) := (x \wedge M) \vee (-M),$$

and assume $\gamma < 1$. Define

$$V := \sup_{M > 0} \sup_{i \geq 1} \left(\text{Var}(\varphi_M(X_i)) + 2 \sum_{j > i} |\text{Cov}(\varphi_M(X_i), \varphi_M(X_j))| \right). \quad (\text{B.77})$$

Then there exist constants $C_0, C_1, C_2, \eta_0 \in (0, \infty)$, depending only on $(b, c, \gamma_1, \gamma_2)$, such that for every $n \geq 4$ and every $y \geq C_0(\log n)^{\eta_0}$,

$$\mathbb{P} \left(\max_{1 \leq j \leq n} \left| \sum_{r=1}^j X_r \right| \geq y \right) \leq (n+1) \exp\left(-\frac{y^\gamma}{C_1}\right) + \exp\left(-\frac{y^2}{C_2 + C_2 n V}\right). \quad (\text{B.78})$$

The next proposition bounds the term V in the strong-mixing Bernstein inequality (B.78) by marginal quantiles and mixing coefficients.

Proposition 32. *Consider the setting of Proposition 31. For $u \in (0, 1]$, define the marginal quantiles*

$$q_{|X_i|}(u) := \inf\{t > 0 : \mathbb{P}(|X_i| > t) \leq u\}, \quad q(u) := \sup_{i \geq 1} q_{|X_i|}(u),$$

Then the strong-mixing covariance inequality (Merlevède, Peligrad and Rio, 2011, see, e.g., Remark 3) implies

$$V \leq \sup_{i \geq 1} \mathbb{E}[X_i^2] + 4 \sum_{k > 0} \int_0^{2\alpha_X(k)} q(u)^2 du. \quad (\text{B.79})$$

Orlicz ψ_2 and ψ_1 norms. Following Vershynin (2018, Definitions 2.5.6 and 2.7.5), define for a real-valued random variable X

$$\|X\|_{\psi_2} := \inf\{t > 0 : \mathbb{E} \exp(X^2/t^2) \leq 2\}, \quad \|X\|_{\psi_1} := \inf\{t > 0 : \mathbb{E} \exp(|X|/t) \leq 2\}.$$

We call X *sub-Gaussian* if $\|X\|_{\psi_2} < \infty$ and *sub-exponential* if $\|X\|_{\psi_1} < \infty$. By Vershynin (2018, Proposition 2.5.2), the ψ_2 norm is equivalent to the standard sub-Gaussian tail and moment conditions; by Vershynin (2018, Proposition 2.7.1), the ψ_1 norm is equivalent to the standard sub-exponential tail and moment conditions.

The next proposition gives maximal bounds for finitely many sub-Gaussian variables, following Vershynin (2018, Proposition 2.5.2 and Exercise 2.5.10).

Proposition 33. *Let $X_1, \dots, X_m$ be real-valued random variables satisfying*

$$\max_{1 \leq j \leq m} \|X_j\|_{\psi_2} \leq K.$$

Then there exist universal constants $C, c \in (0, \infty)$ such that

$$\mathbb{E} \left[\max_{1 \leq j \leq m} |X_j| \right] \leq CK \sqrt{1 \vee \log m}$$

and, for every $u \geq 0$,

$$\mathbb{P} \left(\max_{1 \leq j \leq m} |X_j| > CK \left(\sqrt{\log(2m)} + u \right) \right) \leq 2e^{-cu^2}.$$

In particular,

$$\max_{1 \leq j \leq m} |X_j| = O_p(K \sqrt{1 \vee \log m}).$$

No independence is required.

The next proposition gives the standard sub-Gaussian bound for weighted sums of independent centered variables $X_1, \dots, X_m$, following Vershynin (2018, Proposition 2.6.1).

Proposition 34. *Let $X_1, \dots, X_m$ be independent centered real-valued random variables, and set*

$$K_j := \|X_j\|_{\psi_2}, \quad V_2 := \left(\sum_{j=1}^m K_j^2 \right)^{1/2}.$$

Then there exists a universal constant $C < \infty$ such that

$$\left\| \sum_{j=1}^m X_j \right\|_{\psi_2} \leq CV_2.$$

More generally, for deterministic weights $a_1, \dots, a_m \in \mathbb{R}$,

$$\left\| \sum_{j=1}^m a_j X_j \right\|_{\psi_2} \leq C \left(\sum_{j=1}^m a_j^2 K_j^2 \right)^{1/2}.$$

The next proposition gives the standard product rule turning sub-Gaussian products into sub-exponential variables, following Vershynin (2018, Lemma 2.7.7).

Proposition 35. *Let X and Y be sub-gaussian random variables. Then XY is sub-exponential. Moreover,*

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}.$$

The next proposition gives maximal bounds for finitely many sub-exponential variables, following Vershynin (2018, Proposition 2.7.1).

Proposition 36. *Let $Y_1, \dots, Y_m$ be real-valued random variables satisfying*

$$\max_{1 \leq j \leq m} \|Y_j\|_{\psi_1} \leq K.$$

Then there exist universal constants $C, c \in (0, \infty)$ such that

$$\mathbb{E} \left[\max_{1 \leq j \leq m} |Y_j| \right] \leq CK(1 \vee \log m)$$

and, for every $u \geq 0$,

$$\mathbb{P} \left(\max_{1 \leq j \leq m} |Y_j| > CK(\log(2m) + u) \right) \leq 2e^{-cu}.$$

In particular,

$$\max_{1 \leq j \leq m} |Y_j| = O_p(K(1 \vee \log m)).$$

Again, no independence is required.

C Simulation Results for Partially Studentized Scan Tests

Section 3 of the main draft develops the scan test as a one-sided procedure for detecting forecast outperformance that may occur only over unknown target-month intervals and unknown horizons. The theory justifies the bootstrap scan test asymptotically, but the empirical application has only $n = 215$ target months and uses short interval collections for expert selection. This appendix therefore documents the finite-sample calibration exercises behind the implementation choices in Section 3. The first part calibrates the partial-studentization exponent γ for the horizon-by-horizon scan tests. The second part repeats the calibration for the horizon-pooled pre/post-COVID use of the scan test. The final part benchmarks these choices against the closely related Giacomini–Rossi fluctuation test of Giacomini and Rossi (2010).

C.1 Simulation and the choice of γ

The scan statistic in Section 3 maximizes a partially studentized statistic over many intervals and horizons. This maximization is what lets the test find localized outperformance without specifying its timing or forecast horizon in advance, but it also creates the main finite-sample concern: aggressive studentization can make short, noisy intervals look too extreme. We therefore choose γ by simulating the null in a way that preserves the empirical scale, cross-horizon dependence, and date-specific heterogeneity of the observed loss-differential panels.

The simulation setting is under the null. It fixes an empirical loss-differential panel $D_t^{(h)}$ and draws

$$D_{t,k}^{(h)} = \xi_{t,k} D_t^{(h)}, \quad \xi_{t,k} = \text{sign}(Z_{t,k}),$$

where $Z_{t,k}$ is a stationary Gaussian AR(1) process with autocorrelation ρ . To choose ρ , we fit an AR(1) model to each comparison (sarima–micro, macro–micro, and sarima–macro) and each horizon $h = 1, \dots, 24$. The box plots of the estimates are given below. Thus, we choose $\rho \in \{-0.25, 0, 0.25, 0.5\}$ in our simulations.

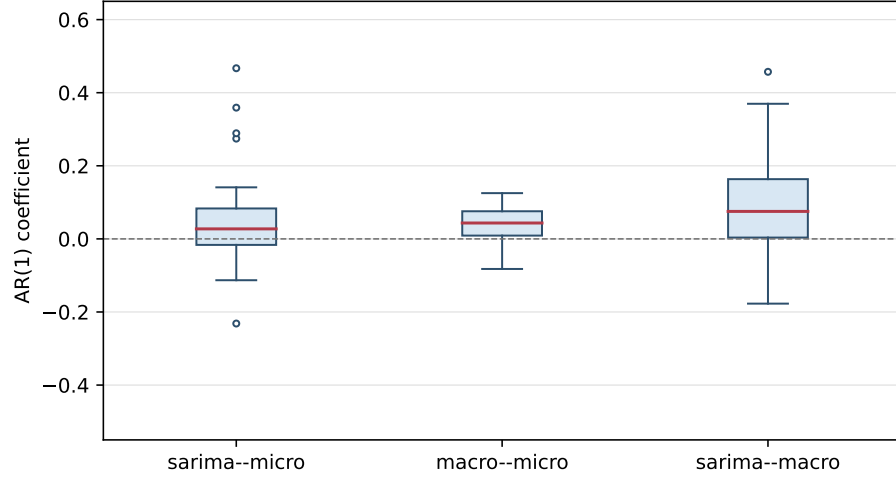


Figure C.1: AR(1) coefficient estimates across horizons for sarima–micro, macro–micro, and sarima–macro.

C.1.1 Simulation results for occasional outperformance

Occasional outperformance is the expert-selection use emphasized in Section 3: the candidate forecast should enter the pool if it improves on the benchmark over some unknown short subperiod and horizon. We use the interval lengths $\{6, 7, \dots, 12\}$ to target half-year to one-year episodes, where a forecast can be useful during a localized inflation episode even if it is not uniformly better over the full sample. This is also the most demanding case for size, because the scan searches over many short windows.

For each setting in Tables C.1 –C.3, we use 1000 Monte Carlo samples and 999 bootstrap draws. The scan intervals are $\{6, 7, \dots, 12\}$ and the target level is denoted by α_0 to avoid conflict with the tuning parameter γ .

Table C.1: Simulation for sarima–micro. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I					(b) Regime D				
γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$	γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
0.00	0.000	0.000	0.003	0.005	0.00	0.003	0.003	0.027	0.056
0.05	0.000	0.002	0.019	0.038	0.05	0.004	0.002	0.026	0.058
0.10	0.003	0.011	0.035	0.075	0.10	0.002	0.002	0.023	0.045
0.15	0.005	0.015	0.038	0.089	0.15	0.000	0.000	0.006	0.011
0.20	0.006	0.016	0.043	0.090	0.20	0.000	0.000	0.000	0.000
0.25	0.005	0.014	0.044	0.101	0.25	0.000	0.000	0.000	0.000
0.30	0.049	0.064	0.098	0.123	0.30	0.002	0.004	0.006	0.002
0.35	0.161	0.159	0.165	0.185	0.35	0.053	0.043	0.044	0.038
0.40	0.235	0.224	0.203	0.188	0.40	0.108	0.097	0.093	0.075
0.45	0.213	0.217	0.180	0.178	0.45	0.185	0.200	0.175	0.171
0.50	0.182	0.194	0.151	0.152	0.50	0.193	0.214	0.185	0.189

Table C.2: Simulation for macro–micro. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I					(b) Regime D				
γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$	γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
0.00	0.002	0.003	0.001	0.003	0.00	0.003	0.003	0.000	0.001
0.05	0.006	0.010	0.010	0.010	0.05	0.005	0.010	0.008	0.008
0.10	0.014	0.026	0.030	0.045	0.10	0.013	0.020	0.024	0.026
0.15	0.068	0.075	0.076	0.089	0.15	0.074	0.067	0.071	0.068
0.20	0.205	0.196	0.210	0.211	0.20	0.199	0.190	0.194	0.191
0.25	0.319	0.310	0.353	0.353	0.25	0.331	0.308	0.340	0.329
0.30	0.394	0.420	0.476	0.484	0.30	0.418	0.422	0.474	0.493
0.35	0.400	0.409	0.452	0.447	0.35	0.424	0.419	0.455	0.457
0.40	0.382	0.382	0.404	0.377	0.40	0.399	0.393	0.416	0.395
0.45	0.319	0.320	0.320	0.302	0.45	0.346	0.337	0.328	0.312
0.50	0.227	0.232	0.209	0.221	0.50	0.241	0.237	0.218	0.224

Table C.3: Simulation for sarima–macro. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I					(b) Regime D				
γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$	γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
0.00	0.000	0.000	0.000	0.000	0.00	0.000	0.000	0.018	0.036
0.05	0.000	0.001	0.003	0.005	0.05	0.003	0.000	0.016	0.039
0.10	0.000	0.004	0.022	0.044	0.10	0.000	0.000	0.012	0.027
0.15	0.004	0.010	0.035	0.075	0.15	0.000	0.000	0.003	0.006
0.20	0.005	0.012	0.036	0.085	0.20	0.000	0.000	0.000	0.000
0.25	0.013	0.027	0.064	0.121	0.25	0.001	0.005	0.003	0.002
0.30	0.049	0.085	0.128	0.183	0.30	0.013	0.025	0.030	0.034
0.35	0.227	0.246	0.279	0.322	0.35	0.086	0.089	0.104	0.085
0.40	0.308	0.316	0.351	0.385	0.40	0.209	0.216	0.244	0.266
0.45	0.309	0.293	0.320	0.322	0.45	0.314	0.323	0.356	0.385
0.50	0.260	0.221	0.258	0.256	0.50	0.300	0.291	0.331	0.343

C.1.2 Simulation results for overall outperformance

Overall outperformance is the companion expert-selection diagnostic from Section 3. Instead of searching for a localized episode, it fixes the interval to the full balanced sample and asks whether one forecast improves on another over the sample as a whole at some horizon. This removes the interval-search noise from the occasional scan and can be more stable when predictive gains are sustained rather than concentrated in a short spell.

For each setting in Tables C.4 –C.6, we use 1000 Monte Carlo samples and 999 bootstrap draws. The scan interval length is $\{215\}$ and the target level is again denoted by α_0 .

Table C.4: Simulation for sarima–micro with length {215}. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I					(b) Regime D				
γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$	γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
0.00	0.008	0.012	0.026	0.042	0.00	0.007	0.006	0.008	0.010
0.05	0.012	0.019	0.036	0.061	0.05	0.011	0.010	0.011	0.019
0.10	0.018	0.025	0.047	0.075	0.10	0.017	0.016	0.017	0.025
0.15	0.025	0.038	0.054	0.094	0.15	0.022	0.023	0.031	0.035
0.20	0.035	0.046	0.063	0.107	0.20	0.037	0.031	0.037	0.047
0.25	0.046	0.052	0.075	0.115	0.25	0.043	0.039	0.047	0.061
0.30	0.054	0.055	0.071	0.105	0.30	0.056	0.040	0.054	0.060
0.35	0.057	0.051	0.077	0.087	0.35	0.058	0.047	0.057	0.058
0.40	0.056	0.049	0.062	0.073	0.40	0.059	0.051	0.057	0.060
0.45	0.055	0.049	0.061	0.063	0.45	0.058	0.052	0.054	0.059
0.50	0.055	0.051	0.057	0.068	0.50	0.056	0.053	0.055	0.059

Table C.5: Simulation for macro–micro with length {215}. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I					(b) Regime D				
γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$	γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
0.00	0.041	0.052	0.050	0.054	0.00	0.039	0.052	0.046	0.053
0.05	0.040	0.057	0.055	0.059	0.05	0.042	0.056	0.049	0.055
0.10	0.043	0.059	0.054	0.059	0.10	0.043	0.058	0.051	0.057
0.15	0.043	0.063	0.057	0.061	0.15	0.048	0.063	0.054	0.060
0.20	0.048	0.070	0.059	0.068	0.20	0.050	0.066	0.057	0.063
0.25	0.049	0.071	0.063	0.075	0.25	0.049	0.067	0.059	0.071
0.30	0.051	0.070	0.064	0.080	0.30	0.052	0.067	0.062	0.074
0.35	0.050	0.074	0.066	0.079	0.35	0.051	0.069	0.061	0.076
0.40	0.048	0.074	0.066	0.073	0.40	0.049	0.070	0.061	0.075
0.45	0.045	0.072	0.065	0.071	0.45	0.047	0.070	0.060	0.076
0.50	0.044	0.072	0.064	0.072	0.50	0.044	0.069	0.058	0.076

Table C.6: Simulation for sarima-macro with length {215}. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I					(b) Regime D				
γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$	γ	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
0.00	0.003	0.012	0.016	0.046	0.00	0.005	0.004	0.001	0.012
0.05	0.005	0.018	0.023	0.056	0.05	0.006	0.007	0.004	0.017
0.10	0.012	0.033	0.032	0.072	0.10	0.010	0.012	0.012	0.024
0.15	0.019	0.044	0.042	0.087	0.15	0.016	0.030	0.016	0.031
0.20	0.023	0.054	0.050	0.094	0.20	0.027	0.037	0.025	0.045
0.25	0.029	0.062	0.053	0.109	0.25	0.034	0.041	0.031	0.054
0.30	0.038	0.057	0.067	0.109	0.30	0.042	0.051	0.039	0.060
0.35	0.037	0.061	0.065	0.093	0.35	0.041	0.056	0.046	0.064
0.40	0.041	0.061	0.056	0.085	0.40	0.041	0.059	0.050	0.066
0.45	0.038	0.062	0.056	0.080	0.45	0.040	0.063	0.052	0.058
0.50	0.033	0.063	0.057	0.074	0.50	0.041	0.063	0.052	0.058

C.1.3 Choice of γ

This subsection converts the two calibration exercises into the tuning rule used in the empirical scan checks. Because Section 3 allows either Regime I or Regime D depending on serial dependence, the decision must control size not only under each fixed multiplier regime but also after the Ljung–Box pre-test chooses between them.

The simulation results suggest $\gamma = 0.1$ avoids over-rejections across the board. So we choose $\gamma = 0.1$ for the scan test. A few rejection probabilities are slightly above the nominal level when $\rho = 0.5$, but the inflation is minor and occurs under the strongest positive serial dependence considered in the simulation grid. We will check robustness by also reporting the empirical results for the adjacent choices $\gamma = 0.05$ and $\gamma = 0.15$.

To confirm that the pre-test does not inflate size, we run the same simulation on the scan test with the Ljung–Box pre-test and the above chosen γ for each comparison. We report the results for both the occasional- and overall-outperformance interval sets.

Table C.7: Simulation for the Ljung–Box pre-test scan test with $\delta = 0.05$ and $\gamma = 0.1$ under both multiplier regimes, using occasional-outperformance lengths $\{6, 7, \dots, 12\}$. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

Comparison	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
sarima–micro	0.003	0.012	0.022	0.063
macro–micro	0.023	0.020	0.032	0.037
sarima–macro	0.005	0.001	0.012	0.035

Table C.8: Simulation for the Ljung–Box pre-test scan test with $\delta = 0.05$ and $\gamma = 0.1$ under both multiplier regimes, using overall-outperformance length $\{215\}$. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

Comparison	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
sarima–micro	0.012	0.032	0.029	0.055
macro–micro	0.045	0.051	0.056	0.061
sarima–macro	0.016	0.027	0.035	0.066

C.1.4 Robustness of occasional and overall performance to the choice of γ

The simulations above motivate the baseline choice $\gamma = 0.1$ by focusing on size control under null designs calibrated to the observed loss-differential panels. The empirical conclusion should not, however, hinge on this single tuning value. We therefore report the actual scan-test p-values for the two expert-selection metrics from Section 3 at $\gamma = 0.1$ and at the adjacent values $\gamma = 0.05$ and $\gamma = 0.15$. The occasional-outperformance metric uses the short interval collection $\{6, 7, \dots, 12\}$ and asks whether a candidate forecast improves on the benchmark during some localized episode and horizon. The overall-outperformance metric uses the full balanced sample length $\{215\}$ and asks whether the improvement is sustained over the sample as a whole at some horizon. Reporting both metrics across nearby values of γ checks that the substantive expert-selection evidence is robust to the degree of partial studentization.

Table C.9: Empirical scan-test p-values with $\gamma = 0.05$. The Ljung–Box pre-test uses level $\delta = 0.05$. Occasional outperformance uses lengths $\{6, 7, \dots, 12\}$; overall outperformance uses length $\{215\}$.

Comparison	Regime	LB p	Occ. p	Overall p
sarima–micro	Regime D	0.0338	0.0352	0.1494
macro–micro	Regime I	0.2260	0.2861	0.0306
sarima–macro	Regime I	0.2234	0.0572	0.0406

Table C.10: Empirical scan-test p-values with $\gamma = 0.10$. The Ljung–Box pre-test uses level $\delta = 0.05$. Occasional outperformance uses lengths $\{6, 7, \dots, 12\}$; overall outperformance uses length $\{215\}$.

Comparison	Regime	LB p	Occ. p	Overall p
sarima–micro	Regime D	0.0338	0.0406	0.1524
macro–micro	Regime I	0.2260	0.2575	0.0256
sarima–macro	Regime I	0.2234	0.0406	0.0324

Table C.11: Empirical scan-test p-values with $\gamma = 0.15$. The Ljung–Box pre-test uses level $\delta = 0.05$. Occasional outperformance uses lengths $\{6, 7, \dots, 12\}$; overall outperformance uses length $\{215\}$.

Comparison	Regime	LB p	Occ. p	Overall p
sarima–micro	Regime D	0.0338	0.0522	0.1544
macro–micro	Regime I	0.2260	0.1870	0.0210
sarima–macro	Regime I	0.2234	0.0286	0.0224

C.2 Simulations for the horizon-pooled scan test

Section 3 also describes a complementary use of the scan test for a pre-specified calendar window, while pooling across horizons. This question is different from expert selection: the window, such as the pre-COVID or post-COVID period, is fixed in advance, and the test asks whether the average loss differential across horizons is positive within that window. Because the statistic is now built from the horizon-pooled series $\bar{D}_t$ rather than from the full interval-horizon panel, we run a separate null calibration for this version of the procedure.

For each empirical panel, we fit $\bar{D}_t = c + \rho \bar{D}_{t-1} + u_t$ to the horizon-pooled series. The estimated AR(1) coefficients are reported in Table C.12.

Table C.12: AR(1) estimates for the horizon-pooled loss-differential series.

Comparison	$\hat{\rho}$	95% confidence interval
(sarima + macro) – three experts	0.050	[−0.086, 0.187]
sarima–micro	0.185	[0.052, 0.317]
sarima–macro	0.160	[0.027, 0.293]

The null DGP is the same sign-AR(1) construction described above, applied to the horizon-pooled series. We run $\gamma \in \{0, 0.05, 0.1, \dots, 0.5\}$, 1000 Monte Carlo samples, and 999 bootstrap draws. Panels (a) and (b) fix Regime I and Regime D, respectively; panel (c) applies the Ljung–Box pre-test with $\delta = 0.05$. For the horizon-pooled calibration tables below, we simulate only $\rho \in \{0, 0.15, 0.3\}$.

C.2.1 Simulation results for (sarima + macro) vs three experts

We show the detailed simulation results for the most important comparison, (sarima+macro) vs three experts, in this subsection. This comparison asks whether adding the micro forecast to the two-expert combination improves the horizon-pooled pre/post-COVID performance. We report the fixed Regime I, fixed Regime D, and pre-test versions here so that the effect of the regime-selection step is visible in the main horizon-pooled calibration. To save space, the next subsection documents the simulation results for $\gamma \in \{0.05, 0.1, 0.15\}$ with the Ljung–Box pre-test for the other comparisons.

Table C.13: Pooled-window simulation for (sarima + macro) – three experts, pre-COVID window. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I				(b) Regime D				(c) Pre-test			
γ	$\rho = 0$	$\rho = .15$	$\rho = .3$	γ	$\rho = 0$	$\rho = .15$	$\rho = .3$	γ	$\rho = 0$	$\rho = .15$	$\rho = .3$
0.00	0.053	0.060	0.046	0.00	0.056	0.062	0.047	0.00	0.050	0.038	0.063
0.05	0.053	0.062	0.046	0.05	0.057	0.061	0.046	0.05	0.051	0.037	0.060
0.10	0.052	0.062	0.046	0.10	0.056	0.062	0.047	0.10	0.048	0.035	0.063
0.15	0.051	0.060	0.045	0.15	0.055	0.062	0.046	0.15	0.048	0.035	0.060
0.20	0.051	0.058	0.044	0.20	0.055	0.061	0.048	0.20	0.048	0.035	0.060
0.25	0.051	0.059	0.045	0.25	0.055	0.063	0.046	0.25	0.048	0.035	0.060
0.30	0.050	0.056	0.044	0.30	0.054	0.061	0.046	0.30	0.049	0.033	0.060
0.35	0.051	0.057	0.043	0.35	0.054	0.061	0.047	0.35	0.048	0.032	0.058
0.40	0.050	0.055	0.043	0.40	0.053	0.060	0.045	0.40	0.048	0.032	0.056
0.45	0.050	0.055	0.042	0.45	0.054	0.059	0.043	0.45	0.048	0.031	0.056
0.50	0.048	0.053	0.041	0.50	0.052	0.058	0.043	0.50	0.048	0.031	0.056

Table C.14: Pooled-window simulation for (sarima + macro) – three experts, post-COVID window. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

(a) Regime I					(b) Regime D					(c) Pre-test				
γ	$\rho = 0$	$\rho = .15$	$\rho = .3$		γ	$\rho = 0$	$\rho = .15$	$\rho = .3$		γ	$\rho = 0$	$\rho = .15$	$\rho = .3$	
0.00	0.054	0.057	0.065		0.00	0.054	0.055	0.065		0.00	0.068	0.055	0.070	
0.05	0.054	0.058	0.065		0.05	0.054	0.054	0.065		0.05	0.069	0.056	0.070	
0.10	0.053	0.058	0.063		0.10	0.054	0.057	0.064		0.10	0.069	0.056	0.069	
0.15	0.053	0.058	0.063		0.15	0.054	0.058	0.065		0.15	0.070	0.054	0.066	
0.20	0.053	0.059	0.065		0.20	0.054	0.057	0.065		0.20	0.071	0.054	0.067	
0.25	0.052	0.059	0.064		0.25	0.053	0.055	0.065		0.25	0.070	0.054	0.069	
0.30	0.051	0.058	0.064		0.30	0.051	0.056	0.061		0.30	0.068	0.051	0.067	
0.35	0.051	0.058	0.064		0.35	0.051	0.056	0.062		0.35	0.068	0.051	0.066	
0.40	0.049	0.057	0.064		0.40	0.052	0.054	0.062		0.40	0.067	0.050	0.064	
0.45	0.048	0.056	0.065		0.45	0.049	0.053	0.058		0.45	0.066	0.050	0.062	
0.50	0.049	0.056	0.065		0.50	0.049	0.053	0.057		0.50	0.064	0.050	0.061	

C.2.2 Other comparisons

For the remaining horizon-pooled comparisons in Table C.12, we focus on the pre-test version with $\gamma \in \{0.05, 0.1, 0.15\}$. These are the specifications most relevant for the empirical tables, because the applied procedure first diagnoses serial dependence and then chooses the corresponding bootstrap regime. The neighboring values of γ show how sensitive the size calibration is around the preferred value $\gamma = 0.1$. Each table fixes one value of γ and reports the pre-COVID rows first, followed by the post-COVID rows, with comparisons ordered as in Table C.12.

Table C.15: Horizon-pooled pre-test simulation for the other comparisons with $\gamma = 0.05$ and $\delta = 0.05$. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

Window	Comparison	$\rho = 0$	$\rho = .15$	$\rho = .3$
Pre-COVID	sarima–micro	0.052	0.046	0.054
Pre-COVID	sarima–macro	0.055	0.059	0.047
Post-COVID	sarima–micro	0.061	0.061	0.069
Post-COVID	sarima–macro	0.054	0.054	0.072

Table C.16: Horizon-pooled pre-test simulation for the other comparisons with $\gamma = 0.10$ and $\delta = 0.05$. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

Window	Comparison	$\rho = 0$	$\rho = .15$	$\rho = .3$
Pre-COVID	sarima-micro	0.053	0.047	0.052
Pre-COVID	sarima-macro	0.055	0.059	0.047
Post-COVID	sarima-micro	0.061	0.061	0.070
Post-COVID	sarima-macro	0.053	0.054	0.073

Table C.17: Horizon-pooled pre-test simulation for the other comparisons with $\gamma = 0.15$ and $\delta = 0.05$. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$.

Window	Comparison	$\rho = 0$	$\rho = .15$	$\rho = .3$
Pre-COVID	sarima-micro	0.053	0.047	0.053
Pre-COVID	sarima-macro	0.054	0.059	0.047
Post-COVID	sarima-micro	0.060	0.061	0.069
Post-COVID	sarima-macro	0.055	0.054	0.072

C.3 Giacomini–Rossi fluctuation test

Section 3 discusses the Giacomini and Rossi (2010) fluctuation test as the closest classical comparison: it also looks for forecast-instability episodes, but it uses fixed window lengths and fixed horizons rather than a joint scan over unknown intervals and horizons. We include it here to make clear what changes when the localized-outperformance problem is treated with the classical Brownian-limit calibration instead of the one-sided multiplier-bootstrap scan used in the main procedure.

For comparison, we implement a one-sided version of this fluctuation test. For a fixed horizon h and a fixed window length m , define

$$G_m^{(h)} = \max_{1 \leq s \leq n-m+1} \frac{\sum_{t=s}^{s+m-1} D_t^{(h)}}{\sqrt{m\hat{\sigma}_h^2}},$$

where $\hat{\sigma}_h^2$ is a full-sample Newey–West long-run variance estimate for the loss differential at horizon h . We use lag $h-1$ in this estimate. The alternative is one-sided: large positive values indicate that the first forecast has larger loss than the second forecast over some fixed-length window.

Under the local equal-predictive-ability null and a functional central limit theorem,

$$\max_s \frac{\sum_{t=s}^{s+m-1} D_t^{(h)}}{\sqrt{m\hat{\sigma}_h^2}} \Rightarrow \sup_s \frac{W(s+m/n) - W(s)}{\sqrt{m/n}},$$

where W is standard Brownian motion. In the simulations below we approximate this null distribution on the observed grid by simulating iid standard-normal increments and taking the maximum rolling sum divided by $\sqrt{m}$.

We run the same sign-AR(1) null simulation described above with $m \in \{6, 7, \dots, 12\}$, 1000 Monte Carlo samples, and 200000 Brownian-limit draws for each value of m . The test is horizon-by-horizon, so each table reports the mean rejection probability across the 24 horizons. This mirrors the occasional-outperformance calibration while respecting the fixed-window, fixed-horizon structure of the Giacomini–Rossi statistic.

Table C.18: Giacomini–Rossi simulation for sarima–micro. Entries are mean rejection probabilities across horizons $h = 1, \dots, 24$ at nominal level $\alpha_0 = 0.05$; brackets give approximate 95% Monte Carlo intervals.

m	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
6	0.41 [0.38, 0.45]	0.41 [0.38, 0.45]	0.43 [0.39, 0.46]	0.42 [0.39, 0.45]
7	0.39 [0.36, 0.42]	0.40 [0.36, 0.43]	0.40 [0.37, 0.43]	0.40 [0.37, 0.43]
8	0.37 [0.34, 0.40]	0.37 [0.34, 0.40]	0.37 [0.34, 0.40]	0.37 [0.34, 0.40]
9	0.34 [0.31, 0.37]	0.33 [0.30, 0.36]	0.34 [0.31, 0.36]	0.33 [0.30, 0.36]
10	0.31 [0.28, 0.34]	0.31 [0.28, 0.34]	0.31 [0.28, 0.34]	0.31 [0.28, 0.34]
11	0.29 [0.26, 0.32]	0.28 [0.25, 0.31]	0.28 [0.25, 0.31]	0.28 [0.25, 0.31]
12	0.27 [0.24, 0.30]	0.26 [0.23, 0.29]	0.27 [0.24, 0.29]	0.26 [0.23, 0.29]

Table C.19: Giacomini–Rossi simulation for macro–micro. Entries are mean rejection probabilities across horizons $h = 1, \dots, 24$ at nominal level $\alpha_0 = 0.05$; brackets give approximate 95% Monte Carlo intervals.

m	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
6	0.27 [0.24, 0.30]	0.27 [0.25, 0.30]	0.27 [0.25, 0.30]	0.27 [0.25, 0.30]
7	0.24 [0.21, 0.27]	0.25 [0.22, 0.27]	0.25 [0.22, 0.27]	0.25 [0.22, 0.28]
8	0.22 [0.20, 0.25]	0.23 [0.20, 0.25]	0.23 [0.20, 0.25]	0.23 [0.20, 0.26]
9	0.20 [0.18, 0.23]	0.20 [0.18, 0.23]	0.21 [0.18, 0.23]	0.22 [0.19, 0.24]
10	0.18 [0.16, 0.21]	0.19 [0.16, 0.21]	0.19 [0.17, 0.21]	0.20 [0.17, 0.22]
11	0.16 [0.14, 0.19]	0.17 [0.15, 0.19]	0.18 [0.15, 0.20]	0.18 [0.16, 0.20]
12	0.16 [0.14, 0.18]	0.17 [0.14, 0.19]	0.17 [0.15, 0.19]	0.17 [0.15, 0.20]

Table C.20: Giacomini–Rossi simulation for sarima–macro. Entries are mean rejection probabilities across horizons $h = 1, \dots, 24$ at nominal level $\alpha_0 = 0.05$; brackets give approximate 95% Monte Carlo intervals.

m	$\rho = -.25$	$\rho = 0$	$\rho = .25$	$\rho = .5$
6	0.51 [0.48, 0.54]	0.51 [0.48, 0.54]	0.50 [0.47, 0.53]	0.50 [0.47, 0.53]
7	0.48 [0.45, 0.51]	0.48 [0.45, 0.51]	0.47 [0.43, 0.50]	0.47 [0.44, 0.50]
8	0.44 [0.41, 0.47]	0.44 [0.41, 0.47]	0.43 [0.40, 0.47]	0.45 [0.41, 0.48]
9	0.41 [0.38, 0.44]	0.41 [0.38, 0.44]	0.40 [0.37, 0.43]	0.41 [0.38, 0.45]
10	0.38 [0.35, 0.41]	0.38 [0.35, 0.41]	0.37 [0.34, 0.40]	0.38 [0.35, 0.41]
11	0.35 [0.32, 0.38]	0.36 [0.33, 0.39]	0.35 [0.32, 0.38]	0.35 [0.32, 0.38]
12	0.33 [0.30, 0.36]	0.33 [0.30, 0.36]	0.32 [0.30, 0.35]	0.33 [0.30, 0.36]

We also apply the Giacomini–Rossi statistic to the horizon-pooled sign-AR(1) simulations used for the pre-COVID and post-COVID horizon-pooled analyses above. The input series is $\bar{D}_t = H^{-1} \sum_{h=1}^{24} D_t^{(h)}$ for (sarima + macro) – three experts. The sign-AR(1) multiplier is generated on the full balanced sample and then restricted to the pre-specified calendar window. Because this series pools horizons up to $H = 24$, the Newey–West long-run variance uses lag 23.

Table C.21: Giacomini–Rossi horizon-pooled simulation for (sarima + macro) – three experts, pre-COVID window. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$; brackets give approximate 95% Monte Carlo intervals.

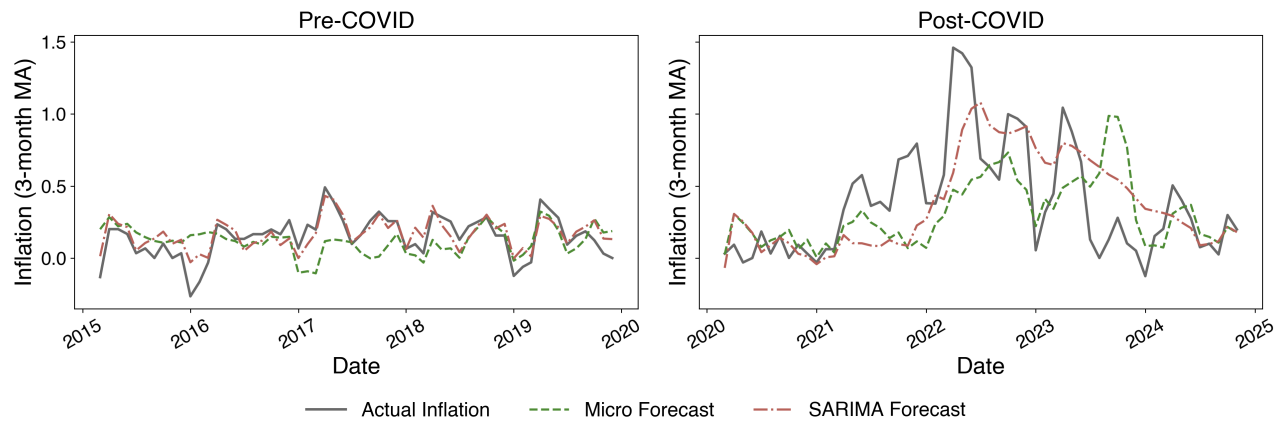
m	$\rho = -.25$	$\rho = 0$	$\rho = .25$
6	0.48 [0.45, 0.51]	0.51 [0.47, 0.54]	0.52 [0.49, 0.55]
7	0.47 [0.44, 0.50]	0.48 [0.45, 0.51]	0.50 [0.47, 0.53]
8	0.46 [0.43, 0.49]	0.46 [0.43, 0.49]	0.46 [0.43, 0.49]
9	0.43 [0.40, 0.46]	0.45 [0.42, 0.48]	0.45 [0.42, 0.48]
10	0.44 [0.41, 0.47]	0.44 [0.41, 0.47]	0.44 [0.41, 0.47]
11	0.39 [0.36, 0.42]	0.40 [0.37, 0.43]	0.40 [0.36, 0.43]
12	0.39 [0.36, 0.42]	0.42 [0.39, 0.45]	0.40 [0.36, 0.43]

Table C.22: Giacomini–Rossi horizon-pooled simulation for (sarima + macro) – three experts, post-COVID window. Entries are rejection probabilities at nominal level $\alpha_0 = 0.05$; brackets give approximate 95% Monte Carlo intervals.

m	$\rho = -.25$	$\rho = 0$	$\rho = .25$
6	0.42 [0.39, 0.45]	0.43 [0.40, 0.46]	0.45 [0.41, 0.48]
7	0.41 [0.38, 0.44]	0.42 [0.39, 0.45]	0.43 [0.40, 0.46]
8	0.41 [0.38, 0.44]	0.40 [0.37, 0.43]	0.41 [0.38, 0.44]
9	0.36 [0.33, 0.39]	0.35 [0.32, 0.38]	0.37 [0.34, 0.40]
10	0.34 [0.31, 0.37]	0.35 [0.32, 0.38]	0.33 [0.30, 0.36]
11	0.32 [0.29, 0.34]	0.33 [0.30, 0.36]	0.33 [0.30, 0.36]
12	0.29 [0.26, 0.32]	0.31 [0.28, 0.34]	0.31 [0.28, 0.33]

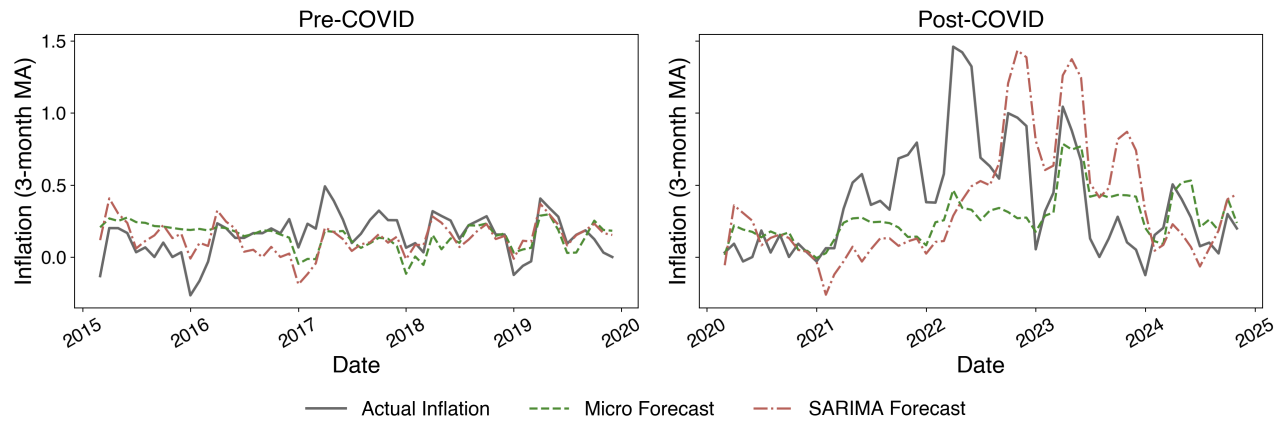
D Appendix Figures

Figure D.1: Micro Forecast vs. Univariate Forecast: $h = 1$



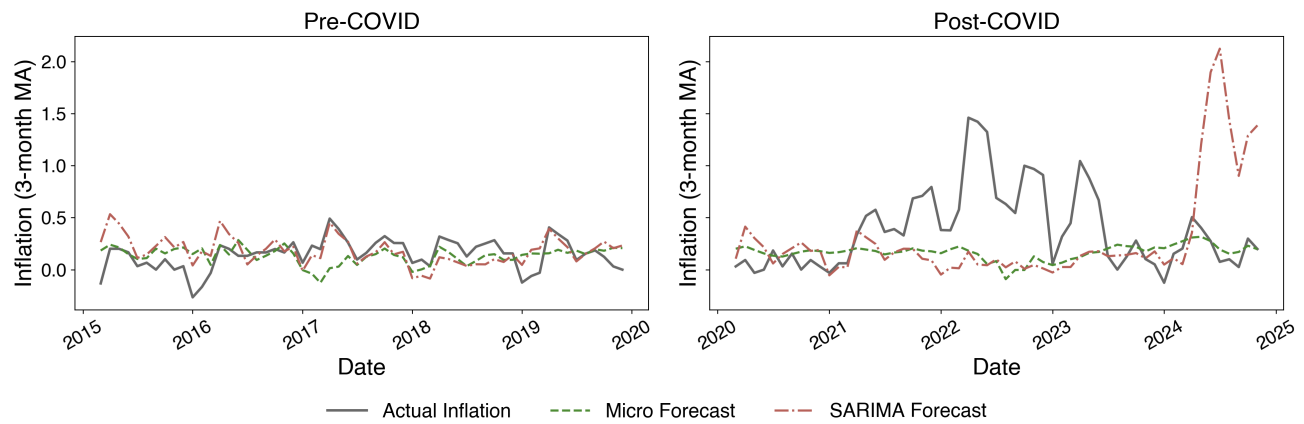
Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the micro 1-month-ahead forecast, and the univariate (SARIMA) 1-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 1$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

Figure D.2: Micro Forecast vs. Univariate Forecast: $h = 6$



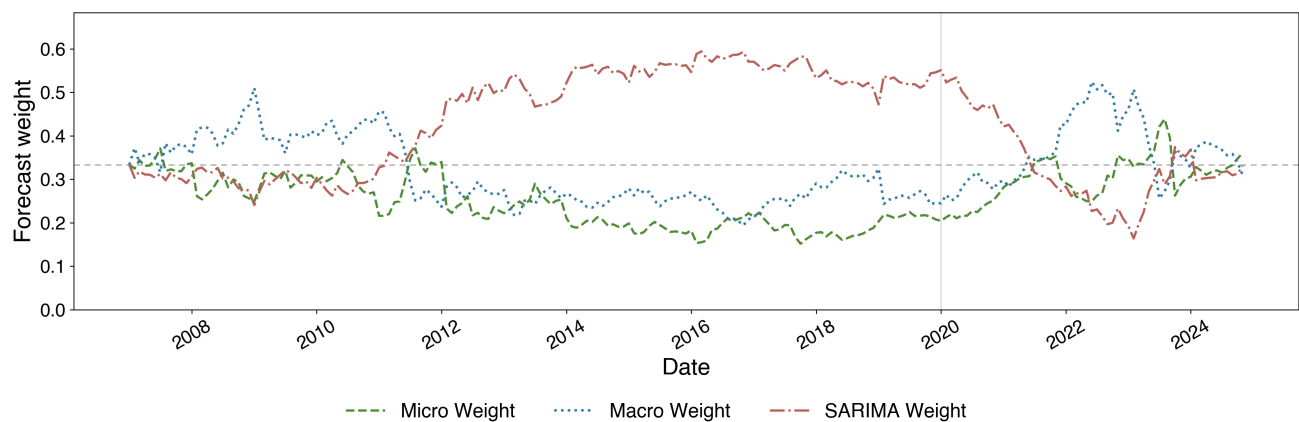
Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the micro 6-month-ahead forecast, and the univariate (SARIMA) 6-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 6$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

Figure D.3: Micro Forecast vs. Univariate Forecast: $h = 24$



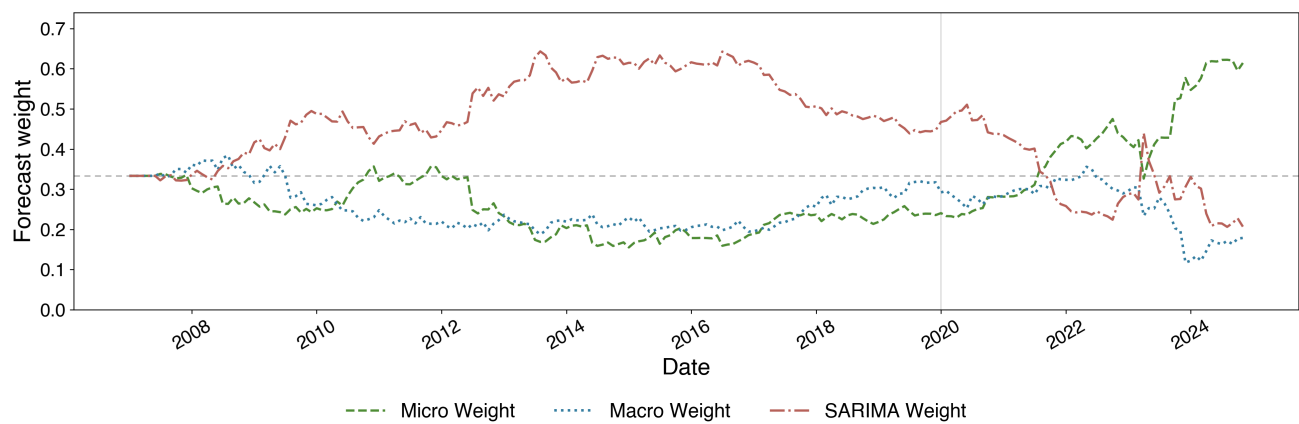
Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the micro 24-month-ahead forecast, and the univariate (SARIMA) 24-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 24$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

Figure D.4: Combination Weights: $h = 1$



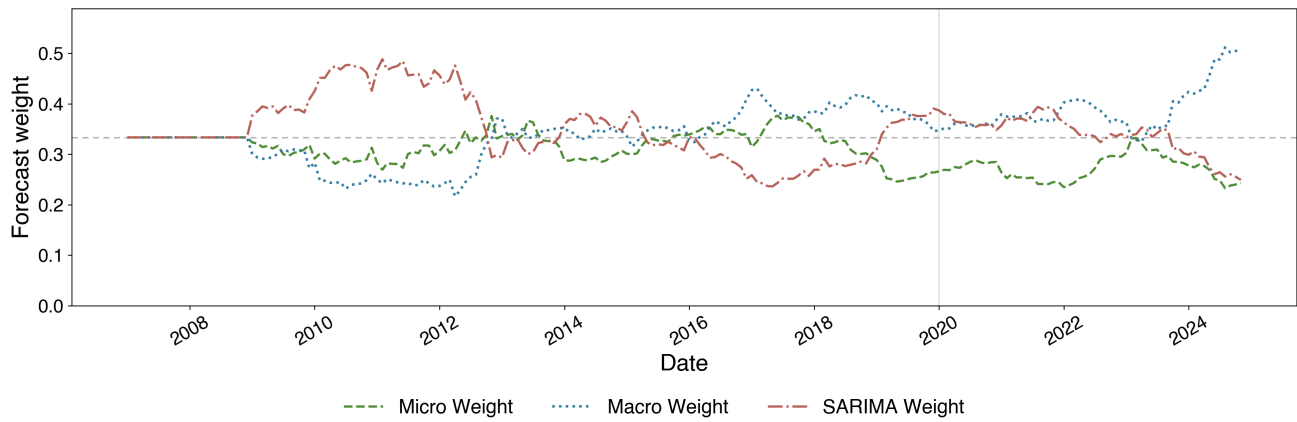
Notes: Combination weights of the Fixed Share baseline ($\eta = 0.5$, $\alpha = 0.02$) on the micro, macro, and univariate experts for the 1-month-ahead forecast, over the full sample. Dates on the x-axis are target months, so date t corresponds to origin $t - 1$. The dashed horizontal line marks the equal weight $1/3$; the vertical line marks 2020.

Figure D.5: Combination Weights: $h = 6$



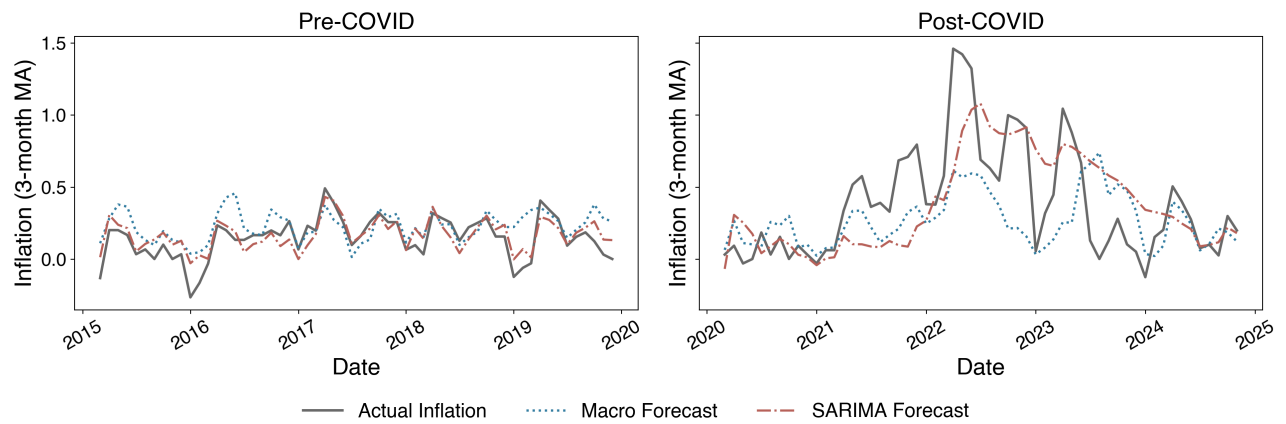
Notes: Combination weights of the Fixed Share baseline ($\eta = 0.5$, $\alpha = 0.02$) on the micro, macro, and univariate experts for the 6-month-ahead forecast, over the full sample. Dates on the x-axis are target months, so date t corresponds to origin $t - 6$. The dashed horizontal line marks the equal weight $1/3$; the vertical line marks 2020.

Figure D.6: Combination Weights: $h = 24$



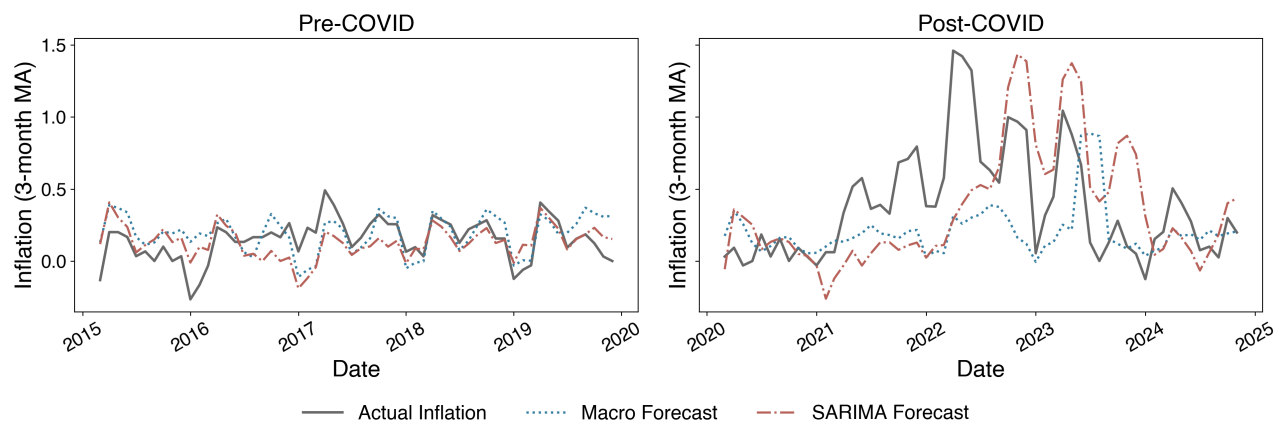
Notes: Combination weights of the Fixed Share baseline ($\eta = 0.5$, $\alpha = 0.02$) on the micro, macro, and univariate experts for the 24-month-ahead forecast, over the full sample. Dates on the x-axis are target months, so date t corresponds to origin $t - 24$. The dashed horizontal line marks the equal weight $1/3$; the vertical line marks 2020.

Figure D.7: Macro Forecast vs. Univariate Forecast: $h = 1$



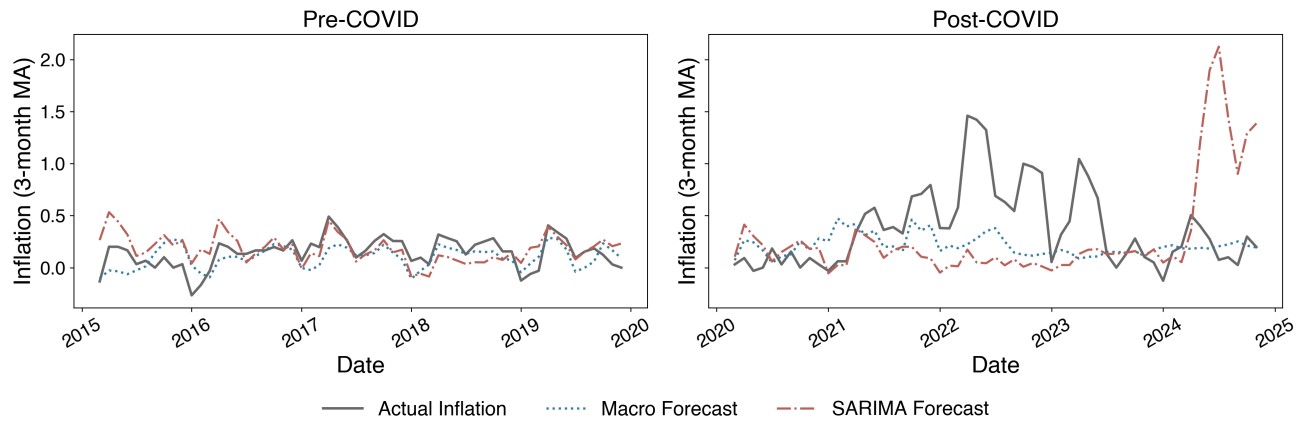
Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the macro 1-month-ahead forecast, and the univariate (SARIMA) 1-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 1$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

Figure D.8: Macro Forecast vs. Univariate Forecast: $h = 6$



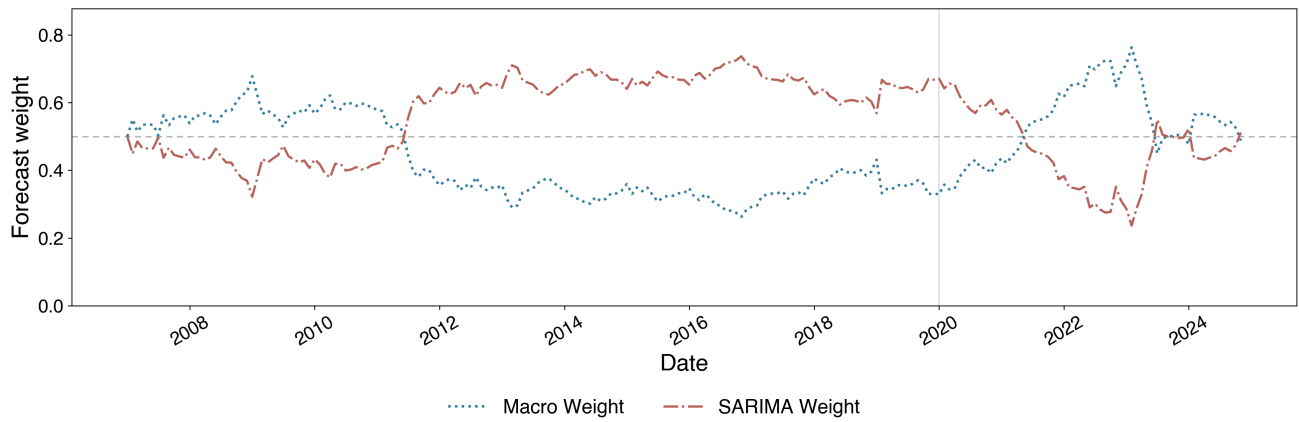
Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the macro 6-month-ahead forecast, and the univariate (SARIMA) 6-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 6$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

Figure D.9: Macro Forecast vs. Univariate Forecast: $h = 24$



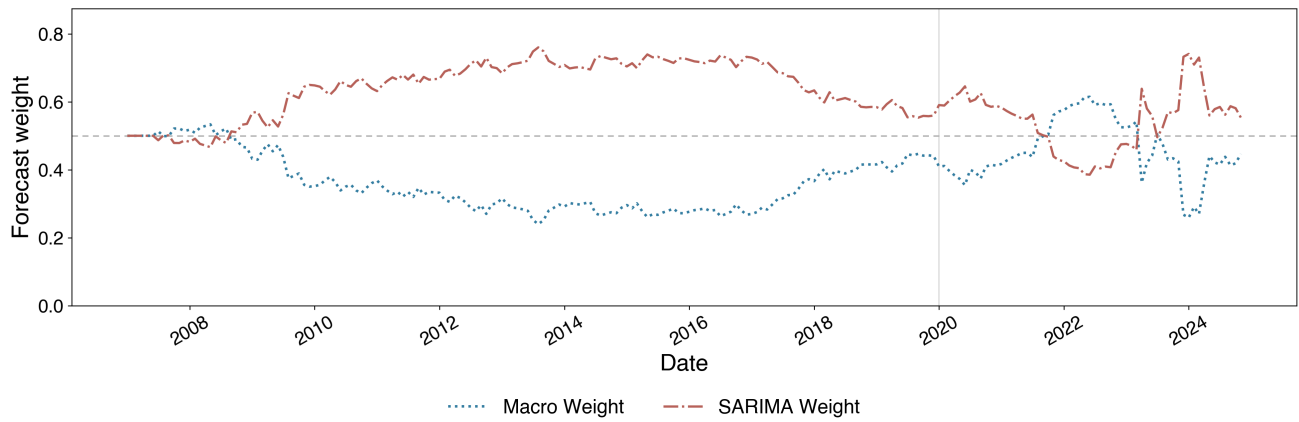
Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the macro 24-month-ahead forecast, and the univariate (SARIMA) 24-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 24$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

Figure D.10: Two-Expert Combination Weights: $h = 1$



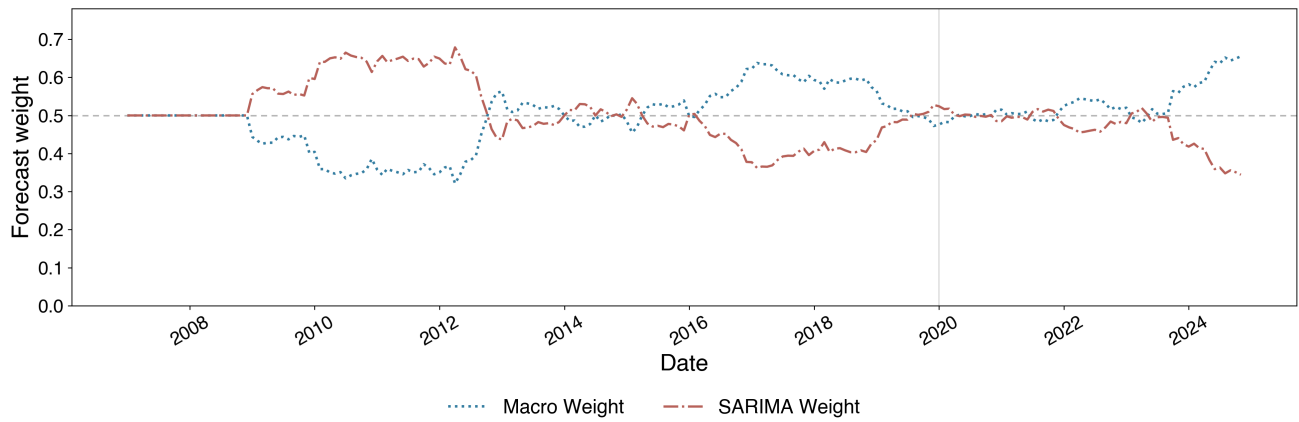
Notes: Combination weights of the two-expert Fixed Share benchmark ($\eta = 0.5$, $\alpha = 0.02$) on the macro and univariate experts for the 1-month-ahead forecast, over the full sample. Dates on the x-axis are target months, so date t corresponds to origin $t - 1$. The dashed horizontal line marks the equal weight $1/2$; the vertical line marks 2020.

Figure D.11: Two-Expert Combination Weights: $h = 6$



Notes: Combination weights of the two-expert Fixed Share benchmark ($\eta = 0.5$, $\alpha = 0.02$) on the macro and univariate experts for the 6-month-ahead forecast, over the full sample. Dates on the x-axis are target months, so date t corresponds to origin $t - 6$. The dashed horizontal line marks the equal weight $1/2$; the vertical line marks 2020.

Figure D.12: Two-Expert Combination Weights: $h = 24$



Notes: Combination weights of the two-expert Fixed Share benchmark ($\eta = 0.5$, $\alpha = 0.02$) on the macro and univariate experts for the 24-month-ahead forecast, over the full sample. Dates on the x-axis are target months, so date t corresponds to origin $t - 24$. The dashed horizontal line marks the equal weight $1/2$; the vertical line marks 2020.

E Notation and Pseudocode

Table E.1 summarizes the main notation used in the forecasting, scan-test, online-learning, and Shapley procedures described in Sections 4, 3, and 6.

Table E.1: Main Notation Used in the Methods and Scan-Test Sections

Symbol	Description
<i>Timing, data, and forecasts</i>	
t	Calendar month index.
τ	Forecast origin: the last month whose information the forecast uses; a forecast formed at origin τ for horizon h targets month $\tau + h$.
h	Forecast horizon in months ahead; $h = 1, \dots, 24$.
H	Maximum forecast horizon; $H = 24$.
π_t	Realized month-on-month inflation in month t .
$\hat{\pi}_{\tau+h \tau}$	Forecast of inflation in target month $\tau + h$ formed from information through origin τ ; equivalently, $\hat{\pi}_{t t-h}$ is the forecast of target t formed at origin $t - h$.
n	Number of evaluated months in the forecast-comparison panel; $n = 215$ at the deployed sample.
<i>Features and cached real-time tuning</i>	
Z_t	Monthly encoding vector observed in month t : the micro distribution statistics for the micro expert, or the macro series for the macro expert.
m_{t+h}	Target-month indicators, $m_{t+h} \in \{0, 1\}^{11}$ (January omitted).
X_t^h	Origin- t feature vector for horizon h : $X_t^h = (Z_{t-11}, \dots, Z_t, m_{t+h})$; every component is observed at origin t .
Γ	XGBoost hyperparameter vector (shrinkage ν and the number of trees; tree depth fixed at 6).
ν	XGBoost learning rate (shrinkage); η is reserved for the combination step.
$\mathcal{G}$	Hyperparameter grid; the baseline is the 7×7 grid $\nu \in \{0.0003, \dots, 0.3\}$, trees $\in \{50, \dots, 1000\}$. (In Section 6, $\mathcal{G}$ instead denotes the set of micro feature groups; see below.)

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Table E.1 continued

Symbol	Description
L	Rolling training window; $L = 120$ months.
W	Trailing validation window: $W = 48$ months for the cached real-time tuning of every expert; $W = 60$ months for the varying-learning-rate rule.
$\hat{\pi}_{t t-h}^{\Gamma}$	Cached real-time forecast of target t from candidate Γ , fitted at origin $t - h$; $e_{t t-h}^{\Gamma}$ is its absolute error, stored once π_t is released.
Γ_{τ}^*	Hyperparameter selected at origin τ by the trailing- W MAE rule; reused (frozen) by every Shapley coalition.
<i>SARIMA benchmark</i>	
Υ	Candidate SARIMA order $(p, d, q)(P, D, Q, 12)$ with $p, d, q, P, D, Q \in \{0, 1\}$.
$\mathcal{T}$	SARIMA candidate set: the 2^6 orders plus the AO-12MA trailing-average candidate.
Θ_c^{Υ}	Maximum-likelihood SARIMA coefficients of order Υ fitted on the rolling window ending at origin c . (The scan bootstrap's covariance matrix is written Θ_n ; see below.)
ε_t	SARIMA innovation term.
<i>Patient Fixed Share combination</i>	
K	Number of experts in the admitted pool; $K = 3$ in the baseline (micro, macro, univariate).
$\hat{\pi}_{t t-h}^k$	Expert k 's forecast of target t formed at origin $t - h$.
$\ell_{t t-h}^k$	Realized absolute loss of expert k for target t : $ \pi_t - \hat{\pi}_{t t-h}^k $.
$v_t^{h,k}$	Internal weight on expert k for horizon h , indexed by the target month and updated only with realized h -step losses (share step first, then loss step).
$w_t^{h,k}$	Forecast weight at origin t : $w_t^{h,k} = (1 - \alpha)^h v_t^{h,k} + (1 - (1 - \alpha)^h)/K$.
$\hat{\pi}_{t+h t}^{\text{FS}}$	Combined (patient Fixed Share) forecast formed at origin t for target $t + h$.
η	Fixed Share learning rate; baseline $\eta = 0.5$.
α	Fixed-share mixing rate; baseline $\alpha = 0.02$, approximately $4/n$.

Continued on next page

Table E.1 continued

Symbol	Description
t_0	Start of the combination sample: the first combined forecasts target January 2007, each formed h months earlier at its forecast origin.
<i>Varying learning rate</i>	
$\mathcal{H}$	Learning-rate grid $\{0.25, 0.30, \dots, 1.25\}$ ($G = 21$ values).
η_τ^*	Learning rate selected at origin τ by the neighbor-only rolling search over the most recent $W = 60$ realized losses.
$\mathcal{I}_\tau$	Candidate index set at origin τ : the incumbent learning rate and its grid neighbors. (The scan test's interval collection is also written $\mathcal{I}$; see below.)
<i>Scan test and Ljung–Box pre-test</i>	
$r_{t,h}$	Absolute-error differential between benchmark method B and competing method M for target month t at horizon h , built from the forecasts $\hat{\pi}_{t t-h}$.
$\bar{r}_h$	Within-horizon sample mean of the loss-differential series; used in the Ljung–Box pre-test.
m	Ljung–Box lag budget; $m = 12$.
Q	Multiplier-bootstrap Ljung–Box statistic, aggregated over horizons.
$D_t^{(h)}$	Loss differential for target month t at horizon h : the benchmark method B absolute error minus the competing method M absolute error.
$I = [s, e]$	Candidate contiguous target-month interval with start month s and end month e .
$\mathcal{I}$	Collection of candidate intervals considered by the scan test.
$L_{\min}$	Minimum admissible interval length in the scan test; $L_{\min} = 6$ for the occasional scan.
$L_{\max}$	Maximum admissible interval length in the scan test; $L_{\max} = 12$ for the occasional scan, $L_{\min} = L_{\max} = n$ for the overall test.
p	Total number of (interval, horizon) pairs searched, $p = H \mathcal{I} $; 34,776 for the deployed occasional scan ($L_{\min} = 6$, $L_{\max} = 12$).

Continued on next page

Table E.1 continued

Symbol	Description
$S_I^{(h)}$	Sum of loss differentials over interval I at horizon h .
$\bar{D}_I^{(h)}$	Mean loss differential over interval I at horizon h .
γ	Partial-studentization exponent; baseline $\gamma = 0.1$.
$Q_I^{(h)}$	Sum of squared deviations of $D_t^{(h)}$ within interval I at horizon h .
$Q_I^{*(h)}$	Bootstrap sum of squared deviations of multiplier-weighted local scores within interval I at horizon h .
$T_I^{(h)}$	Partially-studentized scan statistic for interval I and horizon h .
$T^{\max}$	Maximum partially-studentized scan statistic across all intervals and horizons.
$\xi_t^{(b)}$	Gaussian multiplier in bootstrap draw b for time t .
Θ_n	Covariance matrix of the Gaussian multiplier bootstrap: $\Theta_n = I_n$ in Regime I, kernel-based in Regime D.
$K(\cdot)$	Kernel function of the Regime D covariance matrix.
b_n	Bandwidth of the kernel-based covariance estimator.
B	Number of bootstrap draws; $B = 4,999$.
<i>Grouped Shapley decomposition</i>	
$\mathcal{G}$	Set of micro feature groups (the players); $ \mathcal{G} = G = 5$ in the baseline decomposition.
S	Coalition of feature groups, $S \subseteq \mathcal{G}$.
$v_h(S)$	Coalition value at horizon h : the signed percent MAE reduction of the three-expert forecast built on micro(S) relative to the two-expert benchmark; $v_h(\emptyset) = 0$.
$\phi_{g,h}$	Shapley value of group g at horizon h , in percent of the two-expert benchmark MAE; $\sum_g \phi_{g,h} = v_h(\mathcal{G})$.

Notes: This table collects the notation used in the machine-learning, SARIMA, Fixed Share, scan-test, and Shapley procedures. Forecasts are written in the target-given-origin form: $\hat{\pi}_{a|b}$ is the forecast of inflation in month a formed from information through month b . Loss differentials are written so that positive values favor the competing method M over the benchmark method B .

This appendix also collects the pseudocode for all algorithms used in Sections 4, 3, and 6.

Algorithm 1 PRECOMPUTEROLLINGPREDICTIONS($\mathcal{G}, h, L$)

- 1: **Input:** Hyperparameter grid $\mathcal{G}$; horizon h ; rolling window $L = 120$; features $\{X_t^h\}$ and inflation $\{\pi_t\}$; first usable target $\underline{t}$; origin range $[c_{lo}, c_{hi}]$.
 - 2: **Output:** Cached real-time forecasts and errors $\{\hat{\pi}_{t|t-h}^\Gamma, e_{t|t-h}^\Gamma\}$ for all $\Gamma \in \mathcal{G}$ and reachable targets t .
 - 3: **for** $\Gamma \in \mathcal{G}$, $c = c_{lo}, \dots, c_{hi}$ **do**
 - 4: $s \leftarrow \max\{c - (L - 1), \underline{t}\}$
 - 5: Train ML(Γ) on $\{(X_{j-h}^h, \pi_j)\}_{j=s}^c$ ▷ targets realized by the origin c
 - 6: $\hat{\pi}_{c+h|c}^\Gamma \leftarrow \text{ML}(\Gamma)(X_c^h)$; $e_{c+h|c}^\Gamma \leftarrow |\pi_{c+h} - \hat{\pi}_{c+h|c}^\Gamma|$
 - 7: **end for**
-

Algorithm 2 REALTIMETUNEANDFORECAST(τ, h, W)

- 1: **Input:** Forecast origin τ ; horizon h ; validation window $W = 48$; the cache $\{\hat{\pi}_{t|t-h}^\Gamma, e_{t|t-h}^\Gamma\}$ from Algorithm 1.
 - 2: **Output:** Real-time forecast $\hat{\pi}_{\tau+h|\tau}$ and selected hyperparameter Γ^* .
 - 3: **for** $\Gamma \in \mathcal{G}$ **do**
 - 4: $\text{MAE}(\Gamma) \leftarrow \frac{1}{W} \sum_{t'=\tau-W+1}^{\tau} e_{t'|t'-h}^\Gamma$ ▷ trailing errors, all realized by τ
 - 5: **end for**
 - 6: $\Gamma^* \leftarrow \arg\min_{\Gamma \in \mathcal{G}} \text{MAE}(\Gamma)$; $\hat{\pi}_{\tau+h|\tau} \leftarrow \hat{\pi}_{\tau+h|\tau}^{\Gamma^*}$
-

Algorithm 3 PRECOMPUTEROLLINGSARIMA($\mathcal{T}, h, L$)

- 1: **Input:** Candidate set $\mathcal{T}$; horizon h ; rolling window $L = 120$; inflation $\{\pi_t\}$; first usable month $\underline{t}$; origin range $[c_{lo}, c_{hi}]$.
 - 2: **Output:** Cached real-time forecasts and errors $\{\hat{\pi}_{c+h|c}^Y, e_{c+h|c}^Y\}$ for all $Y \in \mathcal{T}$ and reachable origins c .
 - 3: **for** $Y \in \mathcal{T}$, $c = c_{lo}, \dots, c_{hi}$ **do**
 - 4: $s \leftarrow \max\{c - (L - 1), \underline{t}\}$
 - 5: **if** Y is AO-12MA **then**
 - 6: $\hat{\pi}_{c+h|c}^Y \leftarrow \frac{1}{12} \sum_{j=0}^{11} \pi_{c-j}$ ▷ degenerate candidate, no fit
 - 7: **else**
 - 8: Fit Θ_c^Y for SARIMA(Y) by maximum likelihood on $\{\pi_j\}_{j=s}^c$
 - 9: $\hat{\pi}_{c+h|c}^Y \leftarrow$ iterate SARIMA(Y)(Θ_c^Y) h steps forward from c
 - 10: **end if**
 - 11: $e_{c+h|c}^Y \leftarrow |\pi_{c+h} - \hat{\pi}_{c+h|c}^Y|$ ▷ stored once π_{c+h} is released
 - 12: **end for**
-

Algorithm 4 REALTIMETUNEANDFORECASTSARIMA(τ, h, W)

- 1: **Input:** Forecast origin τ ; horizon h ; validation window $W = 48$; the cache $\{\hat{\pi}_{t|t-h}^Y, e_{t|t-h}^Y\}$ from Algorithm 3.
 - 2: **Output:** Real-time forecast $\hat{\pi}_{\tau+h|\tau}$ and selected candidate Y^* .
 - 3: **for** $Y \in \mathcal{T}$ **do**
 - 4: $\text{MAE}(Y) \leftarrow \frac{1}{W} \sum_{t'=\tau-W+1}^{\tau} e_{t'|t'-h}^Y$ ▷ trailing errors, all realized by τ
 - 5: **end for**
 - 6: $Y^* \leftarrow \operatorname{argmin}_{Y \in \mathcal{T}} \text{MAE}(Y)$; $\hat{\pi}_{\tau+h|\tau} \leftarrow \hat{\pi}_{\tau+h|\tau}^{Y^*}$
-

Algorithm 5 Multiplier-Bootstrap Ljung–Box Test (Aggregated over Horizons)

- 1: **Input:** Residual panel $\{r_{t,h}\}$, $t = 1, \dots, n$, $h = 1, \dots, H$, where $r_{t,h} = |\pi_t - \hat{\pi}_{t|t-h}^B| - |\pi_t - \hat{\pi}_{t|t-h}^M|$ and t indexes target months; maximum lag m ; bootstrap draws B .
 - 2: **Output:** p -value $\hat{p}$.
 - 3: **for** $h = 1, \dots, H$, $k = 1, \dots, m$ **do**
 - 4: $\hat{\rho}_h(k) \leftarrow \frac{\sum_{t=k+1}^n (r_{t,h} - \bar{r}_h)(r_{t-k,h} - \bar{r}_h)}{\sum_{t=1}^n (r_{t,h} - \bar{r}_h)^2}$; $Q_h \leftarrow n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_h(k)^2}{n-k}$
 - 5: **end for**
 - 6: $Q \leftarrow \sum_{h=1}^H Q_h$
 - 7: **for** $b = 1, \dots, B$ **do**
 - 8: Draw $\xi_1^{(b)}, \dots, \xi_n^{(b)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$; form $r_{t,h}^{*(b)} \leftarrow \xi_t^{(b)} r_{t,h}$ for all t, h
 - 9: **for** $h = 1, \dots, H$ **do**
 - 10: Compute $\hat{\rho}_h^{*(b)}(k)$ from $\{r_{t,h}^{*(b)}\}_{t=1}^n$; $Q_h^{*(b)} \leftarrow n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_h^{*(b)}(k)^2}{n-k}$
 - 11: **end for**
 - 12: $Q^{*(b)} \leftarrow \sum_{h=1}^H Q_h^{*(b)}$
 - 13: **end for**
 - 14: $\hat{p} = (1 + \sum_{b=1}^B \mathbf{1}\{Q^{*(b)} > Q\}) / (B + 1)$
-

Algorithm 6 Partially-Studentized Scan Test with Gaussian Multiplier Bootstrap (Regime I: $\Theta_n = I_n$)

- 1: **Input:** Forecast errors $\{e_{t|t-h}^B, e_{t|t-h}^M\}$, indexed by target month t , $t = 1, \dots, n$, $h = 1, \dots, H$; $L_{\min}$; $L_{\max}$; tuning exponent γ ; bootstrap draws B .
 - 2: **Output:** p -value $\hat{p}$.
 - 3: **for** $t = 1, \dots, n$, $h = 1, \dots, H$ **do**
 - 4: $D_t^{(h)} \leftarrow |e_{t|t-h}^B| - |e_{t|t-h}^M|$
 - 5: **end for**
 - 6: $\mathcal{J} \leftarrow \{[s, e] \subset [n] : L_{\min} \leq e - s + 1 \leq L_{\max}\}; \quad T^{\max} \leftarrow -\infty$
 - 7: **for each** $I = [s, e] \in \mathcal{J}$, $h = 1, \dots, H$ **do**
 - 8: $S_I^{(h)} \leftarrow \sum_{t=s}^e D_t^{(h)}$; $\bar{D}_I^{(h)} \leftarrow S_I^{(h)} / |I|$; $Q_I^{(h)} \leftarrow \sum_{t=s}^e (D_t^{(h)} - \bar{D}_I^{(h)})^2$
 - 9: $A_I^{(h)}(\gamma) \leftarrow |I|^{1/2-\gamma} (Q_I^{(h)})^\gamma$
 - 10: $T_I^{(h)} \leftarrow S_I^{(h)} / A_I^{(h)}(\gamma)$ if $A_I^{(h)}(\gamma) > 10^{-12}$, else 0; $T^{\max} \leftarrow \max(T^{\max}, T_I^{(h)})$
 - 11: **end for**
 - 12: **for** $b = 1, \dots, B$ **do**
 - 13: Draw $\xi_1^{(b)}, \dots, \xi_n^{(b)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$; $T^{*,(b),\max} \leftarrow -\infty$
 - 14: **for each** $I = [s, e] \in \mathcal{J}$, $h = 1, \dots, H$ **do**
 - 15: $\bar{D}_I^{(h)} \leftarrow |I|^{-1} \sum_{t=s}^e D_t^{(h)}$; $e_{t,I}^{(h,b)} \leftarrow \xi_t^{(b)} (D_t^{(h)} - \bar{D}_I^{(h)})$ for $t = s, \dots, e$
 - 16: $S_I^{*(h)} \leftarrow \sum_{t=s}^e e_{t,I}^{(h,b)}$; $Q_I^{*(h)} \leftarrow \sum_{t=s}^e (e_{t,I}^{(h,b)})^2 - (S_I^{*(h)})^2 / |I|$
 - 17: $A_I^{*(h)}(\gamma) \leftarrow |I|^{1/2-\gamma} (Q_I^{*(h)})^\gamma$
 - 18: $T_I^{*,(h)} \leftarrow S_I^{*(h)} / A_I^{*(h)}(\gamma)$ if $A_I^{*(h)}(\gamma) > 10^{-12}$, else 0
 - 19: $T^{*,(b),\max} \leftarrow \max(T^{*,(b),\max}, T_I^{*,(h)})$
 - 20: **end for**
 - 21: **end for**
 - 22: $\hat{p} = (1 + \sum_{b=1}^B \mathbf{1}\{T^{*,(b),\max} > T^{\max}\}) / (B + 1)$
-

Algorithm 7 Partially-Studentized Scan Test with Gaussian Multiplier Bootstrap (Regime D: Kernel Θ_n)

- 1: **Input:** Forecast errors $\{e_{t|t-h}^B, e_{t|t-h}^M\}$, indexed by target month t , $t = 1, \dots, n$, $h = 1, \dots, H$; $L_{\min}$; $L_{\max}$; tuning exponent γ ; bootstrap draws B ; kernel K .
 - 2: **Output:** p -value $\hat{p}$.
 - 3: **for** $t = 1, \dots, n$, $h = 1, \dots, H$ **do**
 - 4: $D_t^{(h)} \leftarrow |e_{t|t-h}^B| - |e_{t|t-h}^M|$
 - 5: **end for**
 - 6: **for** $h = 1, \dots, H$ **do**
 - 7: Fit $D_t^{(h)} = c_h + \rho_h D_{t-1}^{(h)} + \varepsilon_{h,t}$ by least squares; obtain $(\hat{\rho}_h, \hat{\sigma}_h^2)$
 - 8: **end for**
 - 9: $\hat{a} \leftarrow \frac{\sum_{h=1}^H 4\hat{\rho}_h^2 \hat{\sigma}_h^4 (1 - \hat{\rho}_h)^{-8}}{\sum_{h=1}^H \hat{\sigma}_h^4 (1 - \hat{\rho}_h)^{-4}}$; $b_n \leftarrow 1.3221 (\hat{a} n)^{1/5}$
 - 10: Compute $\Theta_n \in \mathbb{R}^{n \times n}$ with $\Theta_n(t, u) = K\left(\frac{t-u}{b_n}\right)$ for all t, u
 - 11: $\mathcal{J} \leftarrow \{[s, e] \subset [n] : L_{\min} \leq e - s + 1 \leq L_{\max}\}$; $T^{\max} \leftarrow -\infty$
 - 12: **for each** $I = [s, e] \in \mathcal{J}$, $h = 1, \dots, H$ **do**
 - 13: $S_I^{(h)} \leftarrow \sum_{t=s}^e D_t^{(h)}$; $\bar{D}_I^{(h)} \leftarrow S_I^{(h)} / |I|$; $Q_I^{(h)} \leftarrow \sum_{t=s}^e (D_t^{(h)} - \bar{D}_I^{(h)})^2$
 - 14: $A_I^{(h)}(\gamma) \leftarrow |I|^{1/2-\gamma} (Q_I^{(h)})^\gamma$
 - 15: $T_I^{(h)} \leftarrow S_I^{(h)} / A_I^{(h)}(\gamma)$ if $A_I^{(h)}(\gamma) > 10^{-12}$, else 0; $T^{\max} \leftarrow \max(T^{\max}, T_I^{(h)})$
 - 16: **end for**
 - 17: **for** $b = 1, \dots, B$ **do**
 - 18: Draw $\xi^{(b)} = (\xi_1^{(b)}, \dots, \xi_n^{(b)})^\top \sim \mathcal{N}(0, \Theta_n)$; $T^{*,(b),\max} \leftarrow -\infty$
 - 19: **for each** $I = [s, e] \in \mathcal{J}$, $h = 1, \dots, H$ **do**
 - 20: $\bar{D}_I^{(h)} \leftarrow |I|^{-1} \sum_{t=s}^e D_t^{(h)}$; $e_{t,I}^{(h,b)} \leftarrow \xi_t^{(b)} (D_t^{(h)} - \bar{D}_I^{(h)})$ for $t = s, \dots, e$
 - 21: $S_I^{*(h)} \leftarrow \sum_{t=s}^e e_{t,I}^{(h,b)}$; $Q_I^{*(h)} \leftarrow \sum_{t=s}^e (e_{t,I}^{(h,b)})^2 - (S_I^{*(h)})^2 / |I|$
 - 22: $A_I^{*(h)}(\gamma) \leftarrow |I|^{1/2-\gamma} (Q_I^{*(h)})^\gamma$
 - 23: $T_I^{*,(h)} \leftarrow S_I^{*(h)} / A_I^{*(h)}(\gamma)$ if $A_I^{*(h)}(\gamma) > 10^{-12}$, else 0
 - 24: $T^{*,(b),\max} \leftarrow \max(T^{*,(b),\max}, T_I^{*,(h)})$
 - 25: **end for**
 - 26: **end for**
 - 27: $\hat{p} = (1 + \sum_{b=1}^B \mathbf{1}\{T^{*,(b),\max} > T^{\max}\}) / (B + 1)$
-

Algorithm 8 FIXEDSHAREPATH(η, α, h) (Patient Fixed Share)

- 1: **Input:** $\eta > 0$, $\alpha \in [0, 1]$, horizon h ; realized $\{\pi_t\}_{t=t_0}^n$; expert forecasts $\{\hat{\pi}_{t|t-h}^k\}$ and $\{\hat{\pi}_{t+h|t}^k\}$, $k = 1, \dots, K$.
 - 2: **Output:** Combined forecasts $\{\hat{\pi}_{t+h|t}^{\text{FS}}\}$ and forecast weights $\{w_t^{h,k}\}$.
 - 3: Initialize $v_{t_0}^{h,k} \leftarrow 1/K$ for all k . ▷ internal weights over experts, indexed by target
 - 4: **for** $t = t_0, \dots, n$ **do**
 - 5: **if** $t < t_0 + h$ **then**
 - 6: $v_t^{h,k} \leftarrow 1/K$ for all k ▷ no realized h -step loss yet
 - 7: **else**
 - 8: $\ell_{t|t-h}^k \leftarrow |\pi_t - \hat{\pi}_{t|t-h}^k|$ for all k ▷ realized loss of target t , formed at origin $t - h$
 - 9: $v_t^{h,k} \leftarrow \left[(1 - \alpha) v_{t-1}^{h,k} + \frac{\alpha}{K} \right] \exp(-\eta \ell_{t|t-h}^k)$, then normalize over k ▷ share first, then loss
 - 10: **end if**
 - 11: $w_t^{h,k} \leftarrow (1 - \alpha)^h v_t^{h,k} + \frac{1 - (1 - \alpha)^h}{K}$ for all k ▷ propagate h loss-free share steps
 - 12: $\hat{\pi}_{t+h|t}^{\text{FS}} \leftarrow \sum_{k=1}^K w_t^{h,k} \hat{\pi}_{t+h|t}^k$ ▷ forecast at origin t for target $t + h$
 - 13: **end for**
-

Algorithm 9 Patient Fixed Share with Rolling-Window Neighbor Selection of η

- 1: **Input:** $\alpha \in [0, 1]$, horizon h ; realized $\{\pi_t\}$; expert forecasts $\{\hat{\pi}_{t|t-h}^k\}$, $k = 1, \dots, K$; grid $\mathcal{H} = \{\eta^{(1)} < \dots < \eta^{(G)}\}$; window length W ; initial value $\eta_0 = \eta^{(g_0)}$.
 - 2: **Output:** Combined forecasts $\{\hat{\pi}_{\tau+h|\tau}^{\text{FS}}\}$.
 - 3: **for** each $\eta^{(g)} \in \mathcal{H}$ **do**
 - 4: Run $\text{FIXEDSHAREPATH}(\eta^{(g)}, \alpha, h)$ (Algorithm 8) to obtain the path $\{\hat{\pi}_{\tau+h|\tau}^{\text{FS}, (\eta^{(g)})}\}$
 - 5: **end for**
 - 6: **for** each forecast origin τ , in order **do**
 - 7: **if** fewer than W realized h -step forecast losses exist at τ **then**
 - 8: $g_\tau \leftarrow g_0$
 - 9: **else**
 - 10: $\mathcal{J}_\tau \leftarrow \{\max(1, g_{\tau-1} - 1), g_{\tau-1}, \min(G, g_{\tau-1} + 1)\}$ ▷ incumbent and its grid neighbors
 - 11: $g_\tau \leftarrow \operatorname{argmin}_{g \in \mathcal{J}_\tau} \frac{1}{W} \sum_{s=\tau-W+1}^{\tau} |\pi_s - \hat{\pi}_{s|s-h}^{\text{FS}, (\eta^{(g)})}|$ ▷ ties keep the incumbent
 - 12: **end if**
 - 13: $\eta_\tau^* \leftarrow \eta^{(g_\tau)}$; set $\hat{\pi}_{\tau+h|\tau}^{\text{FS}} \leftarrow \hat{\pi}_{\tau+h|\tau}^{\text{FS}, (\eta_\tau^*)}$
 - 14: **end for**
-

Algorithm 10 FROZENMICROCOALITION(S, h)

- 1: **Input:** Coalition $S \subseteq \mathcal{G}$ of micro feature groups; horizon h ; rolling window $L = 120$; grouped features $\{X_t^h[g]\}$; frozen hyperparameter selections $\{\Gamma_\tau^*\}$ from Algorithm 2; inflation $\{\pi_t\}$; first usable month $\underline{t}$; origin range $[c_{lo}, c_{hi}]$.
 - 2: **Output:** Micro coalition forecasts $\{\hat{\pi}_{t+h|t}^S\}$ (none when $S = \emptyset$).
 - 3: **if** $S = \emptyset$ **then**
 - 4: **return** no forecasts ▷ no micro expert; the combination is macro + univariate only
 - 5: **end if**
 - 6: **for** $c = c_{lo}, \dots, c_{hi}$ **do**
 - 7: $s \leftarrow \max\{c - (L - 1), \underline{t}\}$
 - 8: Train ML(Γ_c^*) on $\{(X_{j-h}^h[S], \pi_j)\}_{j=s}^c$ ▷ frozen Γ_c^* , feature subset S , no re-tuning
 - 9: $\hat{\pi}_{c+h|c}^S \leftarrow \text{ML}(\Gamma_c^*)(X_c^h[S])$ ▷ known at origin c
 - 10: **end for**
-

Algorithm 11 GROUPSHAPLEY($\mathcal{G}, h$) over the micro feature groups

- 1: **Input:** Groups $\mathcal{G}$ with $|\mathcal{G}| = G$; horizon h ; the macro and univariate expert forecasts; rates (η, α) ; evaluation window $\mathcal{E}$ (pre- or post-2020).
 - 2: **Output:** Per-horizon Shapley values $\{\phi_{g,h}\}_{g \in \mathcal{G}}$.
 - 3: $\text{MAE}_{\text{baseline},h} \leftarrow \frac{1}{|\mathcal{E}|} \sum_{t \in \mathcal{E}} |\pi_t - \hat{\pi}_{t|t-h}^{\text{FS},\emptyset}|$, where $\hat{\pi}^{\text{FS},\emptyset}$ runs Algorithm 8 on {macro, univariate} ▷ 2-expert benchmark
 - 4: **for** each nonempty $S \subseteq \mathcal{G}$ **do** ▷ $2^G - 1$ coalitions; $S = \mathcal{G}$ is the full 3-expert combination
 - 5: $\{\hat{\pi}_{\cdot|t}^S\} \leftarrow \text{FROZENMICROCOALITION}(S, h)$
 - 6: $\{\hat{\pi}_{t+h|t}^{\text{FS},S}\} \leftarrow \text{Algorithm 8 on \{micro}(S), \text{macro, univariate}\}$
 - 7: $\text{MAE}_{S,h} \leftarrow \frac{1}{|\mathcal{E}|} \sum_{t \in \mathcal{E}} |\pi_t - \hat{\pi}_{t|t-h}^{\text{FS},S}|$
 - 8: $v_h(S) \leftarrow 100 \times \frac{\text{MAE}_{\text{baseline},h} - \text{MAE}_{S,h}}{\text{MAE}_{\text{baseline},h}}$ ▷ signed % MAE reduction vs. the 2-expert benchmark
 - 9: **end for**
 - 10: $v_h(\emptyset) \leftarrow 0$
 - 11: **for** each group $g \in \mathcal{G}$ **do**
 - 12: $\phi_{g,h} \leftarrow \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|!(G - |S| - 1)!}{G!} [v_h(S \cup \{g\}) - v_h(S)]$
 - 13: **end for**
 - 14: $\sum_g \phi_{g,h} = v_h(\mathcal{G})$ ▷ adding-up: the full 3- vs. 2-expert % gain
-

F Robustness: Alternative Encodings, Aggregations, and Data Filtering

The baseline micro encoding used in the main text retains 11 features per COICOP1 category (fraction of zero price changes, nine deciles of the non-zero price-change distribution, and its mean), aggregates within each category using CPI weights, and uses the full sales-included price quotes, for a total of $11 \times 11 = 121$ statistics per month. This appendix documents four robustness checks against this baseline. Appendix F.1 replaces the baseline encoding with the full augmented encoding (29 features per category, $11 \times 29 = 319$ statistics per month) while keeping CPI-weighted aggregation and sales-included quotes. Appendix F.2 keeps the 11-feature encoding but replaces CPI-weighted aggregation with equal-weighted (unweighted) aggregation across products within each COICOP1 category. Appendix F.3 keeps the 11-feature encoding and CPI-weighted aggregation but excludes ONS-sale-flagged price quotes from the underlying micro panel. Appendix F.4 likewise keeps the encoding and aggregation but replaces temporary sale prices with the regular price inferred by the Nakamura and Steinsson (2008) filter (parameterisation A) before computing the per-category statistics. Each appendix reports the consequences of the alternative choice at the standalone micro stage and inside the three-expert online-learning combination.

F.1 Augmented Encoding vs Baseline

The full augmented encoding, as described in Subsection 4.1, adds the variance (with Bessel’s correction), the standardised third and fifth central moments and the excess kurtosis of the non-zero price-change distribution, the nine deciles of the time-since-last-change distribution, and the five moments of the same distribution, raising the per-category feature count from 11 to 29 and the per-month feature count from 121 to 319. This is the same set of micro variables as the six-group Shapley decomposition (Appendix G.4), which spans exactly this augmented feature set; the decomposition’s coalition refits reuse the baseline expert’s frozen tuning, whereas the augmented expert here is tuned on its own cached real-time schedule.

F.1.1 Standalone Micro Forecast

Table F.1 reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast under both encodings, for the pre-2020 (2015–2019) and post-2020 (2020–2024) windows and all horizons $h = 1, \dots, 24$. Averaging within four horizon groups ($h = 1$ to 6, 7 to 12, 13 to 18, and 19 to 24), the augmented encoding’s percentage MAE improvements relative to the univariate forecast are -35.1% , -25.6% , -20.6% , and -27.1% in the pre-2020 window and -0.4% , 5.4% , -3.4% ,

and 22.6% in the post-2020 window. The corresponding gaps relative to the baseline encoding (augmented minus baseline) are -5.7 , -6.8 , -3.3 , and $+5.5$ percentage points pre-2020 (averaging a 2.5 percentage-point cost) and -9.0 , -5.4 , -11.9 , and -2.7 percentage points post-2020 (averaging a 7.3 percentage-point cost). The augmented encoding therefore underperforms the baseline at the standalone stage in both windows, with the cost now concentrated in the post-COVID period, where the added higher-moment and time-since blocks are noisiest, at short and medium-long horizons. This is consistent with overfitting risk rising in the number of predictors when the univariate forecast itself remains competitive, as discussed in Subsection 4.1.

Table F.1: Micro Forecast Percentage MAE Improvement Relative to the Univariate Forecast: Baseline versus Augmented Encoding

h	Pre-2020		Post-2020	
	Baseline	Augmented	Baseline	Augmented
1	-35.67	-49.21	8.49	14.85
2	-28.92	-51.74	4.46	-3.26
3	-29.10	-37.63	2.14	1.11
4	-30.97	-28.42	9.67	-7.85
5	-32.52	-17.42	6.73	-8.23
6	-19.51	-26.05	20.31	0.70
7	-17.33	-19.33	13.00	-0.44
8	-17.05	-25.52	8.63	8.03
9	-27.49	-39.07	14.63	8.33
10	-18.19	-21.73	10.43	8.84
11	-19.84	-24.42	16.04	10.14
12	-12.81	-23.51	1.79	-2.67
13	-10.32	-7.69	12.95	-0.49
14	-0.13	-13.61	15.28	-3.30
15	-22.62	-28.94	14.27	0.55
16	-21.06	-23.27	11.03	-10.65
17	-31.47	-26.04	-3.33	-3.28
18	-18.48	-23.92	0.82	-3.15
19	-28.04	-17.37	9.07	4.78
20	-26.65	-35.14	9.34	10.56
21	-32.13	-32.66	35.42	31.01
22	-44.26	-33.76	32.27	31.43
23	-36.87	-28.26	39.33	35.81
24	-27.88	-15.45	26.46	21.85

Notes: Each cell reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast at horizon h . The baseline uses the 11-feature encoding; the augmented robustness check uses the 29-feature encoding (adding the higher moments of the price-change distribution and the time-since-last-change block), the same feature universe as the six-group Shapley decomposition. Positive values indicate lower MAE than the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

F.1.2 Three-Expert Online Learning

Table E.2 reports the percentage MAE improvement of the three-expert patient Fixed Share combination (fixed $\alpha = 0.02$, $\eta = 0.5$, the main-text baseline) relative to the univariate forecast, with the micro model trained on the baseline and augmented encodings respectively, holding the macro and univariate forecasts fixed. Averaging within the same four horizon groups, the augmented-encoding three-expert combination's percentage MAE improvements are 0.8%, -2.5%, -2.8%, and -1.7% in the pre-2020 window and 8.0%, 12.1%, 8.1%, and 23.3% in the post-2020 window. The corresponding gaps relative to the baseline-encoding three-expert combination are -0.4, -1.1, -1.1, and +1.3 percentage points pre-2020 (averaging a 0.3 percentage-point cost) and -4.0, -1.6, -3.8, and -1.9 percentage points post-2020 (averaging a 2.8 percentage-point cost). The standalone gap therefore shrinks from 7.3 to 2.8 percentage points once the augmented micro forecast is combined with the macro and univariate forecasts through the Fixed Share combination. The combination is robust to encoding choice because the algorithm reallocates weight away from the data-rich expert when its realized performance deteriorates and keeps a meaningful share on the univariate benchmark in the pre-COVID window.

F.2 Unweighted Aggregation vs Baseline

The unweighted aggregation keeps the 11-feature baseline encoding but constructs each per-category statistic as an equal-weighted average across products rather than weighting by CPI shares. This isolates the contribution of the CPI-share weighting choice in the baseline.

F.2.1 Standalone Micro Forecast

Table E.3 reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast under CPI-weighted (baseline) and unweighted aggregation, for all horizons $h = 1, \dots, 24$. Averaging within four horizon groups ($h = 1$ to 6, 7 to 12, 13 to 18, and 19 to 24), the unweighted aggregation's percentage MAE improvements relative to the univariate forecast are -43.1%, -22.4%, -15.2%, and -23.9% in the pre-2020 window and -1.9%, 7.2%, 8.4%, and 26.4% in the post-2020 window. The corresponding gaps relative to the baseline (unweighted minus baseline) are -13.6, -3.6, +2.1, and +8.7 percentage points pre-2020 (averaging a 1.6 percentage-point cost) and -10.6, -3.5, -0.1, and +1.1 percentage points post-2020 (averaging a 3.3 percentage-point cost). Unweighted aggregation therefore underperforms CPI-weighted aggregation on average in both windows, with the cost concentrated at short horizons, where CPI shares concentrate on the most volatile categories, and a partial offset at the longest horizons.

Table E2: Three-Expert Fixed Share: Percent MAE Improvement Relative to the Univariate Forecast under Baseline versus Augmented Encoding of Micro Features ($\alpha = 0.02, \eta = 0.5$)

h	Pre-2020		Post-2020	
	Baseline	Augmented	Baseline	Augmented
1	-1.58	-0.84	12.03	14.19
2	1.83	-0.29	7.64	5.52
3	4.29	2.68	7.40	5.86
4	0.66	0.85	13.17	5.78
5	-0.14	2.40	13.72	7.63
6	2.37	0.26	17.89	8.76
7	-0.08	-0.38	15.63	11.38
8	-1.64	-3.76	14.73	14.41
9	-3.29	-5.03	18.23	17.20
10	-0.93	-3.97	15.22	15.78
11	-1.85	-0.14	15.89	13.51
12	-0.32	-1.46	2.81	0.43
13	2.33	2.56	12.26	8.38
14	4.33	0.03	15.49	6.99
15	-6.69	-5.66	13.89	10.19
16	-6.57	-7.24	10.52	4.60
17	-5.06	-5.56	8.27	7.99
18	1.55	-1.08	10.88	10.52
19	1.15	0.16	13.64	11.18
20	-5.52	-8.21	15.88	13.97
21	-4.20	-6.14	30.03	26.94
22	-8.20	-4.51	30.22	29.06
23	-4.08	0.96	35.30	34.03
24	3.30	7.75	26.53	24.82

Notes: Each cell reports the percentage MAE improvement of the three-expert patient fixed-share combination (fixed $\alpha = 0.02, \eta = 0.5$) relative to the univariate forecast at horizon h , with the micro expert trained on the 11-feature baseline versus the 29-feature augmented encoding, holding the macro and univariate forecasts fixed. Positive values indicate lower MAE than the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

Table E3: Micro Forecast Percentage MAE Improvement Relative to the Univariate Forecast: Baseline versus Equal-Weighted Aggregation

h	Pre-2020		Post-2020	
	Baseline	Unweighted	Baseline	Unweighted
1	-35.67	-62.02	8.49	-1.74
2	-28.92	-58.46	4.46	-2.48
3	-29.10	-39.90	2.14	-0.98
4	-30.97	-34.31	9.67	-8.29
5	-32.52	-28.48	6.73	-1.42
6	-19.51	-35.14	20.31	3.39
7	-17.33	-27.21	13.00	10.15
8	-17.05	-20.85	8.63	7.05
9	-27.49	-27.80	14.63	14.62
10	-18.19	-15.65	10.43	4.13
11	-19.84	-26.44	16.04	10.33
12	-12.81	-16.23	1.79	-2.91
13	-10.32	-6.35	12.95	4.71
14	-0.13	-1.56	15.28	11.37
15	-22.62	-12.72	14.27	8.40
16	-21.06	-24.60	11.03	9.78
17	-31.47	-27.43	-3.33	8.97
18	-18.48	-18.75	0.82	7.37
19	-28.04	-14.36	9.07	12.16
20	-26.65	-22.72	9.34	11.21
21	-32.13	-38.05	35.42	35.12
22	-44.26	-25.26	32.27	31.55
23	-36.87	-24.48	39.33	38.22
24	-27.88	-18.61	26.46	30.44

Notes: Each cell reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast at horizon h . The baseline uses CPI expenditure shares when aggregating within categories; the equal-weighted robustness check gives each product equal weight within the category. Positive values indicate lower MAE for the micro forecast than for the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

F.2.2 Three-Expert Online Learning

Table F4 reports the percentage MAE improvement of the three-expert patient Fixed Share combination (fixed $\alpha = 0.02$, $\eta = 0.5$, the main-text baseline) relative to the univariate forecast, with the micro model trained on CPI-weighted (baseline) and unweighted aggregations respectively, holding the macro and univariate forecasts fixed. Averaging within the same four horizon groups, the unweighted three-expert combination's percentage MAE improvements are -0.7% , -1.6% , -1.2% , and -1.4% in the pre-2020 window and 9.4% , 12.9% , 11.3% , and 25.5% in the post-2020 window. The corresponding gaps relative to the baseline three-expert combination are -2.0 , -0.3 , $+0.5$, and $+1.6$ percentage points pre-2020 (averaging to zero) and -2.6 , -0.8 , -0.5 , and $+0.2$ percentage points post-2020 (averaging a 0.9 percentage-point cost). The standalone gap shrinks to below one percentage point on average once the equal-share variant is combined with the macro and univariate forecasts through the Fixed Share combination, which is therefore robust to the aggregation-weighting choice.

F.3 Sales-Excluded Micro vs Baseline

The sales-excluded variant keeps the 11-feature encoding and CPI-weighted aggregation but drops ONS-sale-flagged price quotes from the underlying micro panel before computing the per-category statistics. This isolates the contribution of temporary-sale price changes to the baseline forecast.

F.3.1 Standalone Micro Forecast

Table F5 reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast under sales-included (baseline) and sales-excluded micro input, for all horizons $h = 1, \dots, 24$. Averaging within four horizon groups ($h = 1$ to 6, 7 to 12, 13 to 18, and 19 to 24), the sales-excluded variant's percentage MAE improvements relative to the univariate forecast are -43.6% , -18.2% , -16.8% , and -10.4% in the pre-2020 window and -10.3% , -1.7% , 7.3% , and 22.5% in the post-2020 window. The corresponding gaps relative to the baseline (sales-excluded minus baseline) are -14.1 , $+0.6$, $+0.6$, and $+22.2$ percentage points pre-2020 (averaging a 2.3 percentage-point gain) and -19.0 , -12.5 , -1.2 , and -2.8 percentage points post-2020 (averaging an 8.9 percentage-point cost). Dropping sale-flagged quotes therefore helps the standalone forecast on average in the stable pre-COVID period, driven by the longest horizons, but hurts it across all horizon groups in the post-COVID period, most severely at short horizons. Sale-driven price changes add predictive signal during the inflation surge, while in stable times they primarily inject noise that the baseline encoding has to absorb.

Table F4: Three-Expert Fixed Share: Percent MAE Improvement Relative to the Univariate Forecast under Baseline versus Equal-Weighted Aggregation of Micro Features ($\alpha = 0.02$, $\eta = 0.5$)

h	Pre-2020		Post-2020	
	Baseline	Unweighted	Baseline	Unweighted
1	-1.58	-1.11	12.03	9.54
2	1.83	-4.37	7.64	6.25
3	4.29	2.81	7.40	7.63
4	0.66	0.10	13.17	7.57
5	-0.14	-0.49	13.72	10.32
6	2.37	-1.24	17.89	14.88
7	-0.08	-2.25	15.63	13.24
8	-1.64	-1.83	14.73	14.76
9	-3.29	-2.21	18.23	19.77
10	-0.93	-2.51	15.22	16.38
11	-1.85	-1.59	15.89	13.60
12	-0.32	0.56	2.81	-0.12
13	2.33	2.05	12.26	8.40
14	4.33	5.75	15.49	11.44
15	-6.69	-3.83	13.89	12.33
16	-6.57	-4.60	10.52	10.44
17	-5.06	-5.20	8.27	11.79
18	1.55	-1.07	10.88	13.66
19	1.15	0.57	13.64	14.85
20	-5.52	-6.73	15.88	14.65
21	-4.20	-6.06	30.03	30.53
22	-8.20	-3.33	30.22	29.80
23	-4.08	1.13	35.30	34.91
24	3.30	6.25	26.53	27.97

Notes: Each cell reports the percentage MAE improvement of the three-expert patient fixed-share combination (fixed $\alpha = 0.02$, $\eta = 0.5$) relative to the univariate forecast at horizon h , with the micro expert trained on the baseline (CPI-expenditure-weighted) versus equal-weighted aggregation, holding the macro and univariate forecasts fixed. Positive values indicate lower MAE than the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

Table F.5: Micro Forecast Percentage MAE Improvement Relative to the Univariate Forecast: Baseline versus Sales-Excluded Input

h	Pre-2020		Post-2020	
	Baseline	Sales-Excluded	Baseline	Sales-Excluded
1	-35.67	-35.11	8.49	-0.90
2	-28.92	-54.83	4.46	-4.37
3	-29.10	-41.51	2.14	-11.52
4	-30.97	-46.75	9.67	-15.50
5	-32.52	-41.77	6.73	-17.14
6	-19.51	-41.49	20.31	-12.62
7	-17.33	-27.53	13.00	6.65
8	-17.05	-23.47	8.63	3.28
9	-27.49	-34.67	14.63	9.03
10	-18.19	-11.75	10.43	-10.30
11	-19.84	-13.76	16.04	-5.61
12	-12.81	2.21	1.79	-13.47
13	-10.32	-14.37	12.95	4.25
14	-0.13	-6.12	15.28	10.43
15	-22.62	-26.52	14.27	7.94
16	-21.06	-26.98	11.03	7.79
17	-31.47	-22.28	-3.33	3.63
18	-18.48	-4.39	0.82	9.92
19	-28.04	-6.07	9.07	10.54
20	-26.65	-17.87	9.34	7.45
21	-32.13	-10.19	35.42	31.75
22	-44.26	-3.72	32.27	26.02
23	-36.87	-10.60	39.33	35.48
24	-27.88	-13.89	26.46	23.91

Notes: Each cell reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast at horizon h . The baseline uses sales-included price quotes; the sales-excluded robustness check drops ONS-sale-flagged quotes before computing the micro features. Positive values indicate lower MAE for the micro forecast than for the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

F.3.2 Three-Expert Online Learning

Table F.6 reports the percentage MAE improvement of the three-expert patient Fixed Share combination (fixed $\alpha = 0.02$, $\eta = 0.5$, the main-text baseline) relative to the univariate forecast, with the micro model trained on sales-included (baseline) and sales-excluded micro input respectively, holding the macro and univariate forecasts fixed. Averaging within the same four horizon groups, the sales-excluded three-expert combination's percentage MAE improvements are -2.5% , -0.9% , -3.5% , and $+1.9\%$ in the pre-2020 window and 7.8% , 10.4% , 10.5% , and 23.7% in the post-2020 window. The corresponding gaps relative to the baseline three-expert combination are -3.7 , $+0.5$, -1.8 , and $+4.8$ percentage points pre-2020 (averaging a 0.1 percentage-point cost) and -4.2 , -3.4 , -1.4 , and -1.5 percentage points post-2020 (averaging a 2.6 percentage-point cost). The post-COVID standalone gap therefore shrinks from 8.9 to 2.6 percentage points once the sales-excluded micro forecast is combined with the macro and univariate forecasts through the Fixed Share combination. As with the other robustness checks, the combination absorbs much of the standalone difference: when the sales-excluded micro expert weakens, the algorithm reallocates weight toward the macro and univariate forecasts.

F.4 Sale-Filtered Micro (Nakamura–Steinsson Filter A) vs Baseline

The sale-filtered variant keeps the 11-feature encoding and CPI-weighted aggregation but, instead of dropping sale-flagged quotes, replaces temporary sale prices with the regular price inferred by the Nakamura and Steinsson (2008) filter under its baseline parameterisation (Filter A of Table A.2; Appendix A.4 describes the algorithm). Whereas the sales-excluded variant of Appendix E.3 removes the affected quotes, the filter keeps them at an imputed regular price, so the comparison isolates the consequences of imputing rather than deleting sale-driven price changes. Because the filter's rebound and modal-price rules condition on prices up to LL months after t , the filtered panel is a regular-price measurement exercise rather than a strictly real-time forecasting input.

F.4.1 Standalone Micro Forecast

Table F.7 reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast under sales-included (baseline) and sale-filtered micro input, for all horizons $h = 1, \dots, 24$. Averaging within four horizon groups ($h = 1$ to 6, 7 to 12, 13 to 18, and 19 to 24), the sale-filtered variant's percentage MAE improvements relative to the univariate forecast are -36.2% , -42.1% , -26.0% , and -29.4% in the pre-2020 window and -1.8% , -1.4% , 6.7% , and 24.3% in the post-2020 window. The corresponding gaps relative to the baseline (sale-filtered minus baseline) are -6.8 ,

Table F6: Three-Expert Fixed Share: Percent MAE Improvement Relative to the Univariate Forecast under Baseline versus Sales-Excluded Micro Input ($\alpha = 0.02$, $\eta = 0.5$)

h	Pre-2020		Post-2020	
	Baseline	Sales-Excluded	Baseline	Sales-Excluded
1	-1.58	-0.54	12.03	8.76
2	1.83	-4.38	7.64	5.02
3	4.29	-1.16	7.40	4.76
4	0.66	-4.31	13.17	7.88
5	-0.14	-2.17	13.72	9.74
6	2.37	-2.38	17.89	10.52
7	-0.08	-1.65	15.63	12.34
8	-1.64	-3.20	14.73	13.61
9	-3.29	-3.42	18.23	16.54
10	-0.93	0.29	15.22	11.65
11	-1.85	-0.23	15.89	8.72
12	-0.32	2.96	2.81	-0.73
13	2.33	0.14	12.26	9.60
14	4.33	-0.69	15.49	9.87
15	-6.69	-9.64	13.89	12.23
16	-6.57	-7.70	10.52	7.63
17	-5.06	-4.40	8.27	10.56
18	1.55	1.30	10.88	13.24
19	1.15	0.40	13.64	14.61
20	-5.52	-0.49	15.88	14.12
21	-4.20	1.38	30.03	28.54
22	-8.20	2.94	30.22	27.10
23	-4.08	2.52	35.30	33.13
24	3.30	4.73	26.53	24.97

Notes: Each cell reports the percentage MAE improvement of the three-expert patient fixed-share combination (fixed $\alpha = 0.02$, $\eta = 0.5$) relative to the univariate forecast at horizon h , with the micro expert trained on the sales-included (baseline) versus sales-excluded micro input, holding the macro and univariate forecasts fixed. Positive values indicate lower MAE than the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

−23.3, −8.6, and +3.2 percentage points pre-2020 (averaging an 8.9 percentage-point cost) and −10.4, −12.1, −1.8, and −1.0 percentage points post-2020 (averaging a 6.3 percentage-point cost). Unlike the sales-excluded variant, which helps the standalone forecast in the stable pre-COVID period, the filter hurts it in both windows: the imputed regular prices smooth away sale-driven variation that carries signal after 2020, and the imputation itself injects noise even in stable times.

Table F.7: Micro Forecast Percentage MAE Improvement Relative to the Univariate Forecast: Baseline versus Nakamura–Steinsson Filter-A Sales Treatment

h	Pre-2020		Post-2020	
	Baseline	Sale-Filtered	Baseline	Sale-Filtered
1	-35.67	-34.92	8.49	6.77
2	-28.92	-39.51	4.46	-7.33
3	-29.10	-26.05	2.14	3.57
4	-30.97	-38.21	9.67	-8.32
5	-32.52	-34.63	6.73	-5.31
6	-19.51	-44.06	20.31	0.06
7	-17.33	-50.29	13.00	4.85
8	-17.05	-47.84	8.63	3.08
9	-27.49	-52.95	14.63	5.89
10	-18.19	-40.22	10.43	-11.68
11	-19.84	-30.16	16.04	4.99
12	-12.81	-31.29	1.79	-15.45
13	-10.32	-25.54	12.95	8.37
14	-0.13	-15.77	15.28	10.60
15	-22.62	-31.96	14.27	0.97
16	-21.06	-37.25	11.03	3.36
17	-31.47	-24.49	-3.33	6.75
18	-18.48	-20.91	0.82	10.27
19	-28.04	-24.85	9.07	5.90
20	-26.65	-30.30	9.34	8.55
21	-32.13	-30.94	35.42	35.91
22	-44.26	-30.58	32.27	27.96
23	-36.87	-31.13	39.33	39.47
24	-27.88	-28.39	26.46	27.92

Notes: Each cell reports the percentage MAE improvement of the standalone micro forecast relative to the univariate forecast at horizon h . The baseline includes sale-flagged quotes; the sale-filtered robustness check replaces temporary sale prices with the regular price inferred by the Nakamura–Steinsson filter, parameterisation A. Positive values indicate lower MAE than the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

F.4.2 Three-Expert Online Learning

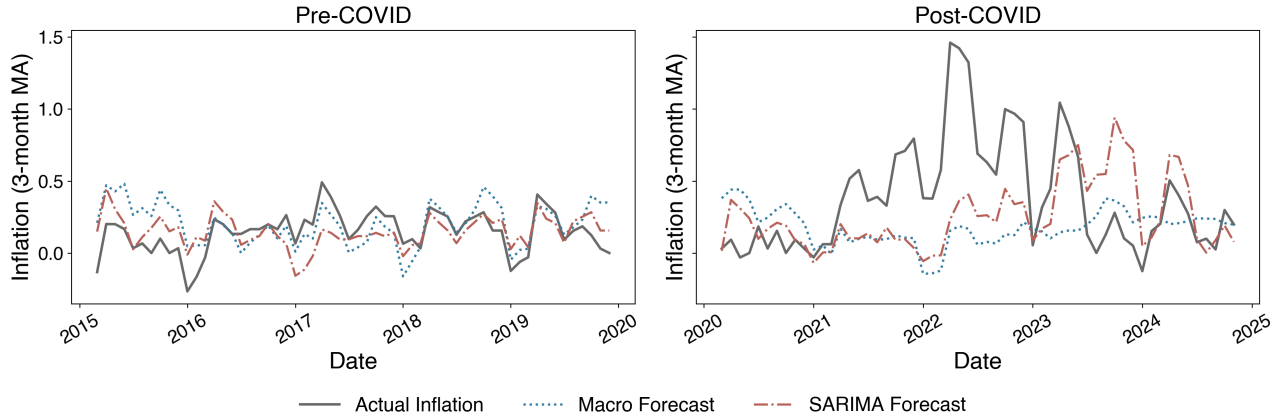
Table F8 reports the percentage MAE improvement of the three-expert patient Fixed Share combination (fixed $\alpha = 0.02$, $\eta = 0.5$, the main-text baseline) relative to the univariate forecast, with the micro model trained on sales-included (baseline) and sale-filtered micro input respectively, holding the macro and univariate forecasts fixed. Averaging within the same four horizon groups, the sale-filtered three-expert combination's percentage MAE improvements are 0.3%, -0.9% , -2.3% , and -0.2% in the pre-2020 window and 8.7%, 10.5%, 10.4%, and 24.9% in the post-2020 window. The corresponding gaps relative to the baseline three-expert combination are -0.9 , $+0.5$, -0.6 , and $+2.7$ percentage points pre-2020 (averaging a 0.4 percentage-point gain) and -3.3 , -3.3 , -1.5 , and -0.3 percentage points post-2020 (averaging a 2.1 percentage-point cost). The post-COVID standalone gap therefore shrinks from 6.3 to 2.1 percentage points once the sale-filtered micro forecast is combined with the macro and univariate forecasts through the Fixed Share combination: when the sale-filtered micro expert weakens, the algorithm reallocates weight toward the macro and univariate forecasts.

Table E8: Three-Expert Fixed Share: Percent MAE Improvement Relative to the Univariate Forecast under Baseline versus Nakamura–Steinsson Filter-A Sales Treatment ($\alpha = 0.02$, $\eta = 0.5$)

h	Pre-2020		Post-2020	
	Baseline	Sale-Filtered	Baseline	Sale-Filtered
1	-1.58	-0.84	12.03	12.29
2	1.83	-0.61	7.64	4.46
3	4.29	4.44	7.40	5.13
4	0.66	0.09	13.17	7.08
5	-0.14	0.41	13.72	12.16
6	2.37	-1.39	17.89	11.10
7	-0.08	-1.54	15.63	12.35
8	-1.64	-3.40	14.73	12.96
9	-3.29	-3.60	18.23	15.56
10	-0.93	-0.82	15.22	13.09
11	-1.85	2.03	15.89	11.65
12	-0.32	2.00	2.81	-2.82
13	2.33	1.88	12.26	9.51
14	4.33	-0.51	15.49	10.33
15	-6.69	-5.83	13.89	9.86
16	-6.57	-6.69	10.52	6.71
17	-5.06	-4.74	8.27	12.13
18	1.55	1.95	10.88	13.92
19	1.15	1.54	13.64	14.06
20	-5.52	-6.62	15.88	14.78
21	-4.20	-1.84	30.03	29.53
22	-8.20	-1.04	30.22	29.06
23	-4.08	1.91	35.30	35.47
24	3.30	4.98	26.53	26.64

Notes: Each cell reports the percentage MAE improvement of the three-expert patient fixed-share combination (fixed $\alpha = 0.02$, $\eta = 0.5$) relative to the univariate forecast at horizon h , with the micro expert trained on sales-included (baseline) versus Nakamura–Steinsson Filter-A sale-filtered micro input, holding the macro and univariate forecasts fixed. Positive values indicate lower MAE than the univariate forecast. Evaluation windows are 2015–2019 and 2020–2024.

Figure G.1: Macro Forecast vs. Univariate Forecast: $h = 12$



Notes: Each panel plots 3-month trailing moving averages of realized monthly inflation, the macro 12-month-ahead forecast, and the univariate (SARIMA) 12-month-ahead forecast. Dates on the x-axis are target months: a forecast dated t was formed at origin $t - 12$. The panels correspond to the 2015–2019 and 2020–2024 evaluation windows.

G Additional Results

G.1 Macro Forecasts versus the Univariate Benchmark

While the primary focus of the paper is on microdata, we also document the performance of the macro forecast, which displays an analogous regime-dependent pattern.

Visual evidence across forecast horizons. Figure G.1 plots both sets of forecasts against realized inflation at horizon $h = 12$. In the pre-2020 panels, the univariate and macro forecasts track realized inflation similarly, with the univariate forecast generally remaining slightly closer. The post-2020 panel shows why $h = 12$ is one of the macro forecast’s weakest post-2020 horizons: neither forecast anticipates the 2021–22 run-up, and the macro forecast stays flat through the surge, so the univariate forecast retains a 9.9% MAE advantage at this horizon. The macro forecast’s gains are instead concentrated at long horizons, where the post-2020 improvements reach 29% to 42% ($h = 21$ to 24; Table G.1), as the univariate forecast overshoots badly in 2023–24 at those horizons while the macro forecast does not. Figures D.7, D.8, and D.9 report the $h = 1$, $h = 6$, and $h = 24$ forecasts.

Aggregate evidence: pre-2020 versus post-2020 windows. Table G.1 reports the percentage MAE improvement of the macro forecast relative to the univariate forecast across the two evaluation windows, pre-2020 (2015–2019) and post-2020 (2020–2024), for all horizons $h = 1, \dots, 24$; Figure G.2 visualizes the same comparison as a heatmap.

Over the pre-2020 window, the macro forecast underperforms the univariate forecast at all 24 horizons; the smallest shortfall is 5.2% at $h = 13$. Averaging within four horizon groups ($h = 1$ to 6, 7 to 12,

13 to 18, and 19 to 24), the percentage MAE losses are 15.1%, 19.4%, 15.9%, and 13.5%. Macro covariates add no predictive value beyond the univariate benchmark in a stable-inflation environment.

After 2020, the ranking changes. The macro forecast improves on the univariate forecast at 19 of the 24 horizons, with the exceptions at $h = 2$ to 5 (between -1.5% and -3.8%) and at $h = 12$ (-9.9%). Averaging within the same four horizon groups, the post-COVID percentage MAE gains are -0.4% , 9.4% , 6.4% , and 26.7% , rising with the horizon and peaking at the longest horizons, where the largest gain reaches 41.9% at $h = 23$. The contrast between the two windows indicates that the value of macro data is state dependent: limited during normal times but substantially higher during periods of macroeconomic stress.

Table G.1: Macro percent MAE improvement over SARIMA, pre- and post-COVID.

h	Pre-COVID % impr.	Post-COVID % impr.
1	-25.19	7.86
2	-22.25	-3.14
3	-14.44	-3.82
4	-11.02	-3.09
5	-5.59	-1.51
6	-11.86	1.10
7	-10.26	14.52
8	-19.80	16.59
9	-25.06	19.53
10	-21.28	10.08
11	-19.13	5.34
12	-20.69	-9.92
13	-5.24	3.59
14	-10.14	2.51
15	-23.74	11.30
16	-25.62	4.18
17	-20.12	7.66
18	-10.26	9.14
19	-13.54	10.09
20	-23.46	11.46
21	-20.07	33.55
22	-10.89	34.12
23	-5.73	41.87
24	-7.28	29.18

G.2 Univariate + Macro Online Learning as a Benchmark without Microdata

To isolate the incremental contribution of microdata to the online-learning step, we consider a two-expert benchmark that combines only the macro and univariate forecasts, using the same patient Fixed Share baseline as in Subsection 4.3 ($\eta = 0.5$, $\alpha = 0.02$). This exercise asks whether a flexible combination of macro and univariate forecasts can already replicate the gains that we attribute to adding microdata.

Results. Table G.2 shows that the two-expert benchmark tracks the univariate forecast before 2020 and improves on it broadly afterwards. In the pre-2020 window, the percentage MAE improvements averaged within four horizon groups ($h = 1$ to 6, 7 to 12, 13 to 18, and 19 to 24) are 1.1%, -1.5% , -3.4% , and 0.4% , with positive gains at 9 of the 24 horizons and an average gap of -0.9% . In the post-2020 window, by contrast, the two-expert combination improves on the univariate forecast at all 24 horizons, with group averages of 8.8%, 12.9%, 9.6%, and 21.6%, averaging 13.2% and strongest at the longest horizons. Fixed Share combination of macro and univariate information therefore captures a substantial share of the post-2020 forecasting signal. Figure G.3 visualizes the two-expert versus univariate forecast comparison as a heatmap.

Visual evidence across forecast horizons. Figure G.4 displays the aggregate forecast path and expert weights for the representative horizon $h = 12$. In the pre-2020 window, the two-expert combination tracks realized inflation roughly as closely as the univariate forecast alone, reflecting modest diversification gains. In the post-2020 window, no forecast anticipates the 2021–22 run-up at this horizon, and the combined forecast tracks the univariate forecast through most of the episode, improving on it only modestly at $h = 12$; as with the macro expert itself, the two-expert combination’s gains are concentrated at the longest horizons (Table G.2). Figures D.10, D.11, and D.12 report the corresponding combination weights for $h = 1$, $h = 6$, and $h = 24$.

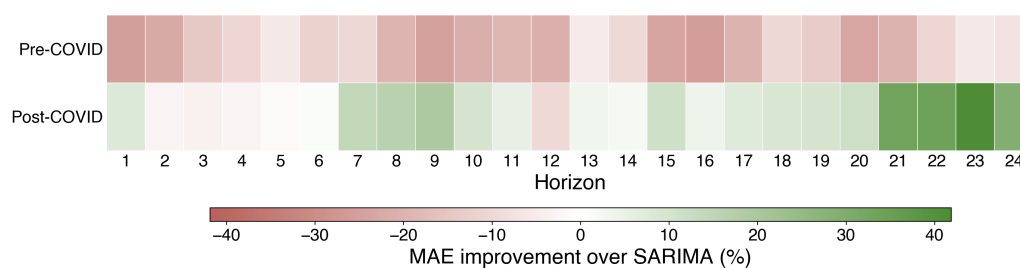
At $h = 1$, the combination leans toward the univariate forecast before 2020, with an average univariate weight of 0.660, but the macro weight rises to 0.539 after 2020 (see Appendix Figure D.10). At $h = 6$, the average post-2020 weights are 0.457 on the macro forecast and 0.543 on the univariate forecast (see Appendix Figure D.11). At $h = 12$, the univariate forecast keeps the majority weight in both windows (0.623 before 2020 and 0.647 after), consistent with $h = 12$ being the macro expert’s weakest post-2020 horizon. At $h = 24$, the weights stay close to balanced, with average macro weights of 0.555 before and 0.532 after 2020 (see Appendix Figure D.12).

Table G.2: Fixed-Share Combination Mean Absolute Error Before and After 2020 (Macro + SARIMA Combination)

h	Pre 2020			Post 2020		
	FS	Macro	SARIMA	FS	Macro	SARIMA
1	0.1390	0.1708	0.1364	0.3487	0.3607	0.3915
2	0.1352	0.1678	0.1373	0.3596	0.3962	0.3842
3	0.1382	0.1629	0.1424	0.3786	0.4137	0.3985
4	0.1523	0.1679	0.1512	0.3624	0.4080	0.3957
5	0.1440	0.1559	0.1476	0.3553	0.4035	0.3975
6	0.1469	0.1681	0.1503	0.3782	0.4221	0.4268
7	0.1472	0.1631	0.1479	0.4104	0.4072	0.4764
8	0.1546	0.1835	0.1532	0.4082	0.4040	0.4843
9	0.1500	0.1843	0.1474	0.4139	0.4079	0.5070
10	0.1579	0.1876	0.1547	0.3773	0.4050	0.4504
11	0.1513	0.1763	0.1479	0.4283	0.4622	0.4883
12	0.1598	0.1878	0.1556	0.4183	0.4656	0.4235
13	0.1820	0.1921	0.1825	0.4374	0.4582	0.4753
14	0.1848	0.2004	0.1820	0.4539	0.4755	0.4878
15	0.1760	0.2042	0.1651	0.4325	0.4340	0.4893
16	0.1757	0.2079	0.1655	0.4439	0.4569	0.4768
17	0.1683	0.1928	0.1605	0.4158	0.4360	0.4722
18	0.1723	0.1868	0.1694	0.4156	0.4303	0.4736
19	0.1844	0.2034	0.1792	0.4087	0.4244	0.4720
20	0.1814	0.2062	0.1670	0.4083	0.4172	0.4712
21	0.1715	0.1988	0.1656	0.5050	0.4339	0.6530
22	0.1678	0.1877	0.1693	0.4607	0.4090	0.6208
23	0.1600	0.1782	0.1686	0.4850	0.4105	0.7061
24	0.1570	0.1907	0.1778	0.4463	0.4113	0.5808

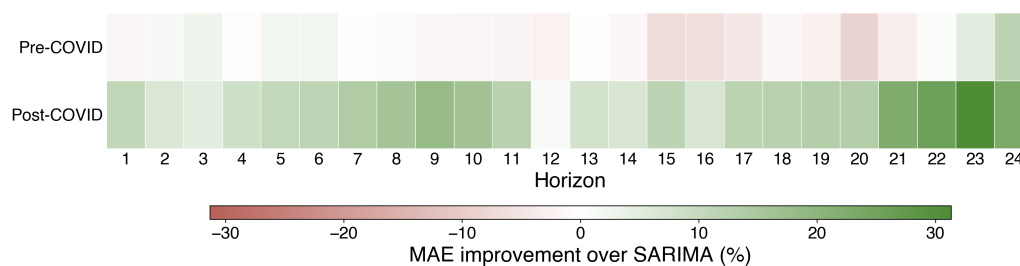
Notes: Mean absolute error of the patient fixed-share combination (FS, $\alpha = 0.02$, $\eta = 0.5$, $\rho = 0$, grid=fine) against its experts (Macro, SARIMA), by forecast horizon. The pre-2020 window is 2015–2019 and the post-2020 window is 2020–2024.

Figure G.2: Macro Forecast MAE Improvement Relative to the Univariate Forecast



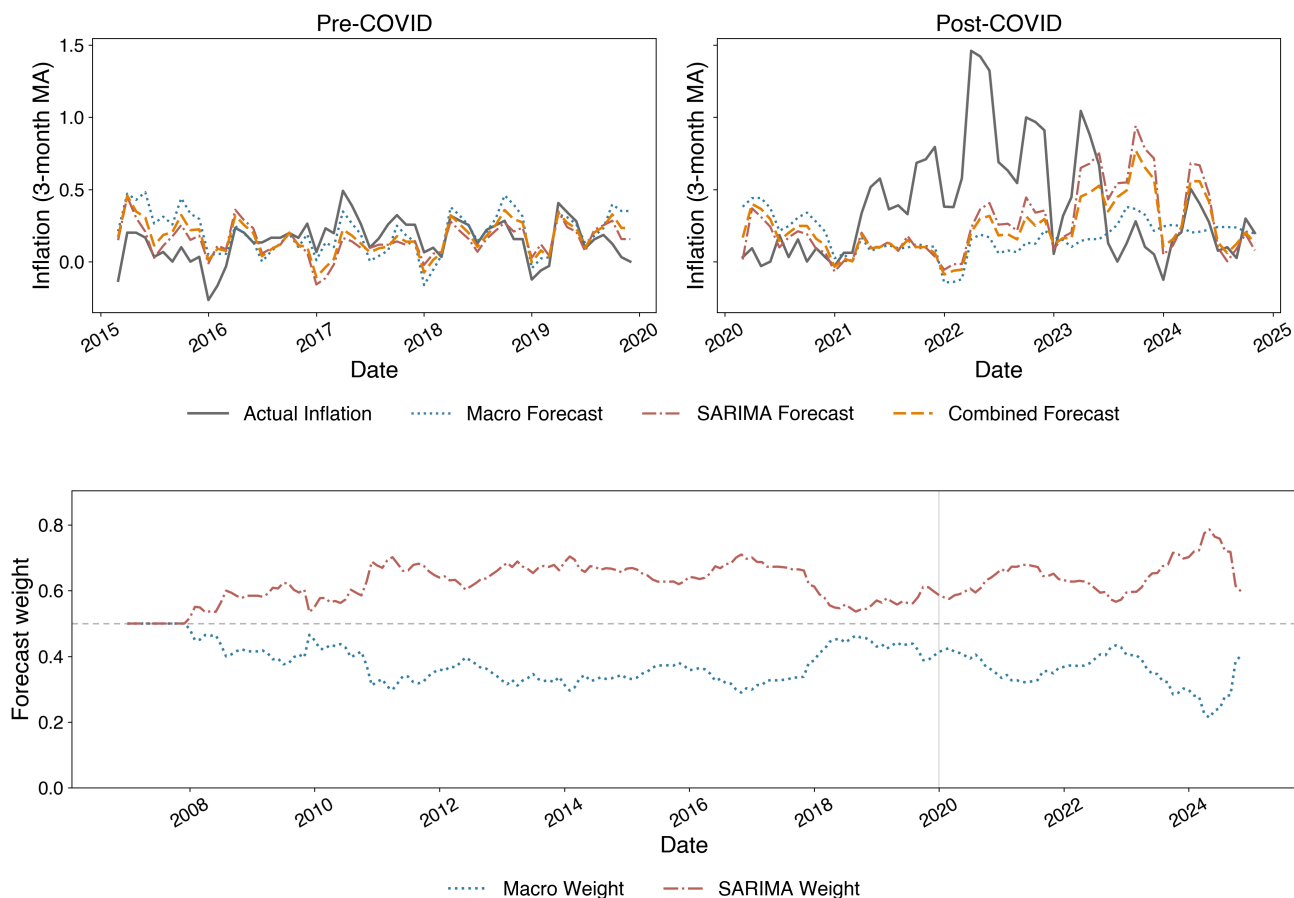
Notes: Cells report the percentage MAE improvement of the macro forecast relative to the univariate forecast by forecast horizon $h = 1, \dots, 24$ and evaluation window. Positive values indicate lower MAE for the macro forecast than for the univariate forecast; negative values indicate the reverse. Green shading denotes positive improvements and red shading denotes negative improvements. Numerical values are reported in Table G.1.

Figure G.3: Univariate + Macro Online-Learning MAE Improvement Relative to the Univariate Forecast



Notes: Cells report the percentage MAE improvement of the two-expert Fixed Share combination relative to the univariate forecast by forecast horizon $h = 1, \dots, 24$ and evaluation window. The two experts are the macro and univariate forecasts. Positive values indicate lower MAE for the combination than for the univariate forecast; negative values indicate the reverse. Numerical values are reported in Table G.2.

Figure G.4: Two-Expert Combined Forecast and Weights: $h = 12$



Notes: The top panel plots 3-month trailing moving averages of realized inflation, the macro and univariate expert forecasts, and the two-expert Fixed Share combined forecast for the pre-2020 (2015–2019) and post-2020 (2020–2024) windows; dates on the x-axis are target months, so a forecast dated t was formed at origin $t - 12$. The bottom panel plots the combination weights on the two experts over the full sample, with the dashed horizontal line at the equal weight $1/2$ and the vertical line marking 2020.

G.3 Three-Expert Combination with a Varying Learning Rate

Table G.3 reports the MAE of the three-expert combination when the learning rate is re-selected at each forecast origin by the neighbor-only rolling search of Algorithm 9 ($W = 60$ months, fixed $\alpha = 0.02$), alongside the fixed $\eta = 0.5$ baseline reported in the main text. The two specifications are nearly indistinguishable: the adaptive rule attains a lower MAE at 9 of the 24 horizons before 2020 and at 12 of the 24 horizons after 2020, and the average gap is within 0.3 percent in both windows. The transparency of a fixed learning rate therefore costs nothing in accuracy.

Table G.3: Adaptive Fixed-Share MAE Before and After 2020

h	Pre 2020		Post 2020	
	Adaptive η	Fixed η	Adaptive η	Fixed η
1	0.1381	0.1385	0.3417	0.3444
2	0.1376	0.1348	0.3506	0.3548
3	0.1384	0.1363	0.3664	0.3690
4	0.1501	0.1502	0.3421	0.3436
5	0.1489	0.1478	0.3416	0.3429
6	0.1472	0.1467	0.3505	0.3505
7	0.1491	0.1481	0.4010	0.4019
8	0.1565	0.1557	0.4138	0.4130
9	0.1494	0.1523	0.4137	0.4145
10	0.1570	0.1562	0.3847	0.3819
11	0.1498	0.1507	0.4123	0.4107
12	0.1567	0.1561	0.4129	0.4116
13	0.1780	0.1783	0.4176	0.4170
14	0.1741	0.1741	0.4123	0.4122
15	0.1770	0.1761	0.4238	0.4213
16	0.1771	0.1763	0.4287	0.4266
17	0.1689	0.1686	0.4332	0.4332
18	0.1661	0.1668	0.4212	0.4221
19	0.1774	0.1771	0.4050	0.4077
20	0.1756	0.1762	0.3953	0.3964
21	0.1730	0.1725	0.4546	0.4569
22	0.1856	0.1831	0.4333	0.4332
23	0.1777	0.1754	0.4580	0.4569
24	0.1703	0.1719	0.4247	0.4267

Notes: Mean absolute error of the varying-learning-rate patient fixed share, which re-selects η at each origin by the neighbor-only rolling search ($W = 60$, fixed $\alpha = 0.02$, grid=fine), against the fixed $\eta = 0.5$ patient baseline, by forecast horizon. Pre-2020 is 2015–2019, post-2020 is 2020–2024.

G.4 6-Group Shapley Values

Construction. We partition the micro variables, computed within each COICOP1 category and then stacked across categories (except the monthly dummies, which are common across categories), into six groups:

1. the fraction of zero price changes;
2. the mean price change;
3. the higher moments of the price-change distribution;
4. the deciles of the price-change distribution;
5. the time since last change features;
6. monthly dummies.

This decomposition is based on the augmented encoding rather than the baseline, so it adds two feature groups that are absent from the main-text decomposition, namely the higher moments of the price-change distribution and the hazard-style statistics based on time since last adjustment. It lets us separate the location of the price-change distribution (the mean), its coarse shape (deciles), its higher moments, and the time-since-last-change features.

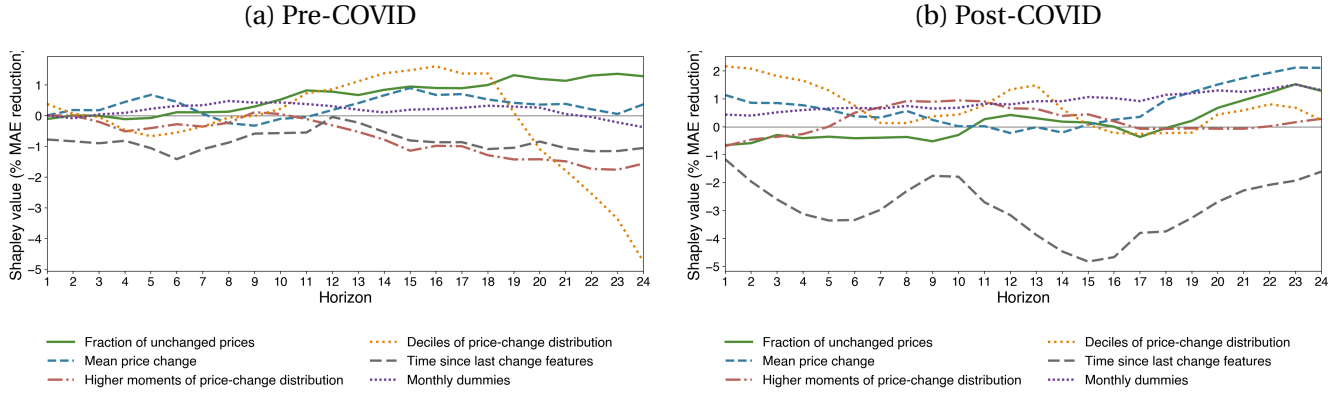
Table G.4: 6-Grouped Shapley Values for the Micro Expert, Averaged over 24 Horizons

Feature group	Pre-2020	Post-2020
Fraction of zero price changes	0.5964	0.0743
Mean price change	0.2853	0.7311
Higher moments of price-change distribution	-0.6774	0.2298
Deciles of price-change distribution	-0.0886	0.7128
Time since last change features	-0.8356	-2.8658
Monthly dummies	0.1804	0.8718

Notes: Entries are grouped Shapley contributions of each feature group to the MAE gain from adding the micro expert to the two-expert online-learning benchmark, which combines the univariate and macro forecasts. Values are in percent of benchmark MAE and are averaged over horizons $h = 1, \dots, 24$ within each evaluation window. Positive entries reduce MAE; negative entries increase it.

Table G.4 decomposes the augmented-encoding micro expert. In the pre-COVID window, no feature group makes a large positive contribution: the fraction of zero price changes is the most positive at 0.60, followed by the mean price change at 0.29 and monthly dummies at 0.18, while the time-since-last-change features (-0.84), the higher moments (-0.68), and the deciles (-0.09) are negative.

Figure G.5: 6-Horizon Moving Average of Per-Horizon 6-Group Shapley Values



Notes: The figure reports 6-group Shapley contributions for the incremental value of the augmented-encoding micro forecast in the three-expert combined forecast relative to the two-expert benchmark that combines the univariate and macro forecasts. Values are smoothed across adjacent horizons using a 6-horizon moving average. Positive values indicate reductions in MAE from the corresponding feature group; negative values indicate increases in MAE. The horizontal line marks a zero contribution.

This again points to limited incremental forecasting content in microdata during the low and stable inflation period.

In the post-COVID window, the positive contributions come from monthly dummies (0.87), the mean price change (0.73), the deciles of the price-change distribution (0.71), and, more modestly, the higher moments (0.23); the fraction of zero price changes is essentially zero (0.07). The time-since-last-change features, however, contribute -2.87 , more than offsetting every positive group. The six contributions therefore sum to the augmented-encoding three-expert improvement over the two-expert benchmark, which is negative after 2020 (-0.25 percent). The augmented encoding therefore underperforms the baseline encoding after COVID, and the shortfall is accounted for by the hazard-style timing block rather than by the distributional statistics, consistent with the encoding-level robustness comparison in Subsection 5.6.

Figure G.5 shows the same pattern at the horizon level. Before COVID, the per-horizon contributions are small at short and medium horizons, and the deciles group deteriorates steeply at the longest horizons, echoing the long-horizon losses of the baseline decomposition. After COVID, monthly dummies are positive at every horizon, the mean price change strengthens toward long horizons, and the deciles are positive at short and medium horizons. The time-since-last-change features are strongly negative at every horizon in the post-COVID window, reaching below -4 at medium horizons, and are negative at most horizons before COVID as well. Taken together, the 6-group decomposition suggests that the post-COVID forecasting value of microdata comes from the location and shape of the price-change distribution together with seasonal timing, while the hazard-style duration statistics actively hurt.

G.5 4-Group Shapley Values

This subsection reports a coarser 4-group Shapley decomposition. It partitions the same baseline feature set as the 5-group specification in Section 6, but instead of separating the center and the tails of the price-change distribution, it keeps the mean price change and the deciles as single groups:

1. the fraction of zero price changes;
2. the mean price change;
3. the deciles of the price-change distribution;
4. monthly dummies.

Table G.5: 4-Group Shapley Values for the Micro Expert, Averaged over 24 Horizons

Feature group	Pre-2020	Post-2020
Fraction of zero price changes	0.1254	-0.0093
Mean price change	-0.2216	0.9672
Deciles of price-change distribution	-0.6009	0.6423
Monthly dummies	0.2993	1.3076

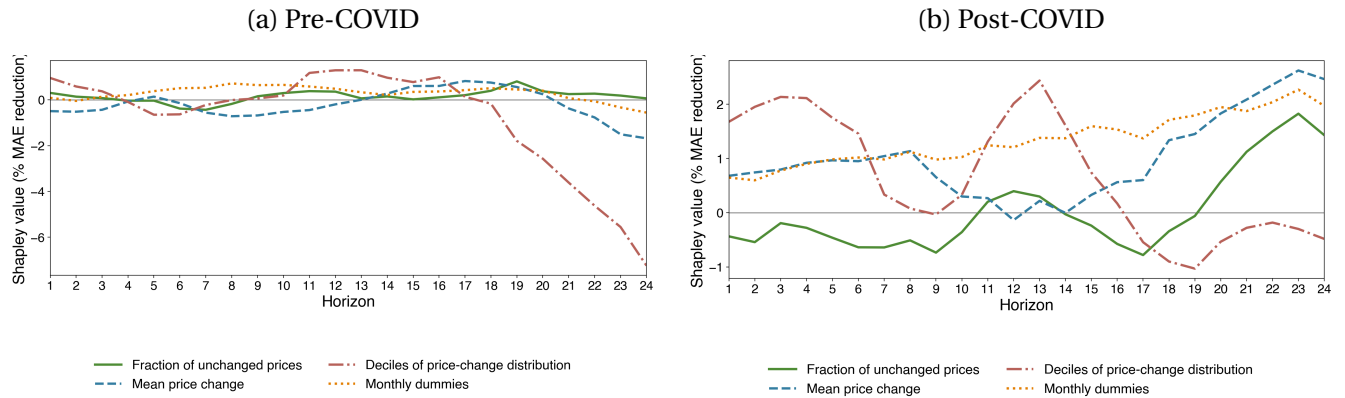
Notes: Entries are grouped Shapley contributions of each feature group to the MAE gain from adding the micro expert to the two-expert online-learning benchmark, which combines the univariate and macro forecasts. Values are in percent of benchmark MAE and are averaged over horizons $h = 1, \dots, 24$ within each evaluation window. Positive entries reduce MAE; negative entries increase it.

Table G.5 shows that, in the pre-COVID window, the contributions of the deciles of the price-change distribution and the mean price change are negative, with values -0.60 and -0.22 , respectively, while monthly dummies (0.30) and the fraction of zero price changes (0.13) are mildly positive. Thus, before COVID the grouped micro contributions are small overall, with no group delivering a large positive contribution. Because the 4-group and 5-group specifications partition the same baseline feature set, their contributions sum to the same totals: -0.40 percent before 2020 and 2.91 percent after.

In the post-COVID window, the largest positive contribution comes from monthly dummies at 1.31 , followed by the mean price change at 0.97 and the deciles of the price-change distribution at 0.64 ; the fraction of zero price changes is essentially zero, at -0.01 . The positive contribution therefore comes from information in the average and distribution of price changes, together with seasonal timing, while the extensive-margin measure contributes essentially nothing on average.

Figure G.6 confirms the same broad message at the horizon level: before COVID the contributions are small at short and medium horizons, with the losses concentrated in the deciles group at the

Figure G.6: 4-Group Shapley Values by Horizon



Notes: The figure reports grouped Shapley contributions for the incremental value of the micro forecast in the three-expert combined forecast relative to the two-expert benchmark that combines the univariate and macro forecasts. Values are smoothed across adjacent horizons using a 6-horizon moving average. Positive values indicate reductions in MAE from the corresponding feature group; negative values indicate increases in MAE. The horizontal line marks a zero contribution.

longest horizons; after COVID, mean price changes, monthly indicators, and, at short and medium horizons, the price-change deciles are the main positive contributors.

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